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CHEMISTRY AT A GLANCE

E

PHYSICAL CHEMISTRY

MOLE CONCEPT

1. Number of moles of neutrons in 1.8 mL of water is :-

(1) 4.8×10^{23} (2) 4.8×10^{22}

(3) 0.8 (4) 0.1

2. The weight of 350 mL of a diatomic gas at 0°C and 2 atm pressure is 1 g. The weight of one atom is :-

(1) $\frac{16}{N_A}$ (2) $\frac{32}{N_A}$

(3) $16\ N_A$ (4) $32\ N_A$

3. 10 mL of gaseous hydrocarbon on complete combustion gives 20 mL of CO_2 and 30 mL of $H_2O(g)$. The hydrocarbon is :-

(1) C_4H_{10} (2) C_2H_6

(3) C_4H_8 (4) C_8H_{10}

4. For the reaction $A + 2B \rightarrow C$, 5 moles of A and 8 moles of B will produce :-

(1) 5 moles of C

(2) 4 moles of C

(3) 8 moles of C

(4) 13 moles of C

5. If the atomic weight of carbon is set at 24 amu, the value of the avogadro constant would be :-

(1) 6.022×10^{23} (2) 12.044×10^{23}

(3) 3.011×10^{23} (4) none of these

6. Boron has two isotopes, B-10 and B-11. The average atomic mass of Boron is found to be 10.80 u. Calculate the percentage abundance of B^{10} isotope :-

(1) 80 (2) 20 (3) 25 (4) 75

7. A compound on analysis gave the following results C = 54.54%, H = 9.09% and vapour density of compound = 88. Determine the molecular formula of the compound:-

(1) $C_8H_{16}O_4$ (2) $C_4H_{16}O_8$

(3) C_2H_4O (4) CH_4O_2

8. Two elements X and Y (atomic mass of X = 75 and Y = 16) combine to give a compound having 76% of X. The formula of compound is :-

(1) XY (2) X_2Y (3) X_2Y_2 (4) X_2Y_3

9. Mass of CO_2 produced on heating 20 g of 40% pure limestone :-

(1) 8 g (2) 8.8 g

(3) 3.52 g (4) none of these

10. Calculate the amount of 50% H_2SO_4 required to decompose 25 g of calcium carbonate :-

(1) 98 g (2) 49 g

(3) 24.5 g (4) 196 g

11. 8 g H_2 32 g O_2 is allowed to react to form water then which of the following statement is correct ?

(1) O_2 is limiting reagent

(2) O_2 is reagent in excess

(3) H_2 is limiting reagent

(4) 40 g water is formed

12. A compound contains 3.4% by weight of sulphur, the minimum molecular weight of compound is :-

(1) 941.176 (2) 944

(3) 945.27 (4) None

13. The maximum number of molecules are present in :-

(1) 15 L H_2 gas at STP

(2) 5 L of N_2 gas at STP

(3) 0.5 g H_2 gas

(4) 10 g of O_2 gas

14. If the weight of metal chloride is x g containing y g of metal, the equivalent weight of metal will be :-

(1) $\frac{x}{y} \times 35.5$ (2) $\frac{8(y-x)}{x}$

(3) $\frac{y}{x-y} \times 35.5$ (4) $E = \frac{8(x-y)}{y}$

15. Equal masses of H_2, N_2 and CH_4 are taken in a container of volume V at temperature 27°C in identical conditions. The ratio of volumes of gases $H_2 : N_2 : CH_4$ would be :-

(1) 56 : 4 : 7 (2) 7 : 4 : 56

(3) 2 : 28 : 16 (4) 8 : 14 : 1

16. For the formation of 3.65 g of HCl gas, what volume of hydrogen gas and chlorine gas are required at NTP condition ?

(1) 1 L, 1 L

(2) 1.12 L, 2.24 L

(3) 3.65 L, 1.83 L

(4) 1.12 L, 1.12 L

17. Which of the following has the highest mass ?

(1) 1 g-atom of C

(2) $\frac{1}{2}$ mole of CH_4

(3) 10 mL of water

(4) 3.011×10^{23} atoms of oxygen

18. For the complete combustion of 4 litre of CO at NTP, the required volume of O_2 at NTP is :-

(1) 4 litre (2) 8 litre

(3) 2 litre (4) 1 litre

19. A gas is found to have molecular formula $(CO)_x$. Its vapour density is 70, the value of x must be :-

(1) 7 (2) 4 (3) 5 (4) 6

20. 1 litre of a hydrocarbon weighs as much as one litre of CO_2. The molecular formula of hydrocarbon is :-

(1) C_3H_8 (2) C_2H_6

(3) C_2H_4 (4) C_3H_6

21. Four one litre flasks are separately filled with the gases hydrogen, helium, oxygen and ozone at same room temperature and in the different flasks the ratio of total atoms would be :-

(1) 1 : 1 : 1 : 1 (2) 1 : 2 : 2 : 3

(3) 2 : 1 : 2 : 3 (4) 2 : 1 : 3 : 2

22. There are two oxides of sulphur. They contain 50% and 60% of oxygen respectively by weights. The weights of sulphur which combine with 1 g of oxygen are in the ratio of:

(1) 1 : 1 (2) 2 : 1

(3) 2 : 3 (4) 3 : 2

23. A gas is found to contain 2.34 g of nitrogen and 5.34 g of oxygen Simplest formula of the compound is :-

(1) N_2O (2) NO

(3) N_2O_3 (4) NO_2

24. 16 cc of CO_2 are passed over red hot coke. The volume of CO evolved is :-

(1) 18 cc (2) 32 cc (3) 16 cc (4) 4 cc

25. 500 mL of a gaseous hydrocarbon when burnt in excess of O_2 gave 2.0 litre of CO_2 and 2.5 litre of water vapour under same conditions. Molecular formula of the hydrocarbon is :-

(1) C_4H_8 (2) C_4H_{10}

(3) C_5H_{10} (4) C_5H_{12}

26. The total number of ions present in 1 mL of 0.1 M barium nitrate solution is :-

(1) 6.02×10^{18} (2) 6.02×10^{19}

(3) $3.0 \times 6.02 \times 10^{19}$ (4) $3.0 \times 6.02 \times 10^{18}$

27. One g equivalent of substance is present in :-

(1) 0.25 mole O_2 (2) 0.5 mole of O_2

(3) 1.00 mole of O_2 (4) 8.00 mole of O_2

28. In a compound A_xB_y :-

(1) Mole of A = mole of B = mole of A_xB_y

(2) eq of A = eq of B = eq of A_xB_y

(3) yx mole of A = yx mole of B = (x + y) mole of A_xB_y

(4) yx mole of A = yx mole of B

29. How many moles of magnesium phosphate $Mg_3(PO_4)_2$ will contain 0.25 mole of oxygen atoms ?

(1) 2.5×10^{-2} (2) 0.02

(3) 3.125×10^{-2} (4) 1.25×10^{-2}

30. 10 g of MnO_2 on reaction with HCl forms 2.24 L of $Cl_2(g)$ at NTP, the percentage impurity of MnO_2 is :-
(Given molar mass of MnO_2 = 87)

$$MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$$

(1) 87% (2) 25% (3) 33.3% (4) 13 %

31. The number of H-atoms present in 5.6 g urea are
(1) 6.02×10^{23} (2) 2.24×10^{23}
(3) 2.24×10^{22} (4) 3.1×10^{22}

32. How many moles of methane are required to produce 22 g of $CO_2(g)$ after combustion
$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$
(1) 1 mol (2) 0.5 mol
(3) 0.25 mol (4) 1.25 mol

33. In a 5.2 molal aqueous solution of methyl alcohol what will be the mole fraction of methyl alcohol?
(1) 0.190 (2) 0.086 (3) 0.050 (4) 0.100

34. The isotopic abundance of C-12 and C-14 is 98% and 2% respectively. What would be the number of atoms of C-14 isotope in 12 g carbon sample
(1) 1.2×10^{22} (2) 3.01×10^{23}
(3) 5.88×10^{23} (4) 6.02×10^{23}

35. The volume of air needed for complete combustion of 1 kg carbon at STP is
(1) 9333.3 litre (2) 933.33 litre
(3) 93.33 litre (4) 1866.67 litre

36. A compound made of two elements A and B is found to contains 25% A (atomic mass 12.5) and 75% B (atomic mass 37.5). The simplest formula of the the compound is
(1) AB (2) A_2B_2 (3) AB_3 (4) A_3B

37. $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$
what will be the amount of $CaCl_2$ when 10 g $CaCO_3$ and 200 mL 0.75 M HCl is used in the reaction?
(1) 83.25 g (2) 16.65 g
(3) 11.1 g (4) 8.325 g

38. A compound is analyzed and found to consist of 50.4% Ce, 15.1 % N and 34.5% O by mass. What is the correct formula for the compound? (Atomic wt. of Ce = 140)
(1) $Ce_2(NO_3)_2$ (2) $Ce_2(NO_2)_3$
(3) $Ce(NO_3)_2$ (4) $Ce(NO_2)_3$

39. What is the mass of oxygen that is required for the complete combustion of 2.8 kg ethylene ?
(1) 2.8 kg (2) 6.4 kg
(3) 96 kg (4) 9.6 kg

40. The density of water is 1 g/mL what is the volume occupied by 1 molecule of water in mL.
(1) 3×10^{-23} (2) 1×10^{-23}
(2) 2×10^{-23} (4) 4×10^{-23}

41. The volume of a drop of water is 0.0018 mL then the number of water molcules present in two drops of water at room temperatue is :-
(1) 12.046×10^{19} (2) 1.084×10^{18}
(3) 4.84×10^{17} (4) 6.023×10^{23}

42. The law of multiple proportion is illustrated by the two compounds :-
(1) Sodium chloride and sodium bromide
(2) Ordinary water and heavy water
(3) Caustic soda and caustic potash
(4) Sulphur dioxide and sulphur trioxide

43. The hydrated salt $Na_2CO_3.xH_2O$ undergoes 63% loss in mass on heating and becomes anhydrous. The value of x is :-
(1) 10 (2) 12
(3) 8 (4) 18

44. Decreasing order (first having highest and then others following it) of mass of pure NaOH in each of the aqueous solution :-
(i) 50 g of 40% (w/W) NaOH
(ii) 50 mL of 50% (w/V) NaOH [$d_{soln.}$ = 1.2 g/mL]
(iii) 50 g of 15 M NaOH [$d_{soln.}$ = 1 g/mL]
(1) i, ii, iii (2) iii, ii, i
(3) ii, iii, i (4) ii, i, iii

45. The molecular mas of a gas is 45. Its density (in g/L) at STP is :-
(1) 22.4 (2) 11.2 (3) 5.7 (4) 2.0

46. 20 mL of methane is completely burnt using 50 mL of oxygen. The volume of the gas left after cooling to room temperature is :-
(1) 80 mL (2) 40 mL
(3) 60 mL (4) 30 mL

47. The mass of $CaCO_3$ required to react completely with 20 mL of 1.0 M HCl as per the reaction, $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$ is :-
(1) 1 g (2) 2 g (3) 10 g (4) 20 g

48. The equivalent weight of divalent metal is W. The molecular weight of its chloride is :-
(1) W + 35.5 (2) W + 71
(3) 2W + 71 (4) 2W + 35.5

49. 1.25 g of a solid dibasic acid is completely neutralised by 25 mL of 0.25 molar $Ba(OH)_2$ solution. Molecular mass of the acid is :-
(1) 100 (2) 150 (3) 120 (4) 200

50. 0.45 g of an acid of mol. wt. 90 is neutralised by 20 mL of 0.5 N caustic potash (KOH). The basicity of acid is :-
(1) 1 (2) 2 (3) 3 (4) 4

51. The equivalent weight of a metal is double than that of oxygen. How many times is the equivalent weight of its oxide than the equivalent weight of the metal ?
(1) 1.5 (2) 2 (3) 3 (4) 4

ATOMIC STRUCTURE

52. FM radio broadcasts at 900 KHz. What wavelength does this correspond to ?
(1) 333 m (2) 3.03×10^{-3}m
(3) 330 cm (4) 3300 m

53. Energy of 1 mole of radio wave photons with a frequency of 909 KHz is :-
(1) 6.02×10^{-28}J (2) 3.62×10^{-4}J
(3) 1.00×10^{-4}J (4) 6.02×10^{-3} J

54. The wavelength of a neutron with a translatory kinetic energy equal to KT at 300 K is :-
(1) 178 pm (2) 200 pm
(3) 17.8 pm (4) 20.0 pm

55. Frequency of a matter wave is equal to :-
(1) $\frac{(KE)}{2h}$ (2) $\frac{2(KE)}{h}$ (3) $\frac{(KE)}{h}$ (4) λ

56. Ionisation energy of He^+ is 19.6×10^{-18} J $atom^{-1}$. The energy of first stationary state (n =1) of Li^{+2} is :-
(1) 4.41×10^{-16} J $atom^{-1}$
(2) -4.41×10^{-17} J $atom^{-1}$
(3) -2.20×10^{-15} J $atom^{-1}$
(4) 8.82×10^{-17} J $atom^{-1}$

57. Kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is (a_0 is Bohr radius):-
(1) $\frac{h^2}{4\pi m a_0^2}$ (2) $\frac{h^2}{16\pi^2 m a_0^2}$
(3) $\frac{h^2}{32\pi^2 m a_0^2}$ (4) $\frac{h^2}{64\pi^2 m a_0^2}$

58. The potential energy of an electron in the second Bohr's orbit of the He^+ ion is :-
(1) –54.4 eV (2) –27.2 eV
(3) –108.8 eV (4) –13.6 eV

59. If the radius of the first bohr orbit is X, the de Broglie wavelength of the electron in the third orbit is nearly.:-
(1) $2\pi X$ (2) $6\pi X$ (3) 9X (4) X/3

60. According to Bohr's theory, the angular momentum of an electron in 5th orbit is :-
(1) $25\frac{h}{\pi}$ (2) $1.0\frac{h}{\pi}$ (3) $10\frac{h}{\pi}$ (4) $2.5\frac{h}{\pi}$

61. The number of d electrons retained in Fe^{+2} (atomic number Fe = 26) ion is :-
(1) 3 (2) 4 (3) 5 (4) 6

62. Which of the following nuclear reactions will generate an isotope ?
(1) Neutron particle emission
(2) Positron emission
(3) α particle emission
(4) β particle emission

63. The wavelength (in nanometer) associated with a proton (mass = 1.67×10^{-27} kg atom^{-1}) at the velocity of 1.0×10^{3} ms^{-1} is :-
(1) 6.032 nm (2) 0.400 nm
(3) 2.500 nm (4) 4.00 nm

64. How many orbitals are possible for n = 3, l = 2 ?
(1) 1 (2) 3 (3) 5 (4) 7

65. Which of the following is violating Hund's rule:-
(1) $\boxed{\uparrow}$ $\boxed{\uparrow}\boxed{\downarrow}\boxed{\uparrow}$
(2) $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow\downarrow}\boxed{}\boxed{}$
(3) $\boxed{\uparrow\downarrow}$ $\boxed{\uparrow}\boxed{\downarrow}\boxed{\downarrow}$
(4) All

66. In hydrogen atom, the energy of second shell electron is :-
(1) -5.44×10^{-19} J (2) -5.44×10^{-19} KJ
(3) -5.44×10^{-19} Cal (4) -5.44×10^{-19} erg

67. 0.5 g particle has 2×10^{-5} m uncertainity in position find the uncertainty in its velocity (m/s)
(1) 3.0×10^{33} (2) 5×10^{-27}
(3) 4×10^{-30} (4) 4×10^{-10}

68. Which of the following has similar spectrum as that of Li^{+} :-
(1) H (2) Na^{10+}
(3) He (4) He^{+}

69. Radius of first Bohr's orbit of hydrogen atom is 0.53 Å then the radius of 3rd Bohr orbit is :-
(1) 0.70 Å (2) 1.59 Å
(3) 3.18 Å (4) 4.77 Å

70. If n = 3 then correct value of ℓ is:-
(1) 0 (2) 1
(3) 2 (4) All of the above

71. What will be the ratio of de-Broglie wavelength of electron accelerated with 400 volt and 100 volt :-
(1) 1 : 2 (2) 2 : 1
(3) 3 : 10 (4) 10 : 3

72. In an atom 2K, 8L, 8M , 2N electrons are present. If m = 0, s = 1/2, then find no. of electrons :-
(1) 6 (2) 2 (3) 8 (4) 16

73. Which of following has maximum possibility to find electrons in d_{xz} orbital :-
(1) Along x axis
(2) Along z axis
(3) Along xz plane (2D)
(4) At an angle of 45° from z & x axis

74. In Bohr's model $\frac{nh}{2\pi}$ shows :-
(1) Momentum (2) Kinetic energy
(3) Potential energy (4) Angular momentum

75. $2P_y$ orbital electron has energy :-
(1) more than $2P_x$ orbital
(2) more than $2P_z$ orbital
(3) same as $2P_x$ and $2P_z$
(4) same as of 2s orbital

76. Orbital angular momentum of d electron is:-
(1) $\sqrt{\frac{3}{2}}\frac{h}{\pi}$ (2) $\sqrt{6}\frac{h}{2\pi}$
(3) $\frac{h}{\sqrt{2}\pi}$ (4) $\frac{\sqrt{3}h}{2\pi}$

77. Which has lowest wavelength :-
(1) Balmer series (2) Brackett series
(3) Humphrey series (4) Lyman series

78. Which of the following ion has magnetic moment is 3.87 BM:-
(1) Ti^{3+} (2) Sc^{+}
(3) Ti^{+} (4) Mn^{+5}

79. No of electrons in 2d orbital are :-

(1) 10 (2) 5 (3) 2 (4) zero

80. Which of the following value of quantum number is not possible for 4f :-

(1) n = 4 (2) $\ell = 2$

(3) m = – 2 to +2 (4) all are possible

81. No of spectral line obtained when electron is provided with a photon of energy of 11 eV (for H atom)

(1) 1 (2) 2 (3) 0 (4) 6

82. Which of the following sets of quantum number is not possible

(1) $n = 3, \ell = +2, m_l = 0, m_s = +1/2$

(2) $n = 3, \ell = 0, m_l = 0, m_s = -1/2$

(3) $n = 3, \ell = 0, m_l = -1, m_s = +1/2$

(4) $n = 3, \ell = 1, m_l = 0, m_s = +1/2$

83. The number of radial nodes in 3s and 2p subshells are respectively

(1) 2 and 0 (2) 1 and 2

(3) 0 and 2 (4) 2 and 1

84. The mass of an electron is 9.1×10^{-31} kg and velocity is 2.99×10^{10} cm s^{-1}. The wavelength of the electron will be

(1) 0.243 Å (2) 0.0243 Å

(3) 4.21 Å (4) 0.421Å

85. For sodium atom number of electrons with m = 0 will be

(1) 2 (2) 7 (3) 9 (4) 8

86. How many electrons in an atom can have $n = 3, \ell = 1, m = -1$ and $s = +½$

(1) 1 (2) 2 (3) 4 (4) 6

87. Arrange the following wavelengths (λ) of given emission lines of H atoms in increasing order

(a) $n = 3 \xrightarrow{\lambda_1} n = 1$ (b) $n = 12 \xrightarrow{\lambda_3} n = 10$

(c) $n=5 \xrightarrow{\lambda_2} n=3$ (d) $n=22 \xrightarrow{\lambda_4} n=20$

Choose the correct option.

(1) $\lambda_4 < \lambda_3 < \lambda_2 < \lambda_1$ (2) $\lambda_4 < \lambda_2 < \lambda_3 < \lambda_1$

(3) $\lambda_1 < \lambda_2 < \lambda_3 < \lambda_4$ (4) $\lambda_1 < \lambda_3 < \lambda_2 < \lambda_4$

88. If ionisation potential of hydrogen atom is 13.6 eV, then ionisation potential of He^+ will be

(1) 54.4 eV (2) 6.8 eV

(3) 13.6 eV (4) 24.5 eV

89. For which of the following sets of quantum numbers of an electron will have the highest energy?

	n	ℓ	m	s
(1)	3	2	1	–½
(2)	4	1	–1	+½
(3)	4	3	–1	+½
(4)	5	0	0	–½

90. When the value of $(n + \ell)$ is not more than 3 which of the following subshells cannot exist?

(1)2s (2)3s

(3)3p (4)2p

91. Given Rydberg constant $R = 10^5$ cm^{-1}. If electron jumps from M shell to K shell of H-atom, the frequency of the radiation emitted in cycle/s would be–

(1) $\frac{8}{9} \times 10^5$ (2) $\frac{8}{3} \times 10^{15}$

(3) $\frac{8}{3} \times 10^{11}$ (4) $\frac{8}{9} \times 10^{15}$

92. Calculate de Broglie wavelength of a neutron (mass, m= 1.6×10^{-27} kg) moving with kinetic energy of 0.04 eV–

(1)146 Å (2)14.6 Å

(3)1460 Å (4)1.46 Å

93. Consider the following statements–

(a) Electron density in XY plane in $3d_{x^2-y^2}$ is zero

(b) Electron density in XY plane in $3d_{z^2}$ orbital is zero

(c) 2s orbital has only one spherical node

(d) For $2P_z$ orbital YZ is the nodal plane

The correct statements are–

(1) (b) and (c) (2)(a), (b), (c), (d)

(3) only (b) (4)only (c)

94. The momentum (in kg-m/s) of photon having 12 MeV (Mega electron volt) energy is :

(1) 6.4×10^{-21} (2) 2.0

(3) 1.6×10^{-21} (4) 3.2×10^{-21}

95. The increasing order for the values of e/m (charge/mass) is :-

(1) e, p, n, α (2) n, p, e, α

(3) n, p, α, e (4) n, α, p, e

96. What is the frequency of revolution of electron present in 2nd Bohr's orbit of H-atom?

(1) $1.016 \times 10^{16}\ s^{-1}$ (2) $4.065 \times 10^{16}\ s^{-1}$

(3) $1.626 \times 10^{15}\ s^{-1}$ (4) $8.2 \times 10^{14}\ s^{-1}$

97. What is the ratio of time periods (T_1/T_2) in second orbit of hydrogen atom to third orbit of He^+ ion?

(1) 8/27 (2) 32/27

(3) 27/32 (4) None of these

98. Number of waves produced by an electron in one complete revolution in n^{th} orbit is :-

(1) n (2) n^2 (3) (n + 1) (4) (2n + 1)

99. Electronic transition in He^+ ion takes from n_2 to n_1 shell such that :

$2n_2 + 3n_1 = 18$

$2n_2 - 3n_1 = 6$

What will be the total number of photons emitted when electrons transit to n_1 shell ?

(1) 21 (2) 15 (3) 20 (4) 10

100. Which of the following expressions represents the spectrum of Balmer series (If n is the principal quantum number of higher energy level) in Hydrogen atom?

(1) $\bar{v} = \frac{R(n-1)(n+1)}{n^2}$ (2) $\bar{v} = \frac{R(n-2)(n+2)}{4n^2}$

(3) $\bar{v} = \frac{R(n-2)(n+2)}{n^2}$ (4) $\bar{v} = \frac{R(n-1)(n+1)}{4n^2}$

101. An excited state of H atom emits a photon of wavelength λ and returns in the ground state, the principal quantum number of excited state is given by :-

(1) $\sqrt{\lambda R(\lambda R-1)}$ (2) $\sqrt{\frac{\lambda R}{(\lambda R-1)}}$

(3) $\sqrt{\lambda R}$ (4) $\sqrt{\frac{(\lambda R-1)}{\lambda R}}$

102. A dye absorbs a photon of wavelenth λ and re-emits the same energy into two photons of wavelengths λ_1 and λ_2 respectively. The wavelength λ is related with λ_1 and λ_2 is :-

(1) $\lambda = \frac{\lambda_1+\lambda_2}{\lambda_1\lambda_2}$ (2) $\lambda = \frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}$

(3) $\lambda = \frac{\lambda_1^2\lambda_2^2}{\lambda_1+\lambda_2}$ (4) $\lambda = \frac{\lambda_1\lambda_2}{(\lambda_1+\lambda_2)^2}$

103. Be^{3+} and a proton are accelerated by the same potential, their de-Broglie wavelengths have the ratio (assume mass of proton = mass of neutron) :-

(1) 1 : 2 (2) 1 : 4 (3) 1 : 1 (4) $1:3\sqrt{3}$

104. The energy of an electron of $2p_y$ orbital is :-

(1) Greater than 3s orbital

(2) Less than $2p_x$ orbital

(3) Equal to 2s orbital

(4) Same as that of $2p_x$ and $2p_z$ orbital

105. Which of the following subshell can accommodate as many as 10 electrons?

(1) 2d (2) 3d (3) $3d_{xy}$ (4) $3d_{z^2}$

CHEMICAL EQUILIBRIUM

106. By dissociation of 4 g mol of PCl_5 produce 0.8 mol of PCl_3 if volume of container is 1 litre the equilibrium constant is :-

(1) 0.2 (2) 0.1

(3) 0.4 (4) 1

107. In a system $P_{(s)} \rightleftharpoons 2Q_{(g)} + R_{(g)}$, at equilibrium if concentration of 'Q' is doubled then how many times the concentration of R at equilibrium will become :-

(1) Double of its original concentration

(2) $\frac{1}{4}$ of its original concentration

(3) $\frac{1}{2}$ of its original concentration

(4) 4 times of its original concentration

108. 1 mol of $XY_{(g)}$ and 0.2 mol of $Y_{(g)}$ are mixed in 1L vessel. At equilibrium 0.6 mol of $Y_{(g)}$ is present. The value of K for the reaction :-

$$XY(g) \rightleftharpoons X(g) + Y(g) \text{ is}$$

(1) 0.04 (2) 0.06 (3) 0.36 (4) 0.40

109. In which of the following reaction is almost completed :-

(1) $K_{eq} = 10$ (2) $K_{eq} = 1$

(3) $K_{eq} = 10^5$ (4) $K_{eq} = 10^{-2}$

110. If the concentration of Y^- ion in the reaction

$$XY_{3(s)} \rightleftharpoons X^{+3}_{(aq)} + 3Y^-_{(aq)} \text{ is}$$

decreased $\frac{1}{2}$ times, then how many times the equilibrium concentration of X^{+3} will increase:

(1) 2 times (2) 4 times

(3) 8 times (4) 16 times

111. For the reaction :-

$$X_{2(g)} + Y_{2(g)} + Z_{2(g)} \rightleftharpoons 2XYZ_{(g)}$$

At equilibrium concentration of X_2 = 3M, Y_2 = 6M, Z_2 = 9M, XYZ = 6M. If the reaction takes place in a sealed vessel at 527°C, then the value of K_C will be :

(1) $\frac{27}{2}$ (2) $\frac{2}{9}$

(3) 36 (4) 24

112. A gaseous mixture was prepared by taking 3 mol of H_2 and 1 mol of CO. If the total pressure of the mixture was found 2 atmosphere then partial pressure of hydrogen (H_2) in the mixture is :-

(1) $\frac{3}{2}$ (2) $\frac{1}{2}$ (3) 1 (4) 2

113. The value of K_c at 900 K temperature for following reaction is 5.

$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ if equilibrium $conc^n$ of N_2 and O_2 is 0.5M then value of K_P is :-

(1) 0.02 (2) 0.2

(3) 5 (4) $\frac{5}{RT}$

114. The equilibrium constant K_C for the following reaction at 842°C is 7.90×10^{-3}. What is K_P at same temperature ?

$$\frac{1}{2}F_{2(g)} \rightleftharpoons F_{(g)}$$

(1) 8.64×10^{-5} (2) 8.26×10^{-4}

(3) 7.90×10^{-2} (4) 7.56×10^{-2}

115. $NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$

The equilibrium pressure at 25°C is 0.660 atm What is K_p for reaction ?

$NH_4NH_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$

(1) 0.109 (2) 0.218

(3) 1.89 (4) 2.18

116. At a certain temperature only 50% HI dissociated at equilibrium in the following reaction

$$2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$$

The equilibrium constant for reaction

(1) 0.25 (2) 1.0 (3) 3.0 (4) 0.5

117. For the reaction $CO_{(g)} + Cl_{2(g)} \rightleftharpoons COCl_{2(g)}$ the value of $\frac{K_C}{K_P}$ is equal to :-

(1) $\sqrt{RT}$ (2) RT (3) $\frac{1}{RT}$ (4) 1

118. The equilibrium constant (K_c) for the reaction

$$2HCl_{(g)} \rightleftharpoons H_{2(g)} + Cl_{2(g)}$$

is 2×10^{-2} at 25°C. What is the equilibrium constant for the reaction $H_{2(g)} + Cl_{2(g)} \rightleftharpoons 2HCl_{(g)}$:-

(1) 2×10^{-2} (2) 50

(3) 5×10^{2} (4) None of these

119. If K_1 and K_2 are the equilibrium constant for the two reactions

$XeF_{6(g)} + H_2O_{(g)} \rightleftharpoons XeOF_{4(g)} + 2HF_{(g)}$; K_1

$XeO_{4(g)} + XeF_{6(g)} \rightleftharpoons XeOF_{4(g)} + XeO_3F_{2(g)}$; K_2

The equilibrium constant for the reaction

$XeO_{4\,(g)} + 2HF_{(g)} \rightleftharpoons XeO_3F_{2(g)} + H_2O_{(g)}$ is

(1) $K_1\ K_2$ (2) K_1/K_2^2

(3) K_2/K_1 (4) K_1/K_2

120. For which of the following gaseous reaction K_p is less than K_C at 298 K temperature :-

(1) $N_2O_4 \rightleftharpoons 2NO_2$

(2) $2HI \rightleftharpoons H_2+F_2$

(3) $2SO_2 + O_2 \rightleftharpoons 2SO_3$

(4) $N_2+O_2 \rightleftharpoons 2NO$

121. At 25°C the equilibrium constant K_1 and K_2 of two reactions are

$2NH_3 \rightleftharpoons N_2+3H_2$; K_1

$\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightleftharpoons NH_3$; K_2

The relation between K_1 and K_2 is :-

(1) $K_1 = K_2$ (2) $K_2 = \frac{1}{K_1^2}$

(3) $K_1 = \frac{1}{K_2^2}$ (4) $K_1 = \frac{1}{K_2}$

122. For the reaction $2SO_3(g) \rightleftharpoons 2SO_{2(g)} + O_{2(g)}$ the value of $\frac{K_C}{K_p}$ will be :-

(1) $(RT)^1$ (2) $(RT)^{-1}$

(3) $\sqrt{RT}$ (4) $(RT)^{-2}$

123. For $N_2+3H_2 \rightleftharpoons 2NH_3$; $\Delta H = -ve$ then :-

(1) $K_P = K_C$ (2) $K_P = K_C\ RT$

(3) $K_P = K_C\ (RT)^{-2}$ (4) $K_P = K_C\ (RT)^{-1}$

124. Consider equilibrium constant for the reaction:

$2CO(g) + 2H_2O_{(g)} \rightleftharpoons 2CO_{2(g)} + 2H_{2(g)}$ eq. const = K_1

$CH_{4(g)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + 3H_{2(g)}$; eq. const = K_2

$CH_{4(g)} + 2H_2O_{(g)} \rightleftharpoons CO_{2(g)} + 4H_{2(g)}$; eq. const = K_3

Which of the following relation is correct ?

(1) $K_3 = \frac{K_1}{K_2}$ (2) $K_3 = \frac{K_1^2}{K_2^2}$

(3) $K_3 = K_1K_2$ (4) $K_3 = \sqrt{K_1} \cdot K_2$

125. One mole of X_2 is mixed with one mole of Y_2 in a flask of volume 1 litre. If at equilibrium 0.5 mole of Y_2 are obtained. Then find out K_p for reaction

$X_{2(g)} + Y_{2(g)} \rightleftharpoons 2XY_{(g)}$

(1) 12 (2) 9 (3) 4 (4) 36

126. For the reaction $2NO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons N_2O_{5(g)}$ if the equilibrium constant is K_p, then the equilibrium constant for the reaction $2N_2O_{5(g)} \rightleftharpoons 4NO_{2(g)} + O_{2(g)}$ would be :-

(1) K_p^2 (2) $\frac{2}{K_p}$ (3) $\frac{1}{K_p^2}$ (4) $\frac{1}{\sqrt{K_p}}$

127. In the reaction $X_{(g)} + Y_{(g)} \rightleftharpoons 2Z_{(g)}$

2 mol of X, 1 mol of Y and 1 mol of Z are placed in a 10 litre vessel and allowed to reach equilibrium. If final conc^n of Z is 0.2M, then K_C for the given reaction is :-

(1) 1.6 (2) 80/3

(3) 16/3 (4) None of these

128. For the reversible reaction

$PCl_{3(g)} + Cl_{2(g)} \rightleftharpoons PCl_{5(g)}$ + heat

The equilibrium will shift in forward direction:-

(1) By increasing conc of $PCl_{5(g)}$

(2) By decreasing pressure

(3) By decreasing conc^n of $PCl_{3(g)}$ and $Cl_{2(g)}$

(4) By increasing pressure and decreasing temp.

129. For a given endothermic rxn, K_P and K_P' are the equilibrium constant at temp T_1 and T_2 respectively. Assuming the heat of reaction is constant in temp. range between T_1 and T_2, it is readily observed that when $T_2 > T_1$ then :-

(1) $K_P > K_P'$ (2) $K_P < K_P'$

(3) $K_P = K_P'$ (4) $K_P = \frac{1}{K'_P}$

130. Which reaction gives more product as a result of increase in pressure :-

(1) $H_2O + CO \rightleftharpoons H_2 + CO_2$

(2) $H_2 + Br_2 \rightleftharpoons 2HBr$

(3) $2SO_2 + O_2 \rightleftharpoons 2SO_3$

(4) $2HI \rightleftharpoons H_2 + I_2$

131. The equilibrium constant for the following reaction is 10 at 500 K. A system at equilibrium has [CO] = 0.25 M and $[H_2]$ = 1M. What is the $[CH_3OH]$?

$$CO_{(g)} + 2H_{2(g)} \rightleftharpoons CH_3OH_{(g)}$$

(1) 0.25 (2) 2.5

(3) 5 (4) 10

132. At certain temperature compound $AB_{2(g)}$ dissociates according to the reaction

$$AB_{2(g)} \rightleftharpoons AB_{(g)} + B_{(g)}$$

with degree of dissociation $\propto$ which is small compared to unity. The expression of K_P in terms of $\propto$ and initial pressure P is :-

(1) $P\propto^3$ (2) $P\propto^2$

(3) $\frac{\propto^3}{P}$ (4) $\frac{\propto^2}{P}$

133. For which of the following reaction is product formation favoured by low pressure and high temperature

(1) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ $\Delta H° = -9.4KJ$

(2) $CO_{2(g)} + C_{(s)} \rightleftharpoons 2CO_{(g)}$ $\Delta H° = 172.5KJ$

(3) $CO_{(g)} + 2H_{2(g)} \rightleftharpoons CH_3OH_{(g)}$ $\Delta H° = -21.7KJ$

(4) $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$ $\Delta H° = 285$ KJ

134. For which reaction $K_P = K_C$:-

(1) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

(2) $2NOCl_{(g)} \rightleftharpoons 2NO_{(g)} + Cl_{2(g)}$

(3) $I_{2(g)} + H_{2(g)} \rightleftharpoons 2HI_{(g)}$

(4) None of these

135. For the reaction

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$$

at a given temperature the equilibrium amount of $CO_2(g)$ can be increased by

(1) adding suitable catalyst

(2) adding inert gas

(3) decreasing the volume of container

(4) increasing the amount of CO(g)

136. The equilibrium constant for the reaction

$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

is 4×10^{-4} at 200 K. In presence of a catalyst equilibrium is attained ten times faster. Therefore, the equilibrium constant in presence of the catalyst at 200 K is

(1) 40×40^{-4}

(2) 4×10^{-4}

(3) 4×10^{-3}

(4) difficult to compute without more data

137. $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

If equilibrium pressure is 3 atm for the above reaction K_P will be

(1) 4 (2) 27 (3) 4/27 (4) 1/27

138. 1 mole of H_2 and 2 mole of I_2 are taken intially in a 2L vessel. The number of moles of H_2 at equilibrium is 0.2 then, the number of moles of I_2 and HI at equilibrium are

(1) 1.2, 1.6 (2) 1.8, 1.0

(3) 0.4, 2.4 (4) 0.8, 2.0

139. In a vessel of 5L, 26 moles of A and 4 moles of B were placed. At equilibrium 1 mole of C was present. The K_C for the reaction :

$A + 2B \rightleftharpoons C$ is

(1) 0.25 (2) 0.50 (3) 2.5 (4) 4.8

140. Consider the following gaseous equilibrium with equilibrium constant K_1 and K_2 respectively

$$SO_2(g) + \frac{1}{2} O_2(g) \rightleftharpoons SO_3(g)$$

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$$

The equilibrium constants are related as

(1) $2K_1 = K_2^2$ (2) $K_1^2 = \frac{1}{K_2}$

(3) $K_2^2 = \frac{1}{K_1}$ (4) $K_2 = \frac{2}{K_1^2}$

141. Out of the following which one is correct for equilibrium state?

(a) $\Delta G = 0$ (b) $\Delta S = 0$ (c) $r_f = r_b$ (d) $E_{cell} = 0$

(1) a,b (2) only b (3) a,c & d (4) only c

142. For a reaction $SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons SO_{3(g)}$

The value of $\frac{K_P}{K_C}$ is equal to

(1) 1 (2) RT

(3) $(RT)^{1/2}$ (4) $(RT)^{-1/2}$

143. For the reaction $N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$ the degree of dissociation is 0.2 at equilibrium and 1 atm pressure then equilibrium constant K_p will be

(1) $\frac{1}{2}$ (2) $\frac{1}{4}$ (3) $\frac{1}{6}$ (4) $\frac{1}{8}$

144. One mole of X and Y are allowed to react in a 2L container when equilibrium is reached the following reaction occurs $2X+Y \rightleftharpoons Z$. If the concentration of Z is 0.2M calculate the equilibrium constant for this reaction

(1) 0.015 (2) 2.22 (3) 6.70 (4) 66.7

145. 28 g N_2 and 6 g H_2 were mixed. At equilibrium 17 g NH_3 was formed. The mass of N_2 and H_2 at equilibrium are respectively

(1) 11 g, zero (2) 1 g, 3 g

(3) 14 g, 3 g (4) 11 g, 3 g

146. For the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, the equilibrium constant will change with

(1) total pressure

(2) catalyst

(3) concentration of H_2 and I_2

(4) temperature

147. What will happen if temperature is increased in the dissolution of CO_2 gas in water ?

(1) solubility decreases

(2) solubility increases

(3) solubility remains unchanged

(4) None of these

148. Naphthalene, a white solid used to make moth balls has a vapour pressure of 0.10 mm Hg at 27°C. Hence K_p and K_c for the equilibrium are:

$$C_{10}H_8\ (s) \rightleftharpoons C_{10}H_8\ (g)$$

(1) 0.10, 0.10

(2) 0.10, 4.1×10^{-3}

(3) 1.32×10^{-4}, 5.34×10^{-6}

(4) 5.36×10^{-6}, 1.32×10^{-4}

149. At a given temperature equilibrium is attained when 50% of each reactant is converted into the products

$$A(g) + B(g) \rightleftharpoons C(g) + D(g)$$

If in other experiment initial amount of B(g) is doubled, percentage of B converted into products will be–

(1) 100% (2) 50%

(3) 66.67% (4) 33.33%

150. In the following equilibrium

I : $A + 2B \rightleftharpoons C$; $K_{eq} = K_1$

II : $C + D \rightleftharpoons 3A$; $K_{eq} = K_2$

III : $6B + D \rightleftharpoons 2C$; $K_{eq} = K_3$

hence

(1) $3K_1 K_2 = K_3$ (2) $K_1^3 K_2^2 = K_3$

(3) $3K_1 K_2^2 = K_3$ (4) $K_1^3 K_2 = K_3$

151. Increase in the pressure for the following equilibrium

$H_2O\ (\ell) \rightleftharpoons H_2O(g)$ result in the

I : Formation of more $H_2O(\ell)$

II : Formation of more $H_2O(g)$

III : Increase in b.p. of $H_2O(\ell)$

IV : Decrease in b.p. of $H_2O\ (\ell)$

Hence, correct statements are–

(1) I, II (2) II, III

(3) I, III (4) I, IV

152. For the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the value of K_C would depend upon

(1) initial concentration of the reactants

(2) pressure

(3) temperature

(4) all of these

153. For the reaction :

$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$;

the degree of dissociation (α) of HI(g) is related to equilibrium constant K_p by the expression

(1) $\frac{1-2\sqrt{K_p}}{2}$ (2) $\sqrt{\frac{1-2K_p}{2}}$

(3) $\sqrt{\frac{2K_p}{1-2K_p}}$ (4) $\frac{2\sqrt{K_p}}{1+2\sqrt{K_p}}$

154. At temperature T, a compound $AB_2(g)$ dissociates according to the reaction

$2AB_2(g) \rightleftharpoons 2AB(g) + B_2(g)$

With degree of dissociation α, which is small compared with unity. The expression of K_p, in terms of α and total pressure P_T is :

(1) $P_T\frac{\alpha^3}{2}$ (2) $P_T\frac{\alpha^3}{3}$

(3) $P_T\frac{\alpha^2}{3}$ (4) $\frac{P_T\alpha}{2}$

155. At constant temperature, K_p for the equilibrium $N_2O_4 \rightleftharpoons 2NO_2$ is expressed by $K_p = \frac{4x^2P}{1-x^2}$, where P = pressure, x = extent of decomposition. Which of the following statement is true ?

(1) K_p increases with increase of P

(2) K_p increases with increase of x

(3) K_p remains constant with change in P and x

(4) K_p increase with decrease of x

IONIC EQUILIBRIUM

156. What is the K_b of weak base that produce one OH^- ion per molecule if a 0.05 M solution is 2.5% ionised ?

(1) 7.8×10^{-8}

(2) 1.6×10^{-6}

(3) 3.1×10^{-5}

(4) 1.2×10^{-3}

157. Solubility of calcium phosphate (molecular mass, M) in water is w g per 100 ml at 25°C, its solubility product at 25°C will be approximately :-

(1) $10^9\left(\frac{w}{M}\right)^5$ (2) $10^7\left(\frac{w}{M}\right)^5$

(3) $10^5\left(\frac{w}{M}\right)^5$ (4) $10^3\left(\frac{w}{M}\right)^5$

158. 0.2 g sample of benzoic acid C_6H_5COOH is titrated with 0.12 M $Ba(OH)_2$ solution, what volume of $Ba(OH)_2$ solution is required to reach the equivalence point ?

(1) 6.83 ml (2) 13.6 ml

(3) 17.6 ml (4) 35.2 ml

159. The Ksp of Ag_2CrO_4 is 1.1×10^{-12} at 298 K, solubility in (mol/L) of Ag_2CrO_4 in 0.1M $AgNO_3$ solution is :-

(1) 1.1×10^{-11} (2) 1.1×10^{-10}

(3) 1.1×10^{-12} (4) 1.1×10^{-9}

160. For a sparingly soluble salt A_pB_q, the relation of its solubility product (L_s) with its solubility (S) is :-

(1) $L_S = S^{P+q}.P^P.q^q$ (2) $L_S = S^{P+q}P^q.q^P$

(3) $L_S = S^{Pq}.P^P.q^q$ (4) $L_S = S^{Pq}.(Pq)^{P+q}$

161. The degree of dissociation of water at 25°C is 1.8×10^{-7} % and density 1.0 g/cm^3, the ionisation constant of water is :-

(1) 1.0×10^{-15} (2) 1.0×10^{-14}

(3) 1.8×10^{-16} (4) 1.0×10^{-6}

162. A certain weak acid has a dissociation constant 1.0×10^{-4}, the equilibrium constant for its reaction with strong base is :-

(1) 1.0×10^{-4} (2) 1.0×10^{-10}

(3) 1.0×10^{10} (4) 1.0×10^{14}

163. A certain buffer solution contains equal concentration of X^- and HX. The K_b for X^- is 10^{-10} the pH of the buffer is :-

(1) 4 (2) 9 (3) 10 (4) 7

164. Find the pH of a solution prepared by mixing 25 ml of a 0.5M solution of HCl, 10 ml of a 0.5 M solution of NaOH and 15 ml of water :-

(1) 4 (2) 3

(3) 0.82 (4) 0.10

165. Calculate the pH of a 10^{-3}M solution of $Ba(OH)_2$ if it undergoes complete ionisation. ($Kw = 1 \times 10^{-14}$) :-

(1) 12.30 (2) 11.30

(3) 10.00 (4) 9.00

166. 0.01 mole of sodium hydroxide is added to 10 litres of water, How much will the pH of water change?

(1) 4 (2) 7

(3) 5 (4) 8

167. The degree of dissociation of acetic acid in a 0.1 M solution is 1.32×10^{-2}, find out the pKa :-

(1) 5.75 (2) 3.75

(3) 4.00 (4) 4.75

168. Calculate the pH of a 10^{-5} M HCl solution if 1ml of it is diluted to 1000 ml, $Kw = 1 \times 10^{-14}$:-

(1) 7.98 (2) 6.98

(3) 7.0 (4) 5

169. Calculate the degree of hydrolysis of the 0.01 M solution of salt (KF $K_{HF} = 6.6 \times 10^{-4}$) :-

(1) 3.87×10^{-6} (2) 3.87×10^{-5}

(3) 3.87×10^{-2} (4) None of these

170. For poly basic acid, the dissociation constant have a different values for each step.

$H_3A \rightleftharpoons H^+ + H_2A^-$; $Keq = Ka_1$

$H_2A^- \rightleftharpoons H^+ + HA^{-2}$; $Keq = Ka_2$

$HA^{2-} \rightleftharpoons H^+ + A^{3-}$; $Keq = Ka_3$

What is the observed trend of dissociation constant in successive stages ?

(1) $Ka_1 > Ka_2 > Ka_3$

(2) $Ka_1 = Ka_2 = Ka_3$

(3) $Ka_1 < Ka_2 < Ka_3$

(4) $Ka_1 = Ka_2 + Ka_3$

171. Equimolar solutions of HF, HCOOH and HCN at 298 K have the values of Ka are 6.8×10^{-4}, 1.8×10^{-4} and 4.8×10^{-9} respectively, what will be the order of their acidic strength ?

(1) HF > HCN > HCOOH

(2) HF > HCOOH > HCN

(3) HCN > HF > HCOOH

(4) HCOOH > HCN > HF

172. NH_4CN is a salt of weak acid HCN ($K_a = 6.2 \times 10^{-10}$) and a weak base NH_4OH ($K_b = 1.8 \times 10^{-5}$) A molar solution of NH_4CN will be :-

(1) Neutral (2) Strongly acidic

(3) Strongly basic (4) Weakly basic

173. The solubility product of $BaCl_2$ is 4×10^{-9}. What will be its solubility in mol/L :-

(1) 4×10^{-3} (2) 3.2×10^{-9}

(3) 10^{-3} (4) 1×10^{-9}

174. Dissociation constant of CH_3COOH and NH_4OH in aqueous solution are 10^{-5} if pH of a CH_3COOH solution is 3, what will be the pH of NH_4OH solution having same concentration as that of CH_3COOH solution ?

(1) 3.0 (2) 4.0

(3) 10.0 (4) 11.0

175. What is the concentration of SO_4^{-2} required start the precipitation of $BaSO_4$ in a solution containing 1×10^{-4} mol L^{-1} of Ba^{2+}

(Ksp of $BaSO_4 = 4 \times 10^{-10}$) :-

(1) 4×10^{-10} M (2) 2×10^{-10} M

(3) 4×10^{-6} M (4) 2×10^{-3} M

176. The solubility product of AgCl is 1.5625×10^{-10} at 25°C its solubility in gram per litre will be :-

(1) 143.5 (2) 108

(3) 1.57×10^{-8} (4) 1.79×10^{-3}

177. Solubility product of radium sulphate is 4×10^{-11}. What will be the solubility of Ra^{2+} in 0.10 M Na_2SO_4 ?

(1) 4×10^{-10} (2) 2×10^{-5} M

(3) 4×10^{-5} M (4) 2×10^{-10} M

178. Which of the following species can act as an acid as well as a base ?

(1) SO_4^{2-} (2) HSO_4^-

(3) PO_4^{3-} (4) OH^-

179. Which of the following salts will give basic solution on hydrolysis ?

(1) NH_4Cl (2) KCl

(3) K_2CO_3 (4) $(NH_4)_2CO_3$

180. In which of the following solvents silver chloride is easily soluble ?

(1) 0.1 mol dm^{-3} $AgNO_3$ solution

(2) 0.1 mol dm^{-3} HCl solution

(3) H_2O

(4) Aqueous NH_3

181. pH of 10^{-6} M HCl (aq.) is :-

(1) just less than 6 (2) exactly equal to 6

(3) just greater than 6 (4) just less than 7

182. How much water must be added to 300 mL of 0.2M solution of CH_3COOH (Ka = 1.8×10^{-5}) for the degree of dissociation to double ?

(1) 600 ml (2) 900 ml

(3) 1200 ml (4) 1500 ml

183. Selenious acid (H_2SeO_3), a diprotic acid has $Ka_1 = 10^{-6}$ and $Ka_2 = 10^{-8}$ respectively. Approximate pH of 0.01M $NaHSeO_3$ is given by :-

(1) $pH = 7 + \frac{pKa_1}{2} + \frac{\log C}{2}$

(2) $pH = 7 - \frac{pKa_1}{2} - \frac{\log C}{2}$

(3) $pH = \frac{pKa_1 + pKa_2}{2}$

(4) $pH = 7 + \frac{pKa_1}{2} - \frac{pKa_2}{2}$

184. Which of the following would increase the solubility of $Pb(OH)_2$:-

(1) Addition of HCl solution

(2) Addition of $Pb(NO_3)_2$ solution

(3) Addition of NaOH solution

(4) Solubility depends on temperature only

185. Solubility of AgCl in 0.2M NaCl is x and that in 0.1 M $AgNO_3$ is y. Then which of the following is correct ?

(1) x = y (2) x > y

(3) x < y (4) Data insufficient

186. What is the pH of saturated solution of $Cu(OH)_2$? (Ksp = 3.2×10^{-20})

(1) 6.4 (2) 7.6 (3) 7.3 (4) 7.9

187. An acid base indicator has $K_{HIn} = 3.0 \times 10^{-5}$. The acid form of the indicator is red and the basic form is blue. The change in $[H^+]$ required to change the indicator from 75% red to 75% blue is :-

(1) 8×10^{-5} M (2) 9×10^{-5} M
(3) 10^{-5} M (4) 3×10^{-4} M

188. pKa of NH_4^+ is 9.26. Hence effective pH range for $NH_4OH - NH_4Cl$ buffer is :-

(1) 8.26 to 10.26 (2) 4.74 to 5.74
(3) 3.74 to 5.74 (4) 8.26 to 9.26

189. The pH of a solution of 0.10M CH_3COOH increases when which of the following substance is added ?

(1) $NaHSO_4$ (2) $HClO_4$
(3) KNO_3 (4) K_2CO_3

190. H_2CO_3 and $NaHCO_3$ constitute buffer system in blood and maintain its pH close to 7.4. An excess of acid entering the blood stream is removed by:-

(1) HCO_3^- (2) H_2CO_3
(3) H^+ (4) CO_3^{2-}

191. The pK_a for acid A is greater than pK_a for acid B, the stronger acid is :-

(1) B (2) A
(3) Both A and B (4) None of these

192. What is the expression for Ksp for $PbCl_2$?

(1) $[Pb^{2+}]\ [2Cl^-]^2$ (2) $[Pb^+]\ /\ [2Cl^-]^2$
(3) $[Pb^{2+}]\ [Cl^-]^2$ (4) $[Pb^{2+}]\ /\ [Cl^-]^2$

193. What is the $[OH^-]$ in final solution prepared by mixing 20 ml of 0.050 M HCl with 30 ml of 0.10 M $Ba(OH)_2$?

(1) 0.10 M (2) 0.40 M
(3) 0.0050 M (4) 0.12 M

194. At 25°C, the solubility product of Hg_2Cl_2 in water is 3.2×10^{-17} mol^3 dm^{-9}. What is the solubility of Hg_2Cl_2 in water at 25°C ?

(1) 1.2×10^{-12} M (2) 3.0×10^{-6} M
(3) 2×10^{-6} M (4) 1.2×10^{-16} M

195. When a buffer solution of CH_3COOH and CH_3COONa is diluted with water

(1) CH_3COO^- ion concentration increases
(2) H^+ ion concentration increases
(3) OH^- ion concentration increases
(4) H^+ ion concentration does not change

196. The solubility products of AgCl & AgI are 1.1×10^{-10} and 1.6×10^{-16} respectively. If $AgNO_3$ is added drop by drop to the solution containing both chloride and iodide ions. The salt which will precipitate first?

(1) AgI
(2) $AgNO_3$
(3) AgCl
(4) both AgCl and AgI simultaneously

197. In the hydrolysis equilibrium

$B^+ + H_2O \rightleftharpoons BOH + H^+$, $K_b = 1 \times 10^{-5}$

The hydrolysis constant is

(1) 10^{-5} (2) 10^{-19} (3) 10^{-10} (4) 10^{-9}

198. Which one of the following is an acidic salt

(1) Na_2HPO_3 (2) NH_4NO_3
(3) NaH_2PO_2 (4) $NaHCO_3$

199. Which of the following is the conjugate base of OH^-

(1) O^{-2} (2) O^- (3) H_2O (4) O_2

200. CH_3NH_2(0.1 mole, $K_b = 5 \times 10^{-4}$) is added to 0.08 moles of HCl and the solution is diluted to one litre. The resulting hydrogen ion concentration is

(1) 1.6×10^{-11} (2) 8×10^{-11}
(3) 5×10^{-5} (4) 8×10^{-2}

201. If Na_2CO_3 is added to the solution of H_2CO_3, the pH of H_2CO_3 solution

(1) decreases (2) remains constant
(3) increases (4) cannot be predicted

202. Calculate the pH of a solution containing 0.1 M CH_3COOH and 0.15 M CH_3COO^-.
(K_a of $CH_3COOH = 1.8 \times 10^{-5}$)

(1) 9.1 (2) 3.9 (3) 10. (4) 4.91

203. In a saturated aqueous solution of AgBr, concentration of Ag^+ ion is 1×10^{-6} mol L^{-1}. If K_{sp} for AgBr is 4×10^{-13}, then concentration of Br^- in the solution is

(1) 1×10^{-6} mol L^{-1} (2) 4×10^{-8} mol L^{-1}

(3) 4×10^{-7} mol L^{-1} (4) 4×10^{-19} mol L^{-1}

204. A solution of an acid has pH = 4.70. Find out the concentration of OH^- ion ($pK_W = 14$)

(1) 5×10^{-10} (2) 6×10^{-10}

(3) 2×10^{-6} (4) 9×10^{-6}

205. Which of the following is not an example of buffer solution?

(1) $HNO_2 + KNO_2$

(2) $HF + KF$

(3) $PhNH_2 + PhNH_3^+Cl^-$

(4) $NH_4Cl + HCl$

206. Which of the following acid will release maximum amount of heat when completely neutralized by strong base NaOH?

(1) 1 M HCl (2)1M HNO_3

(3) 1M $HClO_4$ (4) 1M H_2SO_4

207. Which of the following is an example of hydrolysis reaction?

(1) $NH_4Cl \rightleftharpoons NH_4^+ + Cl^-$

(2) $NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$

(3) $Na^+ + H_2O \rightleftharpoons NaOH + H^+$

(4) $CH_3COOH + HOC_2H_5 \rightleftharpoons CH_3COOC_2H_5 + H_2O$

208. Which of the following salt will have the lowest pH?

(1) Na_2SO_4 (2) $(NH_4)_2SO_4$

(3) Na_2CO_3 (4) NaCl

209. What will be pH of 0.1 M NaCN solution if K_a for HCN is 10^{-6} at 25°C

(1) 6.5 (2) 9.5 (3) 3 (4) 7

210. The conjugate base of strong acid in the reaction

$CH_3COOH + HCl \rightleftharpoons Cl^- + CH_3COOH_2^+$ will be

(1) HCl (2) Cl^-

(3) CH_3COOH (4) $CH_3COOH_2^+$

211. What will be the approximate degree of dissociation of 0.005 M NH_4OH solution. If pK_b for NH_4OH is 4.7

(1) 2% (2) 0.03%

(3) 0.40% (4) 6.32%

212. Which of the following pairs can be used to form a buffer solution?

(1) H_2SO_4 and Na_2SO_4

(2) NaOCl and NaCl

(3) NH_3 and NH_4Cl

(4) NaH_2PO_4 and NaCl

213. What is the molar solubility of MgF_2 in a 0.2 M solution of KF? ($K_{sp} = 8 \times 10^{-8}$)

(1) 8×10^{-8}M (2) 8×10^{-7}M

(3) 2×10^{-6} M (4) 2.7×10^{-4} M

214. If pK_a of acetic acid and pK_b of ammonium hydroxide are 4.76 each. The pH of ammonium acetate at 25°C is

(1) 7 (2) less than 7

(3) more than 7 (4) zero

215. What is the hydronium ion concentration of 0.25 M HA solution? ($K_a = 4 \times 10^{-8}$)

(1) 10^{-4} (2) 10^{-5}

(3) 10^{-7} (4) 10^{-10}

216. The solubility of PbF_2 (molecular wt = 245) is 0.46 g/L. What is solubility product?

(1) 1.1×10^{-10} (2) 2.6×10^{-8}

(3) 1.1×10^{-7} (4) 6.8×10^{-9}

217. What is molar solubility of Ag_2CO_3 ($K_{sp} = 4 \times 10^{-13}$) in 0.1 M Na_2CO_3 solution

(1) 10^{-6} (2) 10^{-7}

(3) 2×10^{-6} (4) 2×10^{-7}

218. When 50 mL of 0.10 M ammonia solution is treated with 50 mL of 0.05 M HCl solution then pH is :- (pK_b of ammonia = 4.74)

(1)8.26 (2)9.26

(3)4.74 (4)None

219. Which salt is most soluble?
(1) CaF_2 ($K_{sp} = 5.3 \times 10^{-9}$)
(2) $BaCrO_4$ ($K_{sp} = 1.2 \times 10^{-10}$)
(3) CaC_2O_4 ($K_{sp} = 4 \times 10^{-9}$)
(4) Hg_2Cl_2 ($K_{sp} = 1.3 \times 10^{-18}$)

220. Molar solubility of $Cd(OH)_2$($Ksp = 2.5 \times 10^{-14}$) in 0.1 M KOH solution is–
(1) 2.5×10^{-7} M (2) 5×10^{-8} M
(3) 5×10^{-7} M (4) 2.5×10^{-12} M

221. What is $\left[NH_4^+\right]$ in a solution that is 0.02 M NH_3 and 0.01 M KOH? [K_b (NH_3) = 1.8×10^{-5}]
(1) 3.6×10^{-5} M (2) 1.8×10^{-5} M
(3) 0.9×10^{-5} M (4) 7.2×10^{-5} M

222. Which of the following has maximum pH ?
(1) 1M H_2S (2) 1M H_2O
(3) 1M H_2Se (4) 1M H_2Te

223. On addition of KF in the solution of HF –
(1) dissociation of HF increases.
(2) concentration of H^+ ions increases.
(3) concentration of H^+ ions decreases.
(4) concentration of F^- ions decreases.

224. $H_3PO_4 + H_2O \rightleftharpoons H_3O^+ + H_2PO_4^-$; $pK_1 = 2.15$
$H_3PO_4^- + H_2O \rightleftharpoons H_3O^+ + HPO_4^{-2}$; $pK_2 = 7.20$
Hence pH of 0.01M NaH_2PO_4 is
(1) 9.35 (2) 4.675
(3) 2.675 (4) 7.350

225. The number of OH^- ion in 1mL of solution having pH = 4 is
(1) 10^{-4} (2) 10^{-10}
(3) 6.02×10^{10} (4) 6.02×10^{13}

226. When some amount of sodium acetate is further added to a mixture of acetic acid and sodium acetate, then pH of solution.
(1) increses
(2) decreases
(3) remains same
(4) none of these can be predicted from given information

227. The equilibrium constant for this reaction is approx 10^{-3}.
$HPO_4^{-2}(aq) + HCO_3^-(aq) \rightleftharpoons H_2PO_4^-(aq) + CO_3^{-2}(aq)$
Which is the strongest conjugate base in this reaction?
(1) $HPO_4^{-2}(aq)$ (2) $HCO_3^-(aq)$
(3) $H_2PO_4^-(aq)$ (4) $CO_3^{-2}(aq)$

228. Which of the following is most soluble in water?
(1) ZnS ($Ksp = 7 \times 10^{-16}$)
(2) $Fe(OH)_3$ ($Ksp = 6 \times 10^{-38}$)
(3) $Ba_3(PO_4)_2$ ($Ksp = 6 \times 10^{-39}$)
(4) Ag_3PO_4 ($Ksp = 1.8 \times 10^{-18}$)

229. Which of the following is not conjugate acid base pair ?
(1) HNO_2, NO_2^-
(2) $CH_3–NH_3^+$, CH_3NH_2
(3) $C_6H_5–COOH$, $C_6H_5–COO^-$
(4) H_3O^+, OH^-

230. Which among the following does not represent the conjugate acid/base pair.
(1) H_3O^+/H_2O (2) H_2SO_4/SO_4^{-2}
(3) HCO_3^-/CO_3^{-2} (4) All of these

231. Which of the following can act as amphoteric species?
(1) HS^- (2) ClO_4^-
(3) NO_3^- (4) HCl

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232. A system absorb 20 KJ heat and does 10 KJ work then internal energy of system will be -
(1) Decreases by 10 KJ
(2) Increases by 10 KJ
(3) Increases by 30 KJ
(4) Decreases by 30 KJ

233. Three moles of an ideal gas expanded spontaneously in to vacuum. Then which is correct?

(1) $W = 0, \Delta G = 0$

(2) $W = 0, \Delta G < 0$

(3) $W = 0, \Delta G > 0$

(4) $W \neq 0, \Delta G = 0$

234. Match the following in List-I with List-II and select the correct option :-

	List-I		List-II
a	$K_p = Q$	i	Always nonspontaneous
b	$T > \Delta H/\Delta S$	ii	Isothermal process
c	$\Delta H = +ve$, $\Delta S = -ve$	iii	Equilibrium
d	$q = -w$	iv	spontaneous and endothermic

(1) a → (iii) b → (iv) , c → (i), d→ (ii)

(2) a → (i) b → (iii) , c → (ii), d→ (iv)

(3) a → (iii) b → (i) , c → (iv), d→ (ii)

(4) a → (iii) b → (ii) , c → (i), d→ (iv)

235. From the following bond energies:

H–H bond energy = 420 KJ mol^{-1}

C≡C bond energy = 601 KJ mol^{-1}

C–C bond energy = 340 KJ mol^{-1}

C–H bond energy = 425 KJ mol^{-1}

Enthalpy for the reaction :

$$C_2H_2 + 2H_2 \rightarrow C_2H_6$$

(1) –599 KJ mol^{-1} (2) –580 KJ mol^{-1}

(3) –625 KJ mol^{-1} (4) –325 KJ mol^{-1}

236. Which of the following is state function (a) q + w, (b) q (c) w (d) heat is isobaric process (e) work in adiabatic process :-

(1) a, b, c (2) a, e

(3) a, d, e (4) a, d

237. Consider the following reaction :-

(a) $H^+_{(aq)} + Cl^-_{(aq)} \rightarrow HCl_{(\ell)}$; $\Delta H = -X_1$ KJ mol^{-1}

(b) $H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl_{(\ell)}$; $\Delta H = -X_2$ KJ mol^{-1}

(c) $\frac{1}{2}H_{2(g)} + \frac{1}{2}Cl_{2(g)} \rightarrow HCl_{(\ell)}$; $\Delta H = -X_3$ KJ mol^{-1}

(d) $H_{(g)} + Cl_{(g)} \rightarrow HCl_{(\ell)}$; $\Delta H = -X_4$ KJ mol^{-1}

Enthalpy of formation of $HCl_{(\ell)}$ is :

(1) $-X_1$ KJ mol^{-1} (2) $-X_2$ KJ mol^{-1}

(3) $-X_3$ KJ mol^{-1} (4) $-X_4$ KJ mol^{-1}

238. Assume each reaction is carried in open container. For which reaction $\Delta H > \Delta E$?

(1) $H_{2(g)} + I_{2(g)} \rightarrow 2HI_{(g)}$

(2) $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$

(3) $PCl_{5(g)} \rightarrow PCl_{3(g)} + Cl_{2(g)}$

(4) $SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow SO_{3(g)}$

239. Given that Bond energies of H_2 and N_2 are 400 KJ mol^{-1} and 240 KJ mol^{-1} respectively and ΔH_f of NH_3 is –120 KJ mol^{-1}. Calculate the bond energy of N–H bond :-

(1) 300 KJ mol^{-1} (2) 250 KJ mol^{-1}

(3) 410 KJ mol^{-1} (4) 280 KJ mol^{-1}

240. The following reactions are given :

$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(g)}$; $\Delta H = -x$ ...(1)

$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$; $\Delta H = -y$ ...(2)

$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(g)}$; $\Delta H = -z$ ...(3)

Calculate the heat of formation of CH_4 ?

(1) x + y + z (2) y + 2z – x

(3) x – y – 2z (4) none of the above

241. The following two reactions are given:-

$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$; $\Delta H = -27.1$ KJ

$2NO_{(g)} + O_{2(g)} \rightleftharpoons 2NO_{2(g)}$; $\Delta H = -53.5$ KJ

Calculate the enthalpy change for the reaction

$$NO_{2(g)} \rightleftharpoons \frac{1}{2}N_{2(g)} + O_{2(g)}$$

(1) –40.3 KJ (2) +80.6 KJ
(3) +40.3 KJ (4) –80.6 KJ

242. ΔU° of combustion of methane is –X KJ mol^{-1}
The value of ΔH° is :-

(1) –X – 596R (2) –X+596R
(3) X+596R (4) X–596R

243. For the reaction

$2Cl_{(g)} \rightarrow Cl_{2(g)}$

What are the sign of ΔH and ΔS ?

(1) $\Delta H > 0$, $\Delta S > 0$
(2) $\Delta H < 0$, $\Delta S < 0$
(3) $\Delta H > 0$, $\Delta S < 0$
(4) $\Delta H < 0$, $\Delta S > 0$

244. The equilibrium constant for a reaction is 100 what will be the value of ΔG° ?
R = 8.314 JK^{-1} mol^{-1} , T=300 K :-

(1) –11488 KJ
(2) –11.488 KJ
(3) –12 KJ
(4) –12000 KJ

245. Standard enthalpy of vapourisation ΔH° for water is 40.66 KJ mol^{-1} . The internal energy of vapourisation of water for its 2 mol will be :-

(1) +43.76 (2) +40.66
(3) +37.56 (4) None of these

246. During adiabatic expansion of ideal gas, which is correct ?

(1) Temperature increases
(2) q = 0
(3) Temperature remain constant
(4) $\Delta E = 0$

247. A reversible process is one in which :-

(1) External pressure is constant throughout
(2) System is in equilibrium at initial & final stage only
(3) Driving force is slightly greater than opposing force
(4) All of the above

248. For the reaction :-

$$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$$

$\Delta U = 2.0$ KCal , $\Delta S = 50$ Cal K^{-1} at 300 K calculate ΔG ?

(1) +12.4 KCal (2) –12.4 KCal
(3) –6.4 KCal (4) +6.4 KCal

249. Consider the following process :-

$A \rightarrow 2B$; $\Delta H = +150$ KJ

$3B \rightarrow 2C + D$; $\Delta H = -125$ KJ

$G + A \rightarrow 2D$; $\Delta H = +350$ KJ

For $B + D \rightarrow G + 2C$; ΔH will be

(1) +525 KJ (2) +325 KJ
(3) –175 KJ (4) –325 KJ

250. Enthalpy change for the reaction

$2N_{(g)} + 6H_{(g)} \rightarrow 2NH_{3(g)}$ is –900 KJ

The dissociation energy of N–H bond is

(1) +450 KJ (2) –450 KJ
(3) +150 KJ (4) –150KJ

251. The heat of combustion of CH_4 is –400 KJ mol^{-1}. Calculate the heat transfer when 40 g of H_2O is formed upon combustion :-

(1) +444.4 KJ (2) +888.8 KJ
(3) –444.4 KJ (4) –888.8 KJ

252. For a spontaneous process :-

(1) $\Delta G = 0$ (2) $\Delta G < 0$
(3) $\Delta G > 0$ (4) Any of the above

253. Which relation is incorrect :-

(1) $\Delta G = -T\,\Delta S_T$
(2) $\Delta G^\circ = -2.303\,RT \log K$
(3) $\Delta H = \Delta U + \Delta n_g RT$
(4) $w_{useful} = \Delta H$

254. Which is correct at equilibrium :-

(1) $\Delta G^\circ = 0$ (2) $\Delta G = 0$

(3) $\Delta S = 0$ (4) $\Delta E = 0$

255. 2 mol of an ideal gas at 27°C expands isothermally and reversibly from a volume of 4L to 40 L. The work for this process (in KJ) is:-

(1) w = – 28.72 KJ (2) w = – 11.488 KJ

(3) w = – 5.736 KJ (4) w = – 4.988 KJ

256. 5 mol of ideal gas expands isothermally and irreversibly from a pressure of 10 atm to 1 atm against constant external pressure of 1 atm work at 300 K will be :-

(1) –15.921 KJ (2) –11.224 KJ

(3) –110.83 KJ (4) none of these

257. Match the column :-

	Column-I		**Column-II**
A	Adiabatic process	P	q = 0
B	Isothermal process	Q	$\Delta H = 0$
C	Isoenthalpic process	R	$\Delta T = 0$
D	Isoentropic process	S	$\Delta S = 0$

(1) A→ P, B→ S, C→ Q, D→ R

(2) A→ Q, B→ P, C→ S, D→ R

(3) A→ P, B→ R, C→ Q, D→ S

(4) A→ P, B→ R, C→ S, D→ Q

258. Match the column :-

A	$C_{(S,\ graphite)} + O_{2(g)} \rightarrow CO_{2(g)}$	P	$\Delta H^\circ_{Combustion}$
B	$CO_{(g)} + 1/2\ O_{2(g)} \rightarrow CO_{2(g)}$	Q	$\Delta H^\circ_{sublimation}$
C	$CH_{4(g)} \rightarrow C_{(g)} + 4H_{(g)}$	R	$\Delta H^\circ_{formation}$
D	$C_{(s)} \rightarrow C_{(g)}$	S	$\Delta H^\circ_{atomisation}$

(1) A→ R, B→ S, C→ P, D→ Q

(2) A→ R, B→ P, C→ Q, D→ S

(3) A→ P, B→ S, C→ Q, D→ R

(4) A→ R, B→ P, C→ S, D→ Q

259. Match the column :-

	Sign of ΔH & ΔS respectively		Nature of reaction
A	– & –	P	Spontaneous only at low temperature
B	– & +	Q	Spontaneous only at high temperature
C	+ & +	R	Spontaneous at all temperature
D	+ & –	S	non spontaneous at all temperature

(1) A → P, B→ R, C→ Q, D→ S

(2) A → R, B→ P, C→ Q, D→ S

(3) A → Q, B→ R, C→ P, D→ S

(4) A → P, B→ Q, C→ R, D→ S

260. Which of the following is correct for free expansion of ideal gas under isothermal condition :-

(1) $q = 0, \Delta T < 0, w < 0$

(2) $q = 0, \Delta T = 0, w = 0$

(3) $q \neq 0, \Delta T = 0, w = 0$

(4) $q \neq 0, \Delta T = 0, w \neq 0$

261. The entropy of fusion of water is 5.260 Cal/mol K calculate the enthalpy of fusion of water ?

(1) 10.52 KCal / mol (2) 0.525 KCal / mol

(3) 2.225 KCal / mol (4) 1.435 KCal / mol

262. $CH_4 + \frac{1}{2}O_2 \rightarrow CH_3OH\ ;\ \Delta H = -ve$

If enthalpy of combustion of CH_4 and CH_3OH is -x & -y respectively which relation is correct?

(1) x > y (2) x < y

(3) x = y (4) x ≥ y

263. The value of ΔH and ΔS for the reaction $C_{graphite} + O_{2(g)} \rightarrow CO_{2(g)}$ are –100 KJ and $-100 JK^{-1}$ respectively. The reaction will be spontaneous at :-

(1) 1000 K (2) 900 K

(3) 1100 K (4) At any temperature

264. The heat of neutralization of LiOH and H_2SO_4 at 25°C is $-69.6\ KJ\ mol^{-1}$. The heat of ionisation of LiOH will be nearly :-

(1) $22.5\ KJ\ mol^{-1}$ (2) $90\ KJ\ mol^{-1}$

(3) $45\ KJ\ mol^{-1}$ (4) $33.6\ KJ\ mol^{-1}$

265. Criteria for spontaneity of process is :-

(1) Maximum Randomness

(2) Maximum energy

(3) Minimum energy and max randomness

(4) Minimum randomness and max. energy

266. Standard entropy of N_2,H_2 and NH_3 is are 60, 40 and 50 $JK^{-1}\ mol^{-1}$ respectively. For the reaction. $\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightleftharpoons NH_3$, $\Delta H = -30$ KJ to be at equilibrium, the temperature should be :-

(1) 500 K (2) 750 K

(3) 1000 K (4) 1250 K

267. The standard enthalpies of formation of CO, NO_2 and CO_2 are $-110.5\ kJ\ mol^{-1}$, $-33.2\ kJ\ mol^{-1}$ and $-393.5\ kJ\ mol^{-1}$ respectively. The standard enthalpy of the reaction is :-

$4CO(g) + 2NO_2(g) \rightarrow 4CO_2(g) + N_2(g)$ is

(1) – 1065.6 kJ (2) – 200 kJ

(3) – 700 kJ (4) 850 kJ

268. 18 g of water is taken to prepare the tea. Find out the internal energy of vaporisation at 100°C. ($\Delta_{vap}H$ for water at 373 K is $40.66\ kJ\ mol^{-1}$)

(1) $37.56\ kJ\ mol^{-1}$

(2) $-37.56\ kJ\ mol^{-1}$

(3) $43.76\ kJ\ mol^{-1}$

(4) $-43.76\ kJ\ mol^{-1}$

269. Given $C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}; \Delta H = -94.2\ K\,Cal$

$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(g)}; \Delta H = -68.3\ K\,Cal$

$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O; \Delta H = -210.8\ K\,Cal$

what will be heat of formation of CH_4 in (K Cal)?

(1) +45.9 (2) –20.0 (3) +47.8 (4) –47.3

270. Which of the following is correct for a spontaneous process?

(1) $\Delta H < 0$, $\Delta S > 0$ at all possible temperature

(2) $\Delta G > 0$

(3) $\Delta S^0 > 0$

(4) $E_{cell} < 0$

271. Which among the following is an extensive property of the system?

(1) temperature (2) volume

(3) refractive index (4) viscosity

272. From the given information, what is the standard enthalpy of formation for $Al_2O_3(s)$?

$2Al_2O_3(s) \rightarrow 4Al(s) + 3O_2(g)$; $\Delta H^0_{rxn} = 3352$ kJ

(1) – 6704 kJ/mol

(2) –3352 kJ/mol

(3) – 1676 kJ/mol

(4) 1676 kJ/mol

273. Which of the following thermodynamic properties must be associated with a reaction spontaneous at only high temperatures

(1) $\Delta H < 0$, $\Delta S < 0$

(2) $\Delta H < 0$, $\Delta S > 0$

(3) $\Delta H > 0$, $\Delta S > 0$

(4) $\Delta H > 0$, $\Delta S < 0$

274. Given enthalpy of formation of $CO_2(g)$ and CaO(s) are –94.0 kJ and –16.0 kJ respectively and the enthalpy of the reaction $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ is 92 kJ. The enthalpy of formation of $CaCO_3(s)$ is

(1) –42 kJ (2) –202 kJ

(3) 202 kJ (4) –288 kJ

275. If the ΔG° of a cell reaction
$AgCl + e^- \rightarrow Ag + Cl^-$ is –21.20 kJ
the standard emf of cell is
(1) 0.329V (2) 0.220V
(3) –0.220V (4) –0.110V

276. Consider the following reactions
$C(s) + O_2(g) \rightarrow CO_2(g) + X$ kJ
$CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g) + Y$ kJ
The heat of formation of CO(g) is
(1) – (X+Y) kJ/mol (2) (X–Y) kJ/mol
(3) (Y–X) kJ/mol (4) (Y+X) kJ/mol

277. The enthalpy change for the following reaction is 368 kJ. Calculate the average O–F bond energy $OF_2(g) \rightarrow O(g) + 2F(g)$
(1) 184 kJ/mol (2) 368 kJ/mol
(3) 536 kJ/mol (4) 736 kJ/mol

278. $A + B \longrightarrow C + D$
$\Delta H = -\ 10{,}000\ J\ mol^{-1}$
$\Delta S = -\ 33.3\ J\ mol^{-1}\ K^{-1}$
At what temperature the reaction will occur spontaneous from left to right?
(1)= 300.3 K (2) > 300.3 K
(3)< 300.3 K (4)None of these

279. 0.16 g of methane was subjected to combustion at 27°C in a bomb calorimeter system. The temperature of the calorimeter system (including water) was found to rise by 0.5°C. Heat of combustion of methane at constant pressure is– (Heat capacity of the calorimeter system is 17.7 KJ K^{-1}).
(1)–890 KJ (2)–885 KJ
(3)+890 KJ (4)+885 KJ

280. For the water gas reaction
$$C(s) + H_2O\ (g) \rightleftharpoons CO(g) + H_2(g)$$
The standard Gibb's energy of reaction (at 1000 K) is –8.1 KJ mol^{-1}. Value of equilibrium constant is-
(1)2.6 (2) 6.2 (3)8.2 (4) 10

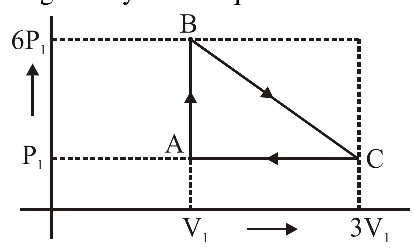
281. Calculate the Gibb's energy change when 1 mole of NaCl is dissolved in water at 25°C. Lattice energy of NaCl = 777.8 KJ mol^{-1} ; ΔS for dissolution = 0.043 KJ mol^{-1} and hydration energy of NaCl = –774.1 KJ mol^{-1}–
(1) –9.114 KJ mol^{-1} (2) –11.4 KJ mol^{-1}
(3) –5.4 KJ mol^{-1} (4) –4.5 KJ mol^{-1}

282. $Na(s) \longrightarrow Na(g)$, the heat of reaction is called as
(1) Heat of vaporisation
(2) Heat of atomisation
(3) Heat of sublimation
(4) Both (2) and (3)

283. $N_2(g) + 2O_2(g) \longrightarrow 2NO_2(g) + xKJ$
$2NO(g) + O_2(g) \longrightarrow 2NO_2(g) + yKJ$
The enthalpy of formation of NO is
(1) x – 2y (2) x – y
(3) ½(y – x) (4) ½(x – y)

284. The entropy of vaporization of benzene is 85 $JK^{-1}\ mol^{-1}$. When 117 g benzene vaporizes at its normal boiling point, the entropy change of surrounding is :-
(1) –85 J/K (2) –85 × 1.5 J/K
(3) 85 × 1.55 J/K (4) None of these

285. An ideal gas is taken around the cycle ABCA is shown in PV diagram. The network done during the cycle in equal to :-



(1) $7P_1V_1$ (2) $6P_1V_1$
(3) $5P_1V_1$ (4) P_1V_1

286. Consider the reaction :
$4NO_{2(g)} + O_{2(g)} \longrightarrow 2N_2O_{5(g)}$ $\Delta_rH = -111$ kJ
If $N_2O_{5(s)}$ is formed instead of $N_2O_{5(g)}$ in the above reaction the Δ_rH value will be
(given ΔH of sublimation of N_2O_5 is 54 kJ mol^{-1})
(1) –219 kJ (2) –165 kJ
(3) 54 kJ (4) 218 kJ

287. Which of the following condition regarding a chemical process ensures spontaneity at all temp. ?
(1) $\Delta H < 0, \Delta S > 0$
(2) $\Delta H > 0, \Delta S < 0$
(3) $\Delta H < 0, \Delta S < 0$
(4) $\Delta H > 0, \Delta S > 0$

288. For a given reaction, $\Delta H = 40$ KJ mol^{-1} and $\Delta S = 80$ JK^{-1}mol^{-1}. The reaction is spontaneous at (assume that ΔH and ΔS do not vary with temperature)
(1) T > 500 (2) T < 500
(3) All temperature (4) T > 298

289. Use the given ΔH_f^o, value to determine to enthalpy of reaction of the following reaction.
$NH_3(g) + 3F_2(g) \longrightarrow NF_3(g) + 3HF(g)$
ΔH_f^o $(NH_3,g) = -46.2$ KJ/mol
$\Delta H_f^o (NF_3,g) = -113.0$ KJ/mol
$\Delta H_f^o (HF,g) = -269.0$ KJ/mol
(1) –335.8 KJ/mol (2) –873.8 KJ/mol
(3) –697.2 KJ/mol (4) –890.4 KJ/mol

290. A reaction occur spontaneously. If
(1) $T\Delta S = \Delta H$ and both ΔH and ΔS are positive
(2) $T\Delta S > \Delta H$ and both ΔH and ΔS are positive
(3) $T\Delta S > \Delta H$ and both ΔH and ΔS are negative
(4) $T\Delta S = \Delta H$ and ΔH is positive and ΔS is negative

291. For vaporization of water at 1 atm pressure, the values of ΔH and ΔS are 40.63 KJ mol^{-2} and 108.8 JK^{-1}mol^{-1} respectively. The temp. at which Gibbs energy change (ΔG) for this transformation will be zero is :-
(1) 273.4 K (2) 393.4 K
(3) 373.4 K (4) 293.4 K

292. Heat of combustion of ethanol at const. pressure and at temp. TK in found to be –q Jmol^{-1}. Hence, heat of combustion in (Jmole^{-1}) of ethanol at the same temp. and const. volume will be :-
(1) RT – q (2) q – RT
(3) q – RT (4) q + RT

293. For the combustion of 1 mole of liquid benzene at 27°C, the heat of reaction at const. pressure in given by

$$C_6H_6(\ell) + \frac{15}{2}O_2(g) \longrightarrow 6CO_2(g) + 3H_2O(\ell)$$

$\Delta H = -78$ KCal. What would be heat of reaction at const. volume?
(1) –78.0 KCal (2) –78.9 KCal
(3) –77.1 KCal (4) 816.0 KCal

294. Calculate $\Delta G°$ for the conversion of oxygen to ozone, $\frac{3}{2}O_{2(g)} \longrightarrow O_{3(g)}$ at 298 K. If K_P for this conversion is 3×10^{-29}.
(1) +175.3 KJ mol^{-1}
(2) +162.7 KJ mol^{-1}
(3) –162.7 KJ mol^{-1}
(4) –140.5 KJ mol^{-1}

295. 2 moles of an ideal is expanded isothermally and reversibly from 1 L to 10 L. Find the enthalpy change in KJ mol^{-1}.
(1) 0 (2) 11.7
(3) –11.7 (4) 25

296. Enthalpies of formation of $C_2H_4(g)$, $CO_2(g)$ and H_2O at 25°C and 1 atm pressure be 52, –394 and –286 KJ mol^{-1} respectively. The enthalpy of combustion of $C_2H_4(g)$ will be :-
(1) +1412 KJ mol^{-1}
(2) –1412 KJ mol^{-1}
(3) +141.2 KJ mol^{-1}
(4) –141.2 KJ mol^{-1}

297. Identify the correct statement regarding entropy.
(1) At 0°C, the entropy of a perfectly crystalline substance is taken to be zero.
(2) At absolute zero temp. the entropy of a perfectly crystalline solid is positive.
(3) At absolute zero temp. the entropy of all crystalline substance is taken be zero
(4) At absolute zero of temp. the entropy of a perfectly crystalline solid is taken to be zero

298. In which of the following neutralization reactions, the heat of neutralization will be the highest ?
(1) NH_4OH and CH_3COOH
(2) NH_4OH and HCl
(3) NaOH and CH_3COOH
(4) NaOH and HCl

299. Which of the following endothermic process is/are spontaneous ?
(1) Melting of ice
(2) Evaporation of water
(3) Heat of combustion
(4) Both (1) and (2)

300. When 0.5 g of sulphur is burnt to SO_2. 4.6 KJ of heat is liberated. What is the enthalpy of formation of sulphur dioxide.
(1) +147.2 KJ (2) –147 KJ
(3) –294.4 KJ (4) +294.4 KJ

301. The species which by definition has zero standard molar enthalpy of formation at 298 K is :-
(1) $Br_{2(g)}$ (2) $Cl_{2(g)}$
(3) $H_2O_{(g)}$ (4) $CH_{4(g)}$

302. For isothermal expansion of ideal gas which is correct ?
(1) $\Delta H = 0$ (2) $\Delta E = 0$ (3) $\Delta T = 0$ (4) All

REDOX

303. In the reaction $SO_2 + H_2S \rightarrow 3S + 2H_2O$ the substance oxidised is :-
(1) S (2) SO_2 (3) H_2S (4) H_2O

304. The oxidation number of phosphorous in $Ba(H_2PO_2)_2$ is :-
(1) +3 (2) +2 (3) +1 (4) –1

305. Whch of the following reaction do not involve oxidation-reduction ?
(I) $2Cs + 2H_2O \rightarrow 2CsOH + H_2$
(II) $2CaI_2 \rightarrow 2Ca + 2I_2$
(III) $NH_4Br + KOH \rightarrow KBr + NH_3 + H_2O$
(IV) $4KCN + Fe(CN)_2 \rightarrow K_4[Fe(CN)_6]$
(1) I, II (2) I, III
(3) I, III, IV (4) III, IV

306. In the reaction
$3Br_2+6CO_3^{2-} + 3H_2O \rightarrow 5Br^{\ominus} + BrO_3^{\ominus} + 6HCO_3^{\ominus}$
(1) Bromine is oxidised and carbonate is reduced
(2) Bromine is reduced and water is oxidised
(3) Bromine is neither reduced non oxidised
(4) Bromine is both reduced and oxidised

307. Which of the following is not a disproportionation reaction :-
(I) $NH_4NO_3 \rightarrow N_2O + H_2O$
(II) $P_4 \rightarrow PH_3 + HPO_2$
(III) $PCl_5 \rightarrow PCl_3 + Cl_2$
(IV) $IO_3^{\ominus} + I^{\ominus} \rightarrow I_2$
(1) I, II (2) I, III, IV
(3) II, IV (4) I, III

308. The number of moles of $K_2Cr_2O_7$ reduced by 1 mol of Sn^{2+} ions will be :-
(1) 1/3 (2) 3
(3) 1/6 (4) 6

309. The number of moles of $KMnO_4$ required to oxidise 2 mol of FeC_2O_4 is acidic medium is :-
(1) 1.2 (2) 3.33 (3) 0.4 (4) 0.8

310. The oxidation number of 'P' in $Mg_2P_2O_7$ is :-
(1) +3 (2) +2 (3) +5 (4) –3

311. Which of the following acid possesses oxidising, reducing and complex forming properties :-
(1) HNO_3 (2) H_2SO_4
(3) HCl (4) HNO_2

312. The number of electrons lost in the following change is :-
$$Fe + H_2O \rightarrow Fe_3O_4 + H^+$$
(1) 2 (2) 4
(3) 6 (4) 8

313. Which gas is evolved when PbO_2 is treated with concentrated HNO_3 :-
(1) NO_2 (2) O_2
(3) N_2 (4) N_2O

314. For decolourisation of 1.5 moles of $KMnO_4$ the moles of H_2O_2 requires

(1) $\frac{3}{2}$ (2) $\frac{9}{4}$ (3) $\frac{15}{4}$ (4) $\frac{21}{4}$

315. In which transfer of five electrons takes place ?

(1) $CrO_4^{2-} \longrightarrow Cr^{+3}$

(2) $MnO_4^- \longrightarrow MnO_2$

(3) $Cr_2O_7^{2-} \longrightarrow 2Cr^{+3}$

(4) $MnO_4^- \longrightarrow Mn^{2+}$

316. In which reaction H_2O_2 acts as reducing agent ?

(1) $Ag_2O + H_2O_2 \rightarrow 2Ag + H_2O + O_2$

(2) $2KI + H_2O_2 \rightarrow 2KOH + I_2$

(3) $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$

(4) $H_2O_2 + SO_2 \rightarrow H_2SO_4$

317. In the equation $NO_2^- + H_2O \rightarrow NO_3^- + 2H^+ + ne^-$, value of n is :-

(1) 1 (2) 2

(3) 3 (4) None of these

318. What mass of HNO_3 is needed to convert 5g of iodine into iodic acid according to the reaction, $I_2 + HNO_3 \rightarrow HIO_3 + NO_2 + H_2O$:-

(1) 12.4g (2) 24.8g

(3) 0.248 g (4) 49.6g

319. In the reduction of dichromate by Fe(II), the number of electrons involved per chronium atom is :-

(1) 2 (2) 1 (3) 3 (4) 4

320. How many grams of I_2 are present in a solution which requires 40 mL of 0.11 N $Na_2S_2O_3$ to react with it :-

$I_2 + S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2I^-$

(1) 12.7g (2) 0.558g

(3) 25.4 g (4) 11.4g

321. In balancing the half reaction, $S_2O_3^{2-} \rightarrow S$, the number of electrons that must be added is :-

(1) 2 on the right (2) 2 on the left

(3) 4 on the left (4) 4 on the right

322. In alkaline medium $KMnO_4$ reacts as follows :-

$2KMnO_4 + 2KOH \rightarrow 2K_2MnO_4 + H_2O + O_2$

Therefore its equivalent mass will be

(1) 31.6 (2) 52.7

(3) 79 (4) 158

323. The oxidation number of C in $K_2C_2O_4$ is

(1) +3 (2) +4

(3) 0 (4) +1

324. $As_2S_3 + HNO_3 \rightarrow H_3AsO_4 + H_2SO_4 + NO$ the element which is oxidised in the reaction is/are

(1) As (2) S

(3) N (4) As and S

325. 0.1 M solution of $KMnO_4$ (in acidic medium) may oxidise

(1) 0.25 M $C_2O_4^{-2}$ (2) 0.8 M Fe^{+2}

(3) 0.1 M FeC_2O_4 (4) 0.6 M $Cr_2O_7^{-2}$

326. In which of the following the oxidation number of oxygen has been arranged in increasing order :-

(1) $OF_2 < KO_2 < BaO_2 < O_3$

(2) $BaO_2 < KO_2 < O_3 < OF_2$

(3) $BaO_2 < O_3 < OF_2 < KO_2$

(4) $KO_2 < OF_2 < O_3 < BaO_2$

327. If it is known that in $Fe_{0.96}O$, Fe is present in +2 and +3 oxidation state. What is the mole fraction of Fe^{2+} in the compound?

(1) 12/25 (2) 25/12

(3) 1/12 (4) 11/12

328. Equivalent weight of FeS_2 in the half reaction, $FeS_2 \rightarrow Fe_2O_3 + SO_2$ is :-

(1) M/10 (2) M/11

(3) M/6 (4) M/1

329. Equivalent weight of H_3PO_2 when it disproportionate into PH_3 and H_3PO_3 is :-

(1) M (2) M/2

(3) M/4 (4) 3M/4

330. When BrO_3^- reacts with Br^- in acidic medium, Br_2 is liberated. The equivalent weight of Br_2 in this reaction is:-

(1) $\frac{5M}{8}$ (2) $\frac{5M}{3}$

(3) $\frac{3M}{5}$ (4) $\frac{4M}{6}$

331. Hydrazine reacts with KIO_3 in presence of HCl as $N_2H_4 + IO_3^- + 2H^+ + Cl^- \rightarrow ICl + N_2 + 3H_2O$ The equivalent masses of N_2H_4 and KIO_3 respectively are :-

(1) 8 and 53.5 (2) 16 and 53.5

(3) 8 and 35.6 (4) 8 and 87

332. Ratio of moles of Fe(II) oxidised by equal volumes of equimolar $KMnO_4$ and $K_2Cr_2O_7$ solutions in acidic medium will be :-

(1) 5 : 3 (2) 1 : 1

(3) 1 : 2 (4) 5 : 6

BEHAVIOUR OF GASES

333. A balloon is filled with hydrogen at room temp. It will burst if, pressure exceeds 0.2 bar. If at 0.1bar pressure the gas occupies 22.7L volume. Up to what volume can the balloon be expanded. :-

(1) < 11.35 L (2) =11.35 L

(3) >11.35 L (4) upto any volume

334. On a ship sailing in pacific ocean where temp. is 23.4°C A balloon is filled with 2L air, what will be the volume of balloon when the ship reaches Indian ocean where temp is 26.1°C :-

(1) 12.01L (2) 6.01L

(3) 2.01L (4) 1.02L

335. At 27°C and 500 mm Hg pressure a gas occupies 400 mL volume. What will be its pressure at a height where temp is 7 °C and volume is 550 mL:-

(1) 340 mm (2) 240 mm

(3) 440 mm (4) 540 mm

336. A methane + H_2 mixture contains 10g methane and 5g Hydrogen. If the pressure of the mixture is 30 atm. what is the partial pressure of H_2 in mixture :-

(1) 6 atm (2) 129 atm

(3) 24 atm (4) 189 atm

337. The temp. at which a real gas obeys ideal gas law over an appreciable range of pressure is called :-

(1) Boiling temp (2) Freezing temp

(3) Boyle's temp (4) Inversion temp

338. The pressure exerted by real gas is :-

(1) lower than ideal gas

(2) more than ideal gas

(3) equal to ideal gas

(4) can't be calculated

339. At critical temp. the valve of z is :-

(1) $\frac{8}{3}$ (2) $\frac{3}{8}$ (3) $\frac{1}{8}$ (4) $\frac{1}{27}$

340. Which one of the following is/are true :-

(i) At critical temp the surface separating two phases disappears

(ii) below critical temperature a gas is called vapour

(iii) A gas can't be liquified below critical temp.

(iv) Higher the critical temp. difficult to liquify a gas

(1) (i), (iii), (iv) (2) (i), (ii), (iv)

(3) (iii), (iv) (4) (i), (ii)

341. Which one of the following is correctly matched:-

(i) z = 1	(A) –ve deviation
(ii) z < 1	(B) +ve deviation
(iii) z > 1	(C) Ideal behaviour
(iv) z_c = 3/8	(D) At critical conditions

(1) (i) –A (ii) –B (iii) –C, (iv) –D

(2) (i) –B (ii) –A (iii) –D, (iv) –C

(3) (i) –C (ii) –A (iii) –B, (iv) –D

(4) (i) –D (ii) –B (iii) –A, (iv) –C

342. A real gas behaves as ideal behaviour when :-

(1) $Z > 1$ (2) $Z < 1$

(3) $Z = 1$ (4) $Z \neq 1$

343. A 1 mol gas occupies 2.4L volume at 27°C and 10 atm pressure then it show :-

(1) +ve deviation (2) –ve deviation

(3) Ideal behaviour (4) None of these

344. The value of vander waal's constant a and b for two gases X and Y are 4.6, 0.04 (for X) and 5.1, 0.06 (for Y). Which can be liquified easily an application of pressure :-

(1) X (2) Y

(3) Both 1 & 2 (4) None of these

345. A gas X is diffused $\frac{1}{\sqrt{2}}$ time slower than SO_2 at same temperature calculate V.D of gas X :-

(1) 64 (2) 128 (3) 32 (4) 16

346. The ratio of rate of diffusion for two gases is 1 : 2 if their molecular masses :-

(1) 16 : 1 (2) 1 : 16 (3) 1 : 4 (4) 4 : 1

347. Which of the following gases always show positive deviation from ideal gas behaviour with increase in pressure?

(1) NH_3 (2) CO_2 (3) H_2 (4) CO

348. If the ratio of the rates of diffusion of two gases A and B is 4:1 the ratio of their density is

(1) 1:16 (2) 1:4 (3) 1:2 (4) 1:8

349. Which of the following gas will have highest value of van der Waal's constant 'a'?

(1) H_2 (2) He (3) CO_2 (4) NH_3

350. At high temperature and low pressure van der Waal's equation becomes

(1) $\left(P+\frac{a}{V^2}\right)=RT$ (2) PV=RT

(3) P(V–b)=RT (4) $\left(P+\frac{a}{V^2}\right)(V-b)=RT$

351. In the van der Waal's equation, the term $\left(\frac{a}{V^2}\right)$ is introduced to correct for

(1) the volume occupied by the molecules themselves

(2) the effect of kinetic energies of molecules

(3) the momentum changes when molecules collide

(4) the effect of forces of attraction between molecules

352. At relatively high pressure, vander Waal's equation reduces to

(1) PV=RT (2) $PV=RT+\frac{a}{V}$

(3) PV=RT +Pb (4) $PV=RT-\frac{a}{V^2}$

353. Which gas can most readly liquified

(1) NH_3 (2) Cl_2

(3) SO_2 (4) CO_2

354. Density of a gas is found to be 5.46 g/L at 27°C and 2 atm pressure. what will be its density at STP

(1) 3 g/L (2) 2 g/L

(3) 1.5 g/L (4) 1 g/L

SOLID STATE

355. The unit cell with parameters $\alpha = \beta = \gamma = 90°$ and $a = b \neq c$ is :-

(1) cubic (2) triclinic

(3) Hexagonal (4) Tetragonal

356. Which of the following crystal lattice has minimum empty space ?

(1) Simple cubic

(2) Body centered cubic

(3) Face centered cubic

(4) All are correct

357. Molybdenum (At. wt = 96 g mol^{-1}) crystallises as BCC crystal. If density of crystal is 10.3 g/cm^3, then radius of Mo atom is :-

(1) 111 pm (2) 314 pm

(3) 135.96 pm (4) None of these

358. In the closest packing of atoms A (radius : r_a), the radius of atom of B that can be fitted into tetrahedral void is :-

(1) 0.155 r_a (2) 0.225 r_a

(3) 0.414 r_a (4) 0.732 r_a

359. In a face centered cubic arrangement of A and B atoms, where A atoms are at the corners of the unit cell and B atoms are at face centers. One of the B atoms missing from one of the face in unit cell. The simplest formula of compound is :-

(1) AB_3 (2) A_8B_5

(3) A_2B_5 (4) $AB_{2/5}$

360. The coordination number of cation and anion in anti-fluorite (Na_2O) is :-

(1) 4 : 8 (2) 8 : 4

(3) 6 : 6 (4) 4 : 4

361. Select the correct statement for CsCl crystal :-

(1) Cs^+ forms simple cubic lattice, Cl^- forms simple cubic lattice

(2) Cl^- occupies the body center of Cs^+

(3) Cs^+ occupies the body center of Cl^-

(4) All are correct

362. The radius of divalent cation A^{+2} is 94 pm and divalent anion B^{-2} is 146 pm. The structure of compound AB is :-

(1) Rock salt (2) Zinc blende

(3) Anti fluorite (4) CsCl type

363. In a solid, oxide ions are arranged in ccp, cations A occupy $\left(\frac{1}{8}\right)^{th}$ of tetrahedral voids and cations B occupy $\left(\frac{1}{4}\right)^{th}$ of octahedral voids. The formula of compound is :-

(1) ABO_4 (2) AB_2O_3

(3) A_2BO_4 (4) AB_4O_4

364. The coordination number and number of carbon atoms in a unit cell of diamond is :-

(1) 4 and 4 (2) 4 and 8

(3) 8 and 4 (4) 6 and 6

365. CsBr has BCC type structure with edge length of 4.3 pm, the shortest interionic distance between Cs^+ and Br^- is :-

(1) 1.86 pm (2) 7.44 pm

(3) 4.3 pm (4) 3.72 pm

366. An element has a body centered cubic (BCC) structure with edge length of 288 pm. The density of the element is 7.2 g/cm^3. How many atoms are present in 208g of the element :-

(1) 6×10^{23} (2) 2.4×10^{23}

(3) 24×10^{23} (4) 12×10^{23}

367. A compound is formed by two elements M and N. The element N forms ccp and atoms of M occupy $\left(\frac{1}{3}\right)^{rd}$ of tetrahedral voids, what is the formula of compound :-

(1) M_3N_2 (2) M_2N_3 (3) M_3N (4) MN_3

368. X-ray diffraction studies of an element shows that it crystallises in FCC unit cell with edge length 3.6×10^{-8} cm. In a separate experiment it is determined that density of element is 8.92 g/cm^3. Calculate the atomic mass of element ?

(1) 156 (2) 63 (3) 40 (4) 23

369. If 'r' stands for radius of atom of the cubic systems like simple cubic, body centered cubic, and face centered cubic, then ratio of edge length of cube in these systems will be respectively :-

(1) $\frac{1}{2}r:\sqrt{3}r:\frac{1}{\sqrt{2}}r$ (2) $\frac{1}{2}r:\frac{\sqrt{3}}{2}r:\frac{\sqrt{2}}{2}r$

(3) $2r:\frac{4r}{\sqrt{3}}:\frac{4r}{\sqrt{2}}$ (4) $2r:\frac{4r}{\sqrt{2}}:\frac{4r}{\sqrt{3}}$

370. The fraction of total volume ocupied by the atoms present in FCC is :-

(1) $\frac{\pi}{6}$ (2) $\frac{\pi}{3\sqrt{2}}$

(3) $\frac{\pi}{4\sqrt{2}}$ (4) $\frac{\pi}{4}$

371. CaS exists in cubic closed packed structure of S^{-2} ions in which Ca^{+2} occupy tetrahedral voids. What is the percentage of tetrahedral voids occupied by Ca^{+2} :-

(1) 25% (2) 50%
(3) 75% (4) 100%

372. The atomic radius of strontium (Sr) is 215 pm and it crystallises in cubic closed packed structure. The edge length of cube is :-

(1) 430 pm (2) 608.2 pm
(3) 496.5 pm (4) None of these

373. Analysis shows that nickel oxide has formula $Ni_{0.98}O$. What fraction of nickel exists as Ni^{+2}:- [If Ni exists in +2 & +3 O.S.]

(1) 0.94 (2) 0.959
(3) 0.98 (4) 0.02

374. Which of the following is not an ionic solid ?

(1) NaCl (2) MgO
(3) CsCl (4) SiC

375. Diamond has a packing efficiency of

(1) 52.4% (2) 34%
(3) 68% (4) 74%

376. A solid compound PQ has CsCl structure. If the radius of cation is 120 pm then find radius of anion?

(1) 88 pm (2) 140 pm
(3) 164 pm (4) 120 pm

377. The vacant space in simple cubic lattice unit cell is :-

(1) 32% (2) 52%
(3) 48% (4) 25%

378. What will be value of $r_{Na^+} + r_{Cl^-}$ in NaCl crystal having edge length 'a' ?

(1) $\frac{a\sqrt{3}}{2}$ (2) $\sqrt{2}a$
(3) $\frac{a}{2}$ (4) $\frac{a}{2\sqrt{2}}$

379. Which of the following can show both Schottky and Frenkel defect :-

(1) NaCl (2) AgCl (3) AgBr (4) KCl

380. The incorrect statement regarding defects in crystalline solid is :-

(1) Density of crystal decreases in Schottky defect
(2) Frenkel defect is dislocation defect
(3) Defects increase conductivity of solid
(4) Density of crystal increases in Frenkel defect

381. Select the incorrect statement ?

(1) Stoichiometry of crystal remain unaffected due to Schottky defect
(2) Frenkel defect is usually shown by ionic compounds having low coordination number
(3) F-center generation is responsible for imparting color to the crystal
(4) Density of crystal always increases due to substitutional impurity defect

382. When AgCl is doped with 0.01 mole $CdCl_2$, What is number of cationic vacancies ?

(1) $0.02\ N_A$ (2) $0.01\ N_A$
(3) 0.01 (4) 0.02

383. Number of tetrahedral void(s) per atom present in cubic closed packed structure is :-

(1) 2 (2) 4 (3) 8 (4) 1

384. If anions (A) form hexagonal closest packing cations (C) occupy $(2/3)^{rd}$ octahedral voids in it, then general formula of compound is :-

(1) CA (2) CA_2 (3) C_2A_3 (4) C_3A_2

385. Total volume of atoms present in a fcc unit cell of metal is (r is atomic radius)

(1) $\frac{20}{3}\pi r^3$ (2) $\frac{24}{3}\pi r^3$
(3) $\frac{12}{3}\pi r^3$ (4) $\frac{16}{3}\pi r^3$

386. The distance between Na^+ and Cl^- ion in NaCl is 281 pm. What is the edge length of the cubic unit cell.

(1) 1405 pm (2) 562 pm
(3) 843 pm (4) 281 pm

387. In a solid, oxide ions are arranged in ccp. Cations A occupy one sixth of THV and cations B occupy one third of the OHV. The empirical formula of compound is

(1) A_2BO_3 (2) $A_2B_2O_3$ (3) ABO_3 (4) AB_2O_3

388. In a compound, atoms A occupy 3/4 of the tetrahedral voids and atoms B form ccp lattice. The emperical formula of the compound is

(1) A_3B_4 (2) A_2B_3 (3) AB (4) A_3B_2

389. Nearest distance between T.H.V. and O.H.V. in F.C.C. unit cell is

(1) $\frac{a}{2}$ (2) $\frac{a}{\sqrt{2}}$ (3) $\frac{\sqrt{3}a}{4}$ (4) $\frac{\sqrt{3}a}{2}$

390. The coordination number of F^- ion in CaF_2 crystalline structure is

(1) 8 (2) 2 (3) 4 (4) 6

391. The number of atoms in 100 g of a fcc crystal with density d = 10 g/cm^3 and cell edge is 100 pm

(1) 1×10^{25} (2) 2×10^{25}
(3) 3×10^{25} (4) 4×10^{25}

392. A face centred cubic is made up of two types of atoms A and B in which A occupies the corner positions and B occupies the face centres. If atoms along one of the body diagonal are removed, empirical formula of remaining solid will be

(1) AB_2 (2) A_3B (3) A_7B_3 (4) AB_4

393. A solid is formed and it has three types of atoms X, Y and Z. X forms a fcc lattice while Y atoms occupying all tetrahedral voids and Z atoms occupying half of octahedral voids. The formula of solid is

(1) X_4YZ_2 (2) X_4Y_2Z
(3) XY_2Z_4 (4) X_2Y_4Z

394. An element X (at. wt. = 80 g/mol) having fcc structure, calculate number of unit cell in 8 g of X

(1) 0.4 N_A (2) 0.025 N_A
(3) 4 N_A (4) 0.2 N_A

395. In the rock salt structure AB, if C is introduced in tetrahedral voids such that no distortion occurs. Then formula of resultant compound is

(1) ABC (2) ABC_2
(3) A_4B_4C (4) ABC_8

396. A compound forms hexagonal close packed structure. How many octahedral voids are present in 0.3 mol of it?

(1) 1.08×10^{23} (2) 1.80×10^{23}
(3) 5.4×10^{23} (4) None of these

397. For graphite which is possible?

(1) $a = b = c, \alpha = \beta = \gamma = 90°$
(2) $a = b \neq c, \alpha = \beta = \gamma = 90°$
(3) $a = b \neq c, \alpha = \beta = 90°, \gamma = 120°$
(4) $a \neq b \neq c, \alpha = \beta = \gamma = 90°$

398. Due to excess of lithium, LiCl crystals appear–

(1) yellow (2) violet
(3) pink (4) white

399. A metal crystallises in FCC lattice. If edge length of unit cell is 4×10^{-8} cm and density is 10.5 g cm^{-3} then the atomic mass of metal is–

(1) 101.2 (2) 50.6
(3) 202.3 (4) None

400. Which of the following is a non-crystalline solid ?

(1) CsCl (2) NaCl
(3) CaF_2 (4) Glass

401. Solid CO_2 is an example of,

(1) Ionic crystal
(2) Covalent crystal
(3) Metallic crystal
(4) Molecular crystal

402. The density of KBr is 2.82 g cm^{-3}. The length of the unit cell is 654 pm. Atomic mass of K = 39, Br = 80. Then the solid is

(1) Face centred cubic
(2) Simple cubic system
(3) Body centred cubic system
(4) None of these

403. Which statement is correct in antifluorite structure of Na_2O

(1) O^{-2} ions have CCP arrangement
(2) Na^+ ions occupy all the tetrahedral sites
(3) It has 4 :8 co-ordination
(4) All are correct

404. When Zn converts from molten state to solid state, it has h.c.p. structure then find out number of nearest neighbours of Zn atoms -

(1) 6 (2) 8
(3) 12 (4) 4

405. A solid AB has ZnS type structure. If the radius of the anion B^- is 100 pm, then the minimum radius of the cation A^+ will be :-

(1) 192.3 pm (2) 414 pm
(3) 22.5 pm (4) 44.4 pm

406. Which of the following statement is incorrect

(1) The co-ordination number of each type of ion in CsCl crystal is 8
(2) A metal that crystallizes in bcc structure has a coordination number of 12
(3) A unit cell of an ionic crystal shares some of its ions with other unit cells
(4) The length of the cell in NaCl is 552 pm (r_{Na^+} = 95 pm ; r_{Cl^-} = 181 pm)

407. The only incorrect statement for the packing of identical spheres in two dimension is :

(1) For square close packing , coordination number is 4.
(2) For hexagonal close packing, coordination number is 6.
(3) The vacant space between atoms in square and hexagonal close packing are called square void and hexagonal void
(4) Hexagonal close packing is more efficiently packed than square close packing.

408.

Column I a = edge length		Column II	
(a)	0.866 a	(p)	Shortest distance between cation & anion in CsCl structure.
(b)	0.5 a	(q)	Shortest distance between cation and anion in NaCl structure.
(c)	0.433 a	(r)	Shortest distance between carbon atoms in diamond.
(d)	1.414 a	(s)	The distance at which second nearest neighbour is present in simple cubic unit cell

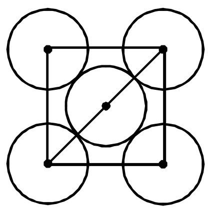
(1) a-s, b-r, c-p, d-q (2) a-p, b-q, c-r, d-s
(3) a-r, b-q, c-p, d-s (4) a-p, b-q, c-s, d-r

409. If the unit cell of a mineral has cubic close packed (ccp) array of oxygen atoms with m fraction of octahedral holes occupied by aluminium ions and n fraction of tetrahedral holes occupied by magnesium ions m and n respectively, are -

(1) $\frac{1}{2},\frac{1}{8}$ (2) $1,\frac{1}{4}$

(3) $\frac{1}{2},\frac{1}{2}$ (4) $\frac{1}{4},\frac{1}{8}$

410. The packing effeciency of the two-dimensional square unit cell shown below is -



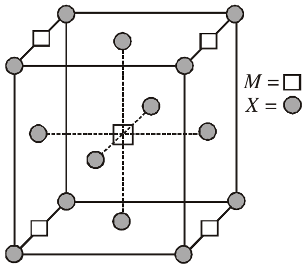
(1) 39.27% (2) 68.02%
(3) 74.05% (4) 78.54%

411. Ammonium chloride crystallizes in a body centred cubic lattice with edge length of unit cell of 390 pm. If the radius of chloride ion is 180 pm, the radius of ammonium ion would be:-

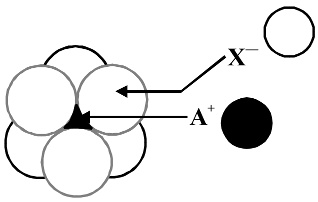
(1) 158 pm (2) 174 pm
(3) 142 pm (4) 126 pm

412. A compound M_pX_q has cubic close packing (ccp) arrangement of X. Its unit cell structure is shown below. The empirical formula of the compound is :

(1) MX
(2) MX_2
(3) M_2X
(4) M_5X_{14}



413. The arrangement of X^- ions around A^+ ion in solid AX is given in the figure (not drawn to scale). If the radius of X^- is 250 pm, the radius of A^+ is



(1) 104 pm (2) 125 pm
(3) 183 pm (4) 57 pm

414. Barium titanate crystallises in cubic structure with barium ions occupying the corners of the unit cell, oxide ions occupying the face centres and titanium ions occupying centres of the unit cell. If Ti^{4+} ion is described as occupying the holes in Ba-O lattice, the type of hole and fraction of these holes occupying by these ions are

(1) 100% of octahedral holes
(2) 25% of tetrahedral holes
(3) 25% of octahedral holes
(4) 50% of tetrahedral holes

415. A sample of wustite is $Fe_{0.93}O$. The percentage of iron present in the form of Fe(III) is

(1) 15.05% (2)25%
(3) 7% (4) 20%

416. Which among following will show anisotropy-

(1) Glass (2) Wood
(3) Barium chloride (4) Paper

417. For tetrahedral co-ordination, the limiting radius ratio (r^+/r^-) should be -

(1) 0.414 – 0.732 (2) > 0.732
(3) 0.156 – 0.225 (4) 0.225 – 0.414

418. Total volume of atoms present in BCC unit cell of metal is (r = atomic radius)

(1) $\frac{20}{3}\pi r^3$ (2) $\frac{24}{3}\pi r^3$
(3) $\frac{12}{3}\pi r^3$ (4) $\frac{8}{3}\pi r^3$

419. What type of crystal defect is indicated in the diagram below ?

Na^+	Cl^-	Na^+	Cl^-	Na^+	Cl^-
Cl^-		Cl^-	Na^+		Na^+
Na^+	Cl^-		Cl^-	Na^+	Cl^-
Cl^-	Na^+	Cl^-	Na^+		Na^+

(1) Frcnkel defect
(2) Schottky defect
(3) Interstitial defect
(4) Frenkel and Schottky defects.

420. In the sodium chloride structure, each Na^+ ion is surrounded by six Cl^- ions nearest neighbors and......... Na^+ ion next nearest neighbors -

(1) 4 (2) 8
(3) 6 (4) 12

421. The correct statement(s) regarding defects in solids is(are)

(1) Frenkel defect is usually favoured by a very small difference in the size of cation and anion
(2) Frenkel defect is a dislocation defect
(3) Trapping of an electron in the interstitial site leads to the formation of F-center
(4) Schottky defects have no effect on the physical properties of solids

SOLUTION

422. In an aqueous solution ethylene glycol has the mass percentage (% w/w) 30% then the mole fraction of ethylene glycol will be :-

(1) 0.428 (2) 0.124
(3) 0.11 (4) 0.889

423. What will be the mass percentage of resulting solution prepared by mixing 15% (w/w) 500 g aqueous solution of urea with 25% (w/w) 400g aqueous solution of it :-

(1) 18% (2) 20%
(3) 25% (4) 15%

424. What will be the molality of solution prepared by dissolving 3.7 g propanoic acid in 80 g benzene ?

(1) 0.77 m (2) 0.625 m
(3) 0.045 m (4) 46.25 m

425. If the sea water is 3.5% (w/w) aqueous solution of NaCl then its molality will be :-

(1) 0.598 (2) 36.27 m
(3) 0.62 m (4) 0.578 m

426. 10% (w/v) solution of glucose is isotonic with 4% (w/v) solution of non-volatile solution then molar mass of non-volatile solute will be :-

(1) 36 (2) 72 (3) 54 (4) 63

427. The vapour pressure of mixture of toluene and xylene at 90°C is 0.5 atm. If at this temperature vapour pressure of pure toluene and xylene are 400 mm and 150 mm respectively then what will be the mole fraction of toluene in mixture?

(1) 0.08 (2) 0.92
(3) 0.88 (4) 0.46

428. What will be the freezing point of 0.2 molal aqueous solution of $MgBr_2$? If salt dissociates 40% in solution and K_f for water is 1.86 KKg mol^{-1}

(1) –3.35°C (2) –0.67°C
(3) –0.6°C (4) –0.45°C

429. Density of 12.25% (w/w) H_2SO_4 solution is 1.052 g/ml then molarity of solution is :-

(1) 1.315 M (2) 2.63 M
(3) 0.657 M (4) 1 M

430. Incorrect relationship for mole fraction is :-

(1) $X < 1$ (2) $0 < X < 1$
(3) $-2 < X < +2$ (4) Always positive

431. 0.1 molal solution of $Hg(NO_3)_2$ freezes at –0.558°C. The cryoscopic constant for water is 1.86 KKg mol^{-1} then what will be the percentage ionisation of salt ?

(1) 33.33% (2) 50%
(3) 75% (4) 100%

432. Which of the following solution has maximum vapour pressure :-

(1) 1 N KNO_3 (2) 1 N $Ba(NO_3)_2$
(3) 1 N $Al_2(SO_4)_3$ (4) 1 N $Ti(NO_3)_4$

433. When HgI_2 is added in KI solution. The freezing point of solution :-

(1) increases
(2) decreases
(3) Remains unchanged
(4) Can't predict

434. A solute dissociates in solution according to reaction $2A \rightarrow 5B$, If solute shows 30% dissociation then van't Haff factor will be :-

(1) 2.2 (2) 1.45 (3) 2.9 (4) 1.9

435. An aqueous solution of urea [6% w/v] is isotonic with NaCl solution then mass-volume percentage (%w/v) of NaCl solution will be :-

(1) 1.46% (2) 5.85%
(3) 2.92% (4) 11.7%

436. The vapour pressure of pure liquid A and liquid B at 350 K are 440 mm and 720 mm of Hg. If total vapour pressure of solution is 580 mm of Hg then the mole fraction of liquid A in vapour phase will be :-

(1) 0.31 (2) 0.38
(3) 0.62 (4) 0.76

437. If T_1, T_2 and T_3 are boiling points of component A, component B and azeotropic mixture then which of the following relation is correct for a minimum boiling azeotrope :-

(1) $T_1 < T_3$ (2) $T_2 < T_3$

(3) $T_3 < T_1$ (4) $T_3 = T_2$

438. Which of the following relationship is correct ?

(1) $K_b = \frac{RT_b^2}{1000 \times \Delta V_v}$ (2) $K_b = \frac{RT_b^2}{1000\ L_v}$

(3) $K_f = \frac{RT_f^2}{1000\ \Delta H_f}$ (4) $K_f = \frac{RT_f^2 \times M}{1000 \times L_f}$

439. If the solubility product of CuS is 6×10^{-6}. Calculate the maximum molarity of CuS is aqueous solution

(1) 2.45×10^{-8} mol L^{-1}

(2) 1.5×10^{-5} mol L^{-1}

(3) 3.5×10^{-2} mol L^{-1}

(4) 2.45×10^{-3} mol L^{-1}

440. Which of the following 0.2m aqueous solutions will show minimum freezing point :-

(1) $CaCl_2$ (2) K_2SO_4

(3) $Al(NO_3)_3$ (4) KBr

441. For which of the following solutions observed boiling point is greater than theoretical boiling point ?

(1) $CHCl_3 + CCl_4$

(2) $CCl_4 + SiCl_4$

(3) $CHCl_3 + CH_3COCH_3$

(4) $C_6H_5CH_3 + C_6H_6$

442. The boiling point of solution obtained by dissolving 0.51 g anthracene in 35 g chloroform increases by 0.32°C then what will be the molar mass of anthracene if for chloroform K_b = 3.9 K Kg mol^{-1} :-

(1) 175.2 (2) 177.6

(3) 178.6 (4) 182.3

443. Which of the following relationship is correct? (A → solvent; B → solute)

(1) $\frac{P^o - P_S}{P^o} = X_A$ (2) $Y_B = \frac{P_B^o - X_B}{P_A^o}$

(3) $P_A = P_A^o X_A$ (4) $P_S = P_A^o X_A + P_B^o X_A$

444. What will be the osmotic pressure of 5% (w/v) aqueous solution of urea at 17°C ?

(1) 19.81 atm (2) 1.16 atm

(3) 1.981 atm (4) 6.61 atm

445. Van't Hoff factor is 1.92 for MgI_2 solution with concentration 0.2M then the degree of dissociation of salt at this concentration is :-

(1) 46% (2) 96%

(3) 30.67% (4) 64%

446. The vapour pressure of any liquid shows following change with temperature :-

(1) Exponential increase

(2) Exponential decrease

(3) Linear increase

(4) Linear decrease

447. During dissolution of gas in liquid solvent :-

(1) Heat is released

(2) Heat is absorbed

(3) No heat transfer takes place

(4) Colour of liquid solvent is always changed

448. 18 g glucose ($C_6H_{12}O_6$) is added to 178.29 of H_2O the vapour pressure of water for this aqueous solution at 100°C.

(1) 759 torr (2) 7.60 torr

(3) 76 torr (4) 752.4 torr

449. The solubility of gases in liquid is maximum at :-

(1) Low temperature and low pressure

(2) Low temperature and high pressure

(3) High temperature and high pressure

(4) High temperature and low pressure

450. If solubility of vinyl chloride (g) is 0.09 M at STP then value of henry's constant will be :-

(1) 0.0016 bar (2) 617.284 bar

(3) 6.17×10^{-2} bar (4) 308.642 bar

451. Which of the following statement is **incorrect**:-

(1) In solution of volatile liquids partial pressure of any component is directly proportional to its mole fraction in solution.

(2) Components of azeotropic mixture can be separated in pure form using fractional distillation

(3) Total vapour pressure of solution of binary volatile liquids, decreases on increasing mole fraction of less volatile component.

(4) At a given pressure, higher the value of K_H, the lower is the solubility of the gas in the liquid.

452. What will be the osmotic pressure of 0.03 N solution of Aluminium sulphate solution at 27°C? If in solution salt dissociates 90% :-

(1) 0.566 atm (2) 0.677 atm

(3) 3.399 atm (4) 4.065 atm

453. What will be the amount of ice separated on cooling solution of 40g ethylene glycol in 400g water upto –9.3° C ?

(K_f for water is 1.86 K Kg mol^{-1}) :-

(1) 177.78 g (2) 129.03 g

(3) 222.22 g (4) 270.97 g

454. Which one of the following solution has maximum vapour pressure at 27°C temperature ?

(1) 1M Na_2SO_4 (2) 1 M $AlCl_3$

(3) 1M KBr (4) 1M $MgCl_2$

455. At 298 K, 1000 cm^3 of a solution containing 4.34 g of solute shows osmotic pressure of 2.55 atm. What is the molar mass of solute?

(R = 0.0821 L atm K^{-1} mol^{-1})

(1) 41.64 g mol^{-1}

(2) 82.73 g mol^{-1}

(3) 58.31 g mol^{-1}

(4) 91.65 g mol^{-1}

456. Which of the following will have maximum depression in freezing point?

(1) 0.5 M Li_2SO_4

(2) 1M KCl

(3) 0.5 M $BaCl_2$

(4) 1M $Al_2(SO_4)_3$

457. 1.00 g of a non electrolyte solute is dissolved in 50g of benzene which lowers the freezing point of benzene by 0.40K. The freezing point depression constant of benzene is 5.12 K kg mol^{-1}. Find the molar mass of the solute

(1) 206 g mol^{-1} (2) 226 g mol^{-1}

(3) 246 g mol^{-1} (4) 256 g mol^{-1}

458. At 40°C the vapour pressure of pure benzene and toluene are 160 mmHg and 60 mmHg respectively. If equimolar above liquids are mixed at same temperature to form an ideal solution. Then vapour pressure of solution will be

(1) 140 mm of Hg

(2) 110 mm of Hg

(3) 220 mm of Hg

(4) 100 mm of Hg

459. Dissolving 120 g of urea (mol wt = 60) in 1000 g of water gave a solution of density 1.15 g mL^{-1}. The molarity of the solution is

(1) 1.78 M (2) 2.00 M

(3) 2.05 M (4) 2.22 M

460. A binary liquid solution is prepared by mixing n-heptane and ethanol which one of the following statement is correct regarding the behaviour of the solution?

(1) the solution formed is an ideal solution

(2) the solution is non-ideal, showing positive deviation from Raoult's law

(3) the solution is non ideal showing negative deviation from Raoult's law

(4) n-heptane shows positive deviation while ethanol shows negative deviation from Raoult's law

461. What is the g-molecular mass of a nonionizing solid if 10 g of this solid dissolved in 100 g of water, forms a solution which froze at –1.22 °C? (K_f = 1.86 K kg mol^{-1})

(1) 265 g/mol (2) 152 g/mol
(3) 130g/mol (4) 65g/mol

462. At 293 K, vapour pressure of pure benzene is 75 mm of Hg and that of pure toluene is 22 mm of Hg. The vapour pressure of the solution which contains 20 mol% benzene and 80 mol % toluene is

(1) 32.6 mm Hg (2) 64.4 mm Hg
(3) 97 mm Hg (4) 3.26 mm Hg

463. The vapour pressure of pure liquid solvent is 0.80 atm. When a non-volatile substance (Z) is added to the solvent, its vapour pressure drops to 0.6 atm. What the mole fraction of the substance (Z) in the solution

(1) 0.75 (2) 0.50
(3) 0.25 (4) 0.33

464. What will be normality of "20V H_2O_2" solution?

(1) 1.78 N (2) 3.56 N
(3) 10N (4) 0.28 N

465. Which of the following solution have highest boiling point. (Assume all salts completely dissociate)

(1) 0.1M $Al_2(SO_4)_3$ (2) 0.1M $BaCl_2$
(3) 0.1M glucose (4) 0.1M $AlCl_3$

466. What will be the freezing point (in °C) of solution obtained by dissolving 0.1 g potassium ferricyanide (mol wt = 329) in 100 g water. If K_f for water is 1.86 K kg mol^{-1}

(1) -2.3×10^{-2} (2) -5.7×10^{-2}
(3) -5.7×10^{-3} (4) -1.2×10^{-2}

467. The van't Hoff factor for 0.1M $Ba(NO_3)_2$ solution is 2.74. The degree of dissociation is

(1) 91.3% (2) 87%
(3) 100% (4) 74%

468. A solution of a substance containing 1.05 g per 100 mL was found to be isotonic with 3% w/v glucose solution. The molecular mass of substance is

(1) 31.5 (2) 6.3 (3) 630 (4) 63

469. The vapour pressure of benzene at 90°C is 1020 torr. A solution of 5g of a solute in 58.5 g benzene has vapour pressure 990 torr. The molecular mass of the solute is

(1) 78.2 (2) 178.2
(3) 200 (4) 220

470. Which of the following has minimum freezing point

(1) 0.1M $K_2Cr_2O_7$ (2) 0.1M NH_4Cl
(3) 0.1M $BaSO_4$ (4) 0.1M $Al_2(SO_4)_3$

471. What is the correct sequence of osmotic pressure of 0.01 M aq solution of

(a) $Al_2(SO_4)_3$ (π_1) (b) Na_3PO_4 (π_2)
(c) $BaCl_2$(π_3) (d) Glucose (π_4)

Choose the correct option.

(1) $\pi_4 > \pi_2 > \pi_3 > \pi_1$ (2) $\pi_3 > \pi_4 > \pi_2 > \pi_1$
(3) $\pi_3 > \pi_4 > \pi_1 > \pi_2$ (4) $\pi_1 > \pi_2 > \pi_3 > \pi_4$

472. A solution contains nonvolatile solute of molecular mass M_2. Which of the following can be used to calculate the molecular mass of solute in terms of asmotic pressure.

(1) $M_2 = \left(\frac{m_2}{\pi}\right)VRT$ (2) $M_2 = \left(\frac{m_2}{V}\right)\frac{RT}{\pi}$

(3) $M_2 = \left(\frac{m_2}{V}\right)\pi RT$ (4) $M_2 = \left(\frac{m_2}{V}\right)\frac{\pi}{RT}$

473. Normal saline solution is

(1) 1N salt solution
(2) 0.9% mass/volume NaCl solution
(3) 1N NaCl solution
(4) Normal salt solution

474. A certain substance 'A' tetramerises in water to the extent of 80%. A solution of 2.5g of A in 100g of water lowers the freezing point by 0.3°C. The molar mass of A is -

(1) 122 (2) 31 (3) 244 (4) 62

475. Compound $PdCl_4.6H_2O$ is a hydrated complex; 1molal aqueous solution of it has freezing point 269.28K. Assuming 100% ionization of complex, calculate the molecular formula of the complex (K_f for water = 1.86 K kg mol^{-1}) -

(1) $[Pd(H_2O)_6]Cl_4$

(2) $[Pd(H_2O)_4Cl_2]Cl_2.2H_2O$

(3) $[Pd(H_2O)_3Cl_3]Cl.3H_2O$

(4) $[Pd(H_2O)_2Cl_4]_4H_2O$

476. Which pair shows a contraction in volume on mixing along with evolution of heat- -

(1) $CHCl_3 + C_6H_6$ (2) H_2O + HCl

(3) $H_2O + HNO_3$ (4) All

477. Decimolar solution of potassium ferricyanide, $K_3[Fe(CN)_6]$ has osmotic pressure of 3.94atm at 27°C. Hence percent ionisation of the solute is -

(1) 10% (2) 20%

(3) 30% (4) 40%

478. A complex containing K^+,Pt (IV) and Cl^- is 100% ionised giving i = 3. Thus, complex is -

(1) $K_2[PtCl_4]$ (2) $K_2[PtCl_6]$

(3) $K_3[PtCl_5]$ (4) $K[PtCl_3]$

479. If $pK_a = -\log K_a = 4$, and $K_a = C\alpha^2$ then van't Hoff factor for weak monobasic acid when C = 0.01 M is -

(1) 0.01 (2) 1.02

(3) 1.10 (4) 1.20

480. In which case van't Hoff factor is maximum? -

(1) KCl, 50% ionised

(2) K_2SO_4 40% ionised

(3) $FeCl_3$,30% ionised

(4) $SnCl_4$, 20% ionised

481. A solution of benzoic acid dissolved in benzene will show a molecular mass closer to -

(1) 122 (2) 244

(3) 61 (4) 366

482. Ratio of $\frac{\Delta T_b}{k_b}$ of 10 gm of AB_2 & 14 gm A_2B in 100 gm of solvent in their respective solution (AB_2 & A_2B both are non - electrolytes) is $1\frac{mol}{kg}$ in both case. Hence atomic wt. of A and B are respectively.-

(1) 100, 40 (2) 60, 20

(3) 20, 60 (4) None of these

483. A solution X of A and B contains 30 moles % of A & is in equilibrium with its vapour that contains 40 mole % of B. the ratio of V.P of pure, A & B will be -

(1) 2 : 7 (2) 7 : 2

(3) 3 : 4 (4) 4 : 3

484. The plots of $\frac{1}{Y_A}$ Vs $\frac{1}{X_A}$ (Where X_A & Y_A are the mole fraction of liquid A in liquid & vap. phase) is linear with slope & intercepts respectively. -

(1) $\frac{P_A^0}{P_B^0}$ & $\frac{P_A^0 - P_B^0}{P_B^0}$ (2) $\frac{P_B^0}{P_A^0}$ & $\frac{P_A^0 - P_B^0}{P_A^0}$

(3) $\frac{P_B^0}{P_A^0}$ & $\frac{P_A^0 - P_B^0}{P_B^0}$ (4) $\frac{P_B^0}{P_A^0}$ & $\frac{P_B^0 - P_A^0}{P_B^0}$

485. A mixture of two volatile liquids A and B for 1 and 3 moles respectively has V.P. of 300 mm at 27°C. If one mole of A is further added to this solution, the vapour pressure becomes 290 mm at 27°C. The vapour pressure of pure A is-

(1) 250mm (2) 316mm

(3) 220mm (4) 270mm

486. If liquid A and B form ideal solution, then :-

(1) $\Delta G_{mix} = 0$

(2) $\Delta H_{mixing} = 0$

(3) $\Delta G_{mix} = 0$, $\Delta S_{mix} = 0$

(4) $\Delta S_{mix} = 0$

487. Which of the following liquid pairs shows a positive deviation from Raoult's law?-

(1) Water - hydrochloric acid

(2) Benzene - methanol

(3) Water - nitric acid

(4) Acetone - chloroform

488. Benzene and toluene form nearly ideal solutions. At 20°C, the vapour pressure of benzene is 75 torr and that of toluene is 22 torr. The partial pressure of benzene at 20°C for a solution containing 78 g of benzene and 46 g of toluene in torr is -

(1) 25 (2) 50

(3) 53.5 (4) 37.5

489. Vapour pressure of pure benzene is 119 torr and that of toluene is 37.0 torr at the same temperature. Mole fraction of toluene in vapour phase which is in equilibrium with a solution of benzene and toluene having a mole fraction of toluene 0.50, will be :

(1) 0.137 (2) 0.205

(3) 0.237 (4) 0.435

490. Consider separate solution of 0.500 M $C_2H_5OH(aq)$, 0.100 M $Mg_3(PO_4)_2(aq)$, 0.250 M KBr(aq) and 0.125 M $Na_3PO_4(aq)$ at 25°C. Which statement is true about these solutions, assuming all salts to be strong electrolytes ?

(1) 0.125 M Na_3PO_4 (aq) has the highest osmotic pressure.

(2) 0.500 M C_2H_5OH (aq) has the highest osmotic pressure.

(3) They all have the same osmotic pressure.

(4) 0.100 M $Mg_3(PO_4)_2$ (aq) has the highest osmotic pressure.

491. The vapour pressure of acetone at 20°C is 185 torr. When 1.2 g of non-volatile substance was dissolved in 100 g of acetone at 20°C, its vapour pressure was 183 torr. The molar mass (g mol^{-1}) of the substance is :

(1) 128 (2) 488

(3) 32 (4) 64

492. An aqueous solution of a salt MX_2 at certain temperature has a van't Hoff factor of 2. The degree of dissociation for this solution of the salt is :

(1) 0.50 (2) 0.80

(3) 0.67 (4) 0.33

493. The freezing point of benzene decreases by 0.45°C when 0.2 g of acetic acid is added to 20 g of benzene. If acetic acid associates to form a dimer in benzene, percentage association of acetic acid in benzene will be :-

(K_f for benzene = 5.12 K kg mol^{-1})

(1) 64.6% (2) 80.4%

(3) 74.6% (4) 94.6%

494. Which statement about the composition of vapour over an ideal 1 : 1 molar mixture of benzene and toluene is correct ? Assume the temperature is constant at 25°C.

Vapour pressure date (25°C)

Benzene 75 mm Hg

Toluene 22 mm Hg

(1) The vapour will contain higher percentage of benzene

(2) The vapour will contain higher percentage of toluene

(3) The vapour will contain equal amount of benzene and toluene

(4) Not enough information is given to make a prediction

495. A solute 'S' undergoes a reversible trimerization when dissolved in a certain solvent. The boiling point elevation of its 0.1 molal solution was found to be identical to the boiling point elevation in case of a 0.08 molal solution of a solute which neither undergoes association nor dissociation. To what percent had the solute 'S' undergone trimerization ?

(1) 30% (2) 40%
(3) 50% (4) 60%

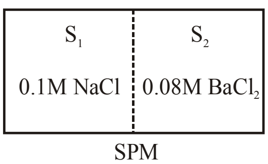
496. Consider a binary mixture of voltile liquids. If a $X_A = 0.4$ the vapour pressure of solution is 580 torr then the mixture could be ($p_A^\circ = 300$ torr, $p_B^\circ = 800$ torr) :-

(1) $CHCl_3–CH_3COCH_3$
(2) $C_6H_5Cl–C_6H_5Br$
(3) $C_6H_6–C_6H_5CH_3$
(4) $nC_6H_{14}–nC_7H_{16}$

497. A 20.0 mL sample of $CuSO_4$ solution was evaporated to dryness, leaving 0.967 g of residue. What was the molarity of the original solution ? (Cu = 66.5) :-

(1) 48.4 M (2) 0.0207 M
(3) 0.0484 M (4) 0.303 M

498. Two solutions S_1 and S_2 containing 0.1M NaCl(aq.) and 0.08M $BaCl_2$(aq.) are separated by semipermeable membrane. Which among the following statement(s) is/are correct -



(1) S_1 and S_2 are isotonic
(2) S_1 is hypertonic and S_2 is hypotonic
(3) S_1 is hypotonic and S_2 is hypertonic
(4) Flow of NaCl will take place from S_1 to S_2

499. The solubility of N_2 in water at 300 K and 500 torr partial pressure is 0.01 g L^{-1}. The solubility (in g L^{-1}) at 750 torr partial pressure and 300K is :

(1) 0.02 (2) 0.005
(3) 0.015 (4) 0.0075

500. 12g of a nonvolatile solute dissolved in 108g of water produces the relative lowering of vapour pressure of 0.1. The molecular mass of the solute is :

(1) 60 (2) 80
(3) 40 (4) 18

CHEMICAL KINETICS

501. Which of the following rate law expression represent zero order reaction :-

(1) $K(A)^{3/2}\ (B)^{-1}\ (C)^{1/2}$
(2) $K(A)^{o}\ (B)^{2}$
(3) $K(A)^{1/2}\ (B)^{1/2}\ (C)^{-1}$
(4) $K(A)^{2}\ (B)^{1}$

502. The ratio of $t_{0.875}$ and $t_{0.75}$ for first order reaction:-

(1) 4 : 3 (2) 3 : 2
(3) 2 : 1 (4) 1 : 2

503. The correct expression for Arrehenius equation:-

(1) $\ell nk = \ell nA - \frac{E}{RT}$

(2) $ke^{-E/RT} = A$

(3) $\log k = \log A - 2.303\frac{E}{RT}$

(4) $K = A^{-1}e^{-E/RT}$

504. For every 10°C increase in temperature the rate becomes twice. If the temperature increases from 10°C to 100°C then rate increases .... times :-

(1) 400 (2) 512
(3) 112 (4) 614

505. The specific rate of reaction for the first order reaction depends upon :-

(1) pressure

(2) temperature

(3) concentration of reactant

(4) concentration of the product

506. The rate contant for zero order reaction is 3×10^{-2} mol L^{-1} sec^{-1}. After 25 sec. If the concentration of reactant is 0.5M then initial concentration of the reactant is :-

(1) 1 M (2) 1.25 M

(3) 0.5 M (4) 0.75

507. The plot of ℓn K versus $\frac{1}{T}$ is linear with slope of:-

(1) $-\frac{E_a}{R}$ (2) $+\frac{E_a}{R}$

(3) $-\frac{E_a}{2.303R}$ (4) $\frac{E_a}{2.303R}$

508. For a first order gaseous reaction A $\rightarrow$ 2B+C, initial pressure is P_i and total pressure after time 't' is P_t then expression of rate constant:-

(1) $K = \frac{2.303}{t} \log \frac{2P_i}{3P_i - P_t}$

(2) $K = \frac{2.303}{t} \log\left(\frac{2P_i}{2P_t - P_i}\right)$

(3) $K = \frac{2.303}{t} \log\left(\frac{P_i}{P_i - P_t}\right)$

(4) None of these

509. In a reaction $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$ rate of appearance of NH_3 is 2.5×10^{-4} mol L^{-1} sec^{-1} then rate of reaction and rate of disappearance of H_2 respectively is:-

(1) 3.75×10^{-4}, 1.25×10^{-4}

(2) 1.25×10^{-4}, 2.5×10^{-4}

(3) 1.25×10^{-4}, 3.75×10^{-4}

(4) 5×10^{-4}, 3.75×10^{-4}

510. Rate of formation of SO_3 in this reaction is 1.6×10^{-3} kg/min

$2SO_2 + O_2 \rightarrow 2SO_3$ then rate of decomposition of $SO_{2(g)}$ is :-

(1) 1.6×10^{-3} kg/min

(2) 8×10^{-4} kg/min

(3) 3.2×10^{-3} kg/min

(4) 1.28×10^{-3} kg/min

511. Calculate the order of the reaction with respect to A and B :-

A (mol/l)	B (mol/l)	Rate
0.05	0.05	1.2×10^{-3}
0.10	0.05	2.4×10^{-3}
0.05	0.10	1.2×10^{-3}

(1) 1 and 0 (2) 1 and 1

(3) 0 and 1 (4) none of these

512. The rate law of reaction A + 2B $\rightarrow$ product is given by

$r = K(A)^2 (B)^1$. If A is taken in excess the order of the reaction will be :-

(1) zero (2) 1

(3) 2 (4) 3

513. The half life for the first order reaction A $\longrightarrow$ B + C is 24 hrs. starting with 20g of A how many grams of A will remain after 72 hours. :-

(1) 1.25 g (2) 0.63 g

(3) 1.77 g (4) 2.5 g

514. Rate of reaction increases with temperature :-

(1) for any reaction

(2) for exothermic reaction only

(3) for endothermic reaction only

(4) for none

515. If concentration of reactant is increased by 2 times then K becomes :-

(1) $\ln\frac{K}{2}$ (2) $\frac{K}{2}$ (3) 2K (4) K

516. K is rate constant at temp T then value of $\lim_{T\to\infty}\log K$ is equal to :-

(1) A/2.303 (2) A

(3) 2.303 A (4) log A

517. The activation energy for a chemical reaction depends upto :-

(1) Temperature

(2) Nature of reacting species

(3) Concentration of the reacting species

(4) Collision frequency

518. For an endomermic reaction, energy of activation is E_a and enthalpy of reaction is ΔH (both of these in KJ/mol) The value of E_a will be :-

(1) equal to zero (2) less than ΔH

(3) equal to ΔH (4) more than ΔH

519. For a reaction $r = K[A]^{3/2}$ then unit of rate of reaction and rate constant respectively :-

(1) $molL^{-1}\ s^{-1}$, $mol^{-1/2}\ L^{1/2}\ s^{-1}$

(2) $mol^{-1}L^{-1}s^{-1}$, $mol^{-1/2}\ L^{1/2}\ s^{-1}$

(3) $molL^{-1}s^{-1}$, $mol^{+1/2}\ L^{1/2}\ s^{-1}$

(4) mol, $mol^{+1/2}\ L^{1/2}\ s$

520. The rate of first order reaction is 1.5×10^{-2} mol $L^{-1}\ min^{-1}$ at 0.5M concentration of the reactant. The half life of the reaction is :-

(1) 7.53 min (2) 0.383 min

(3) 23.1 min (4) 8.73 min

521. The $t_{1/2}$ of a reaction is halved as the initial concentration of the reactant is doubled then order of reaction :-

(1) 1 (2) 0

(3) 2 (4) 3

522. For this reaction

$$N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$$

The relation beteween $\frac{d(NH_3)}{dt}$ and $-\frac{d(H_2)}{dt}$ is :-

(1) $\frac{d(NH_3)}{dt} = -\frac{1}{3}\frac{d(H_2)}{dt}$

(2) $+\frac{d(NH_3)}{dt} = -\frac{2}{3}\frac{d(H_2)}{dt}$

(3) $+\frac{d(NH_3)}{dt} = -\frac{3}{2}\frac{d(H_2)}{dt}$

(4) $\frac{d(NH_3)}{dt} = -\frac{d(H_2)}{dt}$

523. Activation energy of the reaction $A \rightarrow B$ –52KCal is 128KCal. then activation energy of the reaction $B \rightarrow A$ will be :-

(1) –76 KCal (2) +76 KCal

(3) –180 KCal (4) +180 KCal

524. For a reaction rate = $K(A)^{1/2}\ (B)^2$. If concentration of A and B are increased by factor of 4 and 2 respectively then rate is changed by the factor of :-

(1) 4 (2) 6

(3) 8 (4) None of these

525. The reaction 2A+B → product follows the mechanism :-

$$2A \rightleftharpoons A_2\text{(fast)}$$

$$A_2 + B \rightarrow P\text{(slow)}$$

The order of the reaction is

(1) 1.5 (2) 3 (3) 1 (4) 2

526. Which of following is correct for zero order and first order (a → Initial cencentration) :-

(1) $t_{1/2} \propto a,\ t_{1/2} \propto \frac{1}{a}$ (2) $t_{1/2} \propto a,\ t_{1/2} \propto a^{o}$

(3) $t_{1/2} \propto a^{o},\ t_{1/2} \propto a$ (4) $t_{1/2} \propto a,\ t_{1/2} \propto \frac{1}{a^2}$

527. Which of following represents the expression for $\frac{3}{4}$th life of first order reaction :-

(1) $\frac{K}{2.303}\log\frac{4}{3}$ (2) $\frac{2.303}{K}\log\frac{3}{4}$

(3) $\frac{2.303}{K}\log 4$ (4) $\frac{2.303}{K}\log 3$

528. The rate constant for first order reaction whose half life is 480 sec. :-

(1) $1.44 \times 10^{-3}\ sec^{-1}$

(2) $1.44\ sec^{-1}$

(3) $0.72 \times 10^{-3}\ sec^{-1}$

(4) $2.88 \times 10^{-3}\ sec^{-1}$

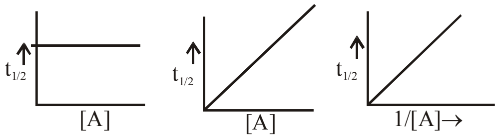
529. $A \xrightarrow{K=2\times10^{-5}} B$; $A \xrightarrow{K=3\times10^{-5}} C$

For this parallel first order reaction then find out percentage of B = ?

(1) 40% (2) 60% (3) 50% (4) 90%

530. consider the plots for a reaction nA → B + C

These plots respectively correspond to the reaction order



(I) (II) (III)

(1) 0,1,2 (2) 1,2,3

(3) 1,0,2 (4) None of these

531. The half life of a first order reaction is 6 hour. How long will it take for the concentration of reactant to change from 0.8 M to 0.25 M?

(1) 1.07 hour (2) 5.1 hour

(3) 2.7 hour (4) 10.07 hour

532. The concentration of a reactant in solution decreases from 0.5 M to 0.25 M in 5 hours and from 1.0 M to 0.25 M is 10 hours. The order of the reaction will be

(1) 2 (2) 1 (3) 3 (4) 0

533. Which of the following is the correct expression for Arrhenius equation?

(1) $\ln\frac{K_2}{K_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

(2) $\log\frac{K_2}{K_1} = \frac{E_a}{2.303}\left(\frac{T_1T_2}{T_2 - T_1}\right)$

(3) $\ln\frac{K_2}{K_1} = \frac{E_a}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

(4) $\frac{K_2}{K_1} = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1T_2}\right)$

534. For a first order reaction A→B, the time taken to reduce to 1/4 of initial concentration is 10 min. The time required to reduce to 1/16 of initial concentration will be

(1) 10 min

(2) 20 min

(3) 4.46 min

(4) 2.24 min

535. For the reaction, $2N_2O_5 \rightarrow 4NO_2 + O_2$ rate and rate constant are 1.02×10^{-4} M sec^{-1} and $3.4 \times 10^{-5}\ sec^{-1}$ respectively, the concentration of N_2O_5, at that time will be

(1) 1.732 M

(2) 3M

(3) 1.02×10^{-4} M

(4) 3.5×10^{5}M

536. For the reaction $N_2O_5(g) \rightarrow N_2O_4(g) + ½O_2(g)$ initial pressure is 114 mm and after 20 seconds the pressure of reaction mixture becomes 133 mm then the average rate of reaction will be

(1) 1.9 atm S^{-1}

(2) 8.75×10^{-3} atm S^{-1}

(3) 2.5×10^{-3} atm S^{-1}

(4) 6.65 atm S^{-1}

537. Reaction A+B →C+D is started with 1M of each A and B and follows the rate law

$$r = k[A]^{1/2}[B]^{1/2}$$

What is the time taken for the concentration of A to drop to 0.1M (K = 2.303×10^{-3} sec^{-1})

(1) 57 sec (2) 100 sec

(3) 434 sec (4) 1000 sec

538. 75% of a first order reaction was found to complete in 32 minute. When will 50% of the same reaction will complete

(1) 24 min (2) 16 min

(3) 8 min (4) 4 min

539. Half lives of a first order and zero order are same. Then the ratio of the initial rates of the first order reaction to that of zero order reaction is [initial conc. of reactant = 1 Molar]

(1) 1/0.693 (2) 2 × 0.693

(3) 2/0.693 (4) 6.93

540. The rate of reaction is expressed

$$+\frac{1}{2}\frac{d[C]}{dt} = -\frac{1}{3}\frac{d[D]}{dt} = +\frac{1}{4}\frac{d[A]}{dt} = \frac{-d[B]}{dt}$$

the reaction is

(1) 4A + B →2C +3D

(2) B + 3D → 4A + 2C

(3) A+B→ C+D

(4) B+D→ A+C

541. Rate constant for first order reaction is 5.78×10^{-5} sec^{-1}. What percentage of initial reactant will react in 10 hours?

(1)12.5% (2) 25% (3) 87.5% (4) 75%

542. The half life period and initial concentration for a reaction are as follows:

Initial concentration	350	540	158
t ½	425	275	941

order of reaction is–

(1)2 (2) 1 (3)0 (4) 4

543. The activation energy for the reaction–

$$H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2$$

is 18 K cal/mol at 300 K. calculate the fraction of molecules of reactonts having energy equal to or greater than activation energy?

Anti log (–13.02) = 9.36×10^{-14}

(1) 9.36×10^{-14} (2) 1.2×10^{-12}

(3) 4.2×10^{-16} (4) 5.2×10^{-15}

544. A first order reaction is 50% completed in 20 minutes at 27°C and 5 minutes at 47°C. The energy of activation of the reaction is–

(1)43.85 KJ (2)55.14 KJ

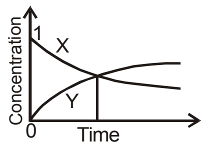
(3)11.97KJ (4) 6.65 KJ

545. The accompanying figure depicts the change in concentration of species X and Y for the reaction X → 3Y as a function of time the point of intersection of the two curves represents.

(1) $t_{1/2}$

(2) $t_{3/4}$

(3) $t_{1/4}$



(4) Data insufficient to predict

546. For the reaction $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ under certain conditions of temperature and partial pressure of the reactants, the rate of formation of NH_3 is 0.001 kg lit^{-1} h^{-1}. The rate of conversion of H_2 under the same condition is ..... kg lit^{-1} h^{-1}.

(1) 0.001 kg lit^{-1} h^{-1}

(2) 0.0015 kg lit^{-1} h^{-1}

(3) 0.00017 kg lit^{-1} h^{-1}

(4) 0.0017 kg lit^{-1} h^{-1}

547. For a first order reaction -

(1) the degree of dissociation is equal to e^{-kt}

(2) a plot of reciprocal concentration of the reactant vs time gives a straight line

(3) the time taken for the completion of 75% reaction is thrice the $t_{1/2}$ of the reaction

(4) the pre-exponential factor in the Arrhenius equation has the dimension of time, T^{-1}

548. In the following reaction : $xA \longrightarrow yB$

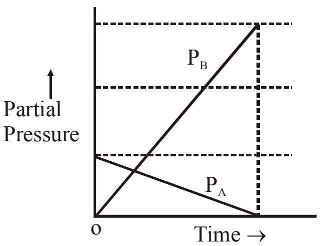
$$\log\left[-\frac{d[A]}{dt}\right] = \log\left[\frac{d[B]}{dt}\right] + 0.3010$$

where –ve sign indicates rate of disappearance of the reactant. Thus, x : y is :

(1) 1 : 2 (2) 2 : 1

(3) 3 : 1 (4) 3 : 10

549. If for a reaction in which A(g) converts to B(g) the reaction carried out at const. V & T results into the following graph.



(1) then the reaction must be $A(g) \rightarrow 3B(g)$ and is a first order reaction

(2) then the reaction must be $A(g) \rightarrow 3B(g)$ and is a second order reaction

(3) then the reaction must be $A(g) \rightarrow 3B(g)$ and is a zero order reaction

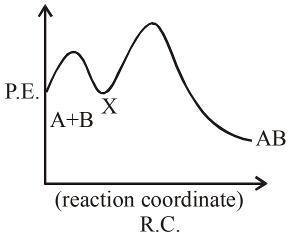
(4) then the reaction must be $A(g) \leftrightarrow 3B(g)$ and is a first order reaction

550. For an exothermic chemical process occurring in two steps as follows

(i) $A + B \rightarrow X$ (slow)

(ii) $X \rightarrow AB$ (fast)

the process of reaction can be best described by :

(1)


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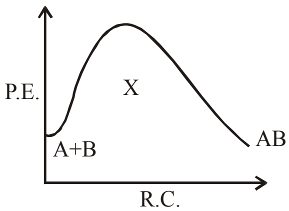
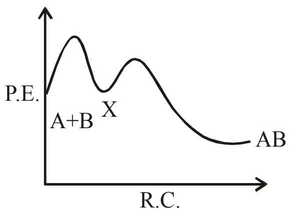
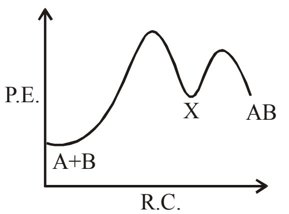


(3)


(4)


551. Arrhenius equation may be written as :–

(1) $\frac{d}{dT}(\ell n\ K) = -\frac{Ea}{RT}$

(2) $\frac{d}{dT}(\ell n\ K) = -\frac{Ea}{RT^2}$

(3) $\frac{d}{dT}(\ell n\ K) = +\frac{Ea}{RT^2}$

(4) $\frac{d}{dT}(\ell n\ K) = \frac{Ea}{RT}$

552. Consider the reaction :

$Cl_2(aq) + H_2S(aq) \rightarrow S(s) + 2H^+(aq) + 2Cl^-(aq)$

The rate equation for this reaction is

rate = $k[Cl_2][H_2S]$

Which of these mechanisms is/are consistent with this rate equation ?

A. $Cl_2 + H_2S \rightarrow H^+ + Cl^- + Cl^+ + HS^-$ (slow)

$Cl^+ + HS^- \rightarrow H^+ + Cl^- + S$(fast)

B. $H_2S \Leftrightarrow H^+ + HS^-$ (fast equilibrium)

$Cl_2 + HS^- \rightarrow 2Cl^- + H^+ + S$(slow)

(1) A only (2) B only

(3) Both A and B (4) Neither A nor B

553. For the first order reaction, which is not correct

$2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$

(1) the concentration of the reactant decreases exponentially with time.

(2) the half-life of the reaction decreases with increasing temperature.

(3) the half-life of the reaction depends on the initial concentration of the reactant.

(4) the reaction proceeds to 99.6% completion in eight half-life duration.

554. Reaction rate between two substances A and B is expressed as following : rate = $k[A]^n [B]^m$ If the concentration of A is doubled and concentration of B is made half of initial concentration, the ratio of the new rate to the earlier rate will be :-

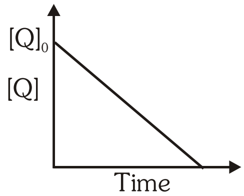
(1) m + n (2) $\frac{1}{2^{(m+n)}}$ (3) $2^{(n-m)}$ (4) n – m

555. An organic compound undergoes first-order decomposition . The time taken for its decomposition of 1/8 and 1/10 of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively. What is the value of $\frac{[t_{1/8}]}{t_{1/10}} \times 10$? (take $\log_{10}2 = 0.3$)

(1) 4 (2) 5 (3) 15 (4) 2

556. In the reaction : $P + Q \longrightarrow R + S$

the time taken for 75% reaction of P is twice the time taken for 50% reaction of P. The concentration of Q varies with reaction time as shown in the figure. The overall order of the reaction is



(1) 2 (2) 3 (3) 0 (4) 1

557. For the reaction $2A + B \rightarrow C + D$, the following kinetic data were obtained in three separate experiments, all at 298 K.

Initial Concentration (A)	Initial Concentration (B)	Initial rate of formation of C (mol L⁻S⁻)
0.1M	0.1M	1.2×10^{-3}
0.1M	0.2M	1.2×10^{-3}
0.2M	0.1M	2.4×10^{-3}

(1) $\frac{d[C]}{dt} = k[A][B]^2$ (2) $\frac{d[C]}{dt} = k[A]$

(3) $\frac{d[C]}{dt} = k[A][B]$ (4) $\frac{d[C]}{dt} = k[A]^2[B]$

558. The reaction of ozone with oxygen atoms in the presence of chlorine atoms as a catalyst can occur by a two step process shown below :

$O_3(g) + Cl^{\bullet}(g) \rightarrow O_2(g) + ClO^{\bullet}(g)$ ......(i)

$k_i = 5.2 \times 10^9$ L mol⁻¹s⁻¹

$ClO^{\bullet}(g) + O^{\bullet}(g) \rightarrow O_2(g) + Cl^{\bullet}(g)$

$k_{ii} = 2.6 \times 10^{10}$ L mol⁻¹s⁻¹ ....(ii)

The closest rate constant for the overall reaction $O_3(g) + O^{\bullet}(g) \rightarrow 2O_2(g)$ is :

(1) 3.1×10^{10} L mol⁻¹s⁻¹

(2) 2.6×10^{10} L mol⁻¹s⁻¹

(3) 5.2×10^{9} L mol⁻¹s⁻¹

(4) 1.4×10^{20} L mol⁻¹s⁻¹

559. Decomposition of hydrogen peroxide as per following reaction $H_2O_2 \rightarrow H_2O + \frac{1}{2}O_2$; follows a first order reaction. In fifty minutes the concentration of H_2O_2 decreases from 0.5 to 0.125 M in one such decomposition. When the concentration of H_2O_2 reaches 0.05 M, the rate of formation of O_2 will be :-

(1) 1.34×10^{-2} M min^{-1}

(2) 6.93×10^{-2} M min^{-1}

(3) 6.93×10^{-4} M min^{-1}

(4) 2.66 L min^{-1} at STP

560. Two reactions R_1 and R_2 have identical pre-exponential factors. Activation energy of R_1 exceeds that of R_2 by 10 kJ mol^{-1}. If k_1 and k_2 are rate constants for reactions R_1 and R_2 respectively at 300 K, then $\ln(k_2/k_1)$ is equal to:- (R = 8.314 J $mol^{-1}K^{-1}$)

(1) 8 (2) 12

(3) 6 (4) 4

561. $\frac{k_{35^0}}{k_{34^0}} > 1$, 1f this means that -

(1) Rate increases with the rise in temperature

(2) Rate decreases with rise in temperature

(3) Rate does not change with rise in temperature

(4) None of the above

562. A radioactive isotope decomposes according to the first order with half life period of 15 hrs. 80% of the sample will decompose in -

(1) 15 × 0.8 hr.

(2) 15 × (log 8) hr.

(3) 15 × (log5 / log2) hr. (or 34.83) hr.

(4) 15 × 10/8 hr.

563. The half-life period for a reaction at initial concentrations of 0.5 and 1.0 mol L^{-1} are 200 sec and 100 sec respectively. The order of the reaction is -

(1) 0 (2) 1

(3) 2 (4) 3

564. Consider the data given below for hypothetical reaction A → X.

Time (sec)	**Rate (mol L^{-1} $sec.^{-1}$)**
0	1.60×10^{-2}
10	1.60×10^{-2}
20	1.60×10^{-2}
30	1.60×10^{-2}

From the above data, the order of reaction is-

(1) 1 (2) 0

(3) 2 (4) Unpredictable

565. For a first order reaction A → products, the concentration of [A] is reduced from 1M to 0.125 M in one hour, the $t_{1/2}$ of this reaction (in sec) is :-

(1) 600 (2) 300

(3) 1200 (4) 0.693/1200

566. For a zero order reaction, the plot of dissociated concentration (x) vs time is linear with -

(1) +ve slope and zero intercept

(2) –ve slope and zero intercept

(3) +ve slope and non-zero intercept

(4) –ve slope and non-zero intercept

567. The rate of reaction at 273 K is R_0. The rate of reaction at 313 K will be (Assuming temperature coefficient equal to 2) -

(1) 16 R_0 (2) 64 R_0

(3) $R_0/32$ (4) $R_0/16$

568. At 100°C, the gaseous reaction A → 2B + C is found to be of first order. Starting with pure A, if at the end of 10 min, the total pressure of the system is 176 mm and the end of reaction, it is 270 mm, the partial pressure of A at the end of 10 min is :-

(1) 94 mm (2) 43 mm

(3) 47 mm (4) 176 mm

569. If for two reaction $E_{a_1} > E_{a_2}$ & TC_1 & TC_2 are temperature coefficient respectively, then which of the following is correct :-

(1) $TC_1 > TC_2$ (2) $TC_1 < TC_2$
(3) $TC_1 = TC_2$ (4) None of these

570. H_2 gas is adsorbed on metal surface like tungsten. This follows ...... order reaction at high pressure.

(1) Third (2) Second
(3) Zero (4) First

571. $t_{1/4}$ can be taken as the time taken for the concentration of a reactant to drop to 3/4 of its initial value. If the rate constant for a first order reaction is K, $t_{1/4}$ can be written as :-

(1) 0.29/K (2) 0.10/K
(3) 0.75/K (4) 0.69/K

572. The E_a of a reaction in the presence of a catalyst is 4.15 kJ mol^{-1} and in the absence of catalyst is 8.3 kJ mol^{-1}. What is the slope of the plot of ℓn k vs $\frac{1}{T}$ in the absence of catalyst :-

(1) +l (2) –l
(3) +1000 (4) –1000

573. For a certain reaction of order 'n' the time for half change $t_{1/2}$ is given by : $t_{1/2} = \frac{2-\sqrt{2}}{K} \times C_0^{1/2}$ where K is rate constant and C_0 is the initial concentration. The value of n is :-

(1) 1 (2) 2 (3) 0 (4) 0.5

ELECTROCHEMISTRY

574. A hypothetical electrochemical cell is shown below

$$Pt(s) | A_2(1atm) | A^-(xM) || B^+(yM) | B(s)$$

The emf measured is 0.30V. The cell reaction is :-

(1) $A_2 + 2B^+ \rightarrow 2A^- + 2B$
(2) $A_2 + 2e^- \rightarrow 2A^-$; $2B^+ + 2e^- \rightarrow 2B$
(3) The cell reaction cannot be predicted
(4) $2A^- + 2B^+ \rightarrow A_2 + 2B$

575. If $E^o_{A^{+2}/A} = -0.30$ V and $E^o_{A^{+3}/A^{+2}} = 0.40$ V, the standard Emf of the reaction : $A + 2A^{+3} \rightarrow 3A^{+2}$ will be :-

(1) 0.30 V (2) 0.40 V
(3) 0.70 V (4) 0.10 V

576. The equilibrium constant of the reaction :

$Zn(s) + 2Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$; $E° = 1.50V$

at 298 K is

(1) 2.6×10^{49} (2) 8.7×10^{51}
(3) 6.1×10^{30} (4) 6.6×10^{50}

577. On the basis of the following E° values, the strongest oxidising agent is :-

$K^+ + e^- \rightarrow K$ $E^0 = -2.923$ V
$Mg^{+2} + 2e^- \rightarrow Mg$ $E^o = -2.337$ V

(1) K^+ (2) Mg^{+2}
(3) K (4) Mg

578. The equivalent conductance of $\frac{M}{20}$ solution of a weak monobasic acid is 10 S cm^2/eq. and at infinite dilution is 200 S cm^2/eq. The dissociation constant of this acid is :-

(1) 1.25×10^{-4}
(2) 1.25×10^{-5}
(3) 1.25×10^{-6}
(4) 6.26×10^{-4}

579. Given $Fe^{+2} + 2e^- \rightarrow Fe(s)$ E° = – 0.447V
$Fe^{+2} \rightarrow Fe^{+3} + e^-$ E° = – 0.771V

Find E° for the reaction $Fe^{+3} + 3e^- \rightarrow Fe(s)$

(1) 0.34 V (2) –0.041V
(3) +0.39 V (4) –0.47 V

580. An increase in molar conductance of a strong electrolyte with dilution is mainly due to :-

(1) Increase in number of ions
(2) Increase in ionic mobility of ions
(3) 100% ionisation of electrolyte at normal dilution
(4) Increase in both i.e. number of ions and ionic mobility of ions.

581. For the reduction of silver ions with copper metals, the standard cell potential was found to be + 0.40V at 25°C. The value of standard Gibb's energy $\Delta G°$ will be ($F = 96500\ C\ mol^{-1}$):-

(1) –70 KJ (2) –77.2 KJ

(3) –88 KJ (4) +90KJ

582. Which relation of emf of an electrochemical cell is correct :-

(1) emf of cell = oxidation potential of anode – reduction potential of cathode

(2) emf of cell = oxidation potential of anode + reduction potential of cathode

(3) emf of cell = reduction potential of anode + reduction potential of cathode

(4) All of these

583. Which of the following expression correctly represents the equivalent conductance at infinite dilution of $Ca_3(PO_4)_2$ Given $\Lambda^o_{Ca^{+2}}$ and $\Lambda^o_{PO_4^{-3}}$ are the equivalent conductance at infinite dilution of the respective ions :-

(1) $\Lambda^o_{Ca^{+2}} + \Lambda^o_{PO_4^{-3}}$ (2) $\left(\Lambda^o_{Ca^{+2}} + \Lambda^o_{PO_4^{-3}}\right) \times 6$

(3) $\frac{1}{2}\Lambda^o_{Ca^{+2}} + \frac{1}{3}\Lambda^o_{PO_4^{-3}}$ (4) $3\Lambda^o_{Ca^{+2}} + 2\Lambda^o_{PO_4^{-3}}$

584. If the E^o_{cell} for a given reaction has a positive value, then which of the following gives the correct relationship for the values of $\Delta G°$ and K_{eq}:-

(1) $\Delta G^o < 0$; $K_{eq} < 1$

(2) $\Delta G^o > 0$; $K_{eq} > 1$

(3) $\Delta G^o > 0$; $K_{eq} < 1$

(4) $\Delta G^o < 0$; $K_{eq} > 1$

585. Standard electrode potential Au^{+3}/Au couple is 1.428 V and that for Na^+/Na couple is –2.714 V. These two couples in their standard state are connected to make a cell. The cell potential will be :-

(1) 4.142 (2) 1.286 V

(3) –1.286 V (4) 3.376 V

586. Standard electrode potential of three metals A,B and C are –1.5V, 0.8V and –2.7V respectively. The reducing power of these metals will be :-

(1) B > C > A (2) B > A > C

(3) C > A > B (4) A > B > C

587. A solution contains Cr^{+2}, Cr^{+3} and I^- ions. This solution was treated with iodine at 35°C. E° for Cr^{+3}/Cr^{+2} is –0.41V and E° for $I_2/2I^-$ = 0.536V, The favourable redox reaction is :-

(1) Cr^{+2} will be oxidised to Cr^{+3}

(2) Cr^{+3} will be reduced to Cr^{+2}

(3) There will be no redox reaction

(4) I^- will be oxidised to I_2

588. Limiting molar conductivity of CH_3COOH (i.e. $\Lambda^0_{m(CH_3COOH)}$ is equal to :-

(1) $\Lambda^o_{m(CH_3COOH)} + \Lambda^o_{m(CH_3COONa)} - \Lambda^o_{m(NaOH)}$

(2) $\Lambda^o_{m(CH_3COONa)} + \Lambda^o_{m(HCl)} - \Lambda^o_{m(NaCl)}$

(3) $\Lambda^o_{m(CH_3COONa)} + \Lambda^o_{m(NaCl)} - \Lambda^o_{m(NaOH)}$

(4) $\Lambda^o_{m(NaOH)} + \Lambda^o_{m(NaCl)} - \Lambda^o_{m(CH_3COONa)}$

589. The weight of copper (At wt=63.5) displaced by a quantity of electricity which displaces 5600 ml of O_2 at STP will be :-

(1) 63.5 g (2) 31.75 g

(3) 15.875 g (4) 127 g

590. When 0.1 mole $Cr_2O_7^{-2}$ is reduced then quantity of electricity required to reduced $Cr_2O_7^{-2}$ to Cr^{+3} completely is :-

(1) 9650 C (2) 96500 C

(3) 57900 C (4) 54900 C

591. A device that converts energy of combustion of fuel like hydrogen and methane directly into electrical energy is known as :-

(1) Electrolytic cell (2) Dynamo

(3) Ni-Cd cell (4) Fuel cell

592. Aqueous solution of which of the following compounds is the best conductor of electric current :-

(1) CH_3COOH (2) $C_{12}H_{22}O_{11}$

(3) $H_2C_2O_4$ (4) NaOH

593. The pressure of H_2 required (in bar) to make the potential of H_2-electrode zero in pure water at 25°C is :-

(1) 10^{-14} (2) 10^{-7} (3) 10^{+14} (4) 10^{+7}

594. The number of electrons delivered at the cathode during electrolysis by a current of 2 ampere in 120 seconds is

(charge on electron = 1.6×10^{-19} C) :-

(1) 3.75×10^{20} (2) 7.48×10^{20}

(3) 1.5×10^{21} (4) 3×10^{20}

595. The molar conducitviy of a 0.1 mol/dm^3 solution of KNO_3 with electrolytic conductivity of 4×10^{-3} Scm^{-1} at 298 K is :-

(1) 11.52 Scm^2 mol^{-1} (2) 20 Scm^2 mol^{-1}

(3) 40 Scm^2 mol^{-1} (4) 13.48 Scm^2 mol^{-1}

596. The resistance of conductivity cell containing 0.001 M KCl solution at 298 K is 1500 ohm. What is the cell constant if the conductivity of 0.001 M KCl solution at 298 K is 0.146×10^{-3} S cm^{-1}

(1) 0.419 cm^{-1} (2) 0.219 cm^{-1}

(3) 0.45 cm^{-1} (4) 0.75 cm^{-1}

597. Calculate the $\Delta G°$ of the following reaction :-

$Fe^{+2}(aq) + Ag^{+}(aq) \rightarrow Fe^{+3}(aq) + Ag(s)$

$E^0_{Ag^+/Ag} = 0.8V$ $E^0_{Fe^{+3}/Fe^{+2}} = 0.77V$

(1) –2895 J mol^{-1}

(2) –3845 J mol^{-1}

(3) –1874 J mol^{-1}

(4) –375 J mol^{-1}

598. Calculate the emf of the cell

$Fe(s) + 2H^{+}(1M) \rightarrow Fe^{+2}(0.001M) + H_2/(g), 1atm)$

(Given : $E^0_{Fe^{+2}/Fe} = -0.44V$)

(1) 0.215 V (2) 0.38 V

(3) –0.48V (4) 0.5285 V

599. A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode ? (Molar mass of Cu = 63.5 g/mol)

(1) 0.492g (2) 0.325g

(3) 0.873g (4) 0.296g

600. The conductance is the :-

(1) Reciprocal of specific resistance

(2) Reciprocal of resistance

(3) Reciprocal of current

(4) Reciprocal of concentration

601. The ion which has lowest ionic mobility in aqueous solution :-

(1) Li^+ (2) Na^+ (3) K^+ (4) Rb^+

602. An electric current of 100 ampere is passed through a molten liquid of sodium chloride per 5 hours. Calculate the volume of chlorine gas liberated at the electrode at NTP :-

(1) 208.91 L (2) 310 L

(3) 40.8 L (4) 150.2 L

603. Zn can displace :-

(1) Mg^{+2} from its aqueous solution

(2) Cu^{+2} from its aqueous solution

(3) Na^+ from its aqueous solution

(4) Al^{+3} from its aqueous solution

604. Which of the following is incorrect regarding salt bridge solution ?

(1) Solution must be of strong electrolyte

(2) Solution should be inert towards both electrodes

(3) Size of cation and anions of salt should be much different

(4) Salt bridge solution is prepared in gelatin or agar-agar to make it semi solid.

605. Cell reaction during discharging of lead storage battery at cathode is :-

(1) $Pb + SO_4^{-2} - 2e^- \rightarrow PbSO_4$

(2) $PbSO_4 + 2e^- \rightarrow Pb + SO_4^{-2}$

(3) $PbO_2 + SO_4^{-2} + 4H^+ + 2e^- \rightarrow PbSO_4 + 2H_2O$

(4) $PbSO_4 + 2H_2O \rightarrow PbO_2 + SO_4^{-2} + 4H^+ + 2e^-$

606. The efficiency of fuel cell is given by :-

(1) $\frac{\Delta G}{\Delta S}$ (2) $\frac{\Delta G}{\Delta H}$ (3) $\frac{\Delta S}{\Delta G}$ (4) $\frac{\Delta H}{\Delta G}$

607. On electrolysis of dilute sulphuric acid using platinum electrode, the product obtained at the anode will be :-

(1) hydrogen (2) oxygen

(3) sulphur dioxide (4) hydrogen sulphide

608. Which metal will dissolve if the cell works $Cu|Cu^{2+} || Ag^{+} | Ag$:-

(1) Cu (2) Ag

(3) Both 1 & 2 (4) None of these

609. On the basis of the electrochemical theory of aeueous corrosion, the reaction occuring at the cathod is :-

(1) $O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O_{(\ell)}$

(2) $Fe_{(s)} \longrightarrow Fe^{+2}_{(aq)} + 2e^-$

(3) $Fe^{+2} \longrightarrow Fe^{+3}_{(aq)} + e^-$

(4) $H_{2(g)} + 20H^-_{(aq)} \longrightarrow 2H_2O_{(\ell)} + 2e^-$

610. When lead storage battery is charged it acts as :-

(1) Fuel cell (2) Electrolytic cell

(3) Galvanic cell (4) Concentration cell

611. The zinc acts as sacrifical or cathodic protection to prevent rusting of iron because :-

(1) E^o_{OP} of Zn < E^o_{OP} of Fe

(2) E^o_{OP} of Zn > E^o_{OP} of Fe

(3) E^o_{OP} of Zn = E^o_{OP} of Fe

(4) Zn is cheaper than Fe

612. The molar conductivity of 0.01 M solution of weak acid is 16.6 Ω^{-1} cm^2 mol^{-1}. Its molar conductivity at infinite dilution is 390.7 Ω^{-1} cm^2 mol^{-1}. The degree of dissociation of weak acid is

(1) 0.24 (2) 0.42

(3) 0.042 (4) 0.024

613. The strongest oxidising agent among the species having SRP values of

In^{+3} (E^0 = –1.34 V), Au^{+3} (E^0 = 1.4V),

Hg^{+2} (E^0 = 0.86 V) and Cr^{+3} (E=–0.74V) is

(1) Cr^{+3} (2) Au^{+3} (3) Hg^{+2} (4) In^{+3}

614. Given the standard electrode potential

(a) K^+/K = – 3.02 V

(b) Cu^{+2}/Cu = + 0.34 V

(c) Hg^{+2}/Hg = 0.92 V

(d) Cr^{+3}/Cr = – 0.74 V

Decreasing order of reducing power of these element is

(1) a > b > c > d (2) a > d > c > b

(3) a > d > b > c (4) c > b > d > a

615. Calculate E_{cell} of the reaction

$Mg(s) + 2Ag^+ \rightarrow Mg^{+2} + 2Ag(s)$

(0.0001M) (0.100M)

If E^0_{cell} = 3.17 V

(1) –2.96 V (2) +2.96V

(3) 3.38 V (4) – 3.38 V

616. Which of the following statements is correct for an electrolytic cell?

(1) electrons flow from cathode to anode through external battery

(2) electrons flow from cathode to anode within the electrolytic solution

(3) Migration of ions along with oxidation reaction at cathode and reduction reaction at anode

(4) migration of ions along with reduction reaction at cathode and oxidation reaction at anode

617. In an electrolysis of acidulated water, 4.48 L of hydrogen at STP was produced by passing a current of 2.14 A. For how many hours was the current passed?

(1) 4 (2) 3

(3) 6 (4) 5

618. When a dilute aq. solution of Li_2SO_4 is electrolysed, the products formed at the anode and cathode are respectively?

(1) H_2 and O_2 (2) O_2 and Li

(3) SO_2 and H_2 (4) O_2 and H_2

619. Based on the data given below, the correct order of reducing power is

$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$; $E^0 = +0.77$ V

$Al^{3+}(aq) + 3e^- \rightarrow Al(s)$; $E^0 = -1.66$ V

$Br_2(aq) + 2e^- \rightarrow 2Br^-(aq)$; $E^0 = +1.08$ V

(1) $Br^- < Fe^{2+} < Al$

(2) $Fe^{2+} < Al < Br^-$

(3) $Al < Br^- < Fe^{2+}$

(4) $Al < Fe^{2+} < Br^-$

620. How much time is required for complete decomposition of 4 mol water using current of 4 Ampere?

(1) 96500 sec

(2) 3.85×10^4 sec

(3) 1.93×10^5 sec

(4) 2.92×10^5 sec

621. The amount of metal deposited at cathode on passing electric current of 0.75 ampere in aqueous ferric sulphate solution for 30 minutes, will be (atomic weight of Fe = 56)

(1) 0.00435 g (2) 0.261 g

(3) 0.783 g (4) 0.522g

622. Pt $|H_2(1atm)|H^+ (0.001M)||H^+ (0.1M)|H_2 (1atm)|$Pt

what will be the value of E_{cell} for this cell

(1) 0.1182 V (2) – 0.1182 V

(3) 0.0591 V (4) – 0.0591 V

623. Consider the following electrode potentials

$Mg^{+2} + 2e^- \rightarrow Mg$ $E^0 = -2.37V$

$V^{+2} + 2e^- \rightarrow V$ $E^0 = -1.18$ V

$Cu^{+2} + e^- \rightarrow Cu^+$ $E^0 = 0.15$ V

Which of the following reactions will proceed from left to right spontaneously?

(1) $Mg^{+2} + V \rightarrow Mg + V^{2+}$

(2) $Mg^{+2} + 2Cu^+ \rightarrow Mg + 2Cu^{+2}$

(3) $V^{+2} + 2Cu^+ \rightarrow V + 2Cu^{+2}$

(4) $V + 2Cu^{+2} \rightarrow V^{2+} + 2Cu^+$

624. A metal bucket is to be electroplated by using $ZnCl_2$ as an electrolyte. How many moles of zinc are deposited in 20 min by a constant current of 10A?

(1) 0.01 (2) 0.03

(3) 0.06 (4) 0.10

625. The quantity of electricity required to release 112 cm^3 of hydrogen at STP from acidified water is

(1) 0.1F (2) 1F

(3) 965C (4) 96500C

626. If $E^0_{Au^+/Au} = 1.69V$ and $E^0_{Au^{3+}/Au} = 1.40V$ then the value of $E^0_{Au^{+3}/Au^+}$ will be

(1) 0.19V (2) 2.945 V

(3) 1.255 V (4) 1.19 V

627. Given the standard electrode potential

$K^+/K = -2.93V$, $Ag^+/Ag = 0.80$ V

$Hg^{2+}/Hg = 0.79V$; $Mg^{2+}/Mg = -2.37V$,

$Cr^{3+}/Cr = -0.74V$

Correct order of their reducing power is –

(1) Ag > Hg > Cr > Mg > K

(2) K > Mg > Cr > Hg > Ag

(3) K > Cr > Mg > Ag > Hg

(4) Cr > Mg > K > Hg > Ag

628. The conductivity of Electrolytic solutions depends on-

(1) Size of the ions produced and their solvation

(2) The nature of the solvent and its viscosity

(3) temperature

(4) All of the above

629. Which statement is incorrect about cell constant

(1) Cell constant depends on the distance between the electrodes and their area of cross section

(2) The dimension of cell constant is $length^{-1}$

(3) The cell constant $G^* = R\kappa$

(4) None of the above

630. The volume of hydrogen gas liberated at STP when a current of 5.36 ampere is passed through dil. H_2SO_4 for 5 hours will be

(1) 5.6L (2) 11.2L

(3) 16.8L (4) 22.4L

631. The electrolysis of silver nitrate solution is carried out using silver electrods. Which the following reaction occurs at the anode?

(1) $Ag \rightarrow Ag^+ + e^-$

(2) $Ag^+ + e^- \rightarrow Ag$

(3) $2H_2O \rightarrow 4H^+ + O_2 + 4e^-$

(4) $4OH^- \rightarrow 2H_2 + O_2 + 4e^-$

632. The equilibrium constant (K) for the reaction $Fe^{2+}(aq) + Ag^+(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$ will be GivenE°(Fe^{3+}/Fe^{2+}) = 0.77V;

E°(Ag^+/Ag) = 0.80V

(1) 100 (2) 3.16

(3) 10 (4) 0.5

633. The ionization constant of a weak electrolyte is 25×10^{-6} while the equivalent conductance of its 0.01M solution is 19.6 $Scm^2\ eq^{-1}$. The equivalent conductance of the electrolyte at infinite dilution (in $Scm^2\ eq^{-1}$) will be

(1) 250 (2) 196 (3) 392 (4) 384

634. The molar ionic conductances at infinite dilution of K^+ and SO_4^{2-} are 73.5 and 160 $Scm^2\ mol^{-1}$ respectively. The molar conductance of solution of K_2SO_4 at infinite dilution will be

(1) 233.5 (2) 307

(3) 153.5 (4) 467

635. How many seconds, 0.2 A current should be passed to deposit 40%, Cu from 200 ml of 0.5M $CuSO_4$ solution ?

(1) 38600 sec (2) 42100 sec

(3) 28200 sec (4) 10500 sec

636. When a solution containing Zn^{++} and Cu^{++} is electrolysed. Cu^{++} is deposited at cathode why.

(1) $(S.R.P.)_{Zn^{++}} > (S.R.P.)_{Cu^{++}}$

(2) $(S.R.P.)_{Zn^{++}} < (S.R.P.)_{Cu^{++}}$

(3) $(S.O.P.)_{Zn^{++}} < (S.O.P.)_{Cu^{++}}$

(4) $(S.R.P.)_{H_2O} > (S.R.P.)_{Cu^{++}}$

637. For half reaction

$$MnO_4^- + 5e^{\ominus} + 8H^+ \rightarrow Mn^{++} + 4H_2O$$

then Nerst equation for E_{RP} is :-

(1) $E_{MnO_4^-/Mn^{2+}} = E^0 + \dfrac{0.059}{5}\log\dfrac{[MnO_4^-][H^+]^8}{[Mn^{2+}]}$

(2) $E_{MnO_4^-/Mn^{2+}} = E^0 - \dfrac{0.059}{5}\log\dfrac{[Mn^{++}][H_2O]^4}{[MnO_4^-]}$

(3) $E_{MnO_4^-/Mn^{2+}} = E^0 - 0.059 \times 5\log\dfrac{[Mn^{++}]}{[MnO_4^-][H^+]^8}$

(4) $E_{MnO_4^-/Mn^{2+}} = E^0 + \dfrac{0.059}{7}\log\dfrac{[MnO_4^-][H^+]^8}{[MnO_4^-][H_2O]^4}$

638. Aqueous sol[n] of $AuCl_3$, $CuSO_4$ & $AgNO_3$ are connected in series, when 3F charge is passed then calculate ratio of moles of metals deposited on cathod.

(1) 2 : 3 : 6 (2) 3 : 4 : 5
(3) 1 : 3 : 5 (4) None

639. Why density of electrolyte in lead storage battery decreases as it is discharge.

(1) Only Pb reacts with H_2SO_4
(2) Only PbO_2 reacts with H_2SO_4
(3) H_2SO_4 is formed
(4) Pb and PbO_2 both reacts with H_2SO_4

640. Which of the following has highest resistance during the passage of current?

(1) 1 N NaCl (2) 0.1 N NaCl
(3) 2N NaCl (4) 0.05 N NaCl

641. 0.36 gram of a metal is deposited on the electrode. When 1.2 A current is passed for 15 min. though its salt. At weight of metal is 96. What will be its valency.

(1) 2 (2) 3 (3) 4 (4) 1

642. If Hg is used as cathode is the electolysis of aqueous NaCl solution, the ions discharged at cathode are :-

(2) H^+ (2) Na^+
(3) OH^- (4) Cl^-

643. Which metal will dissolve if the cell works

$Cu|Cu^{++} \,||\, Ag^+|Ag$

(1) Cu
(2) Ag
(3) both 1 and 2
(4) No metal will dissolve

644. Passage of three faraday of charge through aqueous solution of $AgNO_3$, $CuSO_4$, $Al(NO_3)_3$ and NaCl respectively will deposited metals at cathode in the molar ratio :

(1) 1 : 2 : 3 : 1 (2) 6 : 3 : 2 : 6
(3) 6 : 3 : 0 : 0 (4) 3 : 2 : 1 : 0

645. Which is correct for an electrolytic cell ?

(1) $E_{Cell} < 0$ (2) $E_{Cell} > 0$
(3) $\Delta G = 0$ (4) Both (2) and (3)

646. A quantity of electricity required to reduce 12.3 gm of nitrobenzene to aniline arising 50% current efficiency is :-

(1) 115800 C (2) 579000 C
(3) 231600 C (4) 289500 C

647. How many columbs are required for the oxidaion of 1 mol of H_2O_2 to O_2?

(1) 9.65×10^4 C (2) 93000 C
(3) 1.93×10^5 C (4) 19.3×10^2 C

648. Which of the following equility holds good for the strong electrolytes ?

(1) $\wedge = \wedge^0$ as $C \to 1$
(2) $\wedge = \wedge^0$ as $C \to 0$
(3) $\wedge = \wedge^0$ as $C \to \infty$
(4) $\wedge = \wedge^0$ as $C \to \sqrt{b}$

649. The quantity of electricity required to reduce 0.05 mole of MnO_4^- to Mn^{+2} in acidic medium would be :-

(1) 0.01 F (2) 0.05 F
(3) 0.15 F (4) 0.25 F

SURFACE CHEMISTRY

650. Bredig arc method cannot be used to prepare colloidal solution of which of the following :-

(1) Pt (2) Ag (3) Au (4) Fe

651. Which one of the following is correctly matched:-

(1) Emulsion - curd (2) Foam - mist
(3) Aerosol - smoke (4) Sol - cake

652. The protecting power of Lyophilic colloidal sol is expressed in terms of :-

(1) Coagulation value
(2) Gold number
(3) Critical micelle concentration
(4) Oxidation number

653. According to freundlich adsorption isotherm which of the following is correct ?

(1) $\frac{x}{m} \propto p^{o}$

(2) $\frac{x}{m} \propto p^{1/n}$

(3) $\frac{x}{m} \propto p^{1}$

(4) All the above are correct for different pressure range

654. Among the electrolytes Na_2SO_4 , $CaCl_2$, $Al_2(SO_4)_3$ and NH_4Cl, the most effective coagulating agent for Sb_2S_3 sol is :-

(1) Na_2SO_3 (2) $CaCl_2$

(3) $Al_2(SO_4)_3$ (4) NH_4Cl

655. A plot of log x/m versus log P for the adsorption of a gas on a solid gives a straight line with slope equal to :-

(1) 1/n (2) log K

(3) – log K (4) n

656. In petrochemical industry, alcohols are directly converted to gasoline by passing over heated :-

(1) Platinum (2) ZSM–5

(3) Iron (4) Nickel

657. Which of the following is not a characteristic of chemi-sorption :-

(1) Adsorption is irreversible

(2) ΔH is of the order of 40 KJ

(3) Adsorption is specific

(4) Adsorption increases with increase of surface area

658. Which of the following is an emulsifier ?

(1) soap (2) water

(3) oil (4) NaCl

659. Gold numbers of protective colloids A,B,C and D are 0.50, 0.01, 0.1 and 0.005 respectively. The correct order of their protective power is :-

(1) A < C < B < D

(2) D < A < C < B

(3) C < B < D < A

(4) B < D < A < C

660. The process which is catalysed by one of the product is called :-

(1) Positive catalysis

(2) Negative catalysis

(3) Autocatalysis

(4) None of these

661. Match the column :-

X[Colloids]	Y[Classification]
I Smoke	A Gel
II Gelatin	B Aerosol
III Soap lather	C Emulsion
IV Milk	D Foam

	I	II	III	IV
(1)	A	B	C	D
(2)	A	C	B	D
(3)	B	A	D	C
(4)	B	A	C	D

662. Some of the following are true solutions :-

I : urea solution II : Gelation

III : Glucose solution IV : NaCl solution

V : Butter VI : Blood

Select true solution :

(1) I, III, IV

(2) II, III, IV, V

(3) I, IV, V

(4) II, IV, VI

663. A freshly prepared $Fe(OH)_3$ precipitate is peptized by adding $FeCl_3$ solution. The charge on the colloidal particle is due to preferential adsorption of :-

(1) Cl^- ions (2) Fe^{+++} ions

(3) OH^- ions (4) None of these

664. Match the column :-

	Column-I		Column-II
A	Chemisorption	P	Not specific and decreases with temperature
B	Physisorption	Q	Specific and increases with temperature
C	Desorption of a solute on liquid surface	R	Increases the surface tension of the liquid
D	Adsorption of a solute on a liquid surface	S	Decreases the surface tension of the liquid

	A	B	C	D
(1)	Q	P	R	S
(2)	P	Q	R	S
(3)	S	R	Q	P
(4)	Q	S	P	R

665.

	Column-I		Column-II
A	Liquid dispersed in gas	P	Foam
B	Gas dispersed in liquid	Q	Emulsion
C	Liquid dispersed in solid	R	Aerosol
D	Liquid dispersed in liquid	S	Gel

	A	B	C	D
(1)	R	P	S	Q
(2)	P	Q	R	S
(3)	S	R	Q	P
(4)	Q	R	P	S

666.

	Column-I		Column-II
A	Coagulation	P	Due to presence of charge
B	Electrophoresis	Q	Due to scattering of light
C	Tyndall effect	R	Due to neutralization of charge
D	Brownian movement	S	Due to impact of the molecules of the dispersion medium with colloidal particles

	A	B	C	D
(1)	R	P	Q	S
(2)	P	Q	R	S
(3)	S	Q	P	R
(4)	S	P	Q	R

667. Which property of colloids is not dependent on the charge on colloidal particles :-

(1) Electro osmosis

(2) Tyndall effect

(3) Coagulation

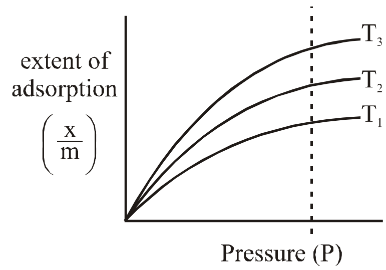
(4) Electrophoresis

668. For the coagulation of 50 ml of ferric hydroxide sol. 10 ml of 0.5 M KCl is required. What is the coagulation value of KCl ?

(1) 5 (2) 10

(3) 100 (4) None of these

669. For the graph below select correct order of tempereture :-



(1) $T_1 > T_2 > T_3$ (2) $T_2 > T_3 > T_1$

(3) $T_3 > T_2 > T_1$ (4) $T_1 = T_2 = T_3$

670. Which of the following is true in respect of chemical adsorption ?

(1) $\Delta H < 0, \Delta S > 0, \Delta G > 0$

(2) $\Delta H < 0, \Delta S < 0, \Delta G < 0$

(3) $\Delta H > 0, \Delta S > 0, \Delta G > 0$

(4) $\Delta H > 0, \Delta S < 0, \Delta G > 0$

671. Which of the following is negatively charged sol.

(1) Metallic sulphides (2) Pt

(3) Acid dye (4) All of these

672. In freundlich adsorption isotherm the value of 1/n is :-

(1) 1 in case of physical adsorption

(2) 1 in case of chemisorption

(3) Between 0 and 1 in all cases

(4) Between 2 and 4 in all cases

673. The Langmuir adsorption isotherm is deduced using the assumption :-

(1) The adsorption sites are equivalent in their ability to adsorb the particles

(2) The heat of adsorption varies with adsorption

(3) The adsorption molecule interact with each other

(4) The adsorption takes place in multilayers

674. Which of the following factors affects the adsorption of a gas on solid ?

(1) Criticle temperature (T_C)

(2) Temperature of gas

(3) Pressure of gas

(4) All of these

675. As the temperature increases, the extent of physisorption of a gas

(1) increases

(2) decreases

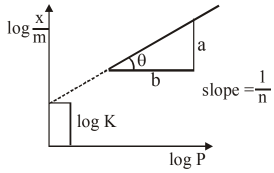
(3) first increases then decreases

(4) remains unchaged

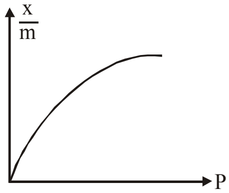
676. Match the expression given in the column-I with the graphs or statement given in the column-II and select the correct option from the codes given below

Column-I Column-II

(A) $\frac{x}{m}$ v/s P (P)



(B) $\log\frac{x}{m}$ v/s log P (Q)



(C) $\frac{x}{m} = KC^{1/n}$ (R) Adsorption from solution phase

Codes	A	B	C
(1)	P	Q	R
(2)	Q	P	R
(3)	R	P	Q
(4)	R	Q	P

677. Which of the following is purification method of colloidal solution?

(1) Coagulation (2) Ultrafiltration

(3) Electroproresis (4) Tyndal effect

678. Which of the following statements is incorrect regarding physisorption

(1) It occurs because of van der Waal's force

(2) easily liquefiable gases are adsorbed readily

(3) under high pressure it may result into multilayered adsorption on adsorbent surface

(4) Enthalpy of adsorption is positive

679. When the tempeature is lowered and pressure is raised, the adsorption of a gas on a solid

(1) decreases

(2) increases

(3) remains unaffected

(4) decreases first than increases

680. The heat evolved in physisorption lies in the range (in kJ/mol) of
(1) 20-40 (2) 40-100
(3) 100-200 (4) 200-400

681. Tyndal effect is observed when–
(a) Diameter of dispersed particles is not much smaller than the wavelength of light used
(b) The refractive indices of the dispersed phase and the dispersion medium differ greatly in magnitude
(c) Diameter of dispersed particles is much larger than the wavelength of the light used
(1) a and b (2) a and c
(3) b and c (4) a, b and c

682. The arsenious sulphide sol has negative charge. The maximum coagulating power for precipitating it is of–
(1) 0.1 N $Zn(NO_3)_2$ (2) 0.1 N Na_3PO_4
(3) 0.1 N $ZnSO_4$ (4) 0.1 N $AlCl_3$

683. The disperse phase in colloidal iron (III) hydroxide and colloidal gold is positively and negatively charged, respectively. Which of the following statement is NOT correct
(1) Magnesium chloride solution coagulates, the gold sol more readily than the iron (III) hydroxide sol
(2) Sodium sulphate solution cause coagulation in both sols
(3) Mixing the sols has no effect
(4) Coagulation in both sols can be brought about by electrophoresis

684. The incorrect statement pertaining to the adsorption of a gas on a solid surface is(are):-
(1) Adsorption is always exothermic
(2) Physisorption may transform into chemisorption at high temperature
(3) Physisorption increases with increasing temperature but chemisorption decreases with increasing temperature
(4) Chemisorption is more exothermic than physisorption, however it is very slow due to higher energy of activation

685. For a linear plot of log(x/m) versus log P in a Freundlich adsorption isotherm, which of the following statements is correct ? (k and n are constants)
(1) log (1/n) appears as the intercept
(2) Both k and 1/n appear in the slope term
(3) 1/n appears as the intercept
(4) Only 1/n appears as the slope

686. A particular adsorption process has the following characteristics: (i) It arises due to van der Waals forces and (ii) it is reversible. Identify the correct statement that describes the above adsorption process:
(1) Enthalpy of adsorption is greater than 100 kJ mol^{-1}
(2) Energy of activation is low.
(3) Adsorption is monolayer
(4) Adsorption increases with increase in temperature.

687. The name aquadag is given to the colloidal sol of-
(1) Copper in water
(2) Platinum in water
(3) Graphite in water
(4) None of the above

688. Flocculation value is expressed in terms of -
(1) Millimole/litre (2) Mole/litre
(3) Grams/litre (4) Mole/millilitre

689. If dispersed phase is a liquid and the dispersion medium is a solid, the colloid is known as -
(1) A sol (2) A gel
(3) An emulsion (4) A foam

690. Colloidal particles of soap sol in water are -
(1) Negatively charged (2) Positively charged
(3) Neutral (4) Unpredictable

691. The curve showing the variation of adsorption with pressure at constant temperature is called-
(1) An isoster (2) Adsorption isotherm
(3) Adsorption isobar (4) All of these

692. The Brownian motion is due to -
(1) Temperature fluctuation within the liquid phase
(2) Attraction and repulsion between charges on the colloidal particles
(3) Impact of molecules of the dispersion medium on the colloidal particles
(4) Convective currents

PHYSICAL CHEMISTRY ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**3**	**1**	**2**	**2**	**2**	**2**	**1**	**4**	**3**	**2**	**1**	**1**	**1**	**3**	**1**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**4**	**1**	**3**	**3**	**1**	**3**	**4**	**4**	**2**	**2**	**3**	**1**	**2**	**3**	**4**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**2**	**2**	**2**	**1**	**1**	**1**	**4**	**4**	**4**	**1**	**1**	**4**	**1**	**2**	**4**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**4**	**1**	**3**	**4**	**2**	**1**	**1**	**2**	**1**	**2**	**2**	**3**	**2**	**2**	**4**
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	**4**	**1**	**2**	**3**	**4**	**1**	**2**	**3**	**4**	**4**	**1**	**1**	**4**	**4**	**3**
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	**2**	**4**	**3**	**4**	**2**	**3**	**3**	**1**	**2**	**2**	**1**	**3**	**1**	**3**	**3**
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	**2**	**4**	**4**	**1**	**4**	**4**	**2**	**1**	**4**	**2**	**2**	**2**	**4**	**4**	**2**
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	**1**	**2**	**4**	**3**	**3**	**2**	**1**	**3**	**4**	**1**	**1**	**2**	**2**	**3**	**3**
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	**3**	**2**	**3**	**4**	**3**	**3**	**3**	**4**	**2**	**3**	**2**	**2**	**2**	**3**	**4**
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	**2**	**1**	**1**	**1**	**2**	**3**	**4**	**3**	**4**	**3**	**4**	**1**	**3**	**4**	**4**
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	**3**	**3**	**4**	**1**	**3**	**3**	**2**	**1**	**2**	**1**	**3**	**3**	**1**	**3**	**2**
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	**1**	**4**	**2**	**2**	**1**	**2**	**4**	**3**	**4**	**3**	**4**	**1**	**2**	**3**	**4**
Que.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
Ans.	**1**	**2**	**3**	**1**	**3**	**2**	**1**	**1**	**4**	**1**	**1**	**3**	**1**	**3**	**4**
Que.	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210
Ans.	**1**	**4**	**4**	**1**	**2**	**3**	**4**	**3**	**1**	**4**	**4**	**2**	**2**	**2**	**3**
Que.	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225
Ans.	**4**	**3**	**3**	**1**	**1**	**2**	**1**	**2**	**1**	**4**	**1**	**2**	**3**	**2**	**3**
Que.	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240
Ans.	**1**	**4**	**4**	**4**	**2**	**1**	**2**	**2**	**1**	**1**	**3**	**3**	**3**	**4**	**3**
Que.	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255
Ans.	**3**	**1**	**2**	**2**	**4**	**2**	**3**	**2**	**4**	**3**	**3**	**2**	**4**	**2**	**2**
Que.	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270
Ans.	**2**	**3**	**4**	**1**	**2**	**4**	**1**	**2**	**1**	**3**	**2**	**1**	**1**	**2**	**1**
Que.	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285
Ans.	**2**	**3**	**3**	**2**	**2**	**3**	**1**	**3**	**1**	**1**	**1**	**4**	**3**	**2**	**3**
Que.	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300
Ans.	**1**	**1**	**1**	**2**	**2**	**3**	**1**	**3**	**2**	**1**	**2**	**4**	**4**	**4**	**3**
Que.	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315
Ans.	**2**	**4**	**3**	**3**	**4**	**4**	**2**	**1**	**1**	**3**	**4**	**4**	**2**	**3**	**4**
Que.	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330
Ans.	**1**	**2**	**1**	**3**	**2**	**3**	**4**	**1**	**4**	**1**	**2**	**4**	**2**	**4**	**3**
Que.	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345
Ans.	**1**	**4**	**2**	**3**	**1**	**3**	**1**	**2**	**4**	**3**	**4**	**2**	**2**	**1**	**4**

Que.	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360
Ans.	**3**	**1**	**4**	**2**	**4**	**3**	**3**	**1**	**4**	**3**	**3**	**2**	**3**	**1**	**4**
Que.	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375
Ans.	**1**	**1**	**2**	**4**	**3**	**2**	**2**	**3**	**2**	**2**	**2**	**2**	**4**	**2**	**3**
Que.	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390
Ans.	**3**	**3**	**3**	**4**	**4**	**2**	**1**	**3**	**4**	**2**	**3**	**4**	**3**	**3**	**4**
Que.	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405
Ans.	**4**	**4**	**2**	**2**	**2**	**3**	**3**	**1**	**4**	**4**	**1**	**4**	**3**	**3**	**2**
Que.	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420
Ans.	**3**	**2**	**1**	**4**	**1**	**2**	**1**	**3**	**1**	**3**	**4**	**4**	**2**	**4**	**2**
Que.	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435
Ans.	**3**	**2**	**2**	**3**	**2**	**2**	**2**	**1**	**3**	**4**	**3**	**1**	**2**	**3**	**2**
Que.	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450
Ans.	**4**	**3**	**2**	**4**	**3**	**3**	**2**	**3**	**1**	**1**	**1**	**1**	**4**	**2**	**2**
Que.	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465
Ans.	**2**	**1**	**4**	**3**	**1**	**4**	**4**	**2**	**3**	**2**	**2**	**1**	**3**	**2**	**1**
Que.	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480
Ans.	**1**	**2**	**4**	**4**	**4**	**4**	**2**	**2**	**4**	**3**	**4**	**2**	**2**	**3**	**3**
Que.	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495
Ans.	**2**	**2**	**2**	**2**	**1**	**2**	**2**	**2**	**3**	**3**	**4**	**1**	**4**	**1**	**1**
Que.	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510
Ans.	**1**	**4**	**3**	**3**	**4**	**3**	**2**	**1**	**2**	**2**	**2**	**1**	**1**	**3**	**4**
Que.	511	512	513	514	515	516	517	518	519	520	521	522	523	524	525
Ans.	**1**	**2**	**4**	**1**	**4**	**4**	**2**	**4**	**1**	**3**	**3**	**2**	**2**	**3**	**2**
Que.	526	527	528	529	530	531	532	533	534	535	536	537	538	539	540
Ans.	**2**	**3**	**1**	**1**	**3**	**4**	**2**	**1**	**2**	**2**	**3**	**4**	**2**	**2**	**2**
Que.	541	542	543	544	545	546	547	548	549	550	551	552	553	554	555
Ans.	**3**	**1**	**1**	**2**	**3**	**3**	**4**	**2**	**3**	**3**	**3**	**1**	**3**	**3**	**3**
Que.	556	557	558	559	560	561	562	563	564	565	566	567	568	569	570
Ans.	**4**	**2**	**3**	**3**	**4**	**1**	**3**	**3**	**2**	**3**	**1**	**1**	**3**	**1**	**3**
Que.	571	572	573	574	575	576	577	578	579	580	581	582	583	584	585
Ans.	**1**	**4**	**4**	**4**	**3**	**4**	**2**	**1**	**2**	**2**	**2**	**2**	**1**	**4**	**1**
Que.	586	587	588	589	590	591	592	593	594	595	596	597	598	599	600
Ans.	**3**	**1**	**2**	**2**	**3**	**4**	**4**	**1**	**3**	**3**	**2**	**1**	**4**	**4**	**2**
Que.	601	602	603	604	605	606	607	608	609	610	611	612	613	614	615
Ans.	**1**	**1**	**2**	**3**	**3**	**2**	**2**	**1**	**1**	**2**	**2**	**3**	**2**	**3**	**2**
Que.	616	617	618	619	620	621	622	623	624	625	626	627	628	629	630
Ans.	**4**	**4**	**4**	**1**	**3**	**2**	**1**	**4**	**3**	**3**	**3**	**2**	**4**	**4**	**2**
Que.	631	632	633	634	635	636	637	638	639	640	641	642	643	644	645
Ans.	**1**	**2**	**3**	**2**	**1**	**2**	**1**	**1**	**4**	**3**	**2**	**2**	**1**	**3**	**1**
Que.	646	647	648	649	650	651	652	653	654	655	656	657	658	659	660
Ans.	**1**	**3**	**2**	**4**	**4**	**3**	**2**	**4**	**3**	**1**	**2**	**2**	**1**	**1**	**3**
Que.	661	662	663	664	665	666	667	668	669	670	671	672	673	674	675
Ans.	**3**	**1**	**2**	**1**	**1**	**1**	**2**	**3**	**1**	**2**	**4**	**3**	**1**	**4**	**2**
Que.	676	677	678	679	680	681	682	683	684	685	686	687	688	689	690
Ans.	**2**	**2**	**4**	**2**	**1**	**1**	**4**	**3**	**3**	**4**	**2**	**3**	**1**	**2**	**1**
Que.	691	692													
Ans.	**2**	**3**													

PHYSICAL CHEMISTRY SOLUTIONS

MOLE CONCEPT

1. Neutrons in 1 molecule of H_2O = 8

no. of moles in 1.8 ml = $\frac{1.8}{18} = 0.1$

no. of molecules = $0.1 \times N_A$

no. of neutrons = $8 \times 0.1 \times N_A$

no. of moles of neutrons = $\frac{8 \times 0.1 \times N_A}{N_A}$ = 0.8

2. PV = nRT

$\frac{2 \times 350}{1000} = \frac{1}{Mw} \times 0.082 \times 273$

Mw = 32

wt of 1 molecule of diatomic molecule $A_2 = \frac{32}{N_A}$

wt of 1 atom of diatomic molecule

$A_2 = \frac{32}{N_A} \times \frac{1}{2} = \frac{16}{N_A}$

3. for unknown hydrocarbon

$C_xH_y + \left(x + \frac{y}{4}\right)O_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$

1mL xm L $\frac{y}{2}$mL

10mL 10xmL $10\frac{y}{2}$mL

10x = 20 and $10\frac{y}{2} = 30$ Hence formula is C_2H_6

x = 2 y = 6

4. $A + 2B \rightarrow C$

Hence B is limiting reagent

2 mole B produce 1 mole C

and 8 mole B produce 4 mole C

5. 1 a.m.u = $\frac{1}{12} \times$ wt of C-12 isotope

1 a.m.u = $\frac{1}{24} \times$ wt of C-12 isotope

1 amu = 2 × 1 amu^1

New amu^1 = $\frac{amu}{2}$

Hence to maintain the weight of 1 mole, value of Avogadro constant gets doubled

Let abondance of B–10 be x and B–11 be 100–x

Avg. mass = $\frac{M_1x_1 + M_2x_2}{x_1 + x_2}$

$\Rightarrow \frac{(10 \times x) + [11 \times (100 - x)]}{100} = 10.80$

$\Rightarrow$ x = 20

7.

Atom	atomic mass	% wt	% / Atomic wt.	simple ratio
C	12	54.54	4.54	2
H	1	9.09	9.09	4
O	16	36.37	2.27	1

Empirical formula = C_2H_4O

Empirical formula weight = 44

Molecular formula weight = 88 × 2 = 176

n = $\frac{176}{44} = 4$

Hence molecular formula = $C_8H_{16}O_4$

8. Formula = $X_{\frac{76}{75}}Y_{\frac{24}{16}}$ = X_2Y_3

9. $CaCO_3 \rightarrow CaO + CO_2$

100 g 56 g 44g

wt of pure limestone = $\frac{20 \times 40}{100} = 8$gm

100 gm pure $CaCO_3$ produce $\rightarrow$ 44g CO_2

8 g pure $CaCO_3$ produce $\rightarrow$ 3.52 g CO_2

10. $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$

100 g 98g

100 g $CaCO_3$ require 98 gm H_2SO_4

25 g $CaCO_3$ require $\frac{98}{100} \times 25$ gm pure H_2SO_4

= 24.5 g pure H_2SO_4

Hence, wt of 50% pure H_2SO_4 will be 49.5 gm

11. $2H_2 + O_2 \longrightarrow 2H_2O$

4gm 32 gm 36gm

given 8gm 32 gm

Hence, O_2 is limiting reagent

12. $\% \text{ wt} = \dfrac{\text{no.of atoms} \times \text{Atomic wt}}{\text{molecular wt}} \times 100$

If minimum molecular wt is to be found then take atomicity = 1

$$3.4 = \frac{1 \times 32 \times 100}{\text{min. molecular wt}}$$

$\Rightarrow$ Minimum molecular wt = 941.176

13. (i) No. of molecules = $\dfrac{15}{22.4} \times N_A$

(ii) No. of molecules = $\dfrac{5}{22.4} \times N_A$

(iii) No. of molecules = $\dfrac{0.5}{2} \times N_A$

(iv) No. of molecules = $\dfrac{10}{32} \times N_A$

Hence 15 L H_2 will have maximum number of molecules

14. Weight of metal chloride = x
weight of metal = y
weight of chlorine = x–y

Equivalent weight = $\dfrac{\text{weight of metal}}{\text{weight of chlorine}} \times 35.5$

$$= \frac{y}{(x-y)} \times 35.5$$

15. Let the mass be 100 g
Ratio of moles = Ratio of volumes

$$= \frac{100}{2} : \frac{100}{28} = \frac{100}{16}$$

= 56 : 4 : 7

16. $H_2 + Cl_2 \longrightarrow 2HCl$ $\quad n_{HCl}\ \dfrac{3.65}{36.5} = 0.1$

2gm 71 gm 73 gm
1mole 1mole 2mole
Here 2 mole HCl is produced by 1 mole H_2
1 mole HCl produce by 0.05 mole H_2

$n_{H_2} = n_{Cl_2} = 0.05$

$V_{H_2} = V_{Cl_2} = 0.05 \times 22.4 = 1.12$ L (at NTP)

17. Weight of
(1) 1g-atom of C=12 g
(2) $\dfrac{1}{2}$ mole CH_4 = 8g
(3) 10 ml water = 10 gm
(4) 3.01×10^{23} atoms of oxygen = 8 g
hence mass is highest in 1g-atom of C

18. $2CO + O_2 \rightarrow 2CO_2$
2L 1L 2L at constant temperature
4L 2L 4L and pressure
Hence 4L CO require 2L O_2

19. Molar mass = 2 × V.D. and x(12 + 16) = 140

= 2 × 70 $\qquad x = \dfrac{140}{28} = 5$

= 140

20. mass of 1L CO_2 = mass of 1L hydrocarbon
mass of 22.4L CO_2 = mass of 22.4L hydrocarbon
44 g = mass of 22.4 L hydrocarbon
mass of 1 mole hydrocarbon = 44 gm
Here C_3H_8 has molar mass 44 g mol^{-1}

21. At same temperature and pressure, same volume and same moles of gases are present

Ratio of moles $nH_2 : nH_e : nO_2 : nO_3$
1 : 1 : 1 : 1

Ratio of no. of atoms $H_2 : He : O_2 : O_3$
2 : 1 : 2 : 3

22. In I[st] oxide
50 g O combine with 50 g S
1 g O combine with 1g S
In II[nd] oxide
60 g O combine with 40 g S
1 g O combine with $\dfrac{2}{3}$ g S
Ratio in which 1 gm O combine with different masses of sulphur is 3 : 2

23.

N	$\dfrac{2.314}{14} = 0.165$	$\dfrac{0.165}{0.165} = 1$
O	$\dfrac{5.34}{16} = 0.333$	$\dfrac{0.333}{0.165} \simeq 2$

$\therefore$ Empirical formula = NO_2

24. $CO_2(g) + C(s) \longrightarrow 2CO(g)$
$\because$ 1cc of CO_2 produces 2cc of CO
$\therefore$ 16 cc of CO_2 produces 32 cc of CO

25. $C_xH_y + \left(x + \dfrac{y}{4}\right)O_2 \rightarrow xCO_2 + \dfrac{y}{2}H_2O$

1mL $\qquad$ xmL $\qquad \dfrac{y}{2}$mL

500 mL $\qquad$ 500 xmL $\quad$ 250 ymL

Hence, and
500 x = 2000 $\qquad$ 250 y = 2500 mL
x = 4 $\qquad$ y = 10 mL
$\therefore$ formula = C_4H_{10}

26. 1 mol $Ba(NO_3)_2$ contain → 3 mole ions

$n_{Ba(NO_3)_2} = \frac{0.1\times1}{1000} = 10^{-4}$

No. of ions = $3 \times 10^{-4} \times 6.02 \times 10^{23}$

$= 3 \times 6.02 \times 10^{19}$

27. 1 g equivalent of O = 8 g

Hence,

0.25 mole O_2 contain 8 g oxygen

29. ∵ 8 mole O is present is → 1 mole $Mg_3(PO_4)_2$

∴ 1 mole O is present in $\frac{1}{8}$ mole $Mg_3(PO_4)_2$

∴ 0.25 mole O is present in

$\Rightarrow \frac{1}{8}\times0.25$ mole $Mg_3(PO_4)_3$

$= 3.125 \times 10^{-2}$ mole $Mg_3(PO_4)_3$

30. $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$

$nCl_2 = \frac{2.24}{22.4} = 0.1$mole

1mole 1mole

87 gm 1mole

1 mole Cl_2 produced by 87 g pure MnO_2

0.1 mole Cl_2 produced by 8.7 g pure MnO_2

% purity = $\frac{8.7}{10}\times100 = 87\%$

% Impurity = 13%

31. No. of atoms

$\frac{5.6}{60}\times N_A\times4 = 2.24\times10^{23}$

32. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

22g

0.5 mol

n_{CH_4} required = 0.5 mol

33. 5.2 mole in 1000 g H_2O

mole fraction x = $\frac{5.2}{5.2+\frac{1000}{18}} = 0.086$

34. $M_{avg} = 12 \times 0.98 + 14 \times 0.2 = 12.04$

$n_{carbon\ sample} = \frac{12}{12.04} = 0.996$

Atoms of $C^{14} = 0.996 \times \frac{2}{100} \times N_A$

$= 0.997 \times 10^{21} = 9.97 \times 10^{20}$

35. $C + O_2 \rightarrow CO_2$

1000g

$\frac{1000}{12}$mole

$n_{O_2\ required} = \frac{1000}{12}$mole

$air_{required} = \frac{1000}{12}\times\frac{1}{0.21}$mole

$= \frac{1000}{12\times0.21}\times22.4L$

= 9333.8 litre

36.

A	B
25	75
$\frac{25}{12.5}$	$\frac{75}{37.5}$

1 : 1 formula = AB

37. $CaCO_3 + 2HCl \rightarrow CaCl_2 + CO_2 + H_2O$

10g $\frac{200\times0.75}{1000}$

0.1mol 0.15mol

(limiting reagent)

n_{CaCl_2} formed = $\frac{0.15}{2}$

$w_{CaCl_2} = \frac{0.15}{2}\times111 = 8.325g$

38.

Ce	N	O
$\frac{50.4}{140}$	$\frac{15.1}{14}$	$\frac{34.5}{16}$
0.36	1.07	2.15
1 :	3 :	6

formula $Ce(NO_2)_3$

39. $C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$

$\frac{2.8}{28}\times10^3$

100 mol

$n_{O_2\ required} = 300$ mol

$w_{O_2\ required} = 300\times32 = 9.6$ kg

40. w_{H_2O} 1 molecule = $18 \times 1.66 \times 10^{-24}$g

v_{H_2O} 1 molecule = $18 \times 1.66 \times 10^{-24}$ mL

$= 3 \times 10^{-23}$ mL

ATOMIC STRUCTURE

52. $E = nh\nu$

$= 6.023 \times 10^{23} \times 6.62 \times 10^{-34} \times 909 \times 10^{3}$

$= 3.62 \times 10^{-4} J$

54. $\lambda = \frac{h}{\sqrt{2mKE}}$

$= \frac{6.63\times10^{-34}}{\sqrt{2\times K\times T\times1.67\times10^{-27}}}$

$= \frac{6.62\times10^{-34}}{\sqrt{2\times\frac{R}{N_A}\times300\times1.67\times10^{-27}}}$

$= 178$ pm

55. $\upsilon = \frac{C}{\lambda} = \frac{C}{\frac{h}{p}} = \frac{PC}{h} = \frac{2K.E}{h}$ (P is momentum)

57. $\frac{E_1}{E_2} = \frac{Z_1^2}{Z_2^2}\times\frac{n_2^2}{n_1^2}$

$\frac{(-19.6\times10^{-18})}{E_2} = \frac{2^2}{3^2}\times\frac{1^2}{1^2}$

$E_2 = -4.41 \times 10^{-17}$ J atom^{-1}

58. $P.E = 2\ T.E \times Z^2$

$= 2 \times (-3.4) \times 2^2$

$= -27.2$ eV

59. $2\pi r = n\lambda$

$2\pi(9X) = 3\lambda$

$l = 6\pi X$

60. $mvr = n\frac{h}{2\pi}$

$= \frac{5h}{2\pi}$

$= 2.5\frac{h}{\pi}$

61. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s\ 3d^6$

63. $\lambda = \frac{h}{mv}$

64. $\ell = 2 \Rightarrow$ d subshell

5 orbitals

66. $E_2 = -3.4$ eV

$= -3.4 \times 1.6 \times 10^{-19}$J

$= -5.44 \times 10^{-19}$ J

67. $\Delta X.\Delta V = \frac{h}{4\pi m}$

$\Delta V = \frac{0.527\times10^{-34}}{5\times10^{-4}\times2\times10^{-5}}$

$= 5.2 \times 10^{-27}$

70. $n = 3$

$\therefore \ell = 0, 1, 2$

71. $\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{V_2}{V_1}} = \sqrt{\frac{100}{400}} = \frac{1}{2}$

72. number of e^- = 2+8 + 8 + 2 = 20

$1s^2 2s^2\ 2p^6\ 3s^2 3p^6\ 4s^2$

$m = 0,\ s = +\frac{1}{2} \Rightarrow$ number of e^- = 6

76. Orbital angular momentum $= \sqrt{\ell(\ell+1)}\frac{h}{2\pi}$

$= \sqrt{6}\frac{h}{2\pi}$

78. Magnetic moment $= \sqrt{n(n+2)}$ B.M

$\therefore$ n = 3 hence Ti^+

84. $\lambda = \frac{h}{mv} = \frac{6.62\times10^{-34}}{9.1\times10^{-31}\times2.99\times10^{-8}} = 0.024$ Å

88. $(I.E)_{He^+} = (I.E.)_H \times Z^2 = 13.6 \times 4 = 54.4$ eV

90. $n + l \le 3$

$n + l = 1$ $n + l = 2$ $n + l = 3$

$1 + 0 \Rightarrow 1s$ $2 + 0 \Rightarrow 2s$ $2 + 1 = 2p$

$3 + 0 = 3s$

91. $\upsilon = RCZ^2\left(\frac{1}{1^2}-\frac{1}{3^2}\right)$

$= 10^7 \times 3 \times 10^8 \times 1 \times \frac{8}{9} = \frac{8}{3}\times10^{15}$

92. $KE = 0.04 \times 1.6 \times 10^{-19}$ J $\Rightarrow \lambda = \frac{h}{\sqrt{2mKE}}$

CHEMICAL EQUILIBRIUM

106. $PCl_5 \rightarrow PCl_3 + Cl_2$

4 0 0

4 – 0.8 0.8 0.8

$K_c = \frac{0.8\times0.8}{3.2} = 0.2$

107. $K_C = [Q]^2[R]$ ....(1)

$K_C = 4\ [Q]^2[R]^1$ ....(2)

from (1) and (2)

(1), (2) $[R]^1 = \frac{1}{4}[R]$

108. $XY_{(g)} \rightleftharpoons X_{(g)} + Y_{(g)}$

t = 0	1 mole	0	0.2 mole
at equilibrium	1-x	x	0.2 + x

$\because$ 0.2 + x = 0.6

$\boxed{x = 0.4\,\text{mole}}$

$K = \frac{0.6 \times 0.4}{0.6} = 0.4$

110. $K_C = [X^{+3}]\,[Y^-]^3$ ....(1)

$K_C = [X^{+3}]^1 \left\{\frac{1}{2}[Y^-]\right\}^3$ ....(2)

(1) = (2)

$\frac{1}{8}[X^{+3}]^1 = [X^{+3}]$

$[X^{+3}]^1 = 8[X^{+3}]$

111. $K_C = \frac{(6)^2}{3 \times 6 \times 9} = \frac{2}{9}$

112. $P_{H_2} = \frac{3}{4} \times 2 = \frac{3}{2}$ atm

113. $K_P = K_C(RT)^{\Delta n_g}$, $\because \Delta n_g = 0$

$\boxed{K_P = 5}$

114. $K_P = (7.9 \times 10^{-3}) \times (0.0821 \times 1115)^{0.5}$

$K_P = 7.56 \times 10^{-2}$

115. $NH_4HS \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$

at equilibrium P P

$P_{Total} = 2\,P = 0.66$

$\boxed{P = 0.33\ \text{atm}}$

116. $2HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)}$

t = 0	1	0	0
t = eq^n	1–0.5	0.25	0.25

$K_C = \frac{(0.25)^2}{(0.5)^2} = 0.25$

147. On T↑ solubility of gas ↓

149. $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$

1-0.5 1-0.5 0.5 0.5

K = 1

now

1–x 2–x x x

$K = \frac{x^2}{(1-x)(2-x)} = 1$

$x^2 = 2 - 3x + x^2$

x = 2/3

% converted = $\frac{2}{3/2} \times 100$ = 33.33%

150. 3 × eq.(i) + eq.(ii) = eq.(iii)

151. On P↑ Eq^n will shift in the backward dir^n

IONIC EQUILIBRIUM

156. One OH^- per molecule means monoacidic base. Like NH_4OH or BOH

$BOH \rightleftharpoons B^+ + OH^-$

$K_b = Ca^2$

$K_b = 0.05 \times \left(\frac{2.5}{100}\right)^2 \Rightarrow 3.1 \times 10^{-5}$

157. $Ca_3(PO_4)_2 \rightleftharpoons 3Ca^{+2} + 2PO_4^{-2}$

S 3S 2S

Solubility (mol L^{-1}) = $\frac{w}{M} \times \frac{1000}{100}$

Now Ksp = $[Ca^{+2}]^3\,[PO_4^{-2}]^2$

$\Rightarrow \left[\frac{w \times 10 \times 3}{M}\right]^3 \left[\frac{w \times 10 \times 2}{M}\right]^2$

Approx solubility = $10^7\left(\frac{w}{M}\right)^5$

158. no. of mole of Benzoic acid = $\frac{0.2}{122}$

eq. of $Ba(OH)_2$ = eq. of C_6H_5COOH

$2 \times \frac{0.12 \times V}{1000} \Rightarrow 1 \times \frac{0.2}{122}$

V = 6.83 ml

160. $A_pB_q \rightleftharpoons P_A^{+q} + qB^{-p}$

Ps qs

Ksp = $[Ps]^p\,[qs]^q$

$\Rightarrow S^{P+q}.\,P^P.\,q^q$

161. $H_2O \rightleftharpoons H^+ + OH^-$ $[H_2O] = 55.5$

$K \Rightarrow \frac{[H^+][OH^-]}{[H_2O]}$ $[H^+] = [OH^-] = 10^{-7}$

at 25°C

$K = \frac{10^{-7} \times 10^{-7}}{55.5}$

$K = 1.8 \times 10^{-16}$

162. $K = \frac{K_a}{K_w} = 10^{10}$

164. meq of HCl = 0.5 × 25 = 12.5

meq of NaOH = 0.5 × 10 = 5

remaining of HCl in mixture = 7.5

Total volume = 25 + 10 + 15 = 50 mL

$[H^+] = \frac{7.5}{50}$

pH = $-\log \frac{7.5}{50} = 0.82$

164. pOH = –log[OH^-], then pH = 14–pOH

165. $[OH^-] = \frac{0.01}{10} = 10^{-3}$ ⇒ pOH ⇒ pH = 14–3 =11

change in pH = 11 –7 = 4

167. Ka = $C\alpha^2$ 0.1 × $(1.32 \times 10^{-2})^2$= 1.74 × 10^{-5}

pKa = 5–log 1.74 = 5 – 0.26

= 4.74

168. $M_1V_1 = M_2V_2$

169. $F^- + H_2O \rightleftharpoons OH^- + HF$

0.01(1–h) 0.01h 0.01h

170. acidic strength α K_a

172. $\boxed{K_b > K_a}$

173. Ksp = $4s^3$ = 4 × 10^{-9} ⇒ S = 10^{-3}

174. Ka = Kb = 10^{-5}

For NH_4OH pOH = 3 then pH = 11

175. $BaSO_4 \rightleftharpoons Ba^{+2} + SO_4^{-2}$

10^{-4} x

Ksp = $[10^{-4}][x]$

$4 \times 10^{-10} = 10^{-4} \times x$ ⇒ x = 4 × 10^{-6}

176. $S = \frac{mol}{Litre}$

Now in gram/litre = S × m.wt

Ksp = 1.56 25 × 10^{-10}

s = 1.25 × 10^{-5} mol/L

s = 143.5 × 1.25 × 10^{-5} gm/L

177. $RaSO_4 \rightleftharpoons Ra^{+2} + SO_4^{-2}$

$Na_2SO_4 \rightarrow 2Na^+ + SO_4^{-2}$

0.20 0.10 + 5

Ksp = [S'] [S' + 0.10]

s' = 4 × 10^{-10}

178. $HSO_4^- \xrightarrow{-H^+} SO_4^{-2}$

$\downarrow +H^+$

H_2SO_4

180. $AgCl + 2NH_3 \rightarrow [Ag[NH_3]_2]Cl$

181. total [H^+] = $10^{-6} + 10^{-7}$

= 10^{-7} [10 + 1]

[H^+] 11× 10^{-7}

pH = –log (11×10^{-7})

pH = 5.98

182. $\alpha\sqrt{v}$

Amphiprotic acid

↑

183. $H_2SeO_3 \underset{}{\overset{Ka_1}{\rightleftharpoons}} HSeO_3^- \overset{Ka_2}{\rightleftharpoons} SeO_3^{2-}$

$$Ka_1 = \frac{[HSeO_3^-][H^+]}{[H_2SeO_3]}; Ka_2 = \frac{[SeO_3^{2-}][H^+]}{[HSeO_3^-]}$$

$$Ka_1 \times Ka_2 = [H^+]^2 \times \frac{[SeO_3^{2-}]}{[H_2SeO_3]} \quad ...(1)$$

At isoelectric point, $[SeO_3^{2-}] \approx [H_2SeO_3]$

then from (1)

$Ka_1 \times Ka_2 = [H^+]^2$

⇒ [H^+] = $(Ka_1 \times Ka_2)^{1/2}$

∴ pH – log $(Ka_1 \times Ka_2)^{1/2}$

⇒ $\boxed{pH = \frac{1}{2}(pKa_1 + pKa_2)}$

184. due to odd ion effect

185. Due to higher conc of NaCl reaction goes more backward than $AgNO_3$

186. $Cu(OH)_2 \rightleftharpoons Cu^{+2} + 2OH^-$

S 2S

Ksp = $4S^3$

Now $\left[\begin{array}{l} 2S = [OH^-] \\ pOH = -\log[OH^-] \end{array}\right.$

$4S^3$ = 32 × 10^{-21}

S^3 = 8 × 10^{-21}

S = 2 × 10^{-7}

[OH^-] = 2S = 4 × 10^{-7}

pOH = 7 – log4

pOH = 7 – 0.6 = 6.4

pH = 14–6.4 = 7.6

187. $[H^+] = K_{In} \times \frac{[HIn]}{[In^-]}$

Case I $[H^+]_1$ = 3 × 10^{-5} × $\frac{0.75}{0.25} = 9 \times 10^{-5}$

Case II $[H^+]_2$ = 3 × 10^{-5} × $\frac{0.25}{0.75}$ = 1× 10^{-5}

Change in [H^+] = 8 × 10^{-5}

188. $pK_a + pK_b = pK_w$

pOH = pK_b ± 1

pH = 14 – pOH

190. Excess of acid entering the blood stream will react with HCO_3^- to produce H_2CO_3 and this newly produced H_2CO_3 will not further dissociate due to common ion effect.

191. $\boxed{pKa = -\log Ka}$ & $\boxed{pKb = -\log Kb}$

↑ pKa Ka↓ (Acidic strength)

193. meq. of HCl = 0.050 × 20 =1

meq. of $Ba(OH)_2$ = 0.10 × 2 × 30 = 6

meq of $Ba(OH)_2$ Left = 6 –1= 5

total volume = 20 + 30 = 50 ml

$$[OH^-] = \frac{5}{50} = 0.1$$

194. $Hg_2Cl_2 \rightleftharpoons Hg_2^{+2} + 2Cl^-$

$Ksp = [Hg_2^{+2}]\,[Cl^-]^2 = 4S^3$

$4S^3 = 32 \times 10^{-18}$

$S^3 = 8 \times 10^{-18}$

$S = 2 \times 10^{-6}$ mol/L

197. $Kn = \frac{K_w}{K_b} = \frac{10^{-14}}{10^{-5}} = 10^{-9}$

200. $CH_3NH_2 + HCl \longrightarrow CH_3NH_3Cl$

0.1	0.08	0	
0.02	0	0.08	Buffer solution

$$pOH = pK_b + \log\left[\frac{salt}{Base}\right]$$

$K_b = 5 \times 10^{-4}$

$pK_b = 4 - \log 5$

$= 4 - 0.7$

$= 3.3$

Now,

$$pOH = 3.3 + \log\frac{0.08}{0.02}$$

$= 3.3 + \log 4$

$= 3.3 + 0.6$

$= 3.9$

pH = 14 – 3.9 = 10.1

$[H^+]$ = Anilog (–10.1)

$= 8 \times 10^{-11}$ mol/L

202. pH of acidic buffer solution is

$$pH = pKa + \log\frac{\{salt\}}{\{Acid\}}$$

$Ka = 1.8 \times 10^{-5}$

$pKa = 5 - \log 1.8$

$= 5 - 0.25$

$= 4.75$

$$pH = 4.75 + \log\frac{0.15}{0.1}$$

$= 4.75 + \log 1.5$

$= 4.75 + 0.15$

$= 4.90$

209. $pH = 7 + \frac{1}{2}pKa + \frac{1}{2}\log C$

$= 7 + 3 - \frac{1}{2} = 9.5$

210. $CH_3COOH + HCl \rightleftharpoons Cl^- + CH_3COOH_2^+$

WB WA SB SA

211. $pK_b = 4.7$

$K_b = 10^{-5} \times 10^{0.3}$

$K_b = 2 \times 10^{-5} = C\alpha^2$

$2 \times 10^{-5} = 5 \times 10^{-3}\,\alpha^2$

$$\alpha^2 = \frac{2\times10^{-5}}{5\times10^{-3}} = 0.4 \times 10^{-2}$$

$\alpha = 6.32 \times 10^{-2} \times 100 = 6.32\%$

213. $Ksp = [Mg^{+2}]\,[F^-]^2$

$Ksp = (s')\,(2s' + c)^2$

$8 \times 10^{-8} = S'\,c^2$

$$s' = \frac{8\times10^{-8}}{4\times10^{-2}}$$

$= 2 \times 10^{-6}M$

214. $pH = 7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b$

pH = 7

215. $[H_3O^+] = \sqrt{K_aC}$

$= \sqrt{4\times10^{-8}\times0.25}$

$= 10^{-4}$

216. $S = \frac{0.46}{245} = 187 \times 10^{-3}$ M

$Ksp = 4S^3$

$= 4 \times (187 \times 10^{-3})^3$

$Ksp = 2.6 \times 10^{-8}$

217. $Ksp = [Ag^{+1}]^2\left[CO_3^{-2}\right]$

$Ag_2CO_3 \rightleftharpoons 2Ag^+ + CO^{-2}$

$\quad 2S' \quad (S'+0.1)$

$Na_2CO_3 \rightarrow 2Na^+ + CO_3^{-2}$

$4 \times 10^{-13} = (2S^1)^2\ (S^1+c)$

$4 \times 10^{-13} = 4\ (S^1)^2\ c$

$$S^1 = \sqrt{\frac{10^{-13}}{10^{-1}}} = 10^{-6}$$

218. $NH_4OH + HCl \longrightarrow NH_4Cl + H_2O$

$5 \qquad 2.5$

$2.5 \qquad 2.5$

$\underbrace{\qquad\qquad}_{\text{Basic Buffer}}$

220. $Cd(OH)_2 \rightleftharpoons Cd^{2+} + 2OH^-$

$\quad s' \quad (2s' + C)$

$KOH \longrightarrow K^+ + OH^-$

$\quad C \quad C$

$K_{SP} = (s')(C)^2$

$$s' = s' = \frac{K_{SP}}{C^2} = \frac{2.5\times10^{-14}}{10^{-2}} = 2.5\times10^{-12}$$

221. $NH_4OH \rightleftharpoons NH_4^+ + OH^-$

$KOH \longrightarrow K^+ + OH^-$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_4OH]}$$

$$1.8\times10^{-5} = \frac{[x][10^{-2}]}{2\times10^{-2}} \Rightarrow x = 3.6\times10^{-5}$$

226. due to common ion effect $[H^+]\downarrow$

THERMODYNAMICS

235. $H-C\equiv C-H + 2H-H \longrightarrow H_3C-CH_3$

$\Delta H_{rxn} = 2$ B.E (C–H) + B.E (C≡C) + 2B.E(H–H) –6 B.E (C–H)+B.E(C–C)

$\Rightarrow 2291–2890 = 2 \times 425 + 601 + 2 \times 240 – 6 \times 425 + 340 = –599$

238. Δn_g shold be greater than zero

239. $\frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \longrightarrow NH_3(g)$

$\Delta H_{rxn} = \Delta_f H(NH_3)$

$= \frac{1}{2}$ B.E.(N≡N) + $\frac{3}{2}$ B.E. (H–H) – 3B.E.(N–H)

$– 120 = \frac{1}{2}\times240 + \frac{3}{2}\times400$ – 3B.E.(N–H)

B.E.(N–H) = 280 KJ/mol.

240. $C(s) + 2H_2(g) \longrightarrow CH_4(g),\ \Delta H_f(CH_4)$

$\Delta H_f(CH_4) = (2) + 2 \times (3) – (1)$

$= –y + 2(–2) – (–x)$

$\Rightarrow x–y – 2z$

241. $(3) = -\frac{1}{2} \times$ eq. (1) – $\frac{1}{2} \times$ eq. (2)

$= -\frac{1}{2}\times(-27.1) - \frac{1}{2}(-53.5)$

$\Delta H. = + 40.3$ KJ

242. $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(\ell)$

$\Delta H° = \Delta U° + \Delta n_g RT$

$\Delta H° = – x + (–2) R \times 298$

$\Delta H° = – x – 596 R$

243. In the given reaction heat will be released in bond formation and randomness decreases. as the number of moles of gases$\downarrow$

244. $\Delta G° = –2.303\ RT \log K_{eq}$

$= – 2.303 \times 8.314 \times 300 \log 100$

$\Delta G° = – 11.488$ KJ

245. $H_2O(\ell) \longrightarrow H_2O(g)$

$\Delta H° = \Delta E° + \Delta n_g RT$

$$40.66 = \Delta E° + \frac{1\times8.314\times373}{1000}$$

$\Delta E° = + 37.56$ KJ/mol

for 2 mole $\Delta E° = 2\times 37.56 = 75.12$ KJ

248. $\Delta H = 2 \times \frac{1\times2\times300}{1000} = 2.6$ KCal

$\Delta G = \Delta H – T\Delta S$

$2.6 - \frac{300\times50}{1000} = –12.4$ KCal.

249. (4) = (1) + (2) – (3)

$\Delta H = 150 + (–125) – 350$

$= –325$ KJ

250. $\Delta H_{rxn} = –900 = –6$ BE. (N–H)

B.E. (N–H) = 150 Kg

251. $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O,\ \Delta H = –400$ KJ/mol

$\because$ for 2 mole or 36 gm water, heat released = –400 KJ/mol

$\therefore$ for 400 gm water, heat released will be

$$= \frac{-400}{36}\times40$$

$= – 444.44$ KJ

255. $W = -2.303\ nRT \log \dfrac{V_2}{V_1}$

$= -\dfrac{2.303 \times 2 \times 8.314 \times 300 \log \frac{40}{4}}{1000}$ KJ

W = –11.488 KJ

256. $W = -P(V_2 - V_1) = -PnRT\left(\dfrac{1}{P_2} - \dfrac{1}{P_1}\right)$

$W = -\dfrac{1 \times 5 \times 8.314 \times 300\left(\frac{1}{1} - \frac{1}{10}\right)}{1000}$ KJ

W = – 11.224 KJ

261. $H_2O(s) \rightleftharpoons H_2O(\ell)$

$\Delta G = \Delta H - T\Delta S$

$O = \Delta H - \dfrac{273 \times 5.26}{1000}$

ΔH = 1.435 KCal/mol

263. $\Delta G = \Delta H - T\Delta S < O$ for spontaneous process

$T > \dfrac{\Delta H}{\Delta S}$

$T < \dfrac{-100 \times 10^3}{-100}$

T < 1000 K

264. $2LiOH + H_2SO_4 \rightarrow Li_2SO_4 + 2H_2O$

$\Delta H_{neut.} = -69.6$ KJ/mol $= 2\Delta H_{ion}(LiOH) + O - 57.3 \times 2$

$\Delta H_{ioni}(LiOH)$ = 22.5 KJ/mol.

266. $\Delta S_{rxn} = 50 - \dfrac{1}{2} \times 60 - \dfrac{3}{2} \times 40 = -40\ JK^{-1}mol^{-1}$

$T = \dfrac{\Delta H}{\Delta S} = \dfrac{-30 \times 10^3}{-40} \Rightarrow 750K$

267. $\Delta H_{rxn} = 4 \times (-393.5) - 4x(-110.5) - 2\times(-33.2)$

ΔH_{rxn} = – 1065.5 KJ

268. 18g water = 1 mol water

$H_2O(\ell) \rightleftharpoons H_2O(g)$

$\Delta H = \Delta E + \Delta n_g RT$

$40.66 = \Delta E + \dfrac{1 \times 8.314 \times 373}{1000}$

ΔE = 40.66 –3.1

ΔE = + 37.56 KJ

269. (1) + 2 × (2) – (3)

$\Delta H_F(CH_4) = \Delta H_1 + 2\Delta H_2 - \Delta H_3$

= – 94.2 + 2 × (–68.3) – (–210.8)

= –20 KCal

272. T.R : $2Al_{(s)} + \dfrac{3}{2}O_{2(g)} \rightarrow Al_2O_{3(s)}$; $\Delta H_r = \Delta H_f$

$\Delta H_f = -\dfrac{3352}{2} = -1676$ KJ/mol

273. $\Delta G = \Delta H - T\Delta S$

= (+ve) – (+ve)

ΔG = (+ve) (–ve)

ΔG = –ve only at high temp.

ΔH = P – R

92 = (–16) + (–94) – x

x = –202

275. $\Delta G° = -nFE°$

–21.20 = – 1× 96500 × E°

$E° = \dfrac{21.20 \times 1000}{96500}$

= 0.22V

276. Target reaction $C_{(s)} + \dfrac{1}{2}O_{2(g)} \rightarrow CO_{(g)}$; $\Delta H_r = \Delta H_f$

Target reaction = (I) + (II) rev

= (–X) + (+Y)

ΔH_r = Y –X

277. average B.E = $\dfrac{368}{2}$ = 184 kJ/mol^{-1}

278. $\Delta G = 0 = \Delta H - T\Delta S$

$T = \dfrac{-10000}{-33.3}$ = 300.3 K

Since ΔH & ΔH both are negative therefore reaction will be spontaneous at T < 300.3 K

279. Heat absorbed by calorimeter = Heat released in combustion of CH_4

$\Downarrow$

$q = C\Delta T$

= 17.7 × 0.5 = 8.85 kJ

0.16 g $\longrightarrow$ 8.85

16 g $\longrightarrow \dfrac{8.85}{0.16} \times 16$ = 885 KJ

280. –8.1 × 1000 = –2.303 × 8.3 × 1000 × log K

log K = 0.42 $\Rightarrow$ K = $10^{0.42}$ = 2.6

281. $\Delta H_{Hyd.} = \Delta H_{Crys.} + \Delta H_{Sol.}$

$-774.1 = -777.8 + \Delta H_{Sol.}$

$\Delta H_{Sol.} = 3.7 \text{ KJ mol}^{-1}$

$\Delta G = \Delta H - T\Delta S$

$= 3.7 - (298 \times 0.043)$

$= -9.114 \text{ KJ mol}^{-1}$

294. $\Delta G^\circ = -2.303 \times nRT \times \log K_P$

REDOX

303. $\overset{+4}{S}O_2 + H_2\overset{-2}{S} \longrightarrow 3\overset{0}{S} + 2H_2O$

304. $Ba^{2+}\ 2(H_2PO_2)$

$2 + x - 4 = -1$

$x = +1$

306. $Br_2 \xrightarrow{\text{Reduction}} Br^{\ominus} + BrO_3^{\ominus}$

Reduction (–1)

oxidation +5

309. $MnO_4^- + FeC_2O_4 \longrightarrow Mn^{+2} + Fe^{+3} + CO_2$

Equivalents of MnO_4^- = Equivalents of FeC_2O_4

$x \times 5 = 2 \times 3$

$x = 1.2$ mol

310. $2 \times 2 + 2x - 14 = 0$

$x = +5$

311. Intermediate oxidation number.

312. $3Fe + 4H_2O \longrightarrow Fe_3O_4 + 8H^+ + 8e^-$

313. $PbO_2 + HNO_3 \longrightarrow Pb(NO_3)_2 + H_2O + \frac{1}{2}O_{2(g)}$

314. Equivalents of $KMnO_4$ = Equivalents of H_2O_2

$1.5 \times 5 = x \times 2$

$x = \frac{1.5 \times 5}{2} = \frac{15}{4}$

318. $\overset{0}{I_2} + H\overset{+5}{N}O_3 \longrightarrow H\overset{+5}{I}O_3 + \overset{+4}{N}O_2 + H_2O$

Equivalents of I_2 = Equivalents of HNO_3

$\frac{5}{254} \times 10 = \frac{x}{63} \times 1$

$x = 12.4$ g

320. Equivalents of I_2 = Equivalents of $S_2O_3^{-2}$

$\frac{X}{254} \times 2 = N \times V(L)$

$\frac{x}{127} = \frac{4}{1000} \times 0.11$

$x = 0.558$ g

325. $MNO_4^- \longrightarrow Mn^{+2}$, v.f. = 5

no. of eq = $0.1 \times 5 \times 1 = 0.5$

Option

(1) $C_2O_4^{2-} \longrightarrow CO_2$, v.f. = $1 \times 2 = 2$

no.f of eq = $0.25 \times 2 \times 1 = 0.5$

(2) $Fe^{2+} \longrightarrow Fe^{3+}$, v.f. = 1

no.f of eq = $0.8 \times 1 \times 1 = 0.8$

(3) $FeC_2O_4 \longrightarrow Fe^{3+} + CO_2$,

v.f. = 3, no. of eq = $0.1 \times 3 \times 1 = 0.3$

(4) $Cr_2O_7^{2-} \longrightarrow Cr^{+3} \Rightarrow$ v.f. = 6

$\Rightarrow$ no. of eq = $0.6 \times 6 \times 1 = 3.6$

BEHAVIOUR OF GASES

333. $P_1V_1 = P_2V_2$

$V_2 = \frac{0.1 \times 22.7}{0.2} = 11.35$

334. $\frac{V_2}{T_2} = \frac{V_1}{T_1}$

$V_2 = \frac{2 \times 299.21}{296.4} = 2.01\text{ L}$

335. $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$P_2 = \frac{500 \times 400 \times 280}{300 \times 550}$

336. $P_{H_2} = \frac{n_{H_2}}{n_{H_2} + n_{CH_4}}$

$P_{H_2} = \frac{\frac{5}{2}}{\frac{5}{2} + \frac{10}{16}} \times 30$

345. $r_{SO_2} = \sqrt{2}\ r_x$

$\frac{r_{SO_2}}{r_x} = \sqrt{2} = \sqrt{\frac{M_{wx}}{Mwso_2}} = \sqrt{\frac{M_{wx}}{64}}$

$M_{wx} = 2 \times 64 = 128$

$V.D = \frac{Mw}{2} = \frac{128}{2} = 64$

346. $\frac{r_A}{r_B} = \frac{1}{2} = \sqrt{\frac{M_{WB}}{M_{WA}}} \Rightarrow \frac{M_{WA}}{M_{WB}} = \frac{4}{1}$

348. $\frac{r_A}{r_B} = \frac{4}{1} = \sqrt{\frac{d_B}{d_A}} \Rightarrow \frac{d_A}{d_B} = \frac{1}{16}$

354. PM = dRT $\Rightarrow \frac{P_1}{d_1T_1} = \frac{R_2}{d_2T_2}$

$\Rightarrow \frac{2}{5.46\times300} = \frac{1}{d\times273}$

SOLID STATE

357. $d = \frac{Z\times M}{a^3\times N_A}$

$10.3 = \frac{2\times96}{a^3\times6\times10^{23}}$

a = 3.14 × 10^{-8} cm = 314 pm

r = $\frac{\sqrt{3}a}{4}$ = 135.96 pm

359. A $\longrightarrow$ corners = $8\times\frac{1}{8}=1$

B $\longrightarrow$ at 5 face centers = $5\times\frac{1}{2}=\frac{5}{2}$

A : B = 1 : $\frac{5}{2}$ = 2 : 5

$\therefore$ A_2B_5

362. $\frac{r_{A^{+2}}}{r_{B^{-2}}} = \frac{94}{146} = 0.64$

The radius ratio lies in between 0.414 to 0.732 so, cation should be in octahadral void

therefore, structure will be Rock salt.

363. $O^{-2} \longrightarrow ccp \rightarrow 4$

A $\longrightarrow \left(\frac{1}{8}\right)^{th}$ T.V. = $\frac{1}{8}\times8=1$

B $\longrightarrow \left(\frac{1}{4}\right)^{th}$ O.V. = $\frac{1}{4}\times4=1$

ABO_4

365. Cl^- is pressent at conurs while Cs^+ is present at Body centre, so minimum interatomic distance

$= \frac{\sqrt{3}a}{2} = 3.72pm$

366. d = $\frac{Z\times M}{a^3\times N}$ M $\rightarrow$ mass of substance; N $\rightarrow$ No of atoms

N = $\frac{2\times208}{(288\times10^{-10})^3\times7.2}$

= 24 × 10^{23}

367. M $\longrightarrow \frac{1}{3}$ of T.V = $\frac{1}{3}\times8=\frac{8}{3}$

N $\longrightarrow$ ccp $\longrightarrow$ 4

M N

$\frac{8}{3}$: 4

2 : 3 $\Rightarrow M_2N_3$

368. d = $\frac{Z\times M}{a^3\times N_A}$

$M = \frac{d\times a^3\times N_A}{2} = \frac{8.92\times(3.6\times10^{-8})^3\times6\times10^{23}}{4}$

= 63

372. For ccp Lattice a = $\frac{4r}{\sqrt{2}} = \frac{4\times215}{\sqrt{2}}$ = 608.2 pm

373. $Ni_{0.98}O$ $\rightarrow$ Ni^{+2} (x), Ni^{+3} (0.98-x)

charge Balancing

x × 2 + (0.98 – x) × 3 = 2

2x + 2.94 – 3x = 2

– x = –0.94

x = 0.94

fraction of Ni^{+2} = $\frac{0.94}{0.98} = 0.959$

376. CsCl $\rightarrow$ cation in cubic void

$\frac{r^+}{r^-} = 0.732$

$r^- = \frac{120}{0.732}$ = 164 pm

382. When AgCl is dopped with $CdCl_2$

No. of Cd^{+2} added = No. of cation vacancies.

384. A $\longrightarrow$ HCP $\longrightarrow$ 6

C $\longrightarrow \left(\frac{2}{3}\right)$ O.V. $\rightarrow \frac{2}{3}\times6 = 4$

C A

4 6

2 : 3

C_2A_3

385. no. of atoms in FCC = 4

volume of 1 atom = $\frac{4}{3}\pi r^3$

Total volume of 4 atoms = 4 × $\frac{4}{3}\pi r^3 = \frac{16}{3}\pi r^3$

386. $r_{Na^+} + r_{Cl^-} = \frac{a}{2}$

281 pm = $\frac{a}{2}$

562 pm = a

387. O^{-2} forms ccp → 4

A occupy $\frac{1}{6}$th of THV → $\frac{1}{6} \times 8 = \frac{4}{3}$

B occupy $\frac{1}{3}$rd of OHV ⇒ $\frac{1}{3} \times 4 = \frac{4}{3}$

Fromula, $A_{\frac{4}{3}}B_{\frac{4}{3}}O_4 \Rightarrow ABO_3$

388. A occupy $\frac{3}{4}$ of THV ⇒ $\frac{3}{4} \times 8 = 6$

B forms ccp ⇒ 4

Formula, $A_6B_4 \Rightarrow A_3B_2$

391. $d = \frac{Z \times M_w}{a^3 \times N_A}$

$10 = \frac{4 \times M_w}{(10^{-8})^3 \times 6 \times 10^{23}}$

$6 = 4 \times M_w$

$\frac{6}{4} = M_w \Rightarrow 1.5g$

$\text{mol} = \frac{\text{Mass}}{M_w} = \frac{100}{105}$

no. of atoms

= mol × N_A

$= \frac{100}{1.5} \times 6 \times 10^{23} = \frac{6}{1.5} \times 10^{25} = 4 \times 10^{25}$

392. A occupies corner ⇒ $8 \times \frac{1}{8}$

B = occupies face centres → $6 \times \frac{1}{2} = 3$

⇒ now, atoms along one body diagonal are removed

$A \to 6 \times \frac{1}{8} = \frac{3}{4}$

Formula

$A_{\frac{3}{4}}B_3 \Rightarrow AB_4$

393. X forms fcc → 4

Y occupies THV → 8

Z occupies $\frac{1}{2}$ OHV → $\frac{1}{2} \times 4$

Formula, $X_4\,Y_8\,Z_2 \Rightarrow X_2Y_4Z$

394. 4 atoms → 1 unit cell

1 atoms → $\frac{1}{4}$ unit cell

0.1 N_A atom → $\frac{0.1}{4}$ N_A unit cell

→ 0.025 N_A unit cell

395. In rock salt

A ⟶ 4 FCC

B ⟶ 4 OHV

C ⟶ 8 THV

Formula,

$A_4B_4C_8$

ABC_2

399. $d = \frac{Z \times M}{N_A \times a^3}$

In FCC → Z = 4

$10.5 = \frac{4 \times M}{6 \times 10^{23} \times (4 \times 10^{-8})^3}$

M = 101.2

SOLUTION

422. 30g ethylene glycol ($HO–CH_2–CH_2–OH$) is in 100 mL solution.

∴ moles of $C_2H_6O_2 = \frac{30}{62} = 0.484$

and moles of $H_2O = \frac{70}{18} = 3.89$

$X_{C_2H_6O_2} = \frac{0.484}{0.484 + 3.89} = \frac{0.484}{4.374} = 0.11$

423. Amount of urea = $\frac{15}{100} \times 500 + \frac{25}{100} \times 400$

= 75 + 100 = 175 g

% w/w = $\frac{175}{900} \times 100 = 19.44\%$

424. moles of acid = $\frac{3.7}{74} = 0.05$

molality of solution = $\frac{w}{m} \times \frac{1000}{w_{(ing)}} = \frac{0.05}{80} \times 1000$

= 0.625m

425. Molality = $\frac{w}{n} \times \frac{1000}{w_{(ing)}}$

$= \frac{3.5}{58.5} \times \frac{1000}{96.5} = 0.62m$

[∵ 3.5% NaCl solution = 3.5 g NaCl in 96.5g H_2O]

426. $\frac{10}{180}\times 10 = \frac{4}{m_w}\times 10 \Rightarrow m_w = 4 \times 18 = 72$

427. $P_S = X_A P_A^\circ + (1 - X_A)P_B^\circ$

428. $\Delta T_f = i \times \frac{1000 K_f\ w}{mW}$

$\Delta T_f = i \times K_f \times \text{molality}$

$MgBr_2 \rightleftharpoons Mg^{+2} + 2Br^-$

$i = 1 - \alpha + 3\alpha$

$= 1 - 0.4 + 3 \times 0.4 = 1.8$

$\therefore \quad \Delta T_f = 1.8 \times 1.86 \times 0.2$

$= 0.67$

and $\quad T_1 = T_f - \Delta T_f = -0.67°C$

429. volume of solution $= \frac{\text{Mass}}{\text{density}} = \frac{100}{1.052}$

$= 95.05\text{mL}$

Molarity $= \frac{12.25}{98} \times \frac{1000}{95.05} \Rightarrow 1.315\text{M}$

431. $i = \frac{(\Delta T_f)_{obs}}{(\Delta T_f)_{Theo}} = \frac{(\Delta T_f)obs}{K_f \times \text{molality}}$

$i = \frac{0.558}{1.86 \times 0.1} = 3$

for $Hg(NO_3)_2 \rightleftharpoons Hg^{+2} + 2\ NO_3^-$

$i = 1 - \alpha + n\alpha$

$\therefore \quad 3 = 1 + \alpha\ (n-1)$

$\alpha = \frac{3-1}{3-1} = 1$ or 100%

434.

	2A $\longrightarrow$	5B
initially	2	0
in solution	$2-2\alpha$	5α

$i = \frac{2-2\alpha+5\alpha}{2} = \frac{2-0.6+1.5}{2} = 1.45$

435. $\pi_1 = \pi_2$

$\therefore\ i_1c_1 = i_2c_2$

$1\times \frac{6}{60}\times 10 = 2\times\frac{w}{58.5}\times 10$

$\therefore\ W = \frac{58.5}{20} = 2.925\text{g}$

436. $P_S = X_A P_A^\circ + X_B P_B^\circ$

$P_S = X_A P_A^\circ + (1 - X_A)P_B^\circ$

$\therefore\ 580 = X_A \times 440 + (1-X_A)\ \ 720$

$140 = 280\ X_A$ and $X_A = 0.5$

$Y_A = \frac{P_A'}{P_S} = \frac{0.5\times 440}{580} = 0.38$

442. $\Delta T_b = \frac{1000 K_b w}{mW}$

and $m = \frac{1000 K_b w}{\Delta T_b w}$

444. $\pi = CRT = \frac{w}{mV}RT$

445. $MgI_2 \longrightarrow Mg^{2+} + 2I^-$

$\alpha = \frac{i-1}{n-1} = \frac{1.92-1}{3-1} = \frac{0.92}{2} = 0.46$

$= 46\%$

450. mole fraction of vinyl chloride $= \frac{n_{v.c}}{n_{v.c} + n_{H_2O}}$

$= \frac{0.09}{55.59} = 0.00162$

$X_{(v.c)} = \frac{P}{K_H}$

$\therefore\ K_H = \frac{P}{X_{(v.c)}} = \frac{1}{0.00162} = 617.284$ bar

452. $Al_2\ (SO_4)_3$

$i = 5\ \therefore\ \alpha = \frac{i=1}{n-1}$

$0.9 = \frac{\alpha-1}{5-1}$

$\alpha = 3.6 + 1 = 4.6$

$\pi = \ i\ CRT$

$= 4.6 \times \left(\frac{0.03}{6}M\right)R(298)$

$= 0.566$ atm

453. $\Delta T_f = 9.3$

$\frac{40}{62}\times\frac{1000}{(400-x)}\times 1.86 = 9.3$

$\frac{40\times 1000}{62\times 9.3}\times 1.86 = 400 - x$

$x = 400 - 129.03$

$\boxed{x = 270.97}$

454. $i \times c$ min then vapour pressure max.

455. $\pi = CRT = \frac{w}{Mw}\times\frac{1000}{v(mL)}\times R\times T$

$2.55 = \frac{4.34}{Mw}\times\frac{1000}{1000}\times 0.082\times 298$

$M_w = 41.64$

456. i c values

$Li_2\,SO_4 = 3 \times 0.5 = 1.5$

$KCl = 2\times 1 = 2$

$BaCl_2 = 3 \times 0.5 = 1.5$

$Al_2(SO_4)_3 = 5 \times 1 = 5$

Max ic is for $Al_2(SO_4)_3$

457. $\Delta T_f = mK_f = \frac{w}{Mw}\times\frac{1000}{W_A}\times K_f$

$$Mw = \frac{1}{0.4}\times\frac{1000}{50}\times 5.12 = 256$$

458. $P_S = P_B^\circ + x_T(P_P^o - P_B^o)$

$= 160 + \frac{1}{2}\ (60 - 160)$

$= 160–50 = 110.$

459. $m = \frac{1000M}{1000d - MMw}$

$$\frac{120}{60}\times\frac{1000}{1000} = \frac{1000M}{1000\times 1.5 - M(60)}$$

$2.30 \times 1000 - 120\ M = 1000\ M$

$$M = \frac{2.30\times 1000}{1120} = 2.05$$

461. $\Delta T_f = 1.22°\ C$

$$\frac{w}{Mw}\times\frac{1000}{W_A}\times K_f = 1.22$$

$$\frac{10}{Mw}\times\frac{1000}{100}\times 1.86 = 1.22$$

$$Mw = \frac{186}{1.22} = 152.4$$

462. $n_{Ben} = 20\% = 0.2$

$n_{Tol} = 80\% = 0.8$

$P_S = X_B P_B^o + X_T\ P_T^\circ$

$P_S = 0.2 \times 75 + 0.8 \times 22$

$= 15.0 + 17.6$

$= 32.6$

463. $X_B = \frac{P^o - P_S}{P^o} = \frac{0.8 - 0.6}{0.8} = \frac{1}{4} = 0.25$

464. $N = \frac{20}{5.6} = 3.56$

467. ic is max. for $Al_2(SO_4)_3$

466. $\Delta T_f = 4\times\frac{0.1}{329}\times\frac{1000}{100}\times 1.86 = 2.26 \times 10^{-2}$

$K_3[Fe(CN)_6]\ \therefore\ i = 4$

467. $i = 2.74$

$$\alpha = \frac{i-1}{n-1} = \frac{2.74-1}{3-1} = \frac{1.74}{2} = 0.87$$

$= 87\%$

468. $\frac{1.05}{Mw}\times 10 = \frac{3}{180}\times 10$

$Mw = 1.05 \times 60 = 63$

469. $\frac{P^o - P_S}{P_S} = \frac{n_B}{n_A} = \frac{W_B / Mw_B}{W_A / Mw_A}$

$$\frac{1020-990}{990} = \frac{5/x}{58.5/18} = \frac{5\times 78}{58.5x}$$

$$\frac{30}{990} = \frac{5\times 78}{58.5x} \Rightarrow x = 220$$

470. ic is max for $Al_2(SO_4)_3$

$\therefore$ min f.P. is for option (4)

471. order of i $\pi_1 > \pi_2 > \pi_3 > \pi_4$

$\therefore$ order of osmotic pressure is also $\pi_1 > \pi_2 > \pi_3 > \pi_4$

CHEMICAL KINETICS

502. $$\frac{t_{0.875}}{t_{0.75}} = \frac{\frac{2.3}{k}\log\left(\frac{A_0}{0.125A_0}\right)}{\frac{2.3}{k}\log\left(\frac{A_0}{0.25A_0}\right)} = \frac{\log 8}{\log 4} = \frac{3}{2}$$

503. $K = Ae^{-\frac{Ea}{RT}}$

$$\text{In } K = \text{In } A - \frac{Ea}{RT}$$

504. $K_2 = K_1\ (m)^{\Delta T/10} = K_1(2)^{90/10} = K_1 \times (2)^9 = 512K_1$

506. $A_0 - A = Kt$

$A_0 - 0.5 = 3\times 10^{-2} \times 25$

$A_0 - 0.5 + 0.75 = 1.25M$

507. $\ell n\ K\ \ell n\ A - \frac{Ea}{RT}$

$$\ell n\ K = \ell n\ A - \frac{Ea}{R}\left(\frac{1}{T}\right)\ \text{slope} = \frac{-Ea}{R}$$

508. $A \longrightarrow 28 + C$

Pi	0	0	$Pi - x + 2x + x = P_t$
Pi – x	2x	x	$x = \frac{P_t - P_i}{2}$

$$K = \frac{2.303}{t}\log\frac{P_i}{P_i - x} = \frac{2.303}{K}\log\frac{2P_i}{3P_i - P_t}$$

509. $ROR = \frac{-d[N_2]}{dt} = \frac{-d[H_2]}{3dt} = \frac{d[NH_3]}{2dt}$

$$ROR = \frac{d[NH_3]}{2dt}$$

$$ROR = \frac{2.5\times 154}{2} = 1.25\times 10^{-4}$$

$$\frac{-d[H_2]}{dt} = \frac{3}{2}\frac{d[NH_3]}{dt} = \frac{3}{2}\times 2.5\times 10^{-4} = 3.75\times 10^{-4}$$

510. $\frac{dSO_3}{dt} = 1.6\times 10^{-3}$ kg/min

$$= \frac{1.6\times 10^{-3}\times 1000}{80}\text{ mol/min}$$

$$\frac{-dSO_2}{2dt} - \frac{-dO_2}{dt} = \frac{dSO_3}{2dt}$$

$$\frac{-dSO_2}{dt} = \frac{dSO_3}{dt} = \frac{1.6}{80}\text{mol/min}$$

$$\frac{-dSO_2}{dt} = \frac{1.6}{80}\text{mol/min}$$

$$\frac{1.6}{80}\times 64\times 10^{-3}\text{kg/mol}$$

$= 1.28 \times 10^{-3}$ mol/min

511. $\frac{Exp(1)}{Exp(2)}\ \frac{1.2\times 10^{-3}}{2.4\times 10^{-3}} = \frac{K(0.05)^x(0.05)^y}{K(0.10)^x(0.05)^y}$

$$\frac{1}{2} = \left(\frac{1}{2}\right)^x (x = 1)$$

$$\frac{Exp(1)}{Exp(3)}\ \frac{1.2\times 10^{-3}}{1.2\times 10^{-3}} = \frac{K(0.05)^x(0.05)^y}{K(0.05)^x(0.10)^y}$$

$$1 = \left(\frac{1}{2}\right)y\quad \boxed{y = 0}$$

513. $20g \xrightarrow{24\,hrs} 10g \xrightarrow{24\,hrs} 5g \xrightarrow{24\,hrs} 2.5g$

515. Rate constant is independent of concentration of Reactant.

516. $\log K = \log A - 10\frac{Ea}{2.3RT}$

$$\lim_{T\to\infty}\log K = \log A - \frac{Ea}{2.3RT}$$

$= \log A$

519. unit of K $\boxed{mol^{1-n}L^{n-1}sec^{-1}}$

520. $r = K(A)^1$

$$t_2 = \frac{0.693}{K}$$

521. $\boxed{t_2 \times \frac{1}{a^{n-1}}}$

523. $\Delta H = (Ea)f - (Ea)b$

$-38 = 20 - Ea_f$

524. $r = K\,(4)^{1/2}\,(2)^2$

$r = 2 \times 2 \times 2 \Rightarrow 8$ times

525. $r = K\,(A_2)(B)\ K\to \frac{(A_2)}{(A)^2}\quad (A_2) = K(A)^2$

$r = KK\,(A)^2(B)^1 \to n = 3$

527. $t = \frac{2.39}{K}\log\frac{9}{9}$

$$t = \frac{2303}{K}\log\frac{9}{9/4} = \frac{2.303}{K}\log 4$$

529. $(B)\% \Rightarrow \frac{K_B}{K_A + K_B}\times 100$

535. $r = k[N_2O_5]^1$

$1.02 \times 10^{-4} = 3.4 \times 10^{-5}[N_2O_5]$

$$\Rightarrow [N_2O_5] = \frac{1.02\times 10^{-4}}{3.4\times 10^{-5}} = 3M$$

536. $N_2O_5 \longrightarrow N_2O_4 + \frac{1}{2}O_2$

t = 0	11MM	O	O
t = 20s	114-x	x	$\frac{x}{2}$

$\therefore\quad 114 + \frac{x}{2} = 133 \Rightarrow x = 38$ mm

$$r = \frac{x}{t} = \frac{38}{20}$$

$r = 1.9$ mm sec^{-1}

$$r = \frac{1.9}{760} = 2.5\times 10^{-3}\text{ atm sec}^{-1}$$

537. $A + B \longrightarrow C + D$

$t = 0 \quad 1M \quad 1M$

$r = k[A]^{1/2}[B]^{1/2}$

if initial concentration of A & B are same

then $r = k\,(A)^1$

$$t = \frac{2.303}{2.303\times10^{-3}}\log\frac{1}{0.1} = 1000 \text{ sec.}$$

539. $r_0 = k_0\,[A]^0 = \frac{r_1}{r_0} = \frac{k_1}{k_0}$ ....(1)

$r_1 = k_1[A]^1$

For zero order $(t_{1/2})_0 = \frac{[A]_0}{2k_0}$

For 1^{st} order $(t_{1/2})_1 = \frac{0.693}{k_1} \Rightarrow (t_{1/2})_0\ (t_{1/2})_1$

$$\frac{[A]_0}{2k_0} = \frac{0.693}{k_1}$$

$$\frac{k_1}{k_0} = \frac{2(0.693)}{(A)_0} = \frac{1.386}{1}$$

541. $5.78\times10^{-5} = \frac{2.303}{10\times3600}\times\log\left(\frac{A_0}{A_t}\right)$

$A_0 = 8A \Rightarrow A_t = \frac{A_0}{8} = 12.5\%\ (A_0)$

542. $t_{1/2} \propto \frac{1}{(a_0)^{n-1}} \Rightarrow n - 1 = 1$

$n = 2$

543. $e^{-\frac{Eq}{RT}} = e - \frac{18000}{2\times300} = e^{-30}$

$= 9.36\times10^{-14}$

544. $\text{Log}\left(\frac{K_2}{K_1}\right) = \frac{Ea}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$

$$\text{Log}4 = \frac{Ea}{2.303\times8}\left(\frac{1}{300} - \frac{1}{320}\right)$$

$$\left\{K_1 = \frac{\ln 2}{20},\ K_2 = \frac{\ln 2}{5}\right\}$$

ELECTROCHEMISTRY

575. Anode reaction $A \rightarrow A^{+2} + 2e^-$

cathode reaction $2A^{+3} + 2e^- \rightarrow 2A^{+2}$

$E^\circ_{ceel} = E^\circ_{cathode} - E^\circ_{Anode}$

$= 0.4 - (-0.3)$

$= 0.7$ V

576. $nFE^\circ = 2.303\ RT \log K$

$$\log K = \frac{nFE^\circ}{2.303RT} = \frac{2\times96500\times1.5}{2.303\times8.3\times298}$$

$K = 6.6\times10^{50}$

577. oxidising power α SRP

578. $\alpha = \frac{\Delta^c_m}{\Delta^o_m};\ k_a = \frac{C\alpha^2}{1-\alpha}$

579. $Fe^{+2} + 2e^- \longrightarrow Fe(s)\ \Delta G^0_1 = -2FE^0_1$

$Fe^{+2} \longrightarrow Fe^{+3} + e^- \quad \Delta G^0_2 = -1\,E^\circ_2$

$Fe^{+3} + 3e^- \longrightarrow Fe(s)\ \Delta G^0_3 = -3FE^\circ_3$

$\Delta G^\circ_3 = \Delta G^\circ_1 - \Delta G^\circ_2$

$-3E^\circ_3 = 2\times0.447 - 1\times0.771$

$E^\circ_3 = -0.041$ V

581. $2Ag^+ + cu \longrightarrow 2Ag + Cu^{+2}$

$\Delta G^\circ = -nFE^\circ_{cell}$

$= -2\times96500\times0.4$

$= -77.2$ KJ

584. E^0_{cell} = positive

$\Delta G^\circ = -nFE^0_{cell}$

ΔG° = Negative

$\Delta G^\circ = -2.303\ RT \log K_{eq}$

$\log K_{eq}$ = +ve value

$K_{eq} > 1$

585. $E^0_{cell} = E^0_{R.P.}\ (\text{cathode}) - E^0_{R.P.}(\text{Anode})$

$= 1.428 - (-2.714) = 4.142$ V

586. Reducing power α S.O.P. $\alpha\ \frac{1}{\text{S.R.P.}}$

587. $E^\circ(I_2/2I^-) > E^\circ(Cr^{+3}/Cr^{+2})$

So I_2 will reduced to I^- and Cr^{+2} will oxidised to Cr^{+3}.

588. According to Kohtrausch law

$$\Lambda^o_{m(CH_3COONa)} + \Lambda^o_{m(HCl)} - \Lambda^o_{m(NaCl)}$$

589. According to faraday's second law

$$\frac{W_{Cu}}{E_{Cu}} = \frac{W_{O_2}}{E_{O_2}} \Rightarrow W_w = 31.75 \text{ g}$$

590. $Cr_2O_7^{-2} \longrightarrow 2Cr^{+3} + 6e^-$

0.1 mole 0.6 mole

Charge required = 0.6 F = 0.6 × 96500

= 57900C

593. $2H^+(aq) + 2e^- \longrightarrow H_2(g)$

$$E = E^\circ - \frac{2.303\times8.314\times323}{2\times96500}\log\frac{P_{H_2}}{[H^+]^2}$$

$$O = O-\frac{2.303\times8.314\times323}{2\times96500}\log\frac{P_{H_2}}{[H^+]^2}$$

$$\frac{P_{H_2}}{10^{-14}}=1$$

$pH_2 = 10^{-14}$ atm

594. Q = ne = It

$$n = \frac{I\times t}{e}=\frac{2\times120}{1.6\times10^{-19}} = 1.5\times10^{21} \text{ electron}$$

597. $E^0_{cell} = 0.8 - 0.77 = 0.03$ V

$\Delta G^\circ = -nFE^0_{cell} = -1 \times 96500 \times 0.03$

= –2895 J/mol

598. $E = 0.44 - \dfrac{0.06}{2}\log\dfrac{10^{-3}\times1}{(1)^2}$

= 0.44 – 0.03 (–3)

= 0.44 + 0.09

= 0.53 V

599. Q = I × t

= 1.5 × 10 × 60

= 900C

Reaction occuring at the cathode is

$Cu^{+2} + 2e^- \rightarrow Cu$

2F = 2 × 96500 deposit 1 mole Cu(63.5g)

⇒ 900 C will deposit = 0.296 g

603. Metal ion having higher SRP is displaced by lower SRP metal.

612. $\alpha=\dfrac{16.6}{390.7}=0.042$

615. $E = 3.17 - \dfrac{0.06}{2}\log\left(\dfrac{10^{-1}}{(10^{-4})^2}\right)$

= 3.17 – 0.03 log (10^7)

= 3.17 – 0.21 = 2.96 V

620. g.eq of H_2O = 4 × 2 = 8

$$\Rightarrow 8 = \frac{I\times t}{96500}$$

$$\Rightarrow 8 = \frac{4\times t}{96500} \Rightarrow t=1.93\times10^5 \text{ sec}$$

621. g.eq of Fe = $\dfrac{0.75\times30\times60}{96500}$

mass of Fe = $\dfrac{g.eq}{3}\times56$

622. $E = 0 - \dfrac{0.06}{2}\log\left\{\dfrac{(10^{-3})^2}{(10^{-1})^2}\right\}$

= –0.03 log (10^{-4})

= 0.03 × 4 = 0.12 V

624. g.eq. of Zn = $\dfrac{20\times10\times60}{96500}$

mole of Zn = $\dfrac{g.eq}{2}=0.06$

625. Faraday = g.eq. of H_2

$$=\left(\frac{112}{22400}\right)\times2=0.01F$$

626. $Au^{3+} + 3e^- \rightarrow Au$ E° = 1.4

$Au \rightarrow Au^+ + e^-$ E° = –1.69

$Au^{3+} + 2e^- \rightarrow Au^+$ $E^\circ = \dfrac{1.4\times3-1.69}{2}=1.25V$

627. Reducing power $\propto \dfrac{1}{E_{RP}}$

630. No. of faraday = no. of equivalents = $\dfrac{I\times t}{96500}$

wt of H_2 = 1g ⇒ mole = $\dfrac{1}{2}$ mole

V = $\dfrac{1}{2}$ × 22.4 = 11.22

632. $E^\circ=\left(E^\circ_{RQ}\right)_{Cat}-\left(E^\circ_{RP}\right)_{Anode}$

= 0.80 – 0.77 = 0.03

n = 1

$\Delta G^\circ = -nFE^\circ = -2.303 \times RT \log K$

log K = 0.5

K = 3.16

633. $K_a = C\alpha^2 \Rightarrow 25 \times 10^{-6} = 10^{-2} \times \alpha^2$

$\alpha = 5 \times 10^{-2}$

$$\alpha = \frac{\lambda_m}{\lambda_m^\infty} \Rightarrow 5\times10^{-2} = \frac{19.6}{\lambda_m^\infty}$$

$$\lambda_m^\infty = \frac{19.6\times100}{5} = 392$$

634. $K_2SO_4 \longrightarrow 2K^+ + SO_4^{2-}$

$$(\lambda_m^o)_{K_2SO_4} = 2(\lambda_m^o)_K + (\lambda_m^o)_{SO_4^{2-}}$$

$= 2 \times 73.5 + 160$

$= 307$

637. $$E_{RP} = E^\circ - \frac{0.059}{5}\log\frac{[M_n^{2+}]}{[MnO_4^-][H^+]^8}$$

$$E_{RP} = E^o + \frac{0.059}{5}\log\frac{[MnO_4^-][H^+]^8}{[Mn^{2+}]}$$

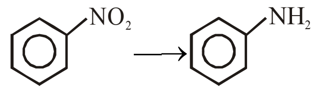
644. Same change is passed through hence deposited g.equivalent of each metal will be same except Al and Na in Aq. solution.

$\because n_{Ag} \times 1 = n_{Cu} \times 2 = 0 \times 3 = 0 \times 1 = 3$

$\Rightarrow n_{Ag} : n_{Cu} : n_{Al} : n_{Na}$

$3 : \frac{3}{2} : 0 : 0$

or $6 : 3 : 0 : 0$

652. 6H + 6e⁻ +  + $2H_2O$

n factor = 6 (change in O.S. of N)

$\because$ 1 mole nitrobenzene require 6 mole of e^-

mole of nitrobenzene = $\frac{12.3}{123} = 0.1$

$\therefore$ mole of e^- required = 0.6

change required = 0.6 F

but if efficiency of current is 50%

$\because Q = 0.6\ F \times \frac{100}{50} = 1.2\ F$

$= 115800\ C$

653. $\underset{1\ mole}{H_2O_2} \longrightarrow O_2$

change in O.S. (n-factor or valency factor) = 2

$\because$ 1 mole H_2O_2 required charge = 2F

$= 2 \times 96500\ C$

$= 1.93 \times 10^5\ C$

657. $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

for 1 mole MnO_4^- change required = 5F

$\therefore$ for 0.05 mole change required = 5 × 0.05

$= 0.25\ F$

IMPORTANT NOTES

INORGANIC CHEMISTRY

PERIODIC TABLE

1. If atomic numbers 117, 120 are discovered then their blocks will be:
(1) s, p (2) p, s (3) p, d (4) d, p

2. Correct order of metallic character will be
(1) P < Si < Be < Mg < Na
(2) P > Si > Be > Mg > Na
(3) P < Si < Be > Mg < Na
(4) P > Si < Be < Mg < Na

3. Which of following statement concerning element with atomic number 10 is false?
(1) Element is monoatomic
(2) It has a almost zero electron affinity
(3) It forms a covalent network solid
(4) It has extremely high value of I.E. in own period

4. Following are configuration of 4 atom :-
P = $(Ne)3s^2 3p^3$
Q = $(Ar)3d^{10} 4s^2 4p^3$
R = $(He)2s^2 2p^5$
S = $(Ne)3s^1$
Incorrect statement is :-
(1) EA : Q > R (EA = electron affinity)
(2) EN : R > P (EN = electro negativity)
(3) IP : P > S (IP = Ionisation potential)
(4) Atomic radius : S > R

5. Among P, S, Cl, F most and least negative ΔH_{eg} will be respectively of:
(1) Cl, P (2) P, Cl
(3) Cl, F (4) F, Cl

6. Which of the following order is valid for Li < B < Be < C < O < N < F < Ne
(1) IE (2) EGE
(3) EN (4) size

7. Which of the following represents the correct order of increasing first ionization enthalpy for Ca, Ba, S, Se and Ar ?
(1) Ca < S < Ba < Se < Ar
(2) S < Se < Ca < Ba < Ar
(3) Ba < Ca < Se < S < Ar
(4) Ca < Ba S < Se < Ar

8.

	Column-I (Elements)		Column-II (Periodic Properties)
(A)	F	(P)	Maximum ionization energy
(B)	Cl	(Q)	Maximum electronegativity
(C)	Fe	(R)	Maximum electron affinity
(D)	He	(S)	Variable oxidation state

Select the correct Match :-
(1) (A) → (P), (B) → (Q), (C) → (R), (D) → (S)
(2) (A) → (R), (B) → (Q), (C) → (P), (D) → (S)
(3) (A) → (Q), (B) → (R), (C) → (S), (D) → (P)
(4) (A) → (Q), (B) → (P), (C) → (S), (D) → (R)

9. Highest floroanion of Boron and Aluminium is:
(1) BF_6^{3-}, AlF_6^{3-} (2) BF_4^-, AlF_6^-
(3) BF_4^-, AlF_6^{3-} (4) BF_4^-, AlF_4^-

10. Which of the following order is wrong :-
(1) $NH_3 < PH_3 < AsH_3$ — Acidic
(2) Li < Be < B < C — First IP
(3) $Al_2O_3 < MgO < Na_2O < K_2O$ — Basic
(4) $Li^+ < Na^+ < K^+ < Cs^+$ — Ionic Radius

11. e^- configuration of $_{90}Th$-
(1) $(n-2)f^2, (n-1)d^1\ ns^1$
(2) $(n-2)f^1, (n-1)d^1\ ns^2$
(3) $(n-2)f^0, (n-1)d^2\ ns^2$
(4) $(n-2)f^2, (n-1)d^0\ ns^2$

12. Which property decreases from left to right across the periodic table and increases from top to bottom?
(i) Atomic radius
(ii) Electronegativity
(iii) Ionisation energy
(iv) Metallic character
(1) (i) only (2) (i), (ii) and (iii)
(3) (i), (iii) and (iv) (4) (i) and (iv)

13. Which of the following statements are not correct:-

(1) The electron affinity of 'Si' is greater than that of C

(2) BeO is amphoteric while B_2O_3 is acidic

(3) The ionisation energy of 'Tl' is less than that of Al

(4) The ionisation energy of elements of 'Cu' group is less than that of the respective elements of Zn-group

14. Which order for atomic radii is incorrect?

(1) $H^- > Li^+ \geq Mg^{+2} > Al^{+3}$

(2) $MnO_2 > KMnO_4$

(3) $O^{-2} > F^- > Na^+ > Mg^{+2}$

(4) $B > Al \simeq Ga < In \simeq Tl$

15. Which is not incorrect for acidic strength?

(1) $H_2S < H_2Se < H_2Te$

(2) $HClO_4 > HClO_3 > HClO_2 > HClO$

(3) $P_4O_{10} > SiO_2$

(4) All are correct

16. Which is incorrect?

(1) Na < Al < Mg < Si IP_1 Order

(2) V < Cr < Fe < Mn IP_3 Order

(3) P < Si < Be < Mg < Na Metallic Character

(4) Yb > Ce > Lu > Sm Order of atomic radius

17. Select the correct matching in column-I with column-II

Column-I (Atomic number)	Column-II (IUPAC official name)
(a) 101	(P) Mendelevium
(b) 105	(Q) Dubnium
(c) 107	(R) Bohrium
(d) 108	(S) Lawrencium

18. Match the column -

Column I	Column II
(a) N_2O	(P) Normal Oxide
(b) Na_2O	(Q) Neutral Oxide
(c) Ga_2O_3	(R) Sub Oxide
(d) C_3O_2	(S) Basic Oxide
(e) V_3O_4	(T) Amphoteric Oxide
	(U) Mixed Oxide

(1) (a) - Q, (b) - P,S (c) - R, (d) - T, (e) - U

(2) (a) - Q (b) - P,S (c) - T, (d) - R, (e) - U

(3) (a) - R,P (b) - Q,S (c) - U, (d) - T, (e) - R

(4) (a) - P,Q (b) - R,S (c) - T, (d) - U, (e) - R

19. Correct statements among the following is/are:-

(1) 2nd electron gain enthalpy is always endothermic for neutral atoms.

(2) Electronegativity is the property of bonded atoms

(3) Al_2O_3 and BeO are amphoteric oxide

(4) None of these are incorrect

20. Which has maximum number of unpaired e^-?

(1) Na (2) Mn

(3) Cr (4) Fe^{+2}

21. In which of the following pairs, first member has higher first IP

(a) N,O (b) B, Be (c) Al, Ga

(d) F, Cl (e) Zn, Ga (f) F^-, Cl^-

Correct option is

(1) a,c,f,d (2) a,d,e

(3) b, d, e, f (4) a, d, e, f

22. Among the following statement identify not correct statement?

(1) O has least electron affinity in its group.

(2) I has largest E.A. among halogen.

(3) S^- has higher I.E then O^-

(4) None of these

23. The electron affinity values in $Kjmol^{-1}$ of halogen x, y and z are respectively –348, –331 and –312 then x, y and z are respectively:-

(1) F, Cl, Br (2) Cl, F, Br

(3) Cl, Br, F (4) Br, Cl, F

24. Which given order is wrong according to given property :-

(1) K_2O > CaO basic nature

(2) $NH_3 > H_2O$ basic nature

(3) $BH_3 > NH_3$ lewis acid nature

(4) KH > NaH acidic nature

25. Incorrect match ?

	I.P.	Reason
(A)	N > O	Half filled configuration
(B)	Zr < Hf	Lanthanoid contraction
(C)	Na > K	Z_{eff}
(D)	Al < Ga	Transition contraction

(1) only A (2) A, B, D

(3) Only C (4) Only C, D

26. The smallest cation and the smallest anion are respectively

(1) H^+ and H^- (2) H^+ and F^-

(3) Li^+ and H^- (4) Li^+ and F^-

27. Which among the following statement is wrong

(1) Electronic configuration of Gd_{64} is $4f^7 5d^1 6s^2$

(2) Ce^{4+} is a good reductant

(3) Actinoids exhibit higher oxidation states than Lanthanoids

(4) Actinoids contraction is greater from element to element than Lanthanoid contraction

28. Which of the following order of IP is incorrect

(1) $Na^+ > Mg^+$ (2) $Mg^{+2} > Mg^+$

(3) $He > Li^+$ (4) $Be > B$

29. Consider the following values of IE(ev) for elements W and X :-

Element	IE_1	IE_2	IE_3	IE_4
W	10.5	15.5	24.9	79.8
X	8	14.8	78.9	105.8

Other two elements Y and Z have outer electronic configuration $ns^2\ np^4$ and $ns^2\ np^5$ respectively. According to given information which of the following compound (s) is/are not possible.

(a) W_2Y_3 (b) X_2Y_3

(c) WZ_2 (d) XZ_2

(1) a, b (2) b, c

(3) c, d (4) a, d

30. Which represents the electronic configuration of the most electropositive element

(1) $[He]2s^1$ (2) $[Xe]6s^1$

(3) $[He]\ 2s^2$ (4) $[Xe]6s^2$

31. IP and EA of 'F' are 17.42 and 3.45 ev/atom resp. EN of 'F' on Pauling scale will be :-

(1) 2.7 (2) 3.7

(3) 4.07 (4) 6.7

32. The correct order of first electron affnity of O, S and Se is

(1) O > S > Se (2) S > O > Se

(3) Se > O > S (4) S > Se > O

33. $A = 1s^2\ 2s^2\ 2p^4$ $B = 1s^2\ 2s^2\ 2p^5$

$C = 1s^2\ 2s^2\ 2p^6$

(One of A/B/C is neutral atom)

A, B & C are atoms/anion of same element. Then:-

(1) $B_{(g)} + e^- \rightarrow C_{(g)}$ is exothermic

(2) $A_{(g)} \rightarrow A^+_{(g)} + e^-$ is exothermic

(3) $C_{(g)} \rightarrow A_{(g)} + 2e$ is exothermic

(4) $B_{(g)} \rightarrow A_{(g)} + e^-$ is exothermic

34. Correct order of increasing atomic size is

(1) N < F < Si < P (2) F > N < P < Si

(3) F < N < P < Si (4) F < N < Si < P

35. The correct order of second IP is :-

(1) Mg > Na > Si > Al

(2) Mg < Si < Al < Na

(3) Na < Mg < Al < Si

(4) Na > Mg > Al > Si

36. Select the correct order of first ionisation potential

(1) $O_2^{2+} > O_2$ (2) $O_2^{2+} < O_2$

(3) $O_2 \approx O_2^+$ (4) None of these

37. The required energy will be maximum for the process

(1) $Ba \rightarrow Ba^{+2}$ (2) $Be \rightarrow Be^{+2}$

(3) $Cs \rightarrow Cs^+$ (4) $Li \rightarrow Li^+$

38. Which of the following molecule has highest E.N. of Xe ?

(1) XeF_2 (2) XeF_4

(3) XeO_3 (4) None of these

39. The compound X – O – H is likely to act as a base, if compared to hydrogen, X has :-

(1) higher ionization potential

(2) lower electron negativity

(3) higher electronegativity

(4) lower atomic radius

40. Among the following hydroxides which is most basic?

(1) $Lu(OH)_3$ (2) $La(OH)_3$

(3) $Pm(OH)_3$ (4) $Yb(OH)_3$

41. Amongst the following elements whose electronic configuration is given below, the one having highest ionisation enthalpy is

(1) $[Ne]3s^2 3p^1$ (2) $[Ne]3s^2 3p^3$

(3) $[Ne]3s^2 3p^2$ (4) $[Ar]3d^{10} 4s^2 4p^3$

42. Group number and period no.of element having configuration $[Kr]4d^{10}5s^{\circ}$ are:

(1) X, 4^{th} (2) VI, 5^{th}

(3) VIII, 4^{th} (4) VIII, 5^{th}

43. Block, group and period number of element, A. respectively. when electronic configuration of A is [Rn] $6d^2 7s^2$:

(1) d-block, IV B, 7^{th}

(2) d-block, II B, 7^{th}

(3) f-block, II B, 6^{th}

(4) f-block, III B, 7th

44.

Column-I (Type of Elements)	**Column-II** (outer electronic configuration)
(A) Inert gas elements	(i) $ns^{1-2}\ np^{0-5}$
(B) Representative elements	(ii) $1s^2$ and $ns^2\ np^6$
(C) Transition elements	(iii) $(n-2)f^{1-14}$ and $(n-1)\ d^{0-1}\ ns^2$
(D) Inner transition elements	(iv) $(n-1)^{1-10}\ ns^{1 or 2}$

(1) A - i, B - ii, C - iii, D - iv

(2) A - ii, B - i, C - iii, D - iv

(3) A - ii, B - i, C - iii, D - iv

(4) A - ii, B - i, C - iv, D - iii

45. Correct order of spin magnetic moment of trivalent Lanthanoids ions is :

(1) $Gd^{+3} < Ce^{+3} < Lu^{+3}$ (2) $Lu^{+3} < Ho^{+3} < Gd^{+3}$

(3) $Gd^{+3} < Yb^{+3} < Tb^{+3}$ (4) $Pr^{+3} > Sm^{+3} > Eu^{+3}$

46. Incorrect order of radius is :

(1) $H^- > Li^+ > Mg^{2+}$ (2) $P^{-3} > S^{-2} > K^+$

(3) $Br^- > S^{-2} > Cl^- > F^-$ (4) Ni > Cu > Zn

47. Incorrect order of ionic radius is :

(1) $La^{+3} > Gd^{+3} > Eu^{+3} > Lu^{+3}$

(2) $Na^+ > Li^+ > Mg^{+2} > Al^{+3} > Be^{+2}$

(3) $In^+ > Sn^{+2} > Sb^{+3}$

(4) $K^+ > Sc^{+3} > V^{+5} > Mn^{+7}$

48. Which of the following order of I.P. is incorrect?

(1) Al > Mg > Be > B (2) S < P < O < N

(3) Sc > Y > La (4) Ni < Pd < Pt

49. Correct trend of first ionisation energy in group-13 is:

(1) B > Al > Ga > In > Tl

(2) B > Al > Ga > Tl > In

(3) B > Tl > Ga > Al > In

(4) B > Ga > Al > In > Tl

50. Which of the following EA order is not correct?

(1) B < Al < C < Si (2) Mg < C < S < F

(3) O < F < S < Cl (4) N < C < Si < S

51. Which of the following is the most basic oxide?

(1) SeO_2 (2) Al_2O_3

(3) Sc_2O_3 (4) Bi_2O_3

52. Considering the elements B, C, N, F and Si, the correct order of their non-metalic character is

(1) B > C > Si > N > F

(2) Si > C > B > N > F

(3) F > N > C > B > Si

(4) F > N > C > Si > B

53. Considering the elements F, Cl, O and N, the correct order of their chemical reactivity in terms of oxidizing property is

(1) F > Cl > O > N (2) F > O > Cl > N

(3) Cl > F > O > N (4) O > F > N > Cl

54. Electronic configurations of some elements are given in column I and their electron gain enthalpies are given in column II. Match the electronic configuration with electron gain enthalpy.

Column (I) Electronic Configuration	**Column (II)** Electron gain enthalpy/kJ mol^{-1}
(i) $1s^2\ 2s^2\ 2p^6$	(A) –53
(ii) $1s^2\ 2s^2\ 2p^6\ 3s^1$	(B) –328
(iii) $1s^2\ 2s^2\ 2p^5$	(C) –141
(iv) $1s^2\ 2s^2\ 2p^4$	(D) +48

(1) i → D, ii → A, iii → B, iv → C

(2) i → A, ii → B, iii → C, iv → D

(3) i → D, ii → B, iii → A, iv → C

(4) i → D, ii → C, iii → A, iv → B

55. In which of the following options order of arrangement does not agree with the variation of property indicated against it?

(1) $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$ (increasing ionic size)

(2) B < C < N < O (increasing first ionisation enthalpy)

(3) I < Br < F < Cl (increasing electron gain enthalpy)

(4) Li < Na < K < Rb (increasing metallic radius)

56. The process requiring the absorption of energy is:

(1) $F \longrightarrow F^{-}$ (2) $H \longrightarrow H^{-}$

(3) $Cl \longrightarrow Cl^{-}$ (4) $O \longrightarrow O^{2-}$

57. Select the correct order of ionic radii:

(1) $Ti^{2+} > Ti^{3+} > Ti^{4+}$ (2) $Ti^{4+} > Ti^{2+}$

(3) $Ti^{3+} > Ti^{2+} > Ti^{4+}$ (4) $Ti^{4+} > Ti^{3+} > Ti^{2+}$

58. Europium belongs to :

(1) s-block (2) p-block

(3) d-block (4) f-block

59. The ionic radius of Cr is minimum in which of the following compounds ?

(1) CrF_3 (2) $CrCl_3$

(3) Cr_2O_3 (4) K_2CrO_4

60. Which of the following is/are Doberiners triad-

(i) P, As, Sb (ii) Cu, Ag, Au

(iii) Fe, Co, Ni (iv) S, Se, Te

Correct answer is -

(1) (i) and (ii) (2) (ii) and (iii)

(3) (i) and (iv) (4) All

61. First, second and third IP values are 100eV, 150eV and 1500eV. Element can be :-

(1) Be (2) B

(3) F (4) Na

62. The increasing thermal stability of the hydrides of group 16 following the sequence :-

(1) H_2O, H_2S, H_2Se, H_2Te

(2) H_2Te, H_2Se, H_2S, H_2O

(3) H_2S, H_2O, H_2Se, H_2Te

(4) H_2Se, H_2S, H_2O, H_2Te

63. Group number and valency has no relation in?

(1) Zero group (2) First group

(3) IIIrd group (4) VII group

64. Order of atomic radius is correct of the elements given below ?

(1) $Fe \simeq Co \simeq Ni$ (2) Ni > Co > Fe

(3) Co > Ni > Fe (4) Co > Fe > Ni

65. Which pair show less similarity in their properties than the other three :-

(1) Li–Mg (2) Be–Al

(3) Na–Ca (4) B–Si

66. Element 'X' having electronic configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$ forms compound with Ca. The compound is :-

(1) Ca_2X_3 (2) Ca_3X

(3) Ca_3X_2 (4) CaX

67. Which of the following in increasing order of paramagnetism ?

(1) Al < Mg < O < N (2) Mg < Al < N < O

(3) Mg < Al < O < N (4) N < O < Al < Mg

68. Set containing isoelectronic species is :-

(1) C_2^{2-}, NO^+, CN^-, O_2^{2+}

(2) CO, NO, O_2, CN

(3) CO_2, NO_2, O_2, N_2O_5

(4) CO, CO_2, NO, NO_2

69. The correct order of second ionization potential of C, N, O and F is :-

(1) C > N > O > F (2) O > N > F > C

(3) O > F > N > C (4) F > O > N > C

70. The correct values of ionization enthalpies (in kJ mol^{-1}) of Si, P, Cl and S respectively are:-

(1) 786, 1012, 999, 1256

(2) 1012, 786, 999, 1256

(3) 786, 1012, 1256, 999

(4) 786, 999, 1012, 1256

71. On the Pauling's electronegativity scale, which element is next to F.

(1) Cl (2) O (3) Br (4) Ne

72. Which of the following sequence regarding the first ionisation potential of coinage metal is correct?
(1) Cu > Ag > Au (2) Cu < Ag < Au
(3) Cu > Ag < Au (4) Ag > Cu < Au

73. The size of the following species increases in the order :-
(1) $Mg^{2+} < Na^+ < F^- < Al$
(2) $F^- < Al < Na^+ > Mg^{2+}$
(3) $Al < Mg^{2+} < F^- < Na^+$
(4) $Na^+ < Al < F^- < Mg^{2+}$

74. Which is not correct order for the stated property?
(1) Ba > Sr > Mg : Atomic radius
(2) F > O > N : First ionisation energy
(3) Cl > F > I : Electron affinity
(4) O > Se > Te : Electronegativity

75. In which of the following arrangements, the sequence is not strictly according to the property written against it ?
(1) $CO_2 < SiO_2 < SnO_2 < PbO_2$: increasing oxidising power
(2) HF < HCl < HBr < HI : increasing acid strength
(3) $NH_3 < PH_3 < AsH_3 < SbH_3$: increasing Lewis basic strength
(4) B < C < O < N : increasing first ionisation enthalpy

76. The atomic radius of elements of which of the following series would be nearly the same :-
(1) Na, K, Rb, Cs (2) Li, Be, B, C
(3) Fe, Co, Ni, Cu (4) F, Cl, Br, I

77. What is the total number of valence electrons in the peroxydisulphate, $S_2O_8^{2-}$, ion ?
(1) 58 (2) 60
(3) 62 (4) 64

78. Which of the following electronic configuration would be associated with the highest spin only magnetic moment ?
(1) d^2 (2) d^4
(3) d^5 (4) d^7

79. In which pair do both speices have the same electronic configuration ?
(1) Se^{2-}, Kr (2) Mn^{2+}, Cr^{3+}
(3) Na^+, Cl^- (4) Ni, Zn^{2+}

80. Comment on the E.N. of Sb in SbF_3 and SbF_5:-
(1) E.N. of Sb (SbF_3) > E.N. of Sb (SbF_5)
(2) E.N. of Sb (SbF_3) < E.N. of Sb (SbF_5)
(3) E.N. of Sb is identical in both cases
(4) No comment can be predicted

CHEMICAL BONDING

81. Which of the following molecule/species have a same bond order as that of O_2^+ ?
(1) NO (2) N_2^-
(3) N_2^+ (4) All of these

82. Which of the following molecule/species are iso-structural with N_3^- ion?
(1) I_3^+ (2) I_3^-
(3) NH_2^- (4) HCO_2^-

83. Which of the following pair of species are iso-electronic?
(1) CN^- & NO^+ (2) N_2^- & N_2^+
(3) $H_2^{\oplus}$ & H_2^- (4) CO & NO^-

84. Which of the following molecule is hypovalent?
(1) AlF_3 (2) ICl_2^-
(3) BCl_3 (4) ICl_2^+

85. Determine the bond order & formal charge on each oxygen atom in HCO_2^- respectively?
(1) 1.5, –0.5 (2) 2, –0.5
(3) 1.33, –1.5 (4) 1.5, –1.33

86. Determine the incorrect order of bond angle?
(1) $NH_3 < NF_3 < NCl_3$
(2) $OF_2 < OH_2 < OCl_2$
(3) $SF_2 < SCl_2 < SBr_2 < SI_2$
(4) $ClO_2 > ClO_2^{-1} > OCl_2$

87. Which of the following molecule is non-polar & planar?

(1) XeF_4 (2) NH_2^- (3) PF_3Cl_2 (4) PCl_3F_2

88. Which of the following is covalent solid.

(1) Solid CO_2 (2) SiO_2
(3) Diamond (4) 2 & 3 both

89. The percentage of p-character in the orbitals forming P–P bonds in P_4 is

(1) 25 (2) 33 (3) 50 (4) 75

90. Correct match is :-

	Ion	Bond order of M-O bond
(a)	ClO_4^-	$\frac{1}{4}$
(b)	PO_4^{-3}	$\frac{3}{4}$
(c)	NO_2^-	$\frac{2}{3}$
(d)	CO_3^{-2}	$\frac{5}{3}$

(1) d (2) a, b (3) c, d (4) None

91. Which is iso-structural?

(1) XeF_2, ICl_2^-, ClF_3 (2) ClF_3, PCl_3, NCl_3

(3) CO_2, XeF_2, I_3^- (4) PCl_5, $XeOF_2$, ICl_5

92. Which of the following molecule have both $p\pi - p\pi$ and $p\pi - d\pi$ bonding?

(1) ClO_2^+ (2) NO_2^+ (3) SO_3^{2-} (4) ClO_4^-

93. Select correct order out of given options :-

(1) $BeCO_3 < BaCO_3 \rightarrow$ Covalent character
(2) $BeO > SrO \rightarrow$ Lattice energy
(3) $Be^{2+} < Li^+ \rightarrow$ Hydration energy
(4) $Be^{2+}_{(aq.)} > Li^+_{(aq.)} \rightarrow$ Ionic Mobility

94. Which of the following statement are true and false ?

(a) In PCl_5 hybridisation is sp^3d and it has a trigonal pyramidal structure.
(b) The angle between the P-Cl bonds is 90º, which is same for all the P and Cl bond present in PCl_5
(c) The bond length of P-Cl in axial position is higher than in equitorial position
(d) PCl_5 have zero dipole moment

Choose the correct option.

(1) TFTT (2) FTTT (3) FFTT (4) TFFT

95. The types of hybrid orbitals of nitrogen in NO_2^+, NO_3^- & NH_4^+ respectively are expected to be

(1) sp, sp^3, sp^2
(2) sp, sp^2, sp^3
(3) sp^2, sp, sp^3
(4) sp^2, sp^3, sp

96. Which halide has highest melting point ?

(1) NaCl (2) LiCl
(3) LiBr (4) NaI

97. If AB_4^n, type species are tetrahedral, then which of the following is incorrectly matched ?

	A	B	n
(1)	Xe	O	Zero
(2)	Se	F	Zero
(3)	P	O	–3
(4)	N	H	+1

98. The species having no $p\pi - p\pi$ bond but its bond order equal to that of O_2^-

(1) ClO_3^- (2) PO_4^{3-}
(3) SO_4^{2-} (4) XeO_3

99. How many π–bond does C_2 have?

(1) 1 (2) 2
(3) 0 (4) 3

100. Which of the following is not true about H_2O molecule ?

(1) The molecule has $\mu = 0$
(2) The molecule can act as a base
(3) Shows abnormally high boiling point in comparison to the hydrides of other elements of oxygen group
(4) The molecule has a bent shape

101. Match the following and choose the correct option given below.

(a) $N_2 \rightarrow N_2^+$ (p) bond order increases
(b) $N_2 \rightarrow N_2^-$ (q) bond order decreases
(c) $O_2 \rightarrow O_2^+$ (r) paramagnetism increases
(d) $O_2 \rightarrow O_2^-$ (s) paramagnetism decreases
(t) No change in bond order

(1) a - (q, r), b - (q, r), c - (p, s), d - (q, s)
(2) a - (q, s), b - (q, s), c - (p, s), d - (q, r)
(3) a - (p, q), b - (q, s), c - (p, r), d - (q, t)
(4) a - (p, s), b - (q, p), c - (q, t), d - (q, t)

102. Which pair(s) has same bond angle?

(a) BF_3, BCl_3 (b) PO_4^{-3}, SO_4^{-2}

(c) BF_3, PF_3 (d) NO_2^+, N_2O

(e) N_3^-, NO_2

correct option are –

(1) a, b, d (2) b, d

(3) b, c, d (4) a, d, e

103. Which is correct?

(1) PbS > ZnS (Solubility)

(2) $Li_2CO_3 > Na_2CO_3$ (Thermal stability)

(3) NaF > KF (Lattice energy)

(4) $BaSO_4 > MgSO_4$ (Solubility)

104. Which among the following attractions is strongest?

(1) HF ....H_2O (2) Na^+....H–Cl

(3) H_2O...Cl_2 (4) Cl–Cl...Cl–Cl

105. Among the following, the pair in which the two species are not iso-structural is

(1) IO_3^- and NH_3 (2) BH_4^- and NH_4^+

(3) PF_6^- and SF_6 (4) SiF_4 and SF_4

106. Which of the following is correct statement ?

(a) $AlCl_3$ is conducting in fused state

(b) Mobility of Li^+ ion in water is greater than Cs^+ ion

(c) MCl_2 is more volatile than MCl_4

(d) $BeSO_4$ is more soluble in water than $BaSO_4$

(1) a, b (2) b, c, d

(3) b, d (4) Only d

107. Which of the following order is not correct :-

(1) $SO_4^{-2} = PO_4^{-3} = ClO_4^-$ Bond angle

(2) $OCl_2 < ClO_2$ Bond angle

(3) $ZnCl_2 < CdCl_2 < HgCl_2$ ionic character

(4) CH_3–Cl > CH_3F > CH_3–Br > CH_3–I Dipole moment

108. **Statement-1 :** p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.

Statement-2 : o-Hydroxybenzoic acid has intermolecular hydrogen bonding.

(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(3) Statement-1 is True, Statement-2 is False.

(4) Statement-1 is False, Statement-2 is False.

109. Hydration energy of Mg^{2+} is higher than

(1) Be^{2+} (2) $Na^{\oplus}$

(3) Al^{3+} (4) All of these

110. Total number of sp-hybridised C-atoms in the following Hydrocarbon will be:

$H_3C-C\equiv C-CH=CH_2$

(1) 5 (2) 4 (3) 2 (4) 1

111. Match the column

Column I	Column II
(a) C_2H_2	(P) sp^3d hybridisation
(b) SO_2	(Q) sp^3 hybridisation
(c) I_3^-	(R) sp^2 hybridisation
(d) NH_4^+	(S) sp hybridisation

(1) (a) S (b) P (c) R (d) Q

(2) (a) P (b) S (c) R (d) Q

(3) (a) S (b) R (c) P (d) Q

(4) (a) R (b) S (c) P (d) Q

112. Match the column

Column I	Column II
(Solid)	(Examples)
(a) Covalent	(P) SiO_2
(b) Molecular	(Q) CaO
(c) Ionic	(R) CCl_4
(d) Metallic	(S) Bronze

(1) (a) P (b) Q (c) R (d) S

(2) (a) R (b) P (c) Q (d) S

(3) (a) S (b) P (c) Q (d) R

(4) (a) P (b) R (c) Q (d) S

113. Match the column

Compound	No. of σ & π Bonds
(a) $H_2S_2O_3$	(P) 6 σ & 2 π
(b) H_2SO_5	(Q) 11 σ & 4 π
(c) $H_2S_2O_8$	(R) 9 σ & 4 π
(d) $H_2S_2O_6$	(S) 7 σ & 2 π

(1) (a) S (b) P (c) Q (d) R
(2) (a) P (b) S (c) Q (d) R
(3) (a) P (b) Q (c) R (d) S
(4) (a) Q (b) S (c) P (d) R

114. Match the column

Compound	Shape
(a) XeO_2F_2	(P) Linear
(b) XeF_5^-	(Q) Square planar
(c) I_3^-	(R) See-saw
(d) XeF_4	(S) Pentagonal planar

(1) (a) R (b) S (c) P (d) Q
(2) (a) R (b) S (c) Q (d) P
(3) (a) P (b) S (c) Q (d) R
(4) (a) S (b) Q (c) P (d) R

115. Which is incorrect?

(1) Dipole moment order $\rightarrow CH_4 < NF_3 < NH_3 < H_2O$
(2) For PCl_5 molecule $\rightarrow B.L._{\text{equatotial}} < B.L._{\text{Axial}}$
(3) Melting point order $\rightarrow H_2O_{(s)} > NH_{3(s)} > HF_{(s)}$
(4) no. of unpaired e^- in $H_2O_2 = 1$

116. Which is correct?

(1) Bond order $\rightarrow CO > CO_3^{2-}$
(2) Bond angle $\rightarrow PH_3 > PF_3$
(3) Bond energy $\rightarrow Cl_2 > Br_2 > I_2 > F_2$
(4) Bond length order $\rightarrow$ C–C < N–N < O–O < F–F

117. Which is not correct?

(1) White vitriol and epsom salt are isomorphous
(2) Thermal stability
$\rightarrow BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3$
(3) Solubility
$\rightarrow NaHCO_3 < KHCO_3 < RbHCO_3 < CsHCO_3$
(4) Melting point $\rightarrow Al_2O_3 < MgF_2$

118. Which molecule does not exist?

(1) MnF_4 (2) SH_6
(3) $(BCl_3)_2$ (4) 2 & 3 both

119. Which is correct?

(1) Ionic mobility in aqueous medium
$\rightarrow Li^+ < Na^+ < K^+ < Rb^+$
(2) Covalant character
$\rightarrow KCl > CaCl_2 > AlCl_3 > SnCl_4$
(3) Boiling point order $\rightarrow H_2O > H_2Se > H_2Te > H_2S$
(4) Dipole – dipole attraction $\rightarrow KCl + H_2O$

120. Which is not incorrect?

(1) Stability order

$$H-O-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-O-H \quad < \quad H-O-\underset{\underset{O}{\downarrow}}{\overset{\overset{O}{\uparrow}}{S}}-O-H$$

(2) In XeF_6(s) hybridisation of anion $\rightarrow sp^3d$
(3) For O_2 molecule bond order is 2.0
(4) Bond angle order $CF_4 < CH_4$

121. Match column-I and Column-II

Column-I	Column-II
(A) SF_4	(1) Tetrahedral
(B) BrF_3	(2) Pyramidal
(C) BrO_3^-	(3) See-saw
(D) NH_4^+	(4) Bent-T

Code :
(1) A (3), B (2), C (1), D (4)
(2) A (3), B (4), C (2), D (1)
(3) A (2), B (4), C (3), D (1)
(4) A (1), B (4), C (2), D (3)

122. Consider the following order

(1) $PH_3 > NH_3 > AsH_3$ (basic character)
(2) $PH_3 > NH_3 > AsH_3$(boiling point)
(3) $HOCl > HClO_2 > HClO_3 > HClO_4$ (oxidising property)
(4) HF > HCl > HBr > HI (acidic)
(5) $H_2O > H_2S > H_2Se$ (bond angle)
(6) $H_2SO_4 > H_3PO_4 > H_2CO_3$ (acidic character)

correct order(s) are

(1) 1,2,4,5 (2) 2,3,5,6
(3) 3,5,6 (4) 4,5,6

123. Which solubility order is correct ?

(1) $BaSO_4 > SrSO_4 > CaSO_4 > MgSO_4$

(2) $ZnS > Na_2S > CoS$

(3) $BaCO_3 > MgCO_3 > Na_2CO_3$

(4) $KOH > NaOH > Mg(OH)_2$

124. What is incorrect about reaction of NH_3 and BF_3?

(1) hyrbridisation of both N & B change

(2) It is an example of redox change

(3) In the final adduct formed, back bonding appears between B & N

(4) All

125. In which molecule / ion there are more than one type of XO bond lengths

(a) NO_3^- (b) $Cr_2O_7^{2-}$

(c) $HCOO^-$ (d) $HClO_3$

(e) PO_4^{3-} (f) SO_4^{2-}

correct code is

(1) a,b,d (2) b,d

(3) b,c,d,f (4) a,b,c,f

126. Correct order of dipole moment is (decreasing order)

(1) CH_3Cl, CH_3Br, CH_3F

(2) CH_3F, CH_3Cl, CH_3Br

(3) CH_3Cl, CH_3F, CH_3Br

(4) CH_3Br, CH_3Cl, CH_3F

127. Correct order of stability of species

N_2, N_2^+, N_2^-

(1) $N_2 > N_2^+ = N_2^-$ (2) $N_2 > N_2^+ > N_2^-$

(3) $N_2 > N_2^- > N_2^+$ (4) $N_2^+ > N_2 > N_2^-$

128. Isostructural species are those which have same shape. Among the following pairs identify isostructural pairs.

(1) [NF_3 & BF_3] (2) [BF_4^- & NH_4^+]

(3) [BCl_3 & $BrCl_3$] (4) [NH_3 & NO_3^-]

129. Which is not stable

(1) KHF_2 (2) KI_3

(3) $CH_3–CH(OH)_2$ (4) $Cl_3C–CH(OH)_2$

130. Which of the following compound will give metal and oxygen gas at high temperature

(1) $NaNO_3$ (2) Ag_2CO_3

(3) K_2CO_3 (4) Li_2CO_3

131. Based on VSEPR theory, the number of 90 degree F–Br–F angles in BrF_5 is

(1) 5 (2) 4

(3) 0 (4) 1

132. Which of the following represent most effective π-bond

(1) 2pπ-3pπ (2) 3dπ-3dπ

(3) 2pπ-3dπ (4) 3dπ-3pπ

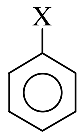
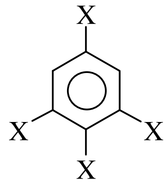
133. In which reaction hybridisation of underlined atom does not change.

(1) $\underline{B}F_3 + F^- \rightarrow BF_4^-$

(2) $\underline{N}H_3 + H^+ \rightarrow NH_4^+$

(3) $\underline{B}F_3 + NH_3 \rightarrow BF_3.NH_3$

(4) $\underline{S}iF_4 + 2F^- \rightarrow SiF_6^{2-}$

134. Dipole moment of  is 1.5 D. The dipole moment of  will be

(1) 1.5 D (2) 3.0 D

(3) 1.0 D (4) 2.35 D

135. Which of the following compound has non-zero dipole moment

(1) XeF_4 (2) B_2H_6 (3) PF_3Cl_2 (4) PCl_3F_2

136. Which of the following molecule is planar due to back bonding

(1) NCl_3 (2) PF_3 (3) BF_3 (4) None

137. Amongst the following, molecule having maximum bond angles of 90° is

(1) XeF_4 (2) XeF_6 (3) SF_6 (4) IF_7

138. Which of the following statement is incorrect

(1) Removal of an electron is easier from O_2 in comparison to O_2^{+2}

(2) In the double bond of C_2 molecule, both are π-bonds

(3) NO is more stable than NO^+

(4) NO_2^+ and CO_2 are isoelectronic and isostructural

139. The coordinate bond is absent in

(1) $NaNO_3$ (2) $CaCO_3$

(3) O_3 (4) KNC

140. Which of the following is least stable
(1) O^- (2) C^- (3) B^- (4) Be^-

141. The true statements from the following are
(1) PH_5 , NCl_5 and $BiCl_5$ do not exist
(2) I_3^- has bent geometry
(3) XeF_4 is polar molecule
(4) O_2 and O_2^{-2} has same bond order

142. The bond order for NO and NO^+, respectively are
(1) 3.0, 2.5 (2) 2.5, 3.0
(3) 3.0, 3.0 (4) 2.5, 2.5

143. Back bonding always changes
(1) bond angle
(2) hybridisation of central atom
(3) planarity
(4) bond length

144. Bond angle in H_2O is
(1) 104.5° (2) 120° (3) 109.5° (4) 107°

145. The correct stability order of N_2 and its given ions is
(1) $N_2 > N_2^+ > N_2^- > N_2^{2-}$
(2) $N_2^- > N_2^+ > N_2^- > N_2^{2-}$
(3) $N_2^+ > N_2^- > N_2 > N_2^{2-}$
(4) $N_2 > N_2^+ = N_2^- > N_2^{2-}$

146. Which of the following have same bond order:-
(I) CO (II) CN^- (III) O_2^+ (IV) NO^+
(1) I, II, III (2) I, II, IV
(3) I, III, IV (4) II, III, IV

147. In XeF_2, XeF_4 and XeF_6 the number of lone pairs of electron on Xe are respectively
(1) 2, 3, 1 (2) 1,2,3 (3) 4,1,2 (4) 3,2,1

148. Which one is most soluble in water
(1) $Mg(OH)_2$ (2) $Sr(OH)_2$
(3) $Ca(OH)_2$ (4) $Ba(OH)_2$

149. The correct order of N–O bond length is
(1) $NO_3^- > NO_2^+ > NO_2^-$
(2) $NO_3^- > NO_2^- > NO_2^+$
(3) $NO_2^+ > NO_3^- > NO_2^-$
(4) $NO_2^- > NO_3^- > NO_2^+$

150. The correct order of A–O–A bond angle of (A=H, F or Cl)
(1) $H_2O > Cl_2O > F_2O$ (2) $Cl_2O > H_2O > F_2O$
(3) $F_2O > Cl_2O > H_2O$ (4) $F_2O > H_2O > Cl_2O$

151. There are some species given below :-
(a) O_2^+ (b) CO
(c) B_2 (d) O_2^-
(e) NO^+ (f) He_2^+
(g) C_2^{+2} (h) CN^-
(i) N_2^-
Total no. of species which have their fractional bond order.
(1) 3 (2) 4 (3) 5 (4) 6

152. Which of the following has fractional bond order
(1) O_2^{2+} (2) O_2^{2-}
(3) F_2^{2-} (4) H_2^-

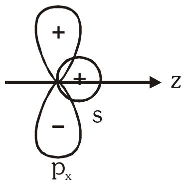
153. When $AgNO_3$ is heated strongly, the product formed are
(1) NO and NO_2 (2) NO_2 and O_2
(3) NO_2 and N_2O (4) NO and O_2

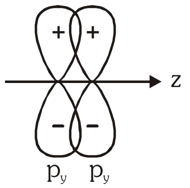
154. Which of the following carbonate of a metals has the least thermal stability
(1) Li_2CO_3 (2) K_2CO_3
(3) Cs_2CO_3 (4) Na_2CO_3

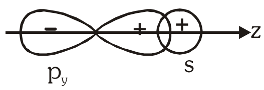
155. Which of the following order is not correct :-
(1) $N_2 < N_2^+$ (Bond length)
(2) $O_2 < O_2^+$ (Bond strength)
(3) $O_2 < O$ (IP)
(4) $NO < NO^+$ (Magnetic moment)

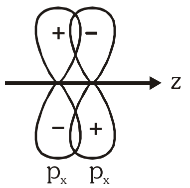
156. The state of hybridisation for the transition state of hydrolysis mechanism of BCl_3 and SF_4 are respectively
(1) sp^2, sp^3d (2) sp^3, sp^2
(3) sp^3, sp^3 (4) sp^3, sp^3d^2

157. Which of the following orbitals overlapping shows zero overlapping :-

(1) 

(2) 

(3) 

(4) 

158. The bond strength in O_2^+, O_2, O_2^- & O_2^{2-} follows the order

(1) $O_2^{2-} > O_2^- > O_2 > O_2^+$

(2) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

(3) $O_2 > O_2^- > O_2^{2-} > O_2^+$

(4) $O_2^- > O_2^{2-} > O_2^+ > O_2$

159. A compound which leaves behind no residue on heating is

(1) $Cu(NO_3)_2$ (2) KNO_3

(3) NH_4NO_3 (4) None of these

160. Which of the following molecule is polar and non-planar

(1) XeF_4 (2) XeF_5^-

(3) CH_2F_2 (4) ClF_3

161. Dipole moment of NH_3 is more than NF_3 because

(1) N–F bond is more polar than N–H bond

(2) NH_3 is pyramidal while NF_3 is planar

(3) In NH_3 orbital dipole due to lone pair is in the same direction as the resultant dipole moment of N–H bonds while in NF_3 orbital dipole due to lone pair is opposite direction of the resultant dipole moment of N–F bonds

(4) None of these

162. Which of the following pairs of ions are isoelectronic and isostructural

(1) CO_3^{-2}, NO_2^- (2) ClO_3^-, CO_3^{-2}

(3) SO_3^{-2}, NO_3^- (4) ClO_3^-, SO_3^{-2}

163. Which of the following pair is having planar structure

(1) SF_4, XeF_4 (2) H_3O^+, SO_2

(3) BF_3, $XeOF_2$ (4) XeF_4, NO_3^-

164. Ammonia is soluble in water but phosphine is insoluble because

(1) phsophine has higher molecular mass than ammonia

(2) ammonia is polar while phosphine is non polar

(3) Ammonia forms inter molecular H-bond with water but phosphine does not

(4) Ammonia is ionic while phosphine is covalent

165. Which of the following resist hydrolysis at room temperature

(1) PCl_3, SF_6 (2) CCl_4, NO_2

(3) PCl_5, XeF_6 (4) SF_6, CCl_4

166. Which of the following is polar

(1) p-dichlorobenzene

(2) trans-1-chloropropene

(3) boron tri fluoride

(4) xenon tetra fluoride

167. Which molecule / ion out of the following does not contain unpaired electrons?

(1) N_2^+ (2) O_2

(3) O_2^{2-} (4) B_2

168. Which of following molecule is having shortest bond length

(1) O_2^+

(2) O_2^{2-}

(3) O_2

(4) All have same bond length

169. Which of the following attraction is strongest?

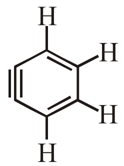
(1) Cl^- .... H --Cl

(2) $CHCl_3$ .... $CHCl_3$

(3) CCl_4 .... H_2O

(4) Cl^- .... H_2O

170. How many sp^2 and sp-hybridised carbon atoms are present respectively in the following compound?



(Benzyne)

(1) 4, 2 (2) 6, 0 (3) 3, 3 (4) 5, 1

171. Phosphorus pentachloride in the solid exists as:

(1) PCl_5 (2) $[PCl_4]^+ [Cl]^-$

(3) $[PCl_4]^+ \cdot [PCl_6]^-$ (4) $PCl_5 \cdot Cl_2$

172. Least stable hydride is :

(1) stannane (2) silane

(3) plumbane (4) germane

173. Solid NaCl is a bad conductor of electricity because:

(1) In solid NaCl there are no ions

(2) Solid NaCl is convalent

(3) In solid NaCl there is no mobility of ions

(4) In solid NaCl there are no electrons

174. Which of the following halides is inert towards hydrolysis at room temperature?

(1) $SiCl_4$ (2) PCl_3

(3) NCl_3 (4) NF_3

175. The correct order of dipole moment is:

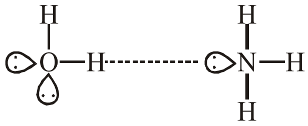
(1) $CH_4 < NF_3 < NH_3 < H_2O$

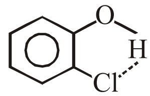
(2) $NF_3 < CH_4 < NH_3 < H_2O$

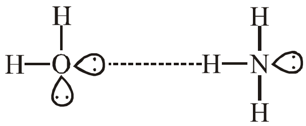
(3) $NH_3 < NF_3 < CH_4 < H_2O$

(4) $H_2O < NH_3 < NF_3 < CH_4$

176. Which of the following is not a best representation of the H-bond?

(1) 

(2) 

(3) 

(4) 

177. Which order are correct?

(I) Thermal stability :

$BeSO_4 < MgSO_4 < CaSO_4 < BaSO_4$

(II) Basic nature :

ZnO > BeO > MgO > CaO

(III) Solubility in water :

LiOH > NaOH > KOH > RbOH > CsOH

(IV) Melting point :

NaCl > KCl > RbCl > CsCl > LiCl

(1) (I), (IV) (2) (I), (II) and (IV)

(3) (II), (III) (4) All correct

178. In which of the following diatomic molecule, bond dissociation energy is maximum?

(1) H_2 (2) F_2 (3) Cl_2 (4) I_2

179. Which of of the following carbonate is maximum stable to heat?

(1) $CaCO_3$ (2) Li_2CO_3

(3) Na_2CO_3 (4) $BaCO_3$

180. The most stable structure of NO is

(1) $:\ddot{N}=\ddot{O}:$ (2) $\underset{\cdot\cdot}{\ddot{N}}=\underset{\cdot}{\ddot{O}}$

(3) $:\underset{\cdot}{N}=\underset{\cdot\cdot}{O}:$ (4) $:N=O:$

181. What is/are true about CO_2 and SO_2?

(1) Both turn acidified $K_2Cr_2O_7$ solution green

(2) Both turn lime water milky

(3) Both are oxidising agents not the reducing agents

(4) All of these

182. Which of the following is most stable?

(1) Na_3N (2) Li_3N (3) Rb_3N (4) K_3N

183. Consider the following statements

I. PCl_3 on hydrolysis in the presence of moisture gives fumes of HCl.

II. PCl_5 exists as $[PCl_4]^{(-)}$ $[PCl_6]^{(+)}$ in solid state.

III. All the five bonds in PCl_5 molecule are equivalent.

Choose the correct statement(s) :

(1) II & III (2) I, II & III

(3) Only I (4) I & II

184. According to Fajan's Rule, ionic character increases for :

(1) Small cation and small charge on cation

(2) Large cation and small anion

(3) Small cation and large anion

(4) Large anion and small charge on anion

185. Which molecule/ion out of the following does not contain unpaired electrons?

(1) N_2^+ (2) O_2 (3) O_2^{2-} (4) B_2

186. In which of the following molecule/ion all the bonds are not equal?

(1) XeF_4 (2) BF_4^- (3) C_2H_4 (4) SiF_4

Para.: The electronic configurations of three elements, A, B and C are given below. Answer the questions 187-190 on the basis of these configurations.

A $1s^2$ $2s^2$ $2p^6$

B $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^3$

C $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^5$

187. Stable form of A may be represented by the formula

(1) A (2) A_2 (3) A_3 (4) A_4

188. Stable form of C may be represented by the formula

(1) C (2) C_2 (3) C_3 (4) C_4

189. The molecular formula of the compound formed from B and C will be

(1) BC (2) B_2C (3) BC_2 (4) BC_3

190. The bond between B and C will be

(1) ionic (2) covalent

(3) hydrogen (4) coordinate.

191. Which of the following order of energies of molecular orbitals of N_2 is correct?

(1) $(\pi 2p_y) < (\sigma 2p_z) < (\pi^* 2p_x) \approx (\pi^* 2p_y)$

(2) $(\pi 2p_y) > (\sigma 2p_z) > (\pi^* 2p_x) \approx (\pi^* 2p_y)$

(3) $(\pi 2p_y) < (\sigma 2p_z) > (\pi^* 2p_x) \approx (\pi^* 2p_y)$

(4) $(\pi 2p_y) > (\sigma 2p_z) < (\pi^* 2p_x) \approx (\pi^* 2p_y)$

192. Formation of PH_4^+ is difficult as compared to NH_4^+ because :-

(1) Lone Pair of Phosphorus is delocalised

(2) Lone Pair of phosphorus resides in almost pure p - orbital

(3) Lone pair of phosphorus resides in sp^3 orbital

(4) Lone pair of phosphorus resides in almost pure s - orbital

193. The incorrect order of solubility in water :-

(1) $Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$

(2) $Li_2CO_3 < Na_2CO_3 < K_2CO_3$

(3) AgF<AgCl<AgBr

(4) $BaSO_4 < MgSO_4$

194. Which of the following statement is incorrect?

(1) In O_2^-, O_2^+, the magnetic behavior is not changed

(2) In N_2^+, N_2^-, the magnetic moment remains unchanged

(3) If z-axis is the overlapping axis, the P_x orbital of one atom and dxy orbital of another atom result antibonding M.O.

(4) None of these

195. Which of the following is planar in monomer as well as in dimer form ?

(1) $AlCl_3$ (2) ICl_3 (3) BH_3 (4) All

196. Which statement is correct in the following :-

(1) $NH_3 > NF_3$: bond angle

(2) $NH_3 > NF_3$: reactivity towards lewis acid

(3) $NH_3 > NF_3$: Dipole moment

(4) All are correct

197. The species having maximum lone pair is :-

(1) BrF_5 (2) XeF_4

(3) SF_4 (4) ClF_3

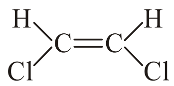
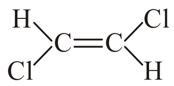
198. Which has a giant covalent structure?

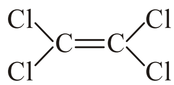
(1) PbO_2 (2) SiO_2 (3) NaCl (4) $AlCl_3$

199. A molecule which cannot exist theoretically is:

(1) SF_4 (2) OF_2 (3) OF_4 (4) O_2F_2

200. Select the isomers given below, which have non-zero dipole moment?

(1)  (2) 

(3)  (4) All of these

201. Select the correct relation:

(1) $\mu_{CH_3OH} = \mu_{CH_3SH}$ (2) $\mu_{CH_3OH} > \mu_{CH_3SH}$

(3) $\mu_{CH_3OH} < \mu_{CH_3SH}$ (4) can't compare

202. The correct order of decreasing polarisability of ions is:

(1) $Cl^- > Br^- > I^- > F^-$

(2) $F^- > I^- > Br^- > Cl^-$

(3) $I^- > Br^- > Cl^- > F^-$

(4) $F^- > Cl^- > Br^- > I^-$

203. Among the following which species has same number σ and π-bonds?

(1) C_2H_6 (2) $C_2(CN)_4$

(3) C_2H_4 (4) $HC\equiv CH$

204. In which of the following pairs, the two species are isostructural?

(1) SO_3^{2-} and NO_3^-

(2) BF_3 and NF_3

(3) BrO_3^- and XeO_3

(4) SF_4 and XeF_4

205. The hybridisation of P in phosphate ion (PO_4^{3-}) is the same as in:

(1) I in ICl_4^- (2) S in SO_3

(3) N in NO_3^- (4) S in SO_3^{2-}

206. Bond order of NO is :

(1) 1.0 (2) 1.5

(3) 2.0 (4) 2.5

207. Intramolecular hydrogen bonding is found in:

(1) Salicylaldehyde (2) Water

(3) Acetaldehyde (4) Phenol

208. Two ice cubes are pressed over each other and unite to form one cube. Which force is responsible for holding them together?

(1) van der Waals' forces

(2) Covalent attraction

(3) Hydrogen bond formation

(4) Dipole-dipole attraction

209. The most suitable method of separation of mixture of ortho and para-nitrophenol in the ratio 1 : 1 is:

(1) Distillation (2) Crystallisation

(3) vaporisation (4) Colour spectrum

210. The forces present in the crystals of naphthalene are:

(1) van der Waal's forces

(2) Electrostatic forces

(3) Hydrogen bonding

(4) Ionic bond

211. If d represents the bond length, then select the correct relation.

(1) $d_{N_2} = d_{N_2^+}$ and $d_{O_2} = d_{O_2^+}$

(2) $d_{N_2} < d_{N_2^+}$ and $d_{O_2} > d_{O_2^+}$

(3) $d_{N_2} < d_{N_2^+}$ and $d_{O_2} < d_{O_2^+}$

(4) $d_{N_2} > d_{N_2^+}$ and $d_{O_2} > d_{O_2^+}$

212. Which of the following conducts electricity:

(1) Diamond (2) KCl

(3) KCl(fused) (4) $BaSO_4$

213. What hybridization is expected on the central atom of each of the following molecules?

(i) BeH_2 (ii) CH_2Br_2

(iii) PF_6^- (iv) BF_3

(1) sp^2, sp, sp^3, sp^2 (2) sp, sp^3, sp^3d, sp^2

(3) sp, sp^3, sp^3d^2, sp^2 (4) sp^2, sp, sp^2, sp^3

214. What is the value of 1 debye in SI units?

(1) 3.336×10^{-30} C.m. (2) 33.36×10^{-30} C.m.

(3) 333.6×10^{-30} C.m. (4) None of these

215. Which of the following molecule has a planar structure?

(1) O_2SF_2 (2) OSF_2

(3) XeF_4 (4) ClO_4^-

216. Which type of shape is found in SF_2 molecule?

(1) V-shaped

(2) Bipyramidal

(3) Linear

(4) Irregular tetrahedron

217. In XeF_2, XeF_4 and XeF_6 the hybridisation of central atom is ?

(1) sp^3d^3, sp^3d^2, sp^3d

(2) sp^3d, sp^3d^3, sp^3d^2

(3) sp^3, sp^3 sp^3

(4) sp^3d, sp^3d^2, sp^3d^3

218. The correct sequence of decrease in bond angles of following hydrides is:

(1) $NH_3 > PH_3 > AsH_3 > SbH_3$

(2) $SbH_3 > AsH_3 > PH_3 > SbH_3$

(3) $SbH_3 > AsH_3 > PH_3 > NH_3$

(4) $PH_3 > NH_3 > AsH_3 > SbH_3$

219. Which of the following is soluble in water?

(1) CS_2 (2) C_2H_5OH

(3) CCl_4 (4) $CHCl_3$

220. Correct order is :-

(1) $SOF_2 > SOCl_2 > SOBr_2$ (B.A.) : $\left(X-\hat{S}-X\right)_{X=F/Cl/Br}$

(2) $SF_4 > XeF_4$ (Dipole moment)

(3) $BF_3 < BCl_3 < BBr_3$ (Bond angle)

(4) $PO_4^{3-} < SO_4^{2-} < ClO_4^-$ (Bond length)

221. Among the following, the molecule with highest dipole moment is:

(1) CH_3Cl (2) CH_2Cl_2

(3) $CHCl_3$ (4) CCl_4

222. Which of the following is least volatile?

(1) HF (2) HCl (2) HBr (4) HI

223. Which of the following has the shortest carbon-carbon bond length?

(1) C_6H_6 (2) C_2H_6

(3) C_2H_4 (4) C_2H_2

224. Carbon atoms in $C_2(CN)_4$ are:

(1) sp-hybridized

(2) Sp^2-hybridized

(3) sp-and sp^2-hybridized

(4) sp, sp^2, and sp^3-hybridized

225. Two elements X and Y have following electronic configurations:

$X = 1s^2, 2s^2\ 2p^6, 3s^2\ 3p^6, 4s^2$ and

$Y = 1s^2, 2s^2\ 2p^6, 3s^2\ 3p^5$

The compound formed by the combination of X and Y is:

(1) XY_2 (2) X_5Y_2

(3) X_2Y_5 (4) XY_5

226. Compare $F - \hat{I} - O$ and $F_{axial} - \hat{I} - F_{axial}$ bond angle in IOF_3 molecule:

(1) $F—\hat{I}—O > F_{axial}—\hat{I}—F_{axial}$

(2) $F_{axial}—\hat{I}—F_{axial} > F—\hat{I}—O$

(3) $F_{axial}—\hat{I}—F_{axial} = F—\hat{I}—O$

(4) None of these

227. How many bondings pairs and lone pairs surround the central atom in the I_3^- ion?

	Bonding pairs	Lone pairs
(1)	2	2
(2)	2	3
(3)	3	2
(4)	4	3

S-BLOCK ELEMENTS

228. Which of the following isn't considered as an alkaline earth metal?

(1) Be (2) Mg

(3) Ca (4) Sr

229. The alkali metals & their salts impart characteristic colour to an:

(1) Oxidising flame (2) Reducing flame

(3) Both a & b (4) None of these

230. The pair of most abundant alkali metals is?

(1) Li & Na (2) Na & K

(3) K & Rb (4) Na & Rb

231. When alkali metals react with liquid ammonia the solution obtained is

(1) Blue & non-conducting

(2) Blue & conducting

(3) Colourless & non-conducting

(4) Colourless & conducting

232. The products obtained on hydrolysis of superoxide

(1) $MO_2 + H_2O \longrightarrow M^+ + OH^- + H_2O_2$

(2) $MO_2 + H_2O \longrightarrow M^+ + OH^- + H_2O$

(3) $MO_2 + H_2O \longrightarrow M^+ + OH^- + H_2O_2 + O_2$

(4) $MO_2 + H_2O \longrightarrow M^+ + OH^-$

233. Milk of magnesia is:

(1) Suspension of $Mg(OH)_2$ in water

(2) Colloid of $Mg(OH)_2$ in water

(3) True solution of $Mg(OH)_2$ in water

(4) Pure $Mg(OH)_2$

234. The tendency to form halide hydrates in group 2 elements?

(1) increases down the group

(2) decreases down the group

(3) remains constant

(4) first decreases then increases down the group

235. For slowing down the process of setting of cement so that it gets sufficiently hard, the compound added is:

(1) Limestone (2) Dicalcium silicate

(3) Gypsum (4) Tricalcium aluminate

236. Which of the following alkali metal doesn't form ethynide on reaction with ethyne?

(1) Li (2) Na (3) K (4) Rb

237. Which of the following compound is thermally most stable?

(1) $LiNO_3$ (2) $NaNO_3$ (3) KNO_3 (4) $RbNO_3$

238. What is the order of relative degree of hydration

(1) $Cs^+ (aq) > Rb^+ (aq) > K^+ (aq) > Na^+(aq) > Li^+(aq)$

(2) $Li^+ (aq) > Na^+(aq) > K^+ (aq) > Rb^+ (aq) > Cs^+ (aq)$

(3) $Na^+(aq) > K^+ (aq) > Rb^+ (aq) > Cs^+ (aq) > Li^+ (aq)$

(4) $Cs^+ (aq) > Na^+ (aq) > Rb^+ (aq) > Li^+(aq) > K^+ (aq)$

239. Least mobile ion is

(1) $[Be(H_2O)_n]^{+2}$ (2) $[Na(H_2O)_n]^+$

(3) $[Mg(H_2O)_n]^{+2}$ (4) $[Li(H_2O)_n]^+$

240. Which is most soluble in water ?

(1) CaF_2 (2) BaF_2

(3) SrF_2 (4) BeF_2

241. A solid compound X on heating gives CO_2 gas and a residue. When mixed with water it forms Y. On passing excess of CO_2 through Y in water a clear solution of Z is obtained. On boiling Z compound X is reformed. Compound X is

(1) $CaCO_3$ (2) Na_2CO_3

(3) K_2CO_3 (4) $Ca(HCO_3)_2$

242. An element of s-block forms an oxide of 'MO' type which is amphoteric in nature. Correct statetement regarding element is

(1) It's hydroxide is most soluble in its group hydroxides

(2) It forms peroxide

(3) Its sulphate is most soluble in its group sulphates

(4) Its carbonate is most stable in its group carbonates

243. Correct order is
(1) LiH < NaH < CsH $\longrightarrow$ ionic character
(2) F–F < H–H < D–D $\longrightarrow$ bond energy
(3) $NH_3 < H_2O < H_2O_2$ $\longrightarrow$ acidic character
(4) all the above

244. Which of the following reacts most vigrously with water ?
(1) Na (2) Be (3) Li (4) Mg

245. Consider the following chemical reaction
$Z + 3LiAlH_4 \rightarrow X + 3LiF + 3AlF_3$
$X + H_2O \rightarrow Y + 6H_2$
$3X + O_2 \xrightarrow{\Delta} B_2O_3 + 3H_2O$
X, Y, Z are respectively
(1) B, BF_3, H_3BO_3 (2) B_2H_6, BF_3, H_3BO_3
(3) B_2H_6, H_3BO_3, BF_3 (4) $Na_2B_4O_7$, B_2H_6

246. Which of the following carbides produces propyne on reaction with water?
(1) CaC_2 (2) Be_2C
(3) Al_4C_3 (4) Mg_2C_3

247. Which one of the following reactions is not associated with the Solvay process of manufacture of sodium carbonate?
(1) $NaCl + NH_4HCO_3 \longrightarrow NaHCO_3 + NH_4Cl$
(2) $2NaOH + CO_2 \longrightarrow Na_2CO_3 + H_2O$
(3) $2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2$
(4) $NH_3 + H_2CO_3 \longrightarrow NH_4HCO_3$

248. The sequence of ionic mobility in aqueous solution is:
(1) $Rb^+ > K^+ > Cs^+ > Na^+$
(2) $Na^+ > K^+ > Rb^+ > Cs^+$
(3) $K^+ > Na^+ > Rb^+ > Cs^+$
(4) $Cs^+ > Rb^+ > K^+ > Na^+$

249. Thermal stability of hydrides of first group elements follows the order is:
(1) LiH > NaH > KH > RbH
(2) LiH > KH > NaH > RbH
(3) LiH > RbH > KH > NaH
(4) LiH > KH > RbH > NaH

250. One mole of magnesium nitride on reaction with an excess of water gives
(1) One mole of ammonia
(2) One mole of nitric acid
(3) Two moles of ammonia
(4) Two moles of nitric acid

251. The chloride that can be extracted with ether is:
(1) NaCl (2) LiCl (3) $BaCl_2$ (4) $CaCl_2$

252. In the manufacture of sodium hydroxide, byproduct obtained is:
(1) O_2 (2) Cl_2
(3) Na_2CO_3 (4) NaCl

253. The compound used in photography is:
(1) Na_2SO_5 (2) $Na_2S_2O_8$
(3) $Na_2S_2O_6$ (4) $Na_2S_2O_3$

254. The ashes of plants contain alkali metals, 90% of which is :
(1) Li (2) K (3) Na (4) Rb

255. The most electropositive element among the alkaline earth metals is :
(1) Be (2) Mg (3) Cs (4) Ba

256. Chile-salt peter is the ore of:
(1) Potassium (K) (2) Bromine (Br)
(3) Sodium (Na) (4) Magnesium (Mg)

257. Which one of the following electrolytes used in Down's process of extracting sodium metal?
(1) NaCl + KCl + KF (2) NaCl
(3) NaOH + KCl + KF (4) NaCl + NaOH

258. Sodium peroxide which is a yellow solid, when exposed to air becomes white due to formation of:
(1) H_2O_2 (2) Na_2O
(3) Na_2O and O_3 (4) NaOH and Na_2CO_3

259. Which of the following is best CO_2 absorber as well as source of O_2 in space capsule?
(1) KO_2 (2) K_2O_2
(3) KOH (4) LiOH

260. A solution of sodium metal in liquid ammonia is strongly reducing due to the presence of:

(1) Sodium hydride

(2) Sodium amide

(3) Sodium

(4) Solvated electrons

261. In the following sequence of reactions. Identify (E):

$$Na_2CO_3 + H_2O + CO_2 \longrightarrow (A) \xrightarrow{\Delta, ZnCl_2} (B)$$

$$\xrightarrow{\Delta} (C) + (D)\uparrow \xrightarrow[\text{excess}]{NaOH} (E)$$

(1) $NaHCO_3$ (2) Na_2O_2

(3) Na_2ZnO_2 (4) $ZnCO_3$

262. By adding gypsum to cement:

(1) Setting time of cement becomes less.

(2) setting time of cement increase

(3) Color of cement becomes light

(4) Shining surface is obtained

263. The reaction of Cl_2 with X gives bleaching powder. X is:

(1) CaO (2) $Ca(OH)_2$

(3) $Ca(OCl_2)$ (4) $Ca(ClO_3)_2$

264. Which of the following reaction/s are correct here?

(I) $B + NaOH \longrightarrow Na_3BO_3 + H_2$

(II) $P_4 + NaOH + H_2O \longrightarrow NaH_2PO_2 + PH_3$

(III) $S + NaOH \longrightarrow Na_2S_2O_3 + Na_2S + H_2O$

(1) I only (2) III only

(3) II and III (4) I, II, and III

265. Deliquescent impurities present in brine solution is :-

(1) $MgCl_2$ (2) NaCl

(3) Na_2SO_4 (4) $CaSO_4$

HYDROGEN & IT'S COMPOUNDS

266. The hydride ion H^- is a stronger base than hydroxide ion. Which of the following reaction will occur if NaH is dissolved in water.

(1) $H^-_{aq} + H_2O_{(\ell)} \longrightarrow H_3O^+_{aq}$

(2) $H^-_{aq} + H_2O_{(\ell)} \longrightarrow OH^-_{aq} + H_{2(g)}$

(3) $H^-_{aq} + H_2O_{(\ell)} \longrightarrow$ no reaction

(4) None of these

267. Hydrogen peroxide is reduced by

(1) Ozone

(2) Barium peroxide

(3) Acidic solution of $KMnO_4$

(4) Lead sulphide

268. Water softening by Clarke's process uses

(1) Calcium bicarbonate

(2) Sodium bicarbonate

(3) Potash alum

(4) Calcium hydroxide

269. Which of the following isotope of hydrogen is radioactive?

(1) ${}_1H^1$ (2) ${}_1H^2$ (3) ${}_1H^3$ (4) Both 2 & 3

270. Which reaction is not used in the preparation of H_2?

(1) Zn + NaOH $\rightarrow$ (2) Mg + NaOH $\rightarrow$

(3) Al + NaOH $\rightarrow$ (4) Be + NaOH $\rightarrow$

271. Which of the following is water gas shift reaction?

(1) $CO + H_2O \rightarrow CO_2 + H_2$

(2) $C + H_2O \rightarrow CO$

(3) $CO + O_2 \rightarrow CO_2$

(4) $CO + H_2 \rightarrow CH_3OH$

272. Which cannot be oxidised by H_2O_2?

(1) Na_2SO_3 (2) PbS

(3) KI (4) O_3

273. Which of the following reaction represents the oxidising property of H_2O_2?

(1) $KMnO_4 + H_2SO_4 + H_2O_2 \longrightarrow K_2SO_4 + MnSO_4 + H_2O + O_2$

(2) $K_3[Fe(CN)_6] + KOH + H_2O_2 \longrightarrow K_4[Fe(CN)_6] + H_2O + O_2$

(3) $PbO_2 + H_2O_2 \longrightarrow PbO + H_2O + O_2$

(4) None of these

274. Permanent hardness can be removed by addding
(1) Cl_2 (2) Na_2CO_3
(3) $CaOCl_2$ (4) K_2CO_3

275. Calgon used as water softner is?
(1) $Na_6P_6O_{18}$ (2) $Na_4P_6O_{18}$
(3) $Na_6P_4O_{18}$ (4) $Na_6P_5O_{10}$

276. Which is not present in clear hard water
(1) $Mg(HCO_3)_2$ (2) $CaCl_2$
(3) $MgSO_4$ (4) $MgCO_3$

277. What is formed when calcium carbide reacts with heavy water ?
(1) C_2D_2 (2) CaD_2
(3) Ca_2D_2O (4) CD_2

278. The adsorption of hydrogen by metals is called:
(1) Dehydrogenation (2) Hydrogenation
(3) Occlusion (4) Absorption

279. Which of the following produces hydrolith with dihydrogen?
(1) Mg (2) Al (3) Cu (4) Ca

280. An ionic compound is dissolved simultaneously in heavy water and simple water. Its solubility is:
(1) Higher in heavy water
(2) Lesser in heavy water
(3) Same in both
(4) Lesser in simple water

281. Hydrogen can be prepared by mixing steam and water gas at 673 K in the presence of Fe_2O_3 and Cr_2O_3. This process is called:
(1) Nelson's process
(2) Serpeck's process
(3) Bosch's process
(4) Parke's process

282. Temporary hardness of water is due to the presence of:
(1) Magnesium bicarbonate
(2) Calcium chloride
(3) Magnesium sulphate
(4) Calcium carbonate

283. Metal hydrides are ionic, covalent or molecular in nature among LiH, NaH, KH, RbH, CsH, the correct order of increasing ionic character is:
(1) LiH > NaH > CsH > KH > RbH
(2) LiH < NaH < KH < RbH < CsH
(3) RbH > CsH > NaH > KH > LiH
(4) NaH > CsH > RbH > LiH > KH

284. Which of the following hydride is electron precise hydride?
(1) B_2H_6 (2) NH_3
(3) H_2O (4) CH_4

285. The compound that gives H_2O_2 on treatment with dilute H_2SO_4 is :
(1) PbO_2 (2) $BaO_2.8H_2O$
(3) MnO_2 (4) TiO_2

286. Which of the following equation depict the oxidizing nature of H_2O_2?
(1) $MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_2$
(2) $2Fe^{3+} + 2H^+ + H_2O_2 \rightarrow 2Fe^{2+} + 2H_2O + O_3$
(3) $2I^- + 2H^+ + H_2O_2 \rightarrow I_2 + 2H_2O$
(4) $KIO_4 + H_2O_2 \rightarrow KIO_3 + H_2O + O_2$

287. In given reaction $H_2O(l)$ act as a –
$H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$
(1) Only acidic (2) Amphoteric
(3) Only basic (4) Neutral

COORDINATION COMPOUND

288. The correct IUPAC name of $[Pt(NH_3)_2Cl_2]$ is
(1) Diamminedichloridoplatinum (II)
(2) Diamminedichloridoplatinum (IV)
(3) Diamminedichloridoplatinum (0)
(4) Dichloridodiammineplatinum (IV)

289. When $CrCl_3.6H_2O$ is treated with excess of $AgNO_3$, 3 Mol of AgCl are obtained. The formula of the complex is:
(1) $[CrCl_3(H_2O)_3] . 3H_2O$
(2) $[CrCl_2(H_2O)_4]Cl . 2H_2O$
(3) $[CrCl(H_2O)_5]Cl_2 . H_2O$
(4) $[Cr(H_2O)_6]Cl_3$

290. Indicate the complex ion which shows geometrical isomerism.
(1) $[Cr(H_2O)_4Cl_2]^+$ (2) $[Pt(NH_3)_3Cl]$
(3) $[Co(NH_3)_6]^{+3}$ (4) $[Co(CN)_5(NC)]^{-3}$

291. Which of the following option is correct for $K_4[Fe(CN)_6]$ complex?
(1) d^2sp^3 Hybridisation, diamagnetic
(2) sp^3d^2 Hybridisation
(3) Paramagnetic
(4) None of these

292. Which of the following complex formed by Cu^{+2} ions is most stable?
(1) $Cu^{+2} + 4NH_3 \longrightarrow [Cu(NH_3)_4]^{+2}$
(2) $Cu^{+2} + 4CN^- \longrightarrow [Cu(CN)_4]^{-2}$
(3) $Cu^{+2} + 2en \longrightarrow [Cu(en)_2]^{+2}$
(4) $Cu^{+2} + 4H_2O \longrightarrow [Cu(H_2O)_4]^{+2}$

293. The compounds $[Co(SO_4)(NH_3)_5]Br$ and $[Co(SO_4)(NH_3)_5]Cl$ represent :-
(1) Linkage isomerism
(2) Ionisation isomerism
(3) Coordination isomerism
(4) no isomerism

294. Which of the following species is not expected to be a ligand?
(1) NO
(2) NH_4^+
(3) $H_2N - CH_2 - CH_2 - NH_2$
(4) CO

295. Which complex is optically active?
(1) $[Co(en)_3]^{+3}$
(2) trans – $[Co(en)_2Cl_2]^+$
(3) Cis – $[Co(en)_2Cl_2]^+$
(4) (1) and (3) both

296. Geometrical shapes of the complexes formed by the reaction of Ni^{+2} with Cl^-, CN^- and H_2O respectively are.
(1) Octahedral, tetrahedral and sq. planar
(2) Tetrahedral. sq. planar and octahedral
(3) Square planar, tetrahedral and octahedral
(4) Octahedral, tetrahedral and square pyrimidal

297. How many EDTA molecules are requried to make an octahedral complex with a Ca^{+2} ion?
(1) 1 (2) 3 (3) 4 (4) 2

298. For the given complex $[Co(en)_2(NH_3)_2]^{+3}$, the number of geometrical isomers, optical isomers and total number of isomers of all type possible respectively are :
(1) 2, 2, 4 (2) 2, 2, 3 (3) 2, 0, 2 (4) 0, 2, 2

299. The value of effective atomic number of Cr in $[Cr(NH_3)_6]Cl_3$ is
(1) 35 (2) 27 (3) 33 (4) 36

300. A complex has a composition corresponding to the formula $CoBr_2Cl.4NH_3$. What is the structural formula if conductance measurements show two ions per formula unit ? Silver nitrate solution given an immediate precipitate of AgCl but no AgBr :-
(1) $[CoBrCl(NH_3)_4]Br$ (2) $[CoCl(NH_3)_4]Br_2$
(3) $[CoBr_2Cl(NH_3)_4]$ (4) $[CoBr_2(NH_3)_4]Cl$

301. How many ions are produced from the complex, $[Co(NH_3)_6]Cl_2$?
(1) 6 (2) 4 (3) 3 (4) 2

302. Ziegler - Naata catalyst is an organometalic compound of metal?
(1) Fe (2) Zr (3) Rh (4) Ti

303. Which of the following is π-acid ligand?
(1) NH_3 (2) CO (3) F^- (4) (en)

304. In Cu-ammonia complex the state of hybridization of Cu^{+2} is
(1) sp^3 (2) spd^2 (3) sp^3d^2 (4) dsp^2

305. The value of n in the carbonyl is
$(CO)_n — Co – Co — (CO)_n$
(1) 4 (2) 5 (3) 6 (4) 8

306. Which of the following doesn't follow effective atomic number rule?
(1) $[Cu(NH_3)_4]^{+2}$ (2) $[Zn(OH)_4]^{-2}$
(3) $[HgI_4]^{2-}$ (4) $Fe(CO)_5$

307. Which complex has highest paramagnetism?
(1) $[Cr(H_2O)_6]^{+3}$ (2) $[Fe(H_2O)_6]^{+2}$
(3) $[Cu(H_2O)_6]^{+2}$ (4) $[Zn(H_2O)_6]^{+2}$

308. Same number of unpaired electron is observed in which of the following complexes
(a) $[MnCl_6]^{3-}$ (b) $[Fe(CN)_6]^{3-}$
(c) $[CoF_6]^{3-}$ (d) $[Ni(NH_3)_6]^{2+}$
(1) a and b (2) a & c
(3) b & d (4) c & d

309. Which of the following statement is correct about ethane 1,2 diamine?

(1) It is a neutral ligand

(2) It is a bidentate ligand

(3) It is a chelating ligand

(4) All of the above

310. Which is correctly matched

Compound	**Total stereoisomer**
(1) $[Co(en)_3]Cl_3$	3
(2) $[Co(en)_2Cl_2]Cl$	2
(3) $[Co(NH_3)_3Cl_3]$	2
(4) $[Pt(NH_3)_2Cl_2]$	3

311. Which of the following complex species is not expected to exhibit optical isomerism

(1) $[Co(en)(NH_3)_2Cl_2]^+$

(2) $[Co(en)_3]^{+3}$

(3) $[Co(en)_2Cl_2]^+$

(4) $[Co(NH_3)_3Cl_3]$

312. When excess of ammonia is added to $CuSO_4$ solution. The deep blue colour complex is formed. The complex is

(1) Tetrahedral & paramagnetic

(2) Tetrahedral & diamagnetic

(3) Square planar & diamagnetic

(4) Square planar & paramagnetic

313. IUPAC name of $[Pt(NH_3)_2Cl(NO_2)]$ is

(1) Platinum diamminechloronitrite

(2) Chloronitrito-N-ammineplatinum(II)

(3) Diamminechloridonitrito-N-Platinum(II)

(4) Diamminechloronitrito-N-Platinum(II)ion

314. Which of the following have maximum number of isomers

(1) $[Co(NH_3)_4Cl_2]^+$ (2) $[Ni(en)(NH_3)_4]^{+2}$

(3) $[Ni(C_2O_4)(en)_2]^0$ (4) $[Cr(SCN)_2(NH_3)_4]^+$

315. Which among the following can exhibit cis-trans isomerism

(1) $CoCl_3.4NH_3$ (2) $CoCl_3.6NH_3$

(3) $CoCl_3.5NH_3$ (4) All of these

316. Out of the following coordination entities which is chiral (optically active)

(1) $[Cr(H_2O)_6]^{3+}$

(2) trans-$[CrCl_2(OX)_2]^{3-}$

(3) cis-$[CrCl_2(NH_3)_4]^+$

(4) cis-$[CrCl_2(OX)_2]^{3-}$

317. Amongst the following the most stable compound is

(1) $[Fe(H_2O)_6]^{3+}$ (2) $[Fe(C_2O_4)_3]^{3-}$

(3) $[Fe(NH_3)_6]^{3+}$ (4) $[FeCl_6]^{3-}$

318. Which of the following is an organometallic compound

(1) cis-platin (2) Zeise salt

(3) Tollen's reagent (4) Sodium nitroprusside

319. Which of the following system has maximum value of μ (only spin magnetic moment)?

(1) d^5 ($\Delta_0 > P$) (2) d^8 (tetrahedral)

(3) d^6 (high spin) (4) d^9 (octahedral)

320. The IUPAC name for $[Pt(NH_3)_3(Br)(NO_2)Cl]Cl$ is

(1) Triamminechlorobromonitroplatinum(IV) chloride

(2) Triamminebromochloronitroplatinum(IV) chloride

(3) Triamminechlorobromoplatinum(IV) chloride

(4) Triamminechloronitrobromoplatinum(IV) chloride

321. Which of the following cannot act as an electrolyte

(1) $CoCl_3.6NH_3$ (2) $CoCl_3.5NH_3$

(3) $CoCl_3.4NH_3$ (4) $CoCl_3.3NH_3$

322. In brown ring complex, the oxidation state of iron will be

(1) +2 (2) +3 (3) +1 (4) 0

323. Hexafluoroferrate(III) ion is an outer orbital complex the number of unpaired electrons present in it is

(1) 1 (2) 5

(3) 4 (4) unpredictable

324. Which of the following does not have optical isomer

(1) $[Co(en)_3]Cl_3$ (2) $[Co(NH_3)_3Cl_3]$

(3) $[Co(en)_2Cl_2]Cl$ (4) $[Co(en)(NH_3)_2Cl_2]Cl$

325. Which of the following species can act as reducing agent ?

(1) $[Co(CO)_4]^-$ (2) $Mn(CO)_6$

(3) $Mn(CO)_5$ (4) $Cr(CO)_6$

326. The geometry of $[Ni(CO)_4]$ and $[Ni(PPh_3)_2Cl_2]$ are
(1) both are square planar
(2) tetrahedral and square planar
(3) both tetrahedral
(4) square planar and tetrahedral

327. Geometrical isomerism can be shown by
(1) $[Ag(CN)(NH_3)]$
(2) $Na_2[Cd(NO_2)_4]$
(3) $[PtCl_4I_2]^{-2}$
(4) $[PtCl(NH_3)_3][Au(CN)_4]$

328. Which of the following will give a pair of enantiomorphs
(1) $[Cr(NH_3)_6][Co(CN)_6]$
(2) $[Co(en)_2Cl_2]Cl$
(3) $[Pt(NH_3)_4][PtCl_6]$
(4) $[Co(NH_3)_4Cl_2]NO_2$

329. Select the correct matching in given table –

	Coordination number	Type of hybridisation	Hybrid orbital in space
(1)	4	dsp^2	square planar
(2)	4	sp^3	square planar
(3)	6	sp^3d	octahedral
(4)	5	sp^3d	octahedral

330. Spin only magnetic moments of a d^8 ion in an octahedral, square planar and tetrahedral complex, respectively are
(1) 2.8 BM, 0.4 B.M., 2.8 BM
(2) 2.8 BM, 0 BM, 2.8 BM
(3) 0, 0 & 0 B.M.
(4) None of these

331. Among the following the colored compound is
(1) CuCl (2) $K_3[Cu(CN)_4]$
(3) CuF_2 (4) $[Cu(CH_3–CN)_4]BF_4$

332. $[NiCl_4]^{-2}$ and $[Ni(CN)_4]^{-2}$ resemble in
(1) geometry (2) magnetic nature
(3) hybridisation (4) oxidation state

333. CFSE of high spin d^4 complex is
(1) $-0.6\ \Delta_0$ (2) $-1.8\ \Delta_0$
(3) $-1.6\ \Delta_0 + P$ (4) $-1.2\ \Delta_0$

334. Total number of possible isomers of complex $[Pd(NH_3)_2(SCN)_2]$
(1) 2 (2) 4 (3) 3 (4) 6

335. Which of the following will have three stereoisomeric forms :-
(i) $[Cr(NO_3)_3(NH_3)_3]$
(ii) $K_3[Co(C_2O_4)_3]$
(iii) $K_3\ [CoCl_2(C_2O_4)_2]$
(iv) $[Co\ BrCl(en)_2]$
(1) iii, iv (2) i, ii and iv
(3) Only iv (4) All

336. The IUPAC name for ionisation isomers of $[Pt(NH_3)_3(Br)(NO_2)(I)]$ Cl is
(1) Triamminebromidochloridonitro platinum (IV) iodide
(2) Triamminebromidochloridoiodido platinum(IV) nitrite
(3) Triamminechloridoiodidonitro platinum (IV) bromide
(4) All are possible

337. Which ion would you expect to have the maximum splitting of d-orbitals
(1) $[Fe(CN)_6]^{-4}$ (2) $[Fe(CN)_6]^{-3}$
(3) $[Fe(H_2O)_6]^{+2}$ (4) $[Fe(H_2O)_6]^{+3}$

338. Indicate the complex ion which shows geometrical isomerism
(1) $[Cr(H_2O)_4Cl_2]^+$ (2) $[Pt(NH_3)_3Cl]^+$
(3) $[Co(NH_3)_6]^{3+}$ (4) $[Co(CN)_5(NC)]^{3-}$

339. **Compound** **Total stereoisomer**
(A) $[Co(en)_2\ Cl_2]^+$ (P) 2
(B) $[Co(en)_3]\ Cl_3$ (Q) 3
(C) $[Pt(NH_3)_2\ Cl_2]$ (R) 4
(D) $[Pt(NH_3)_3\ Cl_3]^+$ (S) 5
Correct code :-
(1) A = R, B = P, C =R, D = R
(2) A = Q, B = P, C = P, D = P
(3) A = P, B = R, C = P, D = P
(4) A = Q' B = Q, C = P, D = P

340. A brown ring is formed in the ring test of NO_3^- ion. It is due to the formation of

(1) $[Fe(H_2O)_5(NO)]^{2+}$

(2) $FeSO_4.NO_2$

(3) $[Fe(H_2O)_4(NO)_2]^{2+}$

(4) $FeSO_4.HNO_3$

341. The correct formula of diammine dichlorodicyano chromate(III) is :-

(1) $[CrCl_2(CN)_2(NH_3)_2]^{3+}$

(2) $[CrCl_2(CN)_2(NH_3)_2]^{3-}$

(3) $[CrCl_2(CN)_2(NH_3)_2]$

(4) $[CrCl_2(CN)_2(NH_3)_2]^-$

342. Select the correct order of E.A.N:

(1) $[Cr(CO)_6] > [Cr(CO)_6]^{\ominus} > [Cr(CO)_6]^{\oplus}$

(2) $[Cr(CO)_6]^+ > [Cr(CO)_6]^{\ominus} > [Cr(CO)_6]$

(3) $[Cr(CO)_6]^{\ominus} > [Cr(CO)_6] > [Cr(CO)_6]^{\oplus}$

(4) $[Cr(CO)_6]^{\ominus} = [Cr(CO)_6] > [Cr(CO)_6]^+$

343. Select the correct I.U.P.A.C. name for $[Cr(C_6H_6)\ (CO)_{3:}]$.

(1) (η^6-benzene) tricarbonylchromate (0)

(2) Tricarbonyl (η^6-benzene) chromate (0)

(3) Tricarbonyl (η^6-benzene) chrominum (0)

(4) (η^6-benzene) tricarbonylchromium (0)

344. Zeise's salt is:

(1) $Fe(\eta^5 - C_5H_5)_2$

(2) $Cr(\eta^6 - C_6H_6)_2$

(3) $K[Pt\ (\eta^2 - C_2H_4)Cl_3]$

(4) $K[Pt(\eta^2 - C_2H_4)_2Cl_2]$

345. Coordination number of calcium is six in :

(1) $[Ca(EDTA)]^{2-}$ (2) CaC_2O_4

(3) $[Ca(C_2O_4)_2]^{2-}$ (4) $CaSO_4.\ 2H_2O$

346. Hybridization, shape, and magnetic moment of $K_3\ [Co(CO_3)_3]$ are:

(1) d^2sp^3, octahedral, 4.9 B.M.

(2) sp^3d^2, octahedral, 4.9 B.M.

(3) dsp^2, square planar, 4.9 B.M.

(4) sp^3, tetrahedral, 4.9 B.M.

347. Consider the following complexes :-

(a) K_2PtCl_6 (b) $PtCl_4.2NH_3$

(c) $PtCl_4.3NH_3$ (d) $PtCl_4.5NH_3$

Their electrical conductance in aqueous solutions are :-

(1) 404, 0, 97, 256 (2) 256, 97, 0, 404

(3) 404, 97, 256, 0 (4) 256, 0, 97, 404

348. The oxidation number of cobalt in $K[Co(CO)_4]$ is :-

(1) –1 (2) +3 (3) +1 (4) –3

349. Which of the following statement is correct?

(1) With d^2sp^3 hybridization $[FeCl(CN)_4(O_2)]^{4-}$ complex is diamagnetic

(2) $[NiCl_4]^{2-}$ complex is more stable than $[Ni(dmg)_2]$

(3) $[V(CO)_6]$ is not very stable and easily reduces to $[V(CO)_6]^-$

(4) Ligands such as CO, CN^-, NO^+ are pie– donar due to the presence of filled p-molecular orbital

350. In the complex $[Pt(O_2)\ (en)_2\ (Br)]^{2+}$, coordination number and oxidation number of platinum are:

(1) 4, 3 (2) 4, 5 (3) 6, 2 (4) 6, 4

351. If $H_x\ [Pt\ y_6]$, y is a monodentate negatively charged ligand then find out the value of x:

(1) 5 (2) 3

(3) 6 (4) None of these

352. The bond length of C—O bond in CO is 1.128 Å. The C—O bond length in $[Fe(CO_5)]$ is :

(1) 1.115 Å (2) 1.128 Å

(3) 1.178 Å (4) 1.150 Å

353. The most stable ion is:

(1) $[Fe(C_2O_4)_3]^{3-}$ (2) $[Fe\ Cl_6]^{3-}$

(3) $[Fe(SCN)_6]^{3-}$ (4) $[Fe(H_2O)_6]^{3+}$

354. The increasing order of the crystal field splitting power of some common ligands is :

(1) $H_2O < NO_2^- < \overset{\ominus}{C}N < NH_3$

(2) $NH_3 < NO_2^- < \overset{\ominus}{C}N < H_2O$

(3) $H_2O < NH_3 < NO_2^- < \overset{\ominus}{C}N$

(4) $H_2O < NH_3 < \overset{\ominus}{C}N < NO_2$

355. Which of the following does not have optical isomers?
(1) $[Co(en)_3]Cl_3$
(2) $[Co(NH_3)_3Cl_3]$
(3) $[Co(en)(NH_3)_2Cl_2]Cl$
(4) None

356. Mg is an important component of which biomolecule occurring extensively in living world?
(1) Hemoglobin (2) Chlorophyll
(3) Vit B_{12} (4) ATP

357. The hybridization and geometry of $[Fe(CO)_4]^{2-}$ are:
(1) sp^3d,TBP (2) sp^3,tetrahedral
(3) dsp^2,square planar (4) sp^3,TBP

358. Which of the following organometallic compound is σ and π-bonded?
(1) $[Fe(\eta^5 - C_5H_5)_2]$ (2) $K[PtCl_3(\eta^2–C_2H_4)]$
(3) $[Co(CO)_5\ NH_3]^{2+}$ (4) $Fe(CH_3)_3$

359. When 0.1 mol $CoCl_3\ (NH_3)_5$ is treated with excess of $AgNO_3$, 0.2 mol of AgCl are obtained. The conductivity of solution will correspond to
(1) 1 : 3 electrolyte (2) 1 : 2 electrolyte
(3) 1 : 1 electrolyte (4) 3 : 1 electrolyte

360. According to crystal field theory, octahedral splitting and tetrahedral splitting of d orbitals caused by the same ligands are related through the expression:
(1) $\Delta_o = \Delta_t$ (2) $4\Delta_o = 9\Delta_t$
(3) $9\Delta_o = 4\Delta_t$ (4) $\Delta_o = 2\Delta_t$

361. Which of the following will show optical isomerism?
(1) $[Cu(NH_3)_4]^{2+}$ (2) $[ZnCl_4]^{2-}$
(3) $[Cr(C_2O_4)_3]^{3-}$ (4) $[Co(CN)_6]^{3-}$

362. $[Cr(H_2O)_6]Cl_3$ (atomic number of Cr = 24) has a magnetic moment of 3.83 B.M. the correct distribution of 3d-electrons in the chromium present in the complex is:
(1) $3d^1_{xy}, 3d^1_{yz}, 3d^1_{zx}$ (2) $3d^1_{xy}, 3d^1_{yz}, 3d^1_{z^2}$
(3) $3d^1_{(x^2-y^2)}, 3d^1_{z^2}, 3d^1_{zx}$ (4) $3d^1_{xy}, 3d^1_{(x^2-y^2)}, 3d^1_{xz}$

363. The total number of coordination isomer for the given compound $[Pt(NH_3)_4Cl_2][PtCl_4]$ is:
(1) 2 (2) 4 (3) 5 (4) 3

364. Pick a poor electrolytic conductor complex in solution.
(1) $K_2[PtCl_6]$ (2) $[Co(NH_3)_3](NO_2)_3$
(3) $K_4\ [Fe(CN)_6]$ (4) $[Co(NH_3)_4]SO_4$

365. $CuSO_4$ solution reacts with excess of KCN solution to form:
(1) $Cu(CN)_2$ (2) $K_2[Cu(CN)_4]$
(3) $K_3[Cu(CN)_4]$ (4) $K[Cu(CN)_2]$

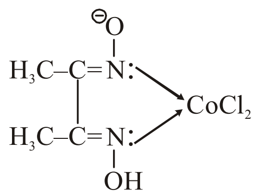
366. The number of ions formed on dissolving one molecule of $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ in water is:
(1) 4 (2) 5 (3) 3 (4) 6

367. $NH_2–NH_2$ may serves as:
(1) monodenatate ligand
(2) chelating ligand
(3) Flexidentate ligand
(4) both (1) and (3)

368. EDTA is a..........ligand :
(1) monodenatate (2) hexadentate
(3) bidentate (4) tridentate

369. In solid $CuSO_4 \cdot 5H_2O$ copper is co-ordinated to :
(1) 4 water molecules
(2) 5 water molecules
(3) one sulphate molecule
(4) one water molecule

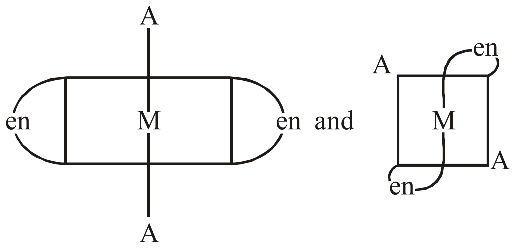
370. The correct IUPAC name of the complex:



is:

(1) dichloridodimethylglyoximatocobalt (II)
(2) bis(dimethylglyoxime) dichlorido cobalt (II)
(3) dimethylglyoximecobalt (II) chloride
(4) dichloridodimethylglyoxime-N,N-cobalt (II)

371. The two complexes given below are:



(1) geometrical isomers(2) ionisation

(3) optical isomers (4) identical

372. Among $[Ni(CN)_4]^{2-}$,$[NiCl_4]^{2-}$ and $[Ni(CO)_4]$:

(1) $[NiCN)_4]^{2-}$ is square planar and $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$ are tetrahedral

(2) $[NiCl_4]^{2-}$ is square planar and $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$ are tetrahedral

(3) $[Ni(CO)_4]$ is square planar and $[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$ are tetrahedral

(4) None of these

373. Which one is the most likely structure of $CrCl_3 \cdot 6H_2O$ if 1/3 of total chlorine of the compound is percipitated by adding $AgNO_3$ to its aqueous solution?

(1) $CrCl_3 \cdot 6H_2O$

(2) $[Cr(H_2O)Cl_3]\cdot(H_2O)_3$

(3) $[CrCl_2(H_2O)_4]Cl \cdot 2H_2O$

(4) $[CrCl(H_2O)_5]Cl_2 \cdot H_2O$

374. The colour of $[Ti(H_2O)_6]^{3+}$ is due to :

(1) transfer of an electron from one Ti to another

(2) excitation of electrons from d $\rightarrow$ d

(3) intramolecular vibration

(4) None

375. Which statement is correct in the case of $[Co(NH_3)_6]^{3+}$ complex?

(1) It is octahedral in shape

(2) It involves d^2sp^3-hybridisation

(3) It has diamagnetic nature

(4) All are correct

376. Which ion is paramagnetic?

(1) $[Ni(H_2O)_6]^{2+}$ (2) $[Fe(CN)_6]^{4-}$

(3) $[Ni(CO)_4]$ (4) $[Ni(CN)_4]^{2-}$

377. Considering H_2O as a weak field ligand, the number of unpaired electrons in $[Mn(H_2O)_6]^{2+}$ will be :

(1) 2 (2) 3 (3) 4 (4) 5

378. For which transition metal ions are low spin complexes possible?

(1) Rh^{3+} (2) Pt^{+2}

(3) Ru^{2+} (4) All are correct

379. Which ligand produces a high crystal field splitting (i.e. are strong ligand field):

(1) CO (2) NO_2^-

(3) CN^- (4) All of these

380. Which order is correct in spectrochemical series of ligands?

(1) $Cl^- < F^- < C_2O_4^{2-} < NO_2^- < CN^-$

(2) $CN^- < C_2O_4^{2-} < Cl^- > NO_2^- < F^-$

(3) $C_2O_4^{2-} < F^- < Cl^- > NO_2^- < CN^-$

(4) $F^- < Cl^- < NO_2^- < CN^- < C_2O_4^{2-}$

381. Which is high spin complex?

(1) $CoCl_6^{3-}$ (2) FeF_6^{3-}

(3) $[Fe(H_2O)_6]^{+3}$ (4) All are correct

382. $K_3[CoF_6]$ is high spin complex. What is the hybrid state of Co-atom in this complex?

(1) sp^3d (2) sp^3d^2

(3) d^2sp^3 (4) dsp^2

383. The structure of iron pentacarbonyl is:

(1) square planar (2) trigonal bipyramidal

(2) triangular (4) None of these

384. Ruthenium carbonyl is :

(1) $Ru(CO)_4$ (2) $Ru(CO)_5$

(3) $Ru(CO)_8$ (4) $Ru(CO)_6$

385. Which of the following is double salt?

(1) Carnallite (2) Mohr's salt

(3) Alum (4) All are correct

386. Carbon donor ligands are strong ligands and usually forms low spin complexes, Among the complexes given below, select the high spin complex:

(1) $Fe(CN)_6]^{3-}$ (2) $[Fe(CN)_6]^{4-}$
(3) $[Fe(C_2O_4)_3]^{3-}$ (4) None of these

387. Crystal field splitting energy (Δ) for transition metals belonging to different transition series lies in the order:

(1) 3d > 4d > 5d (2) 3d≈4d≈5d
(3) 3d < 4d < 5d (4) 3d > 4d < 5d

388. Select the correct order of CFSE (Δ) for the ions given below:

(1) $V^{2+} < Mn^{2+} < Fe^{2+} < Co^{2+} < Ni^{2+}$
(2) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+} < V^{2+}$
(3) $Mn^{2+} < V^{2+} < Co^{2+} < Fe^{2+}\ Ni^{2+}$
(4) $V^{2+} \approx Mn^{2+} \approx Fe^{2+} \approx Co^{2+} \approx Ni^{2+}$

389. The compound $[CoCl_2(NH_3)_2(en)]$ can form :

(1) Linkage isomers
(2) Coordination isomers
(3) Optical isomers
(4) Linkage as well as optical isomers

390. Colour of Halogen is due to :

(1) Polarisation
(2) H.O.M.O. to L.U.M.O. transition
(3) Charge transfer
(4) d-d transition

391. For which of the following species d-d transition does not account for its colour ?

(1) $Cr_2O_7^{-2}$ (2) CrO_4^{-2}
(3) CrO_2Cl_2 (4) All of the above

392. Which of the following ion is not coloured?

(1) $[Ni(DMG)_2]$ (2) $[Co(SCN)_4]^{2-}$
(3) $[Al(OH)_4]^-$ (4) $[Fe(H_2O)_5(SCN)]^{+2}$

393. What will be the correct order of absorption of wavelength of light in the visible region, for the complex, $[Co(NH_3)_6]^{3+}$, $[Co(CN)_6]^{3-}$, $[Co(H_2O)_6]^{3+}$?

(1) $[Co(CN)_6]^{3-} > [Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+}$
(2) $[Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+} > [Co(CN)_6]^{3-}$
(3) $[Co(H_2O)_6]^{3+} > [Co(NH_3)_6]^{3+} > [Co(CN)_6]^{3-}$
(4) $[Co(CN)_6]^{3-} > [Co(NH_3)_6]^{3+} > [Co(H_2O)_6]^{3+}$

394. Which of the following compound is coloured and paramagnetic?

(1) $KMnO_4$ (2) AgI
(3) $CuSO_4 \cdot 5H_2O$ (4) $[Ni(CN)_4]^{2-}$

395. In which of the following complex, cis form is optically active?

(1) $[Co(en)(NH_3)_3Cl]$ Br
(2) $[Pt(NH_3)_4Cl\ Br]SO_4$
(3) $[Pt(NH_3)_2Cl_2Br_2]$
(4) $[Pt(NH_3)_2(NO_2)_2]$

396. AgBr and AgI are coloured because of :

(1) d — d transition
(2) Charge transfer
(3) Polarisation of halide by metal ion
(4) HOMO to LUMO transition

397. Spliting energy of $[Cr(X)_6]^{3+}$, $[Cr(Y)_6]^{3+}$, $[Cr(Z)_6]^{3+}$ are 315, 210 and 184 kJ/mole respectively, then their colour are :

(1) Yellow, Red and Green respectively
(2) Red, Yellow and Green respectively
(3) Green, Red and Yellow respectively
(4) Yellow, Green and Red respectively

398. Na_2S forms violet colour complex when reacts with :

(1) Brown ring complex
(2) Sodium nitroprusside
(3) $K_4[Fe(CN)_6]$
(4) Hypo Solution

399. Which of the following complex can represent both turn bull blue and prussian blue?

(1) $Fe_4[Fe(CN)_6]_3$ (2) $Fe_3[Fe(CN)_6]_2$
(3) $K_4[Fe(CN)_6]$ (4) $KFe[Fe(CN)_6]$

400. Which of the following reactions represent developing in photography?

(1) $AgNO_3 + NaBr \longrightarrow AgBr + NaNO_3$
(2) $AgBr + 2Na_2S_2O_3 \longrightarrow Na_3[Ag(S_2O_3)_2] + NaBr$
(3) $AgBr \xrightarrow{h\nu} Ag^+ + Br^-$
(4) $C_6H_4(OH)_2 + 2AgBr \longrightarrow C_6H_4O_2 + 2HBr + 2Ag$

401. Which is used in cancer-chemotherapy ?

(1) Cis-platin (2) Zeise's salt
(3) both 1 and 2 (4) None of these

P-BLOCK ELEMENTS

402. How many bridging oxygen atoms are present in P_4O_{10}?
(1)6 (2)4 (3) 2 (4) 5

403. Which of the following phosphorus is the most reactive?
(1) Scarlet phosphorus (2) White phosphorus
(3) Red phosphorus (4) Violet phosphorus

404. Which of the following can be hydrolysed?
(1) TeF_6 (2) NCl_3
(3) SF_6 (4) Both 1 & 2

405. Which of the following methyl diboranes does not exist ?
(1) $B_2H_4(CH_3)_2$ (2) $B_2H_5(CH_3)$
(3) $B_2H_2(CH_3)_4$ (4) $B_2H(CH_3)_5$

406. Which of the following is a sesquioxide?
(1) N_2O_4 (2) N_2O_3
(3) N_2O (4) N_2O_5

407. Which of the following does not form during hydrolysis of XeF_6?
(1) XeO_3 (2) $XeOF_4$
(3) XeO_2F_2 (4) $XeOF_3$

408. Mixture used in Holme's signal is?
(1) CaC_2 and $CaCl_2$ (2) $CaCl_2$ and Ca_3P_2
(3) CaC_2 and Ca_3N_2 (4) CaC_2 and Ca_3P_2

409. HNO_2 acts as an/a
(1) acid (2) oxidising agent
(3) reducing agent (4) All

410. The nitrogen oxide that does not contain N-N bond are-
(1) N_2O (2) N_2O_3
(3) N_2O_4 (4) N_2O_5

411. Bleaching action of SO_2 is due to–
(1) reduction (2) oxidation
(3) hydrolysis (4) acidic nature

412. PH_3 can be formed in
(1) hydrolysis of calcium phosphide
(2) heating H_3PO_3
(3) reaction of P_4 & NaOH in inert atmosphere
(4) All

413. A (black compound) $\xrightarrow{\text{halogen acid}}$ B
(green yellow gas)
B(excess) $\xrightarrow{NH_3}$ C (unstable trihalide)
Correct statement is
(1) C is PCl_3 (2) halogen acid is HI
(3) B is Cl_2 (4) A is $KMnO_4$

414. Beryl is a type of
(1) chain silicate (2) cyclic silicate
(3) sheet silicate (4) 3-D silicate

415. Which of the following does not react with oxygen directly :-
(1) Pt (2) Fe (3) Zn (4) Ti

416. Which of the following compound will not give NH_3 on heating
(1) $(NH_4)_2SO_4$ (2) $(NH_4)_2CO_3$
(3) NH_4NO_2 (4) NH_4Cl

417. On hydrolysis CaC_2 gives a gas which on trimerisation gives
(1) C_2H_2 (2) C_6H_6
(3) C_2H_4 (4) C_3H_8

418. $P+Cl_2 \rightarrow A$, P + Excess $Cl_2 \rightarrow B$
Hydrolysis products of A and B are respectively
(1) H_3PO_2, H_3PO_3 (2) H_3PO_4, H_3PO_3
(3) H_3PO_3, H_3PO_4 (4) H_3PO_2, H_3PO_4

419. Which among the following order of given properties is incorrect
(1) $HOCl > HClO_2 > HClO_3 > HClO_4$ – oxidising nature
(2) $Cl > F > Br > I$ – electron affinity
(3) $Cl_2 > Br_2 > I_2 > F_2$ – bond dissociation energy
(4) $HF < HCl < HBr < HI$ – acidic nature

420. Amongst the following the central atoms are directly bonded in
(1) N_2O_5 (2) $S_2O_5^{2-}$
(3) P_4O_{10} (4) Mn_2O_7

421. The chain length of silicones can be controlled by
(1) $(CH_3)_3SiCl$
(2) Addition of Cu powder
(3) Elevation of tempeature
(4) None of these

422. Which of the following statement is incorrect regarding B_2H_6?
(1) On methylation it gives $B_2H(CH_3)_5$
(2) It has two $2e^-3C$ bonds
(3) There can be maximum six atoms in a plane
(4) It has four $2e^-2C$ bonds

423. Red and white phosphorus are similar in
(1) smell (2) solubility in CS_2
(3) Hybridisation of P (4) Stability

424. Which of the following is strongest oxidizing agent
(1) HOCl (2) $HClO_2$
(3) $HClO_3$ (4) $HClO_4$

425. In which species O–O bond is present
(1) $S_2O_8^{-2}$ (2) $S_4O_6^{-2}$
(3) SO_3^{-2} (4) $S_2O_7^{-2}$

426. Glass is soluble in
(1) aqua ragia (2) H_2SO_4
(3) HF (4) $HClO_4$

427. Paramagnetic oxide is
(1) NO (2) N_2O_4
(3) P_4O_6 (4) N_2O_5

428. Which oxide does not act as a reducing agent
(1) NO (2) NO_2
(3) N_2O (4) N_2O_5

429. Borax bead test is not given by
(1) An aluminium salt (2) A cobalt salt
(3) A copper salt (4) A nickel salt

430. Select the endothermic reaction in following :-
(1) $B_2H_6 + O_2 \rightarrow B_2O_3 + 3H_2O$
(2) $4Al + 3O_2 \rightarrow 2Al_2O_3$
(3) $3O_2 \rightarrow 2O_3$
(4) $2SO_2 + O_2 \xrightarrow{V_2O_5} 2SO_3$

431. In which of the following oxy acid of sulphur sulphur atoms has different oxidation states
(1) $H_2S_4O_6$ (2) $H_2S_2O_8$
(3) $H_2S_2O_4$ (4) All

432. Which of the following is most acidic?
(1) Cl_2O_7 (2) SO_3
(3) P_2O_5 (4) SiO_2

433. Which of the following is not peroxide
(1) Na_2O_2 (2) CaO_2
(3) PbO_2 (4) H_2O_2

434. In which of the following reaction, phosphine is not obtained as product
(1) $Ca_3P_2 + HCl \longrightarrow$ (2) $P_4 + NaOH \longrightarrow$
(3) $H_3PO_4 \xrightarrow{\Delta}$ (4) $H_3PO_3 \xrightarrow{\Delta}$

435. Correct statement about boric acid is
(1) boron is sp^3 hybridised
(2) boric acid is triprotic acid
(3) it is used in the treatement of eye infection
(4) It forms covalent network

436. Which of the following oxide is acidic in nature?
(1) B_2O_3 (2) Al_2O_3 (3) Ga_2O_3 (4) In_2O_3

437. The most commonly used reducing agent is
(1) $AlCl_3$ (2) $PbCl_2$ (3) $SnCl_4$ (4) $SnCl_2$

438. Which of the following are peroxy-acid of sulphur
(1) H_2SO_5 and $H_2S_2O_8$
(2) H_2SO_5 and $H_2S_2O_7$
(3) $H_2S_2O_7$ and $H_2S_2O_8$
(4) $H_2S_2O_6$ and $H_2S_2O_7$

439. The element which forms neutral as well as acidic oxide is :-
(1) Sn (2) Si (3) C (4) P

440. P_4O_{10} has short and long P–O bonds. The number of short P–O bonds in this compound is :-
(1) 1 (2) 2 (3) 3 (4) 4

441. Nitrogen dioxide can be obtained by heating:-
(1) KNO_3 (2) $Pb(NO_3)_2$
(3) NH_4NO_3 (4) All of these

442. MF + XeF_4 → 'A' (M – alkali metal) hybridisation of A and shape of A is :-
(1) sp^3d, Trigonal bipuramidal
(2) sp^3d^3, distoted octahedral
(3) sp^3d^3, pentagonal planer
(4) No compound formed at all

443. Which oxide is most acidic :-
(1) Al_2O_3 (2) Na_2O
(3) MgO (4) CaO

444. Cl_2O_6 is an anhydride of :-
(1) $HClO_3$
(2) $HClO_2$
(3) $HClO_4$
(4) Mixed anhydride of $HClO_3$ & $HClO_4$

445. Carbogen is a mixture of
(1) O_2 & H_2 (2) CO_2 & O_2
(3) O_2 & Air (4) O_2 & Ne

446. A black sulphide when treated with ozone becomes white, the white compound is :-
(1) $ZnSO_4$ (2) $CaSO_4$
(3) $BaSO_4$ (4) $PbSO_4$

447. An example of tetrabasic acid is :-
(1) Orthophosphorus acid
(2) Orthophophoric acid
(3) Metaphosphoric acid
(4) Pyrophosphoric acid

448. The reducing power of divalent species decreases in the order is :-
(1) Ge > Sn > Pb (2) Sn > Ge > Pb
(3) Pb > Sn > Ge (4) None of these

449. Basicity of phosphinic acid is
(1) 1 (2) 2 (3) 3 (4) 4

450. Empirical formula of Bleaching powder is :-
(1) $Ca(OH)_2$ (2) $CaOCl_2$
(3) $Ca(OCl)_2$ (4) $CaClO_3$

451. Which of the following silicate is called disilicate?
(1) Orthosilicate (2) Pyrosilicate
(3) Single-chain silicate (4) None of these

452. The cationic part of solid XeF_6 is having the "_______" shape:
(1) Linear (2) Angular
(3) Square pyramidal (4) Tetrahedral

453. In the hydrolysis of ICl, the products are:
(1) HI + HCI (2) HI + HOCI
(3) HCI + HOI (4) HOCI + HOI

454. Which of the following compounds are the common products obtained in the hydrolysis of XeF_6 and XeF_4?
(1) XeO_2F_2 (2) HF
(3) XeO_3 (4) Both (2) and (3)

455. Find the incorrect match:
(1) Al_2Cl_6 : 3C-4e bond is present
(2) $Al_2(CH_3)_6$: All carbon atoms are sp^3-hybridized
(3) I_2Cl_6 : Nonplanar
(4) Al_2Br_6 : Nonpolar

456. Which of the following is not a Lewis acid?
(1) SiF_4 (2) $FeCl_3$ (3) BF_3 (4) C_2H_4

457. Thallium shows different oxidation states because:
(1) Of its high reactivity
(2) Of inert pair of electrons
(3) Of its amphoteric nature
(4) It is a transition metal

458. H_3BO_3 is :
(1) Monobasic and weak Lewis acid
(2) Monobasic and weak bronsted acid
(3) Monobasic acid and strong Lewis acid
(4) Tribasic acid and weak bronsted acid

459. What is formula for carbon suboxide?
(1) CO (2) CO_2
(3) C_2O_4 (4) C_3O_2

460. CCl_4 is used as fire extinguisher because:
(1) Its m.pt. is high
(2) It forms covalent bond
(3) Its b.pt. is low
(4) It gives incombustible vapours

461. Select the least stable oxide in following :-
(1) BrO_2 (2) ClO_2
(3) I_2O_5 (4) Cl_2O_6

462. Carborundum is:
(1) Al_2O_3 (2) SiC (3) BF_3 (4) B_4C

463. The acid which contains a peroxo linkage is:
(1) Sulphurous acid (2) Pyrosulphuric acid
(3) Dithionic acid (4) Caro's acid

464. In SiF_6^{2-} and $SiCl_6^{2-}$, which one is known and why?
(1) SiF_6^{2-} because of small size of F
(2) SiF_6^{2-} because of large size of F
(3) $SiCl_6^{2-}$ because of small size of Cl
(4) $SiCl_6^{2-}$ because of large size of Cl

465. The stability of dihalides of Si, Ge, Sn, and Pb increase stability in the sequence:
(1) $PbX_2 < SnX_2 < GeX_2 < SiX_2$
(2) $GeX_2 < SiX_2 < SnX_2 < PbX_2$
(3) $SiX_2 < GeX_2 < PbX_2 < SnX_2$
(4) $SiX_2 < GeX_2 < SnX_2 < PbX_2$

466. Products formed when $Pb(NO_3)_2$ is heated are:
(1) PbO, N_2,O_2 (2) $Pb(NO_2)_2$,O_2
(3) PbO,NO_2,O_2 (4) Pb, N_2, O_2

467. Silver chloride dissolves in excess of NH_4OH. The cation present in solution is:
(1) Ag^+ (2) $[Ag(NH_3)_4]^+$
(3) $[Ag(NH_3)_2]^+$ (4) $[Ag(NH_3)_6]^+$

468. The catalyst used in the manufacture of ammonia by Haber's process is:
(1) Pt (2) Fe_2O_3
(3) $CuCl_2$ (4) V_2O_5

469. Industrial preparation of nitric acid by Ostwald's process involves:
(1) Oxidation of NH_3
(2) Reduction of NH_3
(3) Hydrogenation of NH_3
(4) Hydrolysis of NH_3

470. When zinc reacts with very dilute nitric acid it produces:
(1) NH_4NO_3 (2) NO
(3) NO_2 (4) H_2

471. Which of the following oxides of nitrogen is the an-hydride of nitrous acid?
(1) NO (2) N_2O_3
(3) N_2O_4 (4) N_2O_5

472. Polar oxide of carbon is :-
(1) CO (2) CO_2
(3) C_3O_2 (4) Both 1 and 3

473. Aqueous solution of ammonia consists of :-
(1) H^+ only (2) OH^- only
(3) NH_4^+ only (4) NH_4^+ and OH^-

474. Phosphine, acetylene and ammonia can be formed by treating water with:
(1) Mg_3P_2, Al_4C_3, Li_3N
(2) Ca_3P_2, CaC_2, Mg_3N_2
(3) Ca_3P_2, Be_2C, NH_4NO_3
(4) Ca_3P_2,Mg_2C_3,NH_4NO_3

475. Atoms in P_4 molecule of white phosphorus are arranged regularly in the following way:
(1) At the corners of a cube
(2) At the corners of an octahedron
(3) At the corners of a tetrahedron
(4) At the center and corners of a tetrahedron

476. Which is formed when $K_2Cr_2O_7$, $CaCl_2$, and conc. H_2SO_4 are heated?
(1) $Cr_2(SO_4)_3$ (2) $CrCl_3$
(3) CrO_2Cl_2 (4) K_2CrO_4

477. Which one of the following reactions does not occur ?
(1) $F_2 + 2Cl^- \longrightarrow 2F^- + Cl_2$
(2) $Cl_2 + 2F^- \longrightarrow 2Cl^- + F_2$
(3) $Br_2 + 2I^- \longrightarrow 2Br^- + I_2$
(4) $Cl_2 + 2Br^- \longrightarrow 2Cl^- + Br_2$

478. BCl_3 does not exist as dimer but BH_3 exists as dimer (B_2H_6) because:

(1) Chlorine is more electronegative than hydrogen

(2) There is $p_\pi - p_\pi$ back bonding in BCl_3, but BH_3 does not contain such multiple bonding

(3) Large sized chlorine atoms do not fit in between the small boron atoms, whereas small-sized hydrogen atoms fit between boron atoms

(4) None of these

479. Red lead is :

(1) PbO (2) Pb_3O_4

(3) PbO_2 (4) HgS

480. The most stable and basic hydride of 15th group is :

(1) NH_3 (2) PH_3

(3) AsH_3 (4) BiH_3

481. C — C bond length is maximum in:

(1) Diamond (2) Graphite

(3) Napthalene (4) Fullerene

482. N forms NCl_3, whereas P can form both PCl_3 and PCl_5. Why?

(1) P has vacant d-orbitals which can be used for bonding but N does not have

(2) N atom is larger in size than P

(3) P is more reactive towards Cl than N

(4) None of the above

483. Which is in the decreasing order of boiling points of VA group hydrides?

(1) $NH_3 > PH_3 > AsH_3 > SbH_3$

(2) $SbH_3 > AsH_3 > PH_3 > NH_3$

(3) $PH_3 > NH_3 > AsH_3 > SbH_3$

(4) $SbH_3 > NH_3 > AsH_3 > PH_3$

484. Which compound acts as an oxidizing as well as a reducing agent?

(1) SO_2 (2) Mn_2O_7

(3) Al_2O_3 (4) CrO_3

485. The most powerful oxidizing agent is :

(1) Fluorine (2) Chlorine

(3) Bromine (4) Iodine

486. Which one of the hydracids does not form any precipitate with $AgNO_3$?

(1) HF (2) HCl

(3) HBr (4) HI

487. Which of the following structure is non-planar?

(1) $Na_3B_3O_6$

(2) I_2Cl_6

(3) Sheet silicates

(4) Inorganic graphite layer

488. In the reaction $LiH + AlH_3 \longrightarrow LiAlH_4$ AlH_3 and LiH act as:

(1) Lewis acid and Lewis base

(2) Lewis base and Lewis acid

(3) Bronsted base and Bronsted acid

(4) None of these

489. Which one is not an acid salt?

(1) NaH_2PO_2 (2) NaH_2PO_3

(3) NaH_2PO_4 (4) None of these

490. The most thermodynamically stable allotropic form of phosphorus is:

(1) Red (2) White

(3) Black (4) Yellow

491. Identify the incorrect statement among the following:

(1) Ozone reacts with SO_2 to give SO_3

(2) Silicon reacts with NaOH(aq.) in the presence of air to give Na_2SiO_3

(3) Cl_2 reacts with excess of NH_3 to give N_2 and NH_4Cl.

(4) Br_2 reacts with hot and strong NaOH solution to give NaBr, $NaBrO_4$ and H_2O.

492. The product of oxidation of I^- with MnO_4^- in alkaline medium is:

(1) IO_3^- (2) I_2 (3) IO^- (4) IO_4^-

493. The chemical formula of feldspar is :
(1) $KAlSi_3O_8$ (2) Na_3AlF_6
(3) $NaAlO_2$ (4) K_2SO_4

494. Which of the following is correct match:
(1) Gun metal $\rightarrow$ Cu, Sn & Zn
(Red brass)
(2) White metal $\rightarrow$ Contains Li
(3) Stainless Steel $\rightarrow$ Fe, Cr, Ni, Carbon
(4) All

495. $(NH_4)_2Cr_2O_7$ on heating liberates a gas. The same gas will be obtained by:
(1) Heating NH_4NO_2
(2) Heating NH_4NO_3
(3) Heating Mg_3N_2
(4) None

496. There is no S—S linkage in
(1) $S_2O_4^{2-}$ (2) $S_2O_5^{2-}$
(3) $S_2O_3^{2-}$ (4) $S_2O_7^{2-}$

497. Total number of lone pairs of electron and P–O–P linkage present in dimer of P_2O_5 are :
(1) 16 (2) 22
(3) 26 (4) 30

498. The number of S—S bonds in polythionic acid ($H_2S_nO_6$):
(1) n (2) n – 1
(3) n – 2 (4) None of these

499. Which of the following halogen oxides is ionic?
(1) I_4O_9 (2) I_2O_5
(3) BrO_2 (4) $C\ell O_3$

500. Antichlor is a compound:
(1) Which absorbs chlorine
(2) Which removes excess of Cl_2 from a material
(3) Which liberates Cl_2 from bleaching powder
(4) Which acts as a catalyst in the manufacture of Cl_2

501. Which of the following statements regarding orthoboric acid (H_3BO_3) is false?
(1) It acts as a weak monobasic acid
(2) It is soluble in hot water
(3) It has a planar structure
(4) It acts as a tribasic acid

502. Which of the following oxides is acidic in nature?
(1) N_2O_3 (2) Al_2O_3
(3) Ga_2O_3 (4) ZnO

503. Catenation i.e., linking of similar atoms depends on size and electronic configuration of atoms. The tendency of catenation in group 14 elements follows the order
(1) C > Si > Ge > Sn (2) C > > Si > Ge ≈ Sn
(3) Si > C > Sn > Ge (4) Ge > Sn > Si > C

504. Cement, the important building material is a mixture of oxides of several elements. Besides calcium, iron and sulphur, oxides of elements of which of the group(s) are present in the mixture?
(1) Group 2
(2) Groups 2, 13 and 14
(3) Groups 2 and 13
(4) Groups 2 and 14

505. The possible oxidation state of Tl are:
(1) + 1 and +2 (2) +2 and +3
(3) +1 and – 1 (4) + 1 and + 3

506. Nitrogen gas is liberated by thermal decomposition of :
(1) NH_4NO_2 (2) NaN_3
(3) $(NH_4)_2Cr_2O_7$ (4) All

507. $BX_3 + NH_3 \xrightarrow{R.T.} BX_3 \cdot NH_3 + \text{Heat}$ of adduct formation (ΔH), ΔH is maximum for (Numerical value)
(1) BF_3 (2) BCl_3 (3) BBr_3 (4) BI_3

508. Which of the following oxyacid contains both P–H and P–P bond simultaneously?
(1) $H_4P_2O_5$ (2) $H_4P_2O_7$
(3) $H_4P_2O_6$ (4) None

509. $Na_2B_4O_7 \cdot 10H_2O \xrightarrow{Heat} X + NaBO_2 + H_2O$,
$X + Cr_2O_3 \xrightarrow{Heat} \underset{\text{(Green coloured)}}{Y}$ X and Y are :
(1) Na_3BO_3 and $Cr(BO_2)_3$
(2) $Na_2B_4O_7$ and $Cr(BO_2)_3$
(3) B_2O_3 and $Cr(BO_2)_3$
(4) B_2O_3 and $CrBO_3$

d and f-block ELEMENTS

510. Which of the following change is possible in Aqueous Medium-
(1) $2\ Cu^{+} \longrightarrow Cu^{2+} + Cu^{0}$
(2) $2Cu^{0} \xrightarrow[-3^{-}]{} Cu^{+} + Cu^{+2}$
(3) $2Cu^{+2} \xrightarrow[+3e^{-}]{} Cu^{0} + Cu^{+}$
(4) $Cu^{+2} + e^{-} \longrightarrow Cu^{+}$

511. Possible oxidation states for titanium can be
(1) +2, +3 & +4 (2) Only +3
(3) +2 & +3 (4) Only +2

512. Colourless ion is :-
(1) Cr^{+3} (2) Sc^{+3}
(3) Cu^{+2} (4) CrO_4^{2-}

513. Ferric Ion + Mono valent anion (element having atomic Number = 53) $\longrightarrow$ x + y
x & y are-
(1) Fe^{0} & Iodate ion
(2) Fe^{+2} & Per iodate ion
(3) Fe^{2+} & I_2
(4) Fe^{+2} & FeI_3

514. Pr, Nd, Tb & Dy show +4 oxidation state in the form of-
(1) Chromates & Halides
(2) Mangnate & Chromate
(3) Both of the above
(4) Oxide

515. Electronic configurations related to Praseodymium, Neodymium & Promethium are respectively-
(1) $4f^{3}6s^{2}$, $4f^{4}6s^{2}$ & $4f^{6}6s^{2}$
(2) $4f^{7}6s^{2}$, $4f^{13}6s^{2}$ & $4f^{6}6s^{2}$
(3) $4f^{3}6s^{2}$, $4f^{4}6s^{2}$ & $4f^{5}6s^{2}$
(4) $4f^{14}5d^{1}6s^{2}$, $4f^{14}6s^{2}$ & $4f^{6}6s^{2}$

516. Which products are formed (respectively) by reaction of lanthanoids with hydrogen oxide, burns in O_2 and heated with sulphur
(1) $Ln(OH)_3$, Ln_2O_3, Ln_2S_3
(2) Ln. xH_2O, Ln. O_2 & heterocyclic sulphides
(3) Ln_2O_3 , $Ln(OH)_3$ & LnS
(4) Macrocyclic ligands containing OH^{-} ions, LnO & Homo cyclic sulphides

517. Catalyst related to polymerisation of monomers having two carbon atoms having one double bond
(1) V_2O_5 + Asbestos + $TiCl_4$
(2) Zeolite + Feldspar
(3) $TiCl_4$ + Tri-methyl aluminium
(4) MnO_2 + $KMnO_4$ + $PdCl_2$

518. Which of the following arrangements does not represent the correct order of the property stated against it?
(1) $V^{2+} < Cr^{2+} < Mn^{2+} < Fe^{2+}$: paramagnetic behaviour
(2) $Ni^{2+} < Co^{2+} < Fe^{2+} < Mn^{2+}$: ionic size
(3) $Co^{3+} < Fe^{3+} < Cr^{3+} < Sc^{3+}$: stability in aqueous solution
(4) Sc < Ti < Cr < Mn : number of oxidation states

519. The yellow colour solution of Na_2CrO_4 changes to orange red on passing CO_2 gas due to the formation of :-
(1) CrO_5 (2) CrO_3
(3) $Na_2Cr_2O_7$ (4) Cr_2O_3

520. What is incorrect about the reactions of $KMnO_4$ and oxalic acid
(1) CO_2 is formed
(2) decolourisation is fast in begining but become slow after some time
(3) Mn^{2+} is autocatalyst
(4) It is a redox change

521. Which statement is correct
(1) Most common oxidation state of lanthanoid is +2
(2) HCl can be used to acidify $KMnO_4$ during redox reaction
(3) In presence of CO_2, orange dichromate solution changes to yellow chromate
(4) To separate Fe_2O_3 and Al_2O_3, NaOH can be used

522. On addition of small amount of $KMnO_4$ to concentrated H_2SO_4, a green oily compound is obtained which is highly explosive in nature the compound is

(1) Mn_2O_7 (2) MnO_2
(3) $MnSO_4$ (4) Mn_2O_3

523. $CrO_4^{2-} \underset{pH=Y}{\overset{pH=X}{\rightleftharpoons}} Cr_2O_7^{-2}$

The pH values of (X) and (Y) are respectively

(1) 4 and 5 (2) 4 and 8
(3) 8 and 4 (4) 8 and 9

524. Which is correct :-

Process	Result
(1) $\left(Ti(H_2O)_6\right)Cl_3$ is heated	Colour is intensified
(2) $CuSO_4$ reacts with KCN	$K_2\left[Cu(CN)_4\right]$ is formed
(3) $KMnO_4$ reacts with HF	MnF_6 is formed
(4) K_2MnO_4 reacts with CO_2	$KMnO_4$ (purple) and MnO_2 (brown) is formed

525. $KMnO_4$ is a strong oxidizing agent in acidic medium. To provide acid medium dil. H_2SO_4 is used instead of HCl. This is because

(1) H_2SO_4 is a stronger acid than HCl
(2) HCl is oxidized by $KMnO_4$ to Cl_2
(3) H_2SO_4 is a dibasic acid
(4) rate is faster in the presence of H_2SO_4

526. In which of the following oxoanions the oxidation state of central atom is not same as that of its group number in periodic table?

(1) MnO_4^- (2) $Cr_2O_7^{-2}$ (3) VO_4^{-3} (4) FeO_4^{-2}

527. Interstitial compounds are formed when small atoms are trapped inside the crystal lattice of metals. Which of the following is not the characteristic property of interstitial compounds?

(1) They have high melting points in comparison to pure metals.
(2) They are very hard
(3) They retain metallic conductivity
(4) They are chemically very reactive

528. When $KMnO_4$ acts as an oxidizing agent and ultimately forms $[MnO_4]^{-2}$, MnO_2, Mn_2O_3, and Mn^{+2}, then the number of electrons transferred in each case, respectively, is:

(1) 4, 3, 1 and 5 (2) 1, 5, 3 and 7
(3) 1, 3, 4 and 5 (4) 3, 5, 7 and 1

529. What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid?

(1) $Cr_2O_7^{2-}$ and H_2O are formed
(2) CrO_7^{2-} is reduced to +3 state of Cr
(3) CrO_7^{2-} is oxidized to +7 state of Cr
(4) Cr^{3+} and $Cr_2O_7^{2-}$ are formed

530. Heating Cu_2O and Cu_2S will give

(1) Cu_2SO_3 (2) CuO + CuS
(3) Cu + SO_3 (4) Cu + SO_2

531. Which of the following statement is not correct?

(1) Copper liberates hydrogen from acid
(2) In its higher oxidation states, Mn form stable compounds with oxygen and fluorine.
(3) Mn^{+3} and Co^{+3} are oxidising agent in aquous solution.
(4) Ti^{+2} and Cr^{+3} are reducing agents in aquous solution.

532. Four Successive members of the first row transition elements are listed below with atomic number. Which one of them is expected to have the highest $E^0_{M^{3+}/M^{2+}}$ Value?

(1) Cr(Z = 24) (2) Mn(Z = 25)
(3) Fe(Z = 26) (4) Co(Z = 27)

533. Chloro compound of Vanadium has only spin magnetic moment of 1.73 BM. This Vanadium chloride has the formula : (at, no. of V = 23)

(1) VCl_4 (2) VCl_3
(3) VCl_2 (4) VCl_5

534. Which of the following is not formed when H_2S reacts with acidic $K_2Cr_2O_7$ solution?

(1) K_2SO_4 (2) $Cr_2(SO_4)_3$
(3) S (4) $CrSO_4$

535. Ammonium dichromate is used in some fireworks. The green-colored powder blown in the air is:
(1) CrO_3 (2) Cr_2O_3
(3) Cr (4) $CrO(O_2)$

536. In the dichromate dianion:
(1) Four Cr — O bonds are equivalent
(2) Six Cr — O bonds are equivalent
(3) All Cr — O bonds are equivalent
(4) All Cr — O bonds are non-equivalent

537. On heating ammonium dichromate, the gas evolved is:
(1) Oxygen (2) Ammonia
(3) Nitrous oxide (4) Nitrogen

538. When MnO_2 is fused with KOH, a colored compound is formed, the product and its color are:
(1) K_2MnO_4, green
(2) $KMnO_4$, purple
(3) Mn_2O_3,brown
(4) Mn_3O_4, black

539. $(NH_4)_2\ Cr_2O_7$ on heating gives a gas which is also given by:
(1) Heating NH_4NO_2
(2) Heating NH_4NO_3
(3) $Mg_3N_2 + H_2O$
(4) Na(compound)+ H_2O_2

540. Which of the following statements is/are not correct, when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with concentrated H_2SO_4?
(1) Deep red vapors are evolved
(2) The vapors when passed into NaOH solution give a yellow solution of Na_2CrO_4
(3) Chlorine gas is evolved
(4) Chromyl chloride is formed

541. In aqueous solution, Eu^{2+} ion acts as:
(1) An oxidizing agent
(2) An reducing agent
(3) Either (1) or (2)
(4) None

542. The actinoids showing +7 oxidation state are:
(1) U,Np (2) Pu, Am
(3) Np, Pu (4) Am, Cm

543. Among the lanthanoids the one obtained by synthetic method is:
(1) Lu (2) Pm
(3) Pr (4) Gd

544. Across the lanthanoid series, the basicity of the lanthanoid hydroxides:
(1) Increases
(2) Decreases
(3) First increases and then decreases
(4) First decreases and then increases

545. The maximum oxidation state exhibited by actinoid elements is:
(1) +5 (2) +4 (3) +7 (4) +8

546. Lanthanoids for which +II and +III oxidation states are common is:
(1) La (2) Nd
(3) Ce (4) Eu

547. Transition elements form binary compounds with halogens. Which of the following elements will form MF_3 type
(1) Zn (2) Co
(3) Cu (4) Hg

548. Which of the following is incorrectly match?

Catalyst	**Process**
(1) V_2O_5	Contact process
(2) $CuCl_2$	deacon process
(3) Finely divided Fe	Vegetable oil to ghee
(4) $TiCl_4 + Al(CH_3)_3$	Zieglar Natta Catalyst

549. K_2MnO_4 can be converted into $KMnO_4$ by :
(1) Passing CO_2 gas
(2) by passing Cl_2
(3) Electrolytic oxidation
(4) All of these

550. Which of the following is not correctly matched?
(1) Fenton reagent $FeSO_4 + H_2O_2$
(2) Adam's catalyst PtO_2
(3) Wilkinson catalyst $[RhCl(PPh_3)_3]$
(4) Zeiglar natta catalyst - $[(C_2H_5)_3B + TiCl_3$

551. Incorrect match is:
(1) Bauxite – $Al_2O_3 \cdot X\ H_2O$
(2) Philospher wool – ZnO
(3) Malachite – Green pigment
(4) None

552. Which of the following is an interstitial carbide?
(1) SiC (2) WC (3) CaC_2 (4) Be_2C

553. Highest fluoride and highest oxide of Mn are respectively:
(1) MnF_7 , Mn_2O_7 (2) MnF_6,MnO_3
(3) MnF_4, Mn_2O_7 (4) MnF_7, MnO_2

554. Which of the following does not contain transition metal?
(1) Haemoglobin (2) Vitamin B_{12}
(3) cis platine (4) Chlorophyll

555. Which of the following does not give disproportionation reaction in acidic or basic medium?
(1) MnO_4^{-2} (2) HNO_2
(3) CrO_4^{-2} (4) Cl_2

556. Acidified $KMnO_4$ act as oxidising agent. It is acidified by:
(1) Conc. H_2SO_4 (2) dil HCl
(3) Conc. HNO_3 (4) dil.H_2SO_4

557. Which of the following compounds does not exist?
(1) FeF_3 (2) BiF_5 (3) CuI_2 (4) CoF_3

METALLURGY

558. In which of the following metallurgy, self reduction is not possible
(1) $ZnS \rightarrow Zn$ (2) $PbS \rightarrow Pb$
(3) $Cu_2S \rightarrow Cu$ (4) $HgS \rightarrow Hg$

559. Which reaction(s) occurs during calcination

(a) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

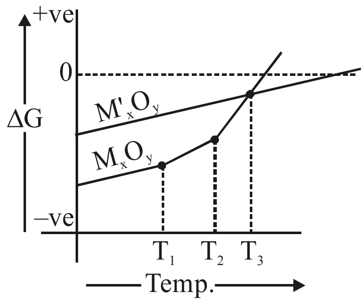
(b) $2FeS_2 + \frac{11}{2}O_2 \xrightarrow{\Delta} Fe_2O_3 + 4SO_2$

(c) $Al_2O_3.x\ H_2O \xrightarrow{\Delta} Al_2O_3 + x\ H_2O$

(d) $ZnS + \frac{3}{2}O_2 \xrightarrow{\Delta} ZnO + SO_2$

correct option are
(1) a and b (2) b and c
(3) a and c (4) b and d

560. Which of the following statement is **incorrect**. On the basis of following structure.



(1) T_1 and T_2 are m.p. and b.p. of metal M respectively
(2) Above T_3 M' is more reducing than M
(3) M_xO_y and M'_xO_y are unstable at $\Delta G > 0$
(4) Below T_3 M is less reducing than M'

561. During the extraction of Ag and Au using a KCN solution and Zn, cyanide ions and Zn react with metal ion as respectively
(1) a reducing Agent, an axidising Agent
(2) a complexing Agent, a reducing Agent
(3) an oxidising Agent, a complexing Agent
(4) a reducing Agent, a complexing Agent

562. Match the column

Column-I	**Column-II**
(a) Zone refining	(P) Ge, Si, Ga
(b) Mond process	(Q) Cu
(c) Van arkel method	(R) Zr, Ti
(d) Electrolytic refining	(S) Ni

(1) (a) P (b) S (c) R (d) Q
(2) (a) S (b) Q (c) P (d) R
(3) (a) R (b) Q (c) P (d) S
(4) (a) Q (b) R (c) P (d) S

563. Match the column-

Column – I	Column – II
(a) Copper pyrites	(P) $ZnCO_3$
(b) Malachite	(Q) $CuCO_3.Cu(OH)_2$
(c) Calamine	(R) $CuFeS_2$
(d) Sphalerite	(S) ZnS

(1) (a) Q (b) R (c) S (d) P
(2) (a) R (b) Q (c) P (d) S
(3) (a) P (b) Q (c) R (d) S
(4) (a) S (b) P (c) R (d) Q

564. Which of the following is incorrectly matched:-
(1) Electrolytic Reduction – Extraction of Al
(2) Cyanide process – Reduction of Pb
(3) Leaching – Extraction of Ag
(4) Zone refining – Ultra pure Ge

565. Sheelite ($CaWO_4$) is an ore of tungsten which contain tungestate ion, Tungestate ion is also present in
(1) Limonite (2) Dolomite
(3) Wolframite (4) Siderite

566. Which of the following pair is incorrectly matched
(1) Kroll's process – Titanium
(2) Froth floatation – Cerussite
(3) Distillation – Zinc
(4) Depressants – NaCN

567. Which of the following metals cannot be extracted by carbon reduction process?
(1) Pb (2) Al
(3) Hg (4) Zn

568. The general formula of bauxite ore is
$$AlO_x(OH)_{3-2x}$$
The select the correct option –
(1) $O \le x < 1$ (2) $O \le x \le 1$
(3) $O < x < 1$ (4) $O < x \le 2$

569. Correct match is

Purification by	Method
(1) Zr	Polling
(2) Zn	Van Arkel
(3) Ni	Distillation
(4) Ge	Zone refining

570. Which one of the following is not used as a refining method during metallurgy –
(1) Leaching (2) Liquation
(3) Distillation (4) Vapour phase

571. Thermite is a mixture of
(1) Zn + Mg (2) Fe + Al
(3) Fe_2O_3 + Al (4) Cu + Mg

572. Sulphide ore is
(1) copper pyrites (2) malachite
(3) haematite (4) magnesite

573. Which of the following term is not related to Al extraction
(1) Serpeck's process
(2) Hall - Heroult process
(3) Thermite process
(4) Hoop's process

574. Which of the following metal is leached by cyanide process
(1) Ag (2) Na
(3) Al (4) Cu

575. Which of the following is concentrated by froth-floatation method?
(1) cassiterite (2) magnetite
(3) malachite (4) galena

576.

List-I	List-II
(a) Cyanide process	(P) Ultra pure 'Ge'
(b) Froth floatation process	(Q) Pine oil
(c) Electrolytic reduction	(R) extraction of Al
(d) Zone refining	(S) extraction of Au

	a	b	c	d
(1)	R	P	S	Q
(2)	S	Q	R	P
(3)	R	Q	S	P
(4)	S	P	R	Q

577. The substance used as froth stablisers in froth-floatation process is :
(1) Copper sulphate
(2) Aniline
(3) Sodium cyanide
(4) Potassium ethyl xanthate

578. In the froth floatation process, for the benefaction of ores, the ore particles float because:
(1) They are light
(2) Their surface is not easily wetted by water
(3) They bear electrostatic charge
(4) They are insoluble

579. Cassiterite is an ore of:
(1) Mn (2) Ni
(3) Sb (4) Sn

580. Pyrolusite is a/an:
(1) Oxide ore (2) Sulphide ore
(3) Carbide ore (4) Not an ore

581. Which of the following metal is not extracted by electrolysis?
(1) Na (2) Mg (3) Al (4) Fe

582. Aluminium is extracted by the electrolysis of:
(1) Bauxite
(2) Alumina
(3) Alumina mixed with molten cryolite
(4) Molten cryolite

583. The function of flux during the smelting of the ore is:
(1) To make the ore porous
(2) To remove gangue
(3) To facilitate reduction
(4) To facilitate oxidation

584. Complex formation method is used for the extraction of :
(1) Zn (2) Ag (3) Hg (4) Cu

585. In alumino-thermite process, aluminium is used as:
(1) Oxidizing agent
(2) Reducing agent
(3) Dehydrating agent
(4) Complex formation agent

586. Generally self-reduction of the sulphide ore takes place during:
(1) Roasting (2) Smelting
(3) Calcination (4) Cupellation

587. Purest form of iron is:
(1) Cast iron (2) Wrought iron
(3) Pig iron (4) None of these

588. In the extraction of nickel by Mond's process, the metal is obtained by:
(1) Electrochemical reduction
(2) Thermal decomposition
(3) Chemical reduction by aluminium
(4) Reduction by carbon

589. Calcination is the process of heating the ore:
(1) In inert gas
(2) In the presence of air
(3) In the absence of air or limited supply of air
(4) In the presence of CaO and MgO

590. The slag obtained during the extraction of copper from copper pyrites is composed of:
(1) Cu_2S (2) $CuSiO_3$
(3) $FeSiO_3$ (4) SiO_2

591. Zone-refining has been employed for preparing ultrapure samples of:
(1) Cu (2) Zn (3) Ge (4) Ag

592. Which method of purification is represented by the equations:

$$\underset{\text{(Impure)}}{Ti} \xrightarrow{500K} TiI_4 \xrightarrow{1675K} \underset{\text{(Pure)}}{Ti} + 2I_2$$

(1) Cupellation (2) Poling
(3) Van Arkel (4) Zone refining

593. When copper ore is mixed with silica in a reverberatory furnace, copper matte produced is?
(1) Sulphides of copper (II) and iron (II)
(2) Sulphides of copper (II) and iron (III)
(3) Sulphides of copper (I) and iron (II)
(4) Sulphides of copper (II) and iron (I)

594. Which of the following reactions is an example of autoreduction?
(1) $Fe_3O_4 + 4CO \rightarrow 3Fe + 4CO_2$
(2) $Cu_2O + C \rightarrow 2Cu + CO$
(3) $Cu^{2+} (aq) + Fe(s) \rightarrow Cu(s) + Fe^{2}$
(4) $Cu_2O + \frac{1}{2} Cu_2S \rightarrow 3Cu + \frac{1}{2} SO_2$

595. Zone refining is based on the principle that_____.
(1) Impurities of low boiling metals can be separated by distillation
(2) Impurities are more soluble in molten metal than in solid metal
(3) Different components of a mixture are differently adosrbed on an adosrbent
(4) Vapours of volatile compound can be decomposed in pure metal

596. Electrolytic refining is used to purify which of the following metals?
(1) Cu and Zn (2) Ge and Si
(3) Zr and Ti (4) Zn and Na

597. Which one of the following ores is best concentrated by froth-flotation method?
(1) Galena (2) Cassiterite
(3) Magnetite (4) Malachite

INORGANIC CHEMISTRY ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**2**	**1**	**3**	**1**	**1**	**1**	**3**	**3**	**3**	**2**	**3**	**4**	**3**	**4**	**4**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**4**	**1**	**2**	**4**	**3**	**2**	**2**	**2**	**4**	**3**	**2**	**2**	**3**	**2**	**2**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**2**	**4**	**3**	**3**	**2**	**1**	**2**	**3**	**2**	**2**	**2**	**4**	**4**	**4**	**2**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**4**	**1**	**1**	**3**	**3**	**4**	**3**	**1**	**1**	**2**	**4**	**1**	**4**	**4**	**3**
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	**1**	**2**	**4**	**1**	**3**	**3**	**3**	**1**	**3**	**3**	**2**	**3**	**1**	**2**	**3**
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	**3**	**3**	**3**	**1**	**2**	**4**	**2**	**1**	**3**	**1**	**1**	**1**	**4**	**4**	**4**
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	**3**	**1**	**2**	**3**	**2**	**1**	**2**	**3**	**2**	**1**	**1**	**1**	**3**	**2**	**4**
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	**4**	**3**	**4**	**2**	**3**	**3**	**4**	**2**	**1**	**4**	**1**	**4**	**4**	**1**	**3**
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	**2**	**3**	**4**	**4**	**2**	**3**	**2**	**2**	**3**	**2**	**3**	**3**	**2**	**1**	**3**
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	**4**	**3**	**3**	**2**	**4**	**1**	**2**	**4**	**1**	**1**	**2**	**4**	**4**	**2**	**2**
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	**2**	**4**	**2**	**1**	**4**	**4**	**1**	**2**	**3**	**3**	**3**	**4**	**4**	**3**	**4**
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	**2**	**3**	**1**	**4**	**2**	**3**	**3**	**3**	**4**	**1**	**3**	**1**	**1**	**3**	**3**
Que.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
Ans.	**2**	**2**	**3**	**2**	**3**	**3**	**1**	**2**	**4**	**2**	**1**	**4**	**3**	**3**	**2**
Que.	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210
Ans.	**4**	**1**	**2**	**3**	**1**	**2**	**3**	**2**	**3**	**4**	**4**	**1**	**3**	**1**	**1**
Que.	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225
Ans.	**2**	**3**	**3**	**1**	**3**	**1**	**4**	**1**	**2**	**2**	**1**	**1**	**4**	**3**	**1**
Que.	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240
Ans.	**2**	**2**	**1**	**1**	**2**	**2**	**3**	**1**	**2**	**3**	**1**	**4**	**2**	**1**	**4**
Que.	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255
Ans.	**1**	**3**	**4**	**1**	**3**	**4**	**2**	**4**	**1**	**3**	**2**	**2**	**4**	**2**	**4**
Que.	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270
Ans.	**3**	**1**	**4**	**1**	**4**	**3**	**2**	**2**	**4**	**1**	**2**	**4**	**4**	**3**	**2**
Que.	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285
Ans.	**1**	**4**	**4**	**2**	**1**	**4**	**1**	**3**	**4**	**2**	**3**	**1**	**2**	**4**	**2**
Que.	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300
Ans.	**3**	**2**	**1**	**4**	**1**	**1**	**3**	**4**	**2**	**4**	**2**	**1**	**2**	**3**	**4**
Que.	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315
Ans.	**3**	**4**	**2**	**4**	**1**	**1**	**2**	**2**	**4**	**3**	**4**	**4**	**3**	**4**	**1**
Que.	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330
Ans.	**4**	**2**	**2**	**3**	**2**	**4**	**3**	**2**	**2**	**2**	**3**	**3**	**2**	**1**	**2**

Que.	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345
Ans.	**3**	**4**	**1**	**4**	**1**	**4**	**2**	**1**	**2**	**1**	**4**	**3**	**4**	**3**	**1**
Que.	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360
Ans.	**2**	**4**	**1**	**3**	**4**	**4**	**4**	**1**	**3**	**2**	**2**	**2**	**3**	**2**	**2**
Que.	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375
Ans.	**3**	**1**	**2**	**4**	**3**	**2**	**1**	**2**	**1**	**1**	**4**	**1**	**3**	**2**	**4**
Que.	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390
Ans.	**1**	**4**	**4**	**4**	**1**	**4**	**2**	**2**	**2**	**4**	**3**	**3**	**1**	**3**	**2**
Que.	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405
Ans.	**4**	**3**	**3**	**3**	**3**	**3**	**1**	**2**	**4**	**2**	**1**	**1**	**2**	**4**	**4**
Que.	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420
Ans.	**2**	**4**	**4**	**4**	**4**	**1**	**4**	**3**	**2**	**1**	**3**	**2**	**3**	**3**	**2**
Que.	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435
Ans.	**1**	**1**	**3**	**1**	**1**	**3**	**1**	**4**	**1**	**3**	**1**	**1**	**3**	**3**	**3**
Que.	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450
Ans.	**1**	**4**	**1**	**3**	**4**	**2**	**3**	**1**	**4**	**2**	**4**	**4**	**1**	**1**	**2**
Que.	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465
Ans.	**2**	**3**	**3**	**4**	**3**	**4**	**2**	**1**	**4**	**4**	**1**	**2**	**4**	**1**	**4**
Que.	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480
Ans.	**3**	**3**	**2**	**1**	**1**	**2**	**1**	**4**	**2**	**3**	**3**	**2**	**3**	**2**	**1**
Que.	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495
Ans.	**1**	**1**	**4**	**1**	**1**	**1**	**3**	**1**	**1**	**3**	**4**	**1**	**1**	**4**	**1**
Que.	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510
Ans.	**4**	**3**	**2**	**1**	**2**	**4**	**1**	**2**	**2**	**4**	**4**	**4**	**4**	**3**	**1**
Que.	511	512	513	514	515	516	517	518	519	520	521	522	523	524	525
Ans.	**1**	**2**	**3**	**4**	**3**	**1**	**3**	**1**	**3**	**2**	**4**	**1**	**2**	**4**	**2**
Que.	526	527	528	529	530	531	532	533	534	535	536	537	538	539	540
Ans.	**4**	**4**	**3**	**1**	**4**	**1**	**4**	**1**	**4**	**2**	**2**	**4**	**1**	**1**	**3**
Que.	541	542	543	544	545	546	547	548	549	550	551	552	553	554	555
Ans.	**2**	**3**	**2**	**2**	**3**	**4**	**2**	**3**	**4**	**4**	**4**	**2**	**3**	**4**	**3**
Que.	556	557	558	559	560	561	562	563	564	565	566	567	568	569	570
Ans.	**4**	**3**	**1**	**3**	**4**	**2**	**1**	**2**	**2**	**3**	**2**	**2**	**3**	**4**	**1**
Que.	571	572	573	574	575	576	577	578	579	580	581	582	583	584	585
Ans.	**3**	**1**	**3**	**1**	**4**	**2**	**2**	**2**	**4**	**1**	**4**	**3**	**2**	**2**	**2**
Que.	586	587	588	589	590	591	592	593	594	595	596	597			
Ans.	**1**	**2**	**2**	**3**	**3**	**3**	**3**	**3**	**4**	**2**	**1**	**1**			

INORGANIC CHEMISTRY SOLUTIONS

14. Radius order

$B < \underbrace{Al \simeq Ga}_{\text{due to Transition Contraction}} < \underbrace{In \simeq T\ell}_{\text{due to Lanthanide Contraction}}$

20. $Cr = 4s^1 5d^5$

no. of U.P. $e^- = 6$

42. $Pd \rightarrow [Kr]5s^o\ 4d^{10}$

Very important exception of periodic table.

43. $[Rn]\ 6d^2\ 7s^2$

$\downarrow \quad \downarrow \quad \downarrow$

$86 + 2 + 2 \rightarrow 90$

f-block.

45. Learn the configuration of Lu^{+3}, Gd^{+3}, Ho^{+3} etc very important for f-block chapter also.

52. Consider electron negativity

56. $O \longrightarrow O^-\ O \longrightarrow O^{-2}$

after gaining e^- on further addition of e^-, repulsion will create. So process is endothermic.

57. +ve charge↑ Ionic radius↓ (For same atoms and iso electronic species)

60. **Dobernier Triad :** Generally all 3 elements belongs to same group & there at no. difference must be same.

62. Consider H is small.

65. Diagonal Relationship

66. Ca^{+2}, $x = 3s^2\ 3p^3 = P = x^{-3}$

$Ca^{+2} P^{-3} \rightarrow Ca_3P_2$

72. 3^{rd} Group IP $3d > 4d > 5d$.

4, 5, 6, 10 (IP) $3d < 4d < 5d$

Remaing (IP) $5d > 3d > 4d$.

73. Radius of Al can'nt be predicted with ions because it is database.

76. $Fe \simeq Co \simeq Ni \leq Cu$ (Atomic Radius)

77. $S_2O_8^{-2}$

$\downarrow \quad \downarrow \quad \downarrow$

$12 + 48 + 2 = 62$

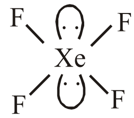
80. +3 +5

SbF_3 SbF_5

⊕ Charge ↑ EN ↑

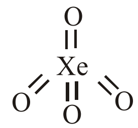
87. XeF_4 $= 4\sigma + 2\ell p$

= square planar



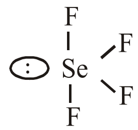
$\mu = 0$ Non polar and planar

97. XeO_4



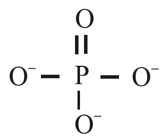
Tetrahedral

SeF_4



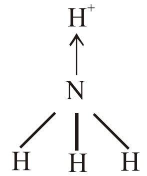
see saw shape

PO_4^{-3}



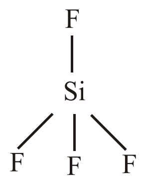
Tetrahedral

NH_4^+

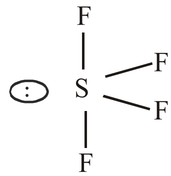


Tetrahedral

105.

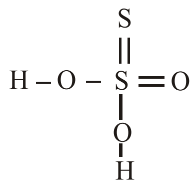


Tetrahedral



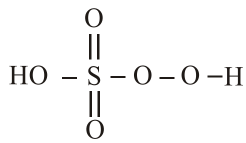
see saw

113. $H_2S_2O_3$



6σ & 2π

H_2SO_5



7σ & 2π

169. ion-dipole attraction (H_2O has more polarity than HCl)

174. due to absence of vacant orbital

175. CH_4 has zero dipole moment

176. O is more electronegative than N

178. due to smallest size

179. Na_2CO_3 does not decompose on heating

181. SO_2 can acts as Reducing agent

182. Due to high lattice energy

186. C_2H_4 has two type of bond, one is C= C & another is C – H.

192. Hybrid orbitals are more directional than unhybrid orbitals (Pure orbitals)

195. ICl_3 hyb. sp^3d with two lp. hence 'T' shape molecule

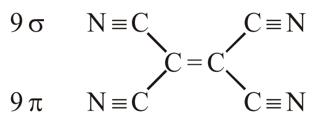
I_2Cl_6 (Dimer of ICl_3)

hyb. sp^3d^2 with two lp. hence sq. planar molecule.

199. Oxygen cannot extend their covalency due to absence of vacant orbital.

202. Polarisability α size of anion

203. $C_2(CN)_4$



204. BrO_3^- & XeO_3 both have pyramids shape.

209. Because difference in boiling point.

210. v.w.f

222. HF has higher boiling point so least volatile.

223. due to triple bond

248. ionic mobility $\alpha \dfrac{1}{\text{size of hydrated ion}}$

250. $Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_3 + 2\ NH_3$

253. Hypo($Na_2S_2O_3$) is used in photography flim to remove undecomposed AgBr

255. Electro positive charactor $\alpha \dfrac{1}{\text{I.P.}}$

256. $NaNO_3$

261. $A \rightarrow NaHCO_3$

$B \rightarrow ZnCO_3$

$C \rightarrow ZnO$

$D \rightarrow CO_2$

$E \rightarrow Na_2ZnO_2$

267. $\underbrace{H_2\overset{-1}{O_2} + PbS \longrightarrow PbSO_4 + H_2\overset{-2}{O}}_{\text{reduction}}$

270. Zn, Al & Be are amphoteric, hence react with NaOH to release H_2 whereas Mg does not.

279. Hydrolith, salt like Binary compound formula is (CaH_2)

used as Reducing agent and source of hydrogen

280. Dielectric constant of heavy water is less than simple water.

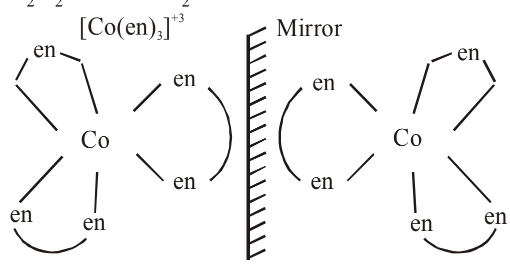
283. Cation size increase, then ionic character also increases.

285. $BaO_2 + H_2SO_4 \rightarrow H_2O_2 + BaSO_4$

286. $2I^- \longrightarrow I_2$

$H_2O_2^{-1} \longrightarrow H_2O^{-2}$

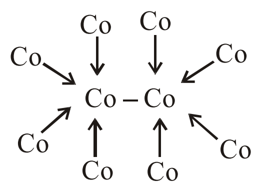
295.



$[Co(en)_3]^{+3}$ show optical isomer

304. $[Cu(NH_3)_4]SO_4$

305.



Co = Z → 27

Number of co-ordinate bond → 4

Number of delta bond → 1

So value of 'n' in $(Co)_n – Co – Co – (Co)_n = 4$

342. Apply

EAN = Z – O·S + 2 × Coord·no·

Cr charge → –1, O, +1

345. $[\underset{+2}{Ca}(EDTA)]^{-2}$

$\therefore$ $(EDTA)^{-4} \rightarrow$ Hexadentate.

346. $K_3[Co(CO_3)_3]$

Charge on Co → +3

CO_3^{-2} weak field ligand (No pairing.)

349. $[V(CO)_6]^-$ stable, EAN = 36

350. $[Pt(O_2)(en)_2\ Br]^{+2}$

$x - 1 + O - 1 = +2$

$x = +4$

354. CFSE $\propto$ SFL

357. $[Fe(CO)_4]^{-2} \rightarrow Fe^{-2} \rightarrow 3d^6$

386. $Fe^{+3} \rightarrow 3d^5$

$C_2O_4^{-2} \rightarrow$ weak field ligand

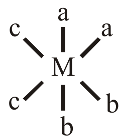
No Pairing

$\therefore\ sp^3d^2$

388. 3d series L $\rightarrow$ R, Zeff $\uparrow$, CFSE $\uparrow$

393. Consider strength of ligand.

395. Option C, $[M\ a_2b_2c_2]$



397. Splitting energy $\uparrow$ Absorbed wave length $\downarrow$ Obseved wave length

398. $Na_2S + Na_2[Fe(CN)_5\ NO] \rightarrow Na_4\ [Fe(CN)_5NOS]$

405. Maximum 4 hydrogen can be replaced

413. A $\rightarrow MnO_2$, B $\rightarrow Cl_2$, C $\rightarrow NCl_3$

428. In N_2O_5 nitrogen present in highest O.S.(+5)

452. $XeF_6\ (s) \rightarrow \underset{\text{(Cationic part)}}{XeF_5^{\oplus}} + F^-$

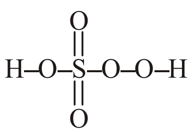
454. $6XeF_4 + 12\ H_2O \rightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$

$XeF_6 + 3H_2O \rightarrow XeO_3 + 6HF$

455. Hyb. of I in $I_2\ Cl_6$ is sp^3d^2 with 2 lp that's why square planar geometry

456. due to absence of vacant orbital

463. Caro's acid (H_2SO_5)



465. On moving top to bottom stability of lower oxidation state increases.

469. In Ostwald process NH_3 convert into NO.

$$\underset{\text{(From air)}}{4NH_3+5O_2} \xrightarrow[\text{500K, 9 bar}]{\text{Pt/Rh}} 4NO + 6H_2O$$

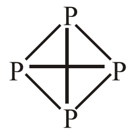
471. $\underset{\text{(Nitrous acid)}}{2HNO_2} \xrightarrow{-H_2O} N_2O_3$

474. $Ca_3P_2 + H_2O \rightarrow Ca(OH)_2 + PH_3$

$Ca_3C_2 + H_2O \rightarrow Ca(OH)_2 + C_2H_4$

$Mg_3N_2 + H_2O \rightarrow Mg(OH)_2 + NH_3$

475.



476. Chromyl chloride test

477. F^- cannot be oxidised by Cl_2

(F_2 is stronger O.A. than Cl_2)

481. Diamond has single C–C bond.

483. Vander wall force of SbH_3 dominates over H–Bonding of NH_3,SbH_3 has high molecular mass

484. In SO_2 sulphur present in +4 O.S.

485. Oxidising nature order $F_2 > Cl_2 > Br_2 > I_2$

486. AgF is soluble in water

AgCl $\rightarrow$ White ppt

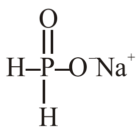
AgBr $\rightarrow$ Pale yellow ppt

AgI $\rightarrow$ Yellow ppt

487. In silicates hyb. of Si is sp^3, that's why silicates are non-planar

488. AlH_3 accept H^- ion from LiH

489. H_3PO_2 is monobasic acid, contains only one acidic hydrogen, so NaH_2PO_2 does not have any acidic hydrogen.

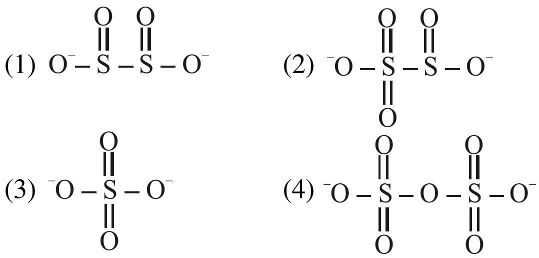


490. stability order white phosphorous < Red phosphorous < Black phosphorous

491. $3Br_2 + 6NaOH$(Hot & conc.)

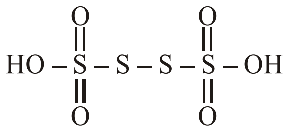
$\rightarrow 5NaBr + NaBrO_3 + 3H_2O$

495. $(NH_4)_2Cr_2O_7$ & NH_4NO_2 give N_2 gas on heating.

496. 

498. H_2SnO_6

for eg n = 4 ; $H_2S_4O_6$



Number of S $\Rightarrow$ 4

Number of S–S bond $\Rightarrow$ 3

$\therefore$ Number of S–S $\Rightarrow$ n–1

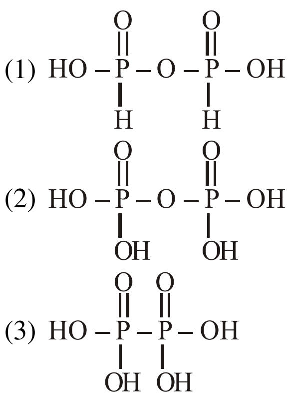
499. I^{+3} & $3IO_3^-$ (I_4O_9)

503. Catenation α Bond energy

507. Lewis acidic nature order

$BF_3 < BCl_3 < BBr_3 < BI_3$

due to back bonding BF_3 has least lewis acidic character.

508. 

520. $2MnO_4^- + 5H_2C_2O_4 + 6H^+ \rightarrow 2Mn^+ + 10CO_2 + 8H_2O$

529. $\underset{\text{(orange)}}{Cr_2O_7^{-2}} \underset{H^+}{\overset{OH^-}{\rightleftharpoons}} \underset{\text{(yellow)}}{CrO_4^{-2}}$

530. Self reduction process

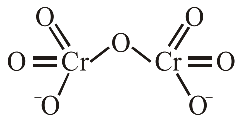
531. Cu + dil $H_2SO_4 \longrightarrow$ No reaction

Cu + con. $H_2SO_4 \longrightarrow CuSO_4 + H_2O + SO_2$

532. Value based

533. V^{+4} ion has one unpaired electron

536. $Cr_2O_7^{-2}$



540. Chromyl chloride test

541. Common & stable O.S. for lanthanoids is +3

555. In CrO_4^{2-}, Cr present in +6 oxidation state which is highest for Cr

562. Zone Refining – Ge, Si, Ga

Mond Process – Ni

Van arkel Method – Zr, Ti

Electrolytic refining – Cu

566. Kroll's Process – Ti

Froth Floatation – Cerussite ($PbCO_3$) not a sulphide ore

Distillation – Zn

Dipressants – NaCN

578. Because :

Surface of ore particles wetted by oil.

Which is water repellent.

579. Cassiterite $\rightarrow SnO_2$

580. Pyrolusite ore $\rightarrow MnO_2$

581. Fe is less reactive metals, It may be extracted by reduction with carbon or carbon monoxide (Blast furnace)

582. $\underset{\text{Alumina}}{Al_2O_3} + \underset{\text{Cryolite}}{\underline{Na_3AlF_6}} + \underset{\text{Flouspar}}{CaF_2}$

585. Main equation

$Cr_2O_3 + 2Al \longrightarrow 2Cr + Al_2O_3$

588. $Ni + 4CO \xrightarrow[\text{process}]{\text{Mond}} [Ni(CO)_4]$

$\xrightarrow[\text{Thermal decomposition}]{200°} Ni + CO$

590. $\underset{\boxed{\text{Copper Pyrite}}}{CuFeS_2} + O_2 \longrightarrow \underset{\text{Molten matte}}{(Cu_2S + FeS)} + SO_2$

$\downarrow$ transfered to Bessemer converter lined with MgO or SiO_2

In Bessemer, the Air oxidises Fes to FeO, which combine with silica forming a slag.

(1) $2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$

(2) $FeO + SiO_2 \longrightarrow FeSiO_3$ Slag.

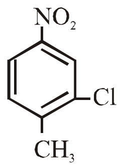
597. Galena is sulphide ore of lead. formula : PbS: That's why froth floatation is used.

IMPORTANT NOTES

ORGANIC CHEMISTRY

NOMENCLATURE

1. IUPAC Name of following compound is:



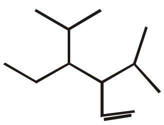
(1) 4-Methyl-5-Chloro-nitrobenzene
(2) 2-Chloro-1-methyl-4-nitrobenzene
(3) 1-Chloro-2-methyl-5-nitrobenzene
(4) 1-Methyl-2-chloro-4-nitrobenzene

2. IUPAC Name for following compound

$CH_2{=}CH{-}CH_2{-}\underset{|}{\underset{OH}{CH}}{-}C{\equiv}CH$

(1) Hex-1-en-5-yn-4-ol
(2) Hex-5-en-1-yn-3-ol
(3) 4-Hydroxy hex-1-en-5-yne
(4) 3-Hydroxy hex-5-en-1-yne

3. IUPAC Name for given comp:-



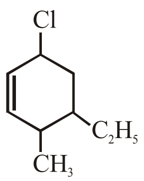
(1) 3-Ethyl-4-isopropylhexa-1-ene
(2) 3, 4-Diisopropylhexa-1-ene
(3) 4-Ethyl-3-isopropyl-5-methylhex-1-ene
(4) 3-Ethyl-4-isopropyl-2-methylhex-5-ene

4. IUPAC name for given compound is :-

$CH_2{-}COOH$
|
$CH{-}OH$
|
$CH{-}COOH$
|
$CH_2{-}COOH$

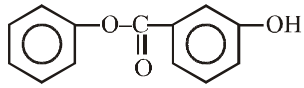
(1) 3-Hydroxybutane-1, 3, 5-trioic acid
(2) 3-Hydroxy-4-carboxy hexan-1, 6- dioic acid
(3) 2-Hydroxybutane-1,3,4-tricarboxylic acid
(4) 3-Hydroxybutane-1,2,4-tricarboxylic acid

5. IUPAC Name of:



(1) 3-Chloro-5-Ethyl-5-methylcyclohex-1-ene
(2) 2-Chloro-4-Ethyl-5-methylcyclohexene
(3) 6-Chloro-4-Ethyl-3-methylcyclohex-1-ene
(4) 5-Chloro-3-Ethyl-2-methylcyclohex-1-ene

6. IUPAC Name of:

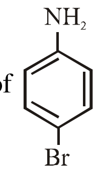


(1) Phenyl m-hydroxybenzoate
(2) 3-Phenoxycarbonylphenol
(3) Phenyl 3-hydroxybenzoate
(4) None of these

7. Which of the following is incorrectly matched:

	Common name	***IUPAC name***
(1)	Crotonaldehyde	: But-2-enal
(2)	Cinnamic acid	: 3-Phenylprop-2-enoic acid
(3)	Lactic acid	: 2-Hydroxypropanoic acid
(4)	Acrylonitrile	: But-2-enenitrile

8. IUPAC name of $(CCl_3)_3CCl$ is-
(1) tris - trichloromethyl-1-chloromethane
(2) 2-Chloro-2-(trischloromethyl) propane
(3) 1,1,1,2,3,3,3-Heptachloro-2-trichloromethylpropane
(4) 1, 1, 1-trichloromethyl-1-chloromethane

9. IUPAC Name of  :

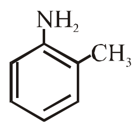
(1) p-Bromoaniline
(2) 4-Bromobenzenamine
(3) 4-Bromoaniline
(4) Both (2) & (3)

10. IUPAC Name of $C_6H_5COC_2H_5$
(1) Ethylphenylmethanone
(2) 1-Phenylpropan-1-one
(3) 3-Phenylpropanone
(4) None of these

11. Incorrectly matched with their common names

(1) $CH_3{-}\underset{|}{\underset{Br}{CH}}{-}CH_2{-}\underset{||}{\underset{O}{C}}{-}H$ β-Bromobutyraldehyde

(2) $CH_3{-}CH_2{-}\underset{|}{\underset{CH_3}{CH}}{-}\underset{||}{\underset{O}{C}}{-}OH$ isovaleric acid

(3)  o-Toluidine

(4) $NH_2{-}CH_2{-}CH{=}CH_2$ Allylamine

12. IUPAC Name for:
$(CH_3)_2C(C_2H_5)_2$
(1) Diethyldimenthylmethane
(2) 2-Ethyl-2-methylbutane
(3) 3,3-Dimethylpentane
(4) None of these

13. IUPAC Name of:

$CH_2=CH-\underset{\underset{CH_2}{\|}}{C}-CH_2-C\equiv CH$

(1) 2-Propynylbuta-1, 2-diene
(2) 4-Methylenehex-5-en-1-yne
(3) 3-Methylenehex-1-en-5-yne
(4) 2-Vinylpent-1-en-4-yne

14. IUPAC Name of the given compound

$CH_3-\underset{\underset{O}{\|}}{C}-CH_2-\underset{\underset{CN}{|}}{CH}-CH_2-CH_3$

(1) 4-Cyano hex-2-one
(2) 2-Ethyl-4-oxopentanenitrile
(3) 2-Oxo-4-cyanohexane
(4) 2-Oxo-hexane-4-nitrile

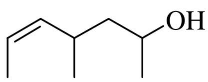
15. IUPAC Name for given compound:

$CH_3-\underset{\underset{CH_3}{|}}{CH}-\underset{\underset{O}{\|}}{C}-\underset{\underset{O}{\|}}{C}-O-CH_2-Cl$

(1) Chloromethyl 2-oxo-3-methylbutanoate
(2) 1-Chloromethoxy-3-methyl butan-1,2-dione
(3) 2-Methyl-3-oxo-1-chloromethylbutanoate
(4) None of these

16. IUPAC name of the following compound is
$CH_3-CH=CH-CH_2-C\equiv CH$
(1) Hex–2-en–5-yne (2) Hex-5-yn-2-ene
(3) Hex-4-en-1-yne (4) Hex-1-yn-4-ene

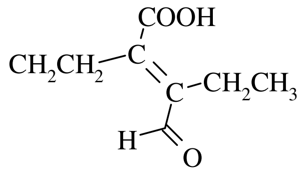
17. IUPAC name of given compound is



(1) 1,3-Dimethylhex-4-ene-1-ol
(2) 4,6-Dimethylhex-2-en-6-ol
(3) 4-Methylhept-2-en-6-ol
(4) 4-Methylhept-5-en-2-ol

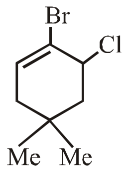
18. Which of the following IUPAC name is incorrect
(1) 3-Bromo-2-chloropentane
(2) Pent-2-en-4-yne
(3) 3-Bromo-butan-2-ol
(4) 1-Aminopentane-2-thiol

19. Number of carbon in parent carbon chain in the following compound



(1) 4 (2) 5 (3) 6 (4) 3

20. The correct IUPAC name of following compound is :

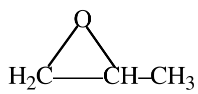


(1) 2-Bromo-3-chloro-5,5-dimethylcyclohex-1-ene
(2) 1-Bromo-6-chloro-4,4- dimethylcyclohex-1-ene
(3) 4-Bromo-5-chloro-1,1-dimethylcyclohex-1-ene
(4) 1-Bromo-2-chloro-4,4-dimenthylcyclohex-1-ene

21. $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-\underset{\underset{CH_3}{|}}{CH}-$

Radical has IUPAC name–
(1) 4-methyl pentyl
(2) 1,3-Dimethyl butyl
(3) 2,4-Dimethyl butyl
(4) 3-methyl pentyl

22. The IUPAC name for the compound



is :-

(1) Propylene oxide
(2) 1,2-Oxopropane
(3) 1,2-Epoxypropane
(4) 1,2-Propoxide

23. One among the following is the correct IUPAC name for the compound

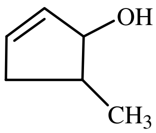
$CH_3CH_2-\overset{\overset{H}{|}}{N}-\overset{\overset{H}{|}}{C}=O$

(1) N-Formylaminoethane
(2) N-Ethylformylamine
(3) N-Ethylmethanamide
(4) Ethylaminomethanal

24. The correct IUPAC name of $HOOC-\underset{\underset{COOH}{|}}{CH}-COOH$ is

(1) Tricarboxymethane
(2) Propanetrioic acid
(3) Tributanoic acid
(4) Methanetricarboxylic acid

25. The IUPAC name of the compound



is

(1) 4-Methylcyclopent-1-en-2-ol
(2) 2-Methylcyclopent-4-en-1-ol
(3) 3-Methylcylopent-1-en-2-ol
(4) 5-Methylcyclopent-2-en-1-ol

26. The correct IUPAC name of the

$H-\overset{\overset{O}{||}}{C}-\overset{\overset{O}{||}}{C}-\overset{\overset{O}{||}}{C}-OH$ is

(1) 2-oxopropanoic acid
(2) 2,3-Dioxopropanoic acid
(3) 1-Hydroxy propane-1, 2, 3-trione
(4) 2-Aldo-2-Keto methanoic acid

ISOMERISM

27. How many cyclic structural isomers of C_5H_{10} are possible.

(1) 5 (2) 3 (3) 4 (4) 6

28. Among these chain isomers are:

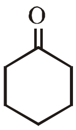
(I) $CH_3-CH_2-CH_2-CH_2-OH$

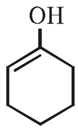
(II) $CH_3-CH_2-\underset{\underset{OH}{|}}{CH}-CH_3$

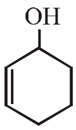
(III) $CH_3-\overset{\overset{OH}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3$

(1) I only (2) I and III
(3) I and II (4) None of these

29. Tautomer of I is

(I) 

(II) 

(III) 

(1) II (2) III
(3) Both II and III (4) None of these

30. Number of structural isomers possible from molecular formula C_3H_6O?

(1) 2 (2) 5 (3) 9 (4) 10

31. Number of structural isomers possible from molecular formula C_4H_6 is

(1) 7 (2) 8 (3) 9 (4) 10

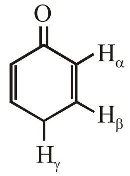
32. The number of structural isomers possible from molecular formula C_3H_9N?

(1) 2 (2) 3 (3) 4 (4) 5

33. The number of esters possible from molecular formula $C_4H_8O_2$ is:

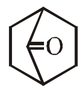
(1) 5 (2) 6 (3) 8 (4) 4

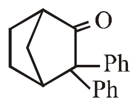
34. Which of the following hydrogen involved in enolisation of given molecule.

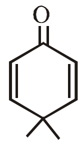


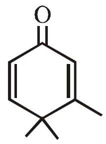
(1) H_α (2) H_β
(3) H_γ (4) Cannot be enolized

35. Which of the following can exhibit tautomerism:

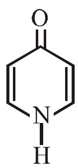
(1) 

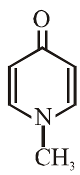
(2) 

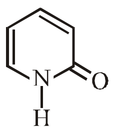
(3) 

(4) 

36. Which among these can exhibit tautomerism?

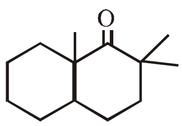
(a) 

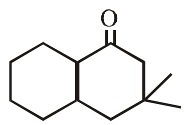
(b) 

(c) 

(1) a only (2) b only
(3) c only (4) a and c

37. Which of following can exhibit tautomerism?

(a) 

(b) 

(1) a only (2) b only
(3) a & b both (4) None of these

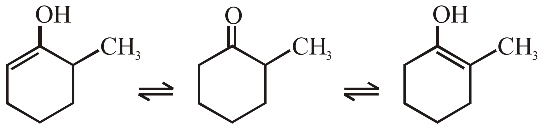
38. Which tautomer isomer is more stable :

$$\underset{(I)}{CH_3-\overset{\overset{O}{\|}}{C}-H} \rightleftharpoons \underset{(II)}{CH_2=C\begin{smallmatrix}OH\\H\end{smallmatrix}}$$

(1) I (2) II

(3) I = II (4) None of these

39. Stability order of these tautomer is:-



(I) (II) (III)

(1) I > II > III (2) III > II > I

(3) II > I > III (4) II > III > I

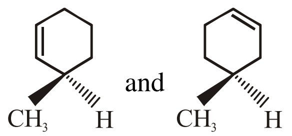
40. Stability order among these tautomer is:

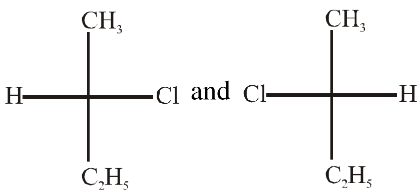
$$\underset{(I)}{CH_2=\overset{\overset{OH}{|}}{C}-CH_2-CH_3} \rightleftharpoons \underset{(II)}{CH_3-\overset{\overset{O}{\|}}{C}-CH_2-CH_3} \rightleftharpoons \underset{(III)}{CH_3-\overset{\overset{OH}{|}}{C}=CH-CH_3}$$

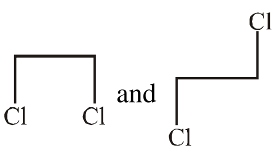
(1) I > II > III (2) III > II > I

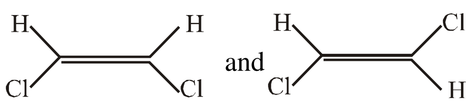
(3) II > I > III (4) II > III > I

41. Which of the following pairs of isomers represents a pair of conformational isomers?

(1) 

(2) 

(3) 

(4) 

42. Geometrical isomers are:

(1) Structural isomers

(2) Conformational isomers

(3) Configurational isomers

(4) None of these

43. Which of the following compound will exhibit geometrical isomerism?

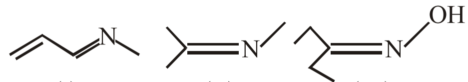
(1) $CH_2 = CH - CH = CH_2$

(2) $CH_2 = CH-CH=CH-CH_3$

(3) $CH_2 = CH-CH_2-CH=CH_2$

(4) $CH_2 = CH - C \equiv C - CH = CH_2$

44. Which of these will exhibit geometrical isomerism?

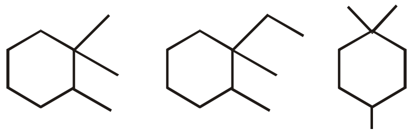


(I) (II) (III)

(1) I (2) II

(3) III (4) I and II

45. Which of these will exhibit geometrical isomerism?

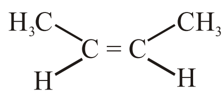


(I) (II) (III)

(1) I (2) II

(3) III (4) All of these

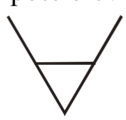
46. This compound can be named as:



(1) Cis –2-butene (2) (Z)–2–butene

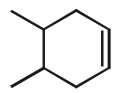
(3) (1) and (2) (4) R–2–butene

47. How many geometrical isomers of this compound are possible?



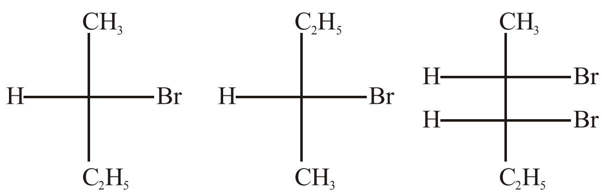
(1) 0 (2) 2 (3) 3 (4) 4

48. How many geometrical isomers of this compound are possible:



(1) 0 (2) 2 (3) 3 (4) 4

49. Which of these molecule is optically active?

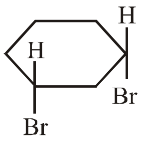


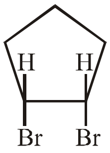
(I) (II) (III)

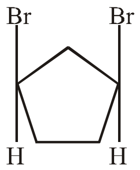
(1) I (2) II

(3) III (4) All of these

50. Which of the following compounds can have non-superimposable mirror images?

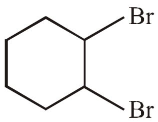
(1) 

(2) 

(3) 

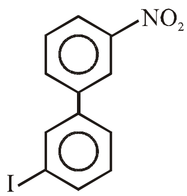
(4) None of the above

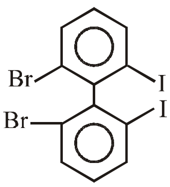
51. How many stereoisomers are possible of this compound?

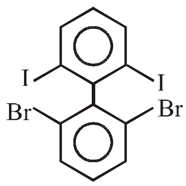


(1) 2 (2) 3 (3) 4 (4) 5

52. Which of the following biphenyl is optical active:

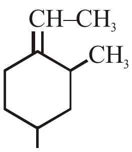
(1) 

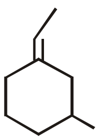
(2) 

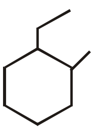
(3) 

(4) None

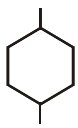
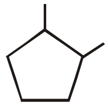
53. Which of the following is optically active

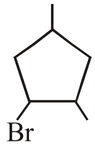
(1) 

(2) 

(3) 
(4) All of these

54. Which of the following can be a meso compound?

(1) 
(2) 

(3) 

(4) All of these

55. Most stable conformation of n-butane is:

(1) Fully eclipsed (2) Anti staggered
(3) Partially eclipsed (4) Gauche

56. Which of the following can show conformational isomerism:

(1) CD_3-H (2) $CH_2(OH)\ CH_2OH$
(3) CH_4 (4) All of these

57. Number of stereoisomers of the given compound is

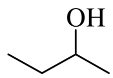
$$CH_3-CH=CH-\underset{\underset{OH}{|}}{CH}-CH_3$$

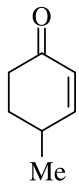
(1) 2 (2) 3 (3) 4 (4) 6

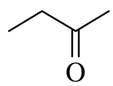
58. How many structures of aldehydes are possible for molecular formula $C_5H_{10}O$?

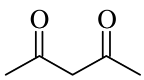
(1) 4 (2) 7 (3) 6 (4) 8

59. Which of the following can show both tautomerism and optical isomerism

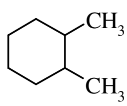
(1) 

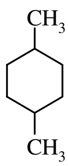
(2) 

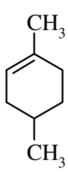
(3) 

(4) 

60. Which of the following compound will not show optical isomerism

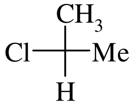
(1) 

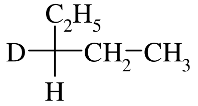
(2) 

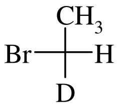
(3) 

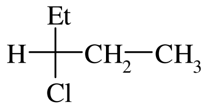
(4) all of these

61. Which of the following molecule has non-superimposable mirror images?

(1) 

(2) 

(3) 

(4) 

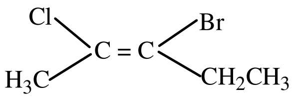
62. What is relation between following compounds

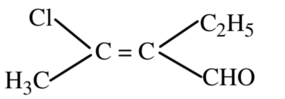
CH_3-CH_2-CH(CN)-CH_2-CHO & CH_3-CH(CN)-CH_2-CH_2-CHO

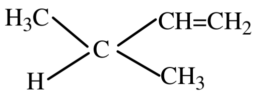
(1) positional isomers (2) chain isomers

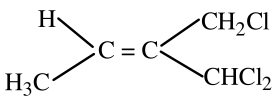
(3) optical isomers (4) metamers

63. The E-isomer from amongst the following is

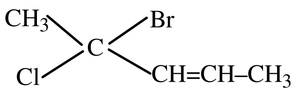
(1) 

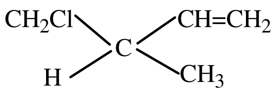
(2) 

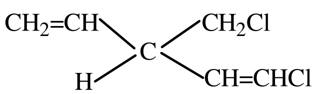
(3) 

(4) 

64. One among the following will not show geometrical isomerism

(1) 

(2) 

(3) 

(4) $CH_3CH = CHCH_2CH_3$

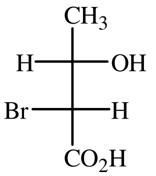
65. The optical inactivity of meso-tartaric acid is because of

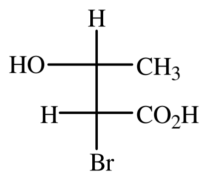
(1) absence of chiral carbon atoms
(2) External compensation
(3) internal compensation
(4) presence of asymmetrical carbon atoms

66. Which of the following will not be able to show optical isomerism?

(1) 1,2-Propadiene (2) 2,3-Pentadiene
(3) sec-Butyl alcohol (4) 2, 3-Butanediol

67. The molecules represented by the following two structures are





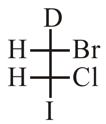
(1) Identical (2) Enantiomers
(3) Diastereomers (4) Epimers

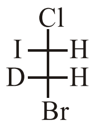
68. Total number of stereoisomer of compound is given below

CH_3–CH=CH–CH(CH_3)–CH=CH–C_2H_5

(1) 2 (2) 4 (3) 6 (4) 8

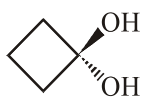
69. The two compounds given below are :

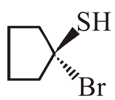


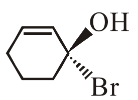


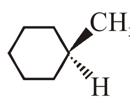
(1) enantiomer (2) identical
(3) meso compound (4) diastereomers

70. Which of the following will be optically active?

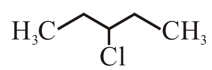
(1) 

(2) 

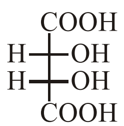
(3) 

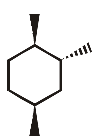
(4) 

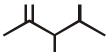
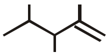
71. Which is optically active ?

(1) 

(2) 

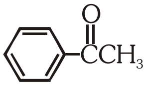
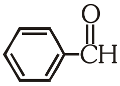
(3) 

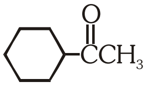
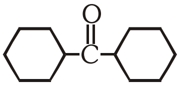
(4) 

72.  and  are :-

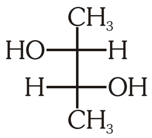
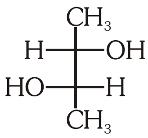
(1) Positional isomer (2) Chain isomer
(3) Identical (4) Metamer

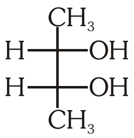
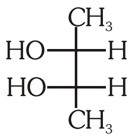
73. Which carbonyl compound forms an oxime does not show *syn-anti* geometrical isomerism?

(1)  (2) 

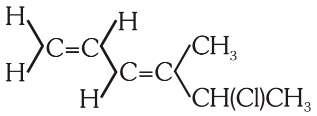
(3)  (4) 

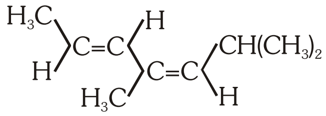
74. (2R, 3R)-2,3-butanediol is :-

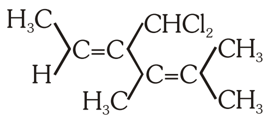
(1)  (2) 

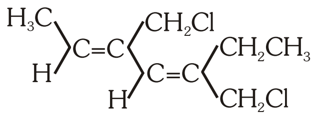
(3)  (4) 

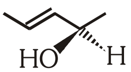
75. One among the following will show optical activity:-

(1) 

(2) 

(3) 

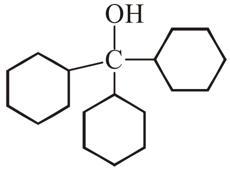
(4) 

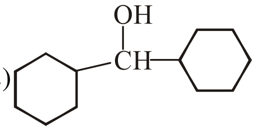
76.  is named as :-

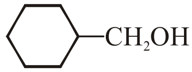
(1) (2R, 3E)-pent-3-en-2-ol
(2) (2S, 3E)-pent-3-en-2-ol
(3) (2E, 3R)-pent-2-en-3-ol
(4) (2E, 3S)-pent-2-en-3-ol

GOC-I

77. Correct order of pKa is :

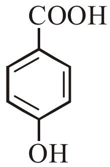
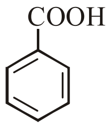
(1) 

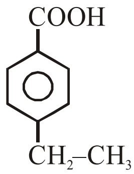
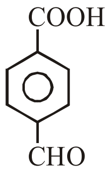
(2) 

(3) 

(1) 3 > 2 > 1 (2) 2 > 3 > 1
(3) 1 > 2 > 3 (4) 1 > 3 > 2

78. Correct order of acidic strength :

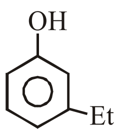
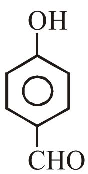
(1)  (2) 

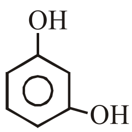
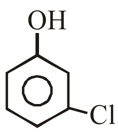
(3)  (4) 

(1) 1 > 2 > 3 > 4 (2) 4 > 3 > 2 > 1
(3) 4 > 2 > 3 > 1 (4) 2 > 4 > 3 > 1

79. K_a increases in benzoic acid if substituent X bonded at para position then X is :

(1) $-OCH_3$ (2) $-CH_3$
(3) $-CN$ (4) $-NH_2$

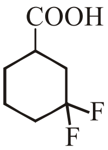
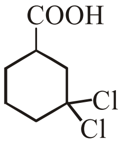
80. (1)  (2) 

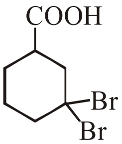
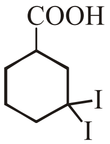
(3)  (4) 

Decreasing order of pKa is :

(1) 1 > 2 > 3 > 4
(2) 1 > 3 > 4 > 2
(3) 2 > 4 > 1 > 3
(4) 4 > 3 > 2 > 1

81. Which of the following present the correct order of acidity in the given compound :

(1)  (2) 

(3)  (4) 

(1) 2 > 1 > 3 > 4 (2) 4 > 3 > 2 > 1
(3) 1 > 2 > 3 > 4 (4) 3 > 2 > 1 > 4

82. The correct order of K_a is :

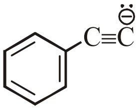
(i) $CH_3–CH_2–COOH$

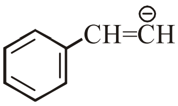
(ii) $OHC–CH_2–COOH$

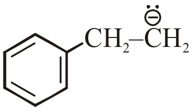
(iii) $Ph–CH_2–COOH$

(1) iii > ii > i (2) i > ii > iii
(3) ii > iii > i (4) i > iii > ii

83. Basic strength order will be :

(1) 

(2) 

(3) 

(1) 1 > 2 > 3 (2) 2 > 1 > 3
(3) 3 > 1 > 2 (4) 3 > 2 > 1

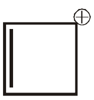
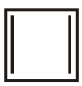
84. Decreasing order of stability of the following carbocation ?

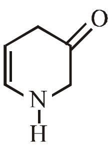
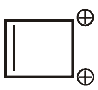
(A) m-(HO)–$C_6H_4–\overset{\oplus}{C}H_2$ (B) p-(HO)–$C_6H_4–\overset{\oplus}{C}H_2$

(C) p-(CHO)-$C_6H_4–\overset{\oplus}{C}H_2$ (D) $C_6H_5–\overset{\oplus}{C}H_2$

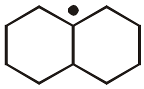
(1) B > A > D > C (2) B > D > A > C
(3) B > D > C > A (4) D > C > B > A

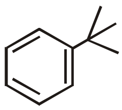
85. Which of the following is Aromatic?

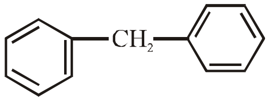
(1)  (2) 

(3)  (4) 

86. Which of the following compound will show Hyperconjugation?

(1) 

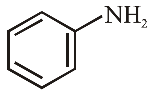
(2) 

(3) 

(1) 2 only (2) 1 and 3 only
(3) 1 and 2 (4) All of these

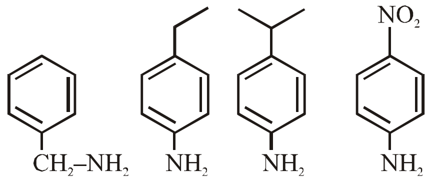
87. Correct order of pK_b is

(i) $CH_3–CH_2–NH_2$ (ii) $Ph–CH_2–NH_2$

(iii) $H–\underset{\underset{O}{\|}}{C}–NH–\underset{\underset{O}{\|}}{C}–H$ (iv) 

(1) i > ii > iii > iv (2) iii > iv > i > ii
(3) iii > iv > ii > i (4) ii > i > iii > iv

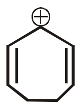
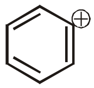
88. Correct Decreasing order of pK_b is:

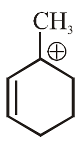
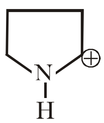


(1) (2) (3) (4)

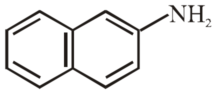
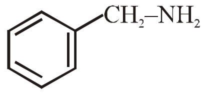
(1) 1 > 2 > 3 > 4 (2) 4 > 3> 2 > 1
(3) 2 > 1 > 3 > 4 (4) 1 > 2 > 4 > 3

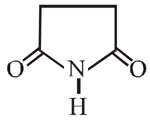
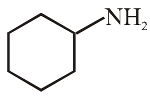
89. Which of the following carbocation is most stable

(1)  (2) 

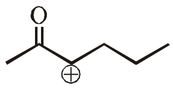
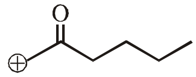
(3)  (4) 

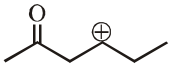
90. Which of the following will be least Basic?

(1)  (2) 

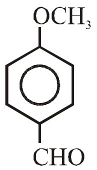
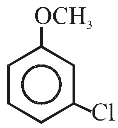
(3)  (4) 

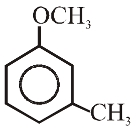
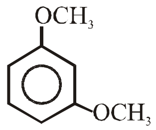
91. Which one of the following Carbocation is most stable:

(1)  (2) 

(3)  (4) $\overset{\oplus}{C}H_2-NO_2$

92. The correct decreasing order of Reactivity for following molecule towards E.S.R.

(1)  (2) 

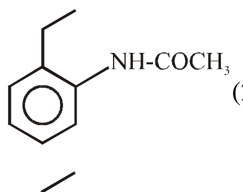
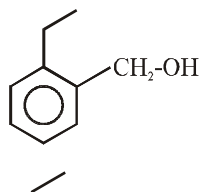
(3)  (4) 

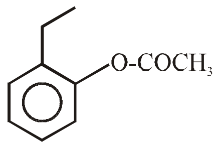
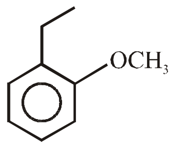
(1) 3 > 4 > 2 > 1 (2) 4 > 3 > 2 > 1
(3) 1 > 2 > 3 > 4 (4) 1 > 3 > 2 > 4

93. Nitrobenzene can be prepared from Benzene by using mixture of conc. HNO_3 and conc. H_2SO_4. In this mixture, H_2SO_4 act as a

(1) Catalyst (2) Reducing Reagent
(3) Acid (4) Base

94. Which one is most Reactive towards electrophilic Reagent:

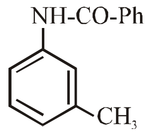
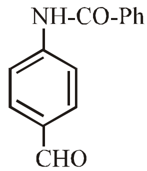
(1)  (2) 

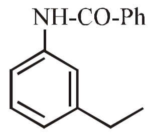
(3)  (4) 

95. Which of the following compound will not give Friedel craft Reaction?

(1) Chlorobenzene (2) Toluene
(3) Benzaldehyde (4) Xylene

96. Correct Reactivity order of E.S.R:

(i)  (ii) 

(iii) 

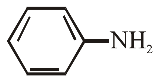
(1) i > ii > iii (2) iii > ii > i
(3) i > iii > ii (4) ii > i > iii

97. Most reactive towards E.S.R:

(1) Benzene (2) Chloro Benzene
(3) Phenyl Acetate (4) Nitro Benzene

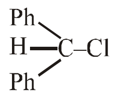
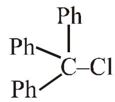
98. Correct Decreasing order of pK_b value of Following Compounds is:

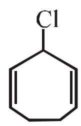
(i) $Et-NH_2$ (ii) $Et-\underset{H}{\underset{|}{N}}-Et$

(iii) $Et-\underset{Et}{\underset{|}{N}}-Et$ (iv) 

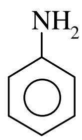
(1) i > ii > iii > iv (2) iv > i > iii > ii
(3) iv > iii > ii > i (4) ii > iii> i > iv

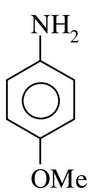
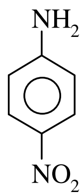
99. In which of the following C-Cl Bond ionisation shall give most stable ion?

(1)  (2) 

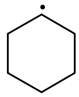
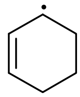
(3)  (4) Ph $-CH_2-Cl$

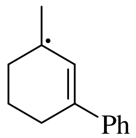
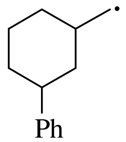
100. Which of following is most basic

(1) CH_3-NH_2 (2) 

(3)  (4) 

101. Which one of the following is most stable

(1)  (2) 

(3)  (4) 

102. Which of the following is insoluble in $NaHCO_3$?

(1) Benzoic acid (2) Benzene sulphonic acid

(3) o-Nitrophenol (4) Picric acid

103. Which is the correct decreasing order of basic strength in aqueous medium

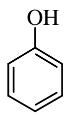
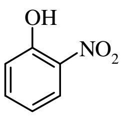
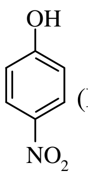
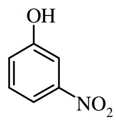
(1) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2 > NH_3$

(2) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

(3) $(CH_3)_2NH > (CH_3)_3N > CH_3NH_2 > NH_3$

(4) $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N > NH_3$

104. Which of the following is correct order of acidic strength

(I)  (II)  (III)  (IV) 

(1) I > II > III > IV (2) III > II > IV > I

(3) II > III > IV > I (4) IV > II > III > I

105. Correct order of basic strength in gas phase is

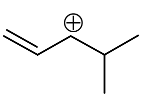
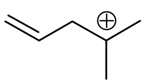
(I) $CH_3–NH_2$ (II) $(CH_3)_2NH$

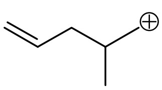
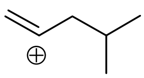
(III) $(CH_3)_3N$ (IV) NH_3

(1) I > II > III > IV (2) II > I > III > IV

(3) III > II > I > IV (4) II > III > I > V

106. Which carbocation is most stable

(1)  (2) 

(3)  (4) 

107. Which of the following group has maximum hyperconjugation effect but minimum inductive effect

(1) $–CH_3$ (2) $–CH_2–CH_3$

(3) $–CD_3$ (4) $-\underset{CH_3}{\overset{CH_3}{C}}-CH_3$

108. What is the correct order of decreasing stability of the following cations

$H_3C-\overset{\oplus}{C}H-CH_3$ (I) $H_3C-\overset{\oplus}{C}H-OMe$ (II)

$H_3C-\overset{\oplus}{C}H-CH_2-OMe$ (III)

(1) II > I > III (2) II > III > I

(3) III > I > II (4) I > II > III

109. Phenol and carboxylic acid can be distinguished by

(1) Na (2) $NaHCO_3$

(3) Litmus test (4) all of these

110. Arrange the following in order of their leaving group tendency?

Br^- (I) $H_3C-\overset{O}{\overset{\|}{C}}-O^-$ (II) $CH_3CH_2O^-$ (III) $Ph-O^-$ (IV)

(1) I > II > III > IV

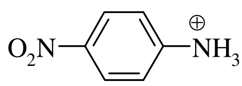
(2) I >III > II > IV

(3) I > II > IV > III

(4) I > IV > III > II

111. Arrange the following in order of their decreasing acidic strength?

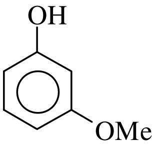
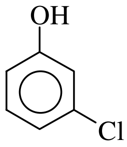
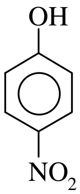
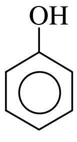
$H_3C-\overset{\oplus}{N}H_3$ (I) $C_6H_5-\overset{\oplus}{N}H_3$ (II)

$H_2N-\overset{\overset{\oplus}{N}H_2}{\overset{\|}{C}}-NH_2$ (III)  (IV)

(1) I > II > III > IV (2) IV > II > I > III

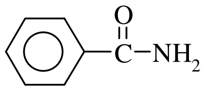
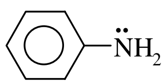
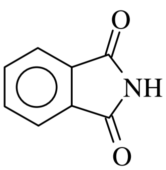
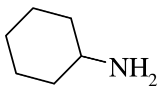
(3) IV > III > II > I (4) IV > I > II > III

112. Arrange the following towards their reactivity for ESR

(I) 
(II) 
(III) 
(IV) 

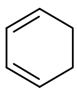
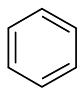
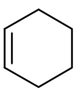
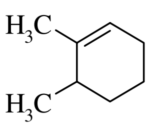
(1) I > IV > II > III (2) I > II > IV > III
(3) I > II > III > IV (4) II > I > III > IV

113. Arrange the following in order of their decreasing basic strength

(I) 
(II) 
(III) 
(IV) 

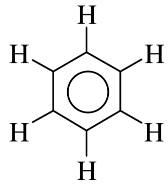
(1) IV > I > II > III (2) IV > I > III > II
(3) IV > II > I > III (4) I > II > III > IV

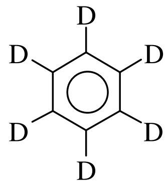
114. Arrange the following in decreasing order of their C=C bond length

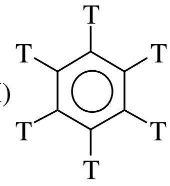
(I) 
(II) 
(III) 
(IV) 

(1) II > IV > I > III (2) II > III > IV > I
(3) II > I > IV > III (4) None

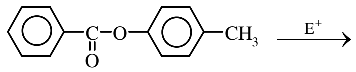
115. Arrange the following in order of their reactivity towards nitration

(I) 
(C_6H_6)

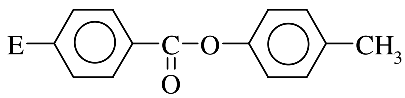
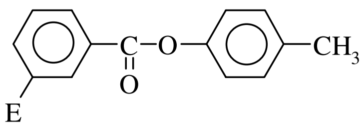
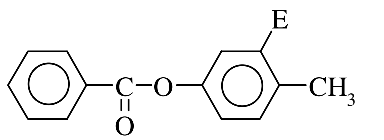
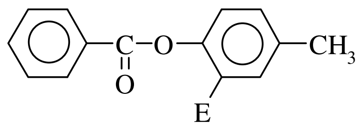
(II) 
(C_6D_6)

(III) 
(C_6T_6)

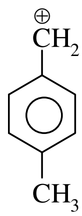
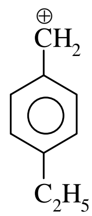
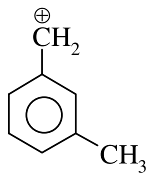
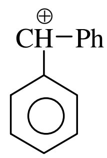
(1) I > III > II (2) I ≈ II ≈ III
(3) I > II > III (4) III > II > I

116.  major product

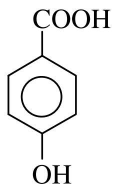
major product will be

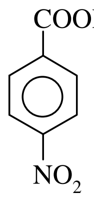
(1) 
(2) 
(3) 
(4) 

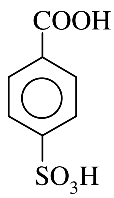
117. Which of the following carbocation is maximum stable

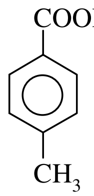
(1) 
(2) 
(3) 
(4) 

118. Which of the following is strongest acid

(1) 

(2) 

(3) 

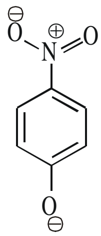
(4) 

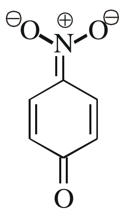
119. Which of the following is not an electrophile

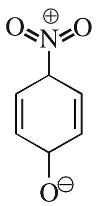
(1) H_3O^+ (2) BH_3

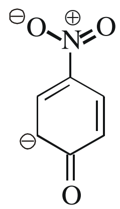
(3) CH_3^+ (4) $ZnCl_2$

120. The most unlikely representation of resonance structure of p-nitrophenoxide is

(1) 

(2) 

(3) 

(4) 

121. Which of the following carbanion is most stable?

(1) $H_3C-\overset{\ominus}{C}H_2$ (2) $H_2C=CH-\overset{\ominus}{C}H_2$

(3) $H_3C-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{C}H_2$ (4) $H_3C-\overset{NH}{\overset{\|}{C}}-\overset{\ominus}{C}H_2$

122. Polarisation of σ-bond caused by polarisation of adjacent σ-bond is referred as–

(1) Mesomeric effect

(2) Electromeric effect

(3) Inductive effect

(4) Hyperconjugation

123. Select the molecule with more polar bond from each of the indicated bonds among given set of molecules:

(a) H_3C-H, H_3C-Br
(I) (II)

(b) H_3C-NH_2, H_3C-OH
(I) (II)

(c) H_3C-Cl, H_3C-F
(I) (II)

(1) I, I, I (2) II, II, II

(3) II, II, I (4) II, I, II

124. Which of the following does not represent a correct set of nucleophile–

(1) $HS^{\ominus}$, $C_2H_5O^{\ominus}$

(2) $(CH_3)_3N$, $\overset{\ominus}{\ddot{N}}H_2$, $\overset{\ominus}{C}N$,

(3) SO_3, H_2O, $\overset{\ominus}{\ddot{C}}H_3$

(4) CH_3-S-CH_3, $\overset{\ominus}{N}O_2$, $\overset{\ominus}{I}$

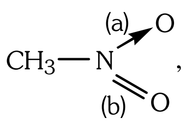
125. Which of the following is correct regarding formation of reactive intermediate in given heterolytic bond cleavage:

(1) $CH_3-S-CH_3 \longrightarrow \overset{\ominus}{\ddot{C}}H_3 + CH_3-\overset{\oplus}{S}$

(2) $CH_3-Cu \longrightarrow \overset{\ominus}{\ddot{C}}H_3 + \overset{\oplus}{Cu}$

(3) $CH_3-Cl \longrightarrow \overset{\ominus}{\ddot{C}}H_3 + \overset{\oplus}{Cl}$

(4) 1 & 2 both

126. In nitromethane  ,

two N – O bond lengths are:

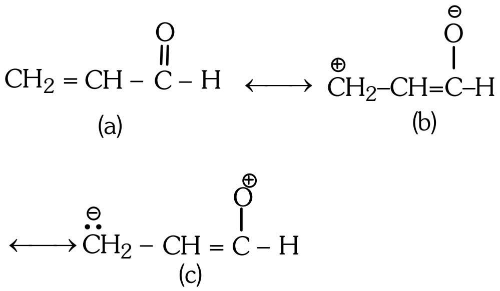
(1) a = b

(2) a > b

(3) b > a

(4) Can not be compared

127. Correct order of stability of the following resonating structure is



(1) a > b > c (2) a > c > b
(3) b > a > c (4) c > b > a

128. Which of the following shows +R effect:

(1) $>C=O$ (2) – X
(3) – COOH (4) – CONHR

129. Which of the following compounds does not show electromeric effect

(1) CH_3– CN (2) CH_3–CHO
(3) $CH_2 = CH_2$ (4) CH_3– Cl

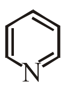
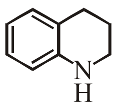
130. Which of the following is not an example of +E effect:

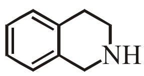
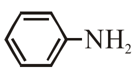
(1) Hydrohalogenation of Propene
(2) Hydroboration of propyne
(3) Formation of acetaldehydecyanohydrine
(4) Hydration of ethene

131. Cyclooctatetraene is not aromatic because

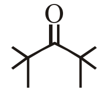
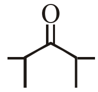
(1) It has four carobn-carbon double bonds
(2) It doesn't have (4n + 2)π electrons (where n = 0, 1, 2, 3, ... etc) and moreover it is non-planar
(3) It is planar in structure
(4) It decolourises potassium permagnate solution

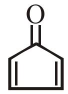
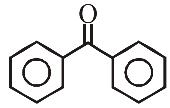
132. Which of the following is most basic ?

(1)  (2) 

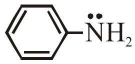
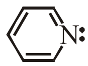
(3)  (4) 

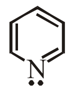
133. Tautomerism is shown by :-

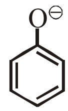
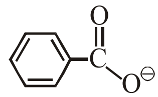
(1)  (2) 

(3)  (4) 

134. Select the correct statement :-

(1)  is more basic than 

(2)  is more basic than 

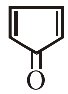
(3)  is more basic than 

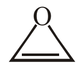
(4) All of them

135. Which of the following is the correct order of stability of free radicals ?

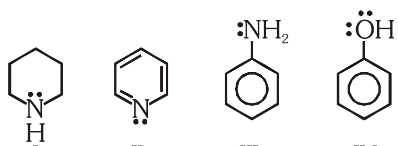
(1) benzyl > allyl > 2° > 1°
(2) allyl > benzyl > 2° > 1°
(3) allyl > 2° > 1° > benzyl
(4) benzyl > 2° > 1° > allyl

136. Which is not anti aromatic ?

(1)  (2) 

(3)  (4) 

137. Consider the following molecules

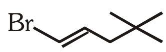


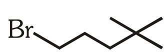
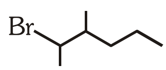
I II III IV

What is the correct order of their basicity ?

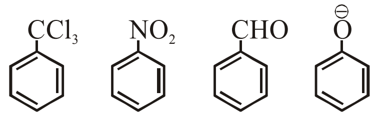
(1) I > II > III > IV (2) IV > II > III > I
(3) I > III > IV > II (4) IV > II > I > III

138. Which one of the following undergoes S_N reaction with NaCN at the slowest rate ?

(1)  (2) 

(3)  (4) 

139. Electrophile $\overset{\oplus}{N}O_2$ attacks the following

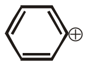


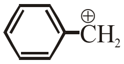
I II III IV

In which cases $\overset{\oplus}{N}O_2$ will be at *meta*-position ?

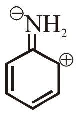
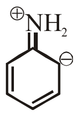
(1) II and IV (2) I, II and III
(3) II and III (4) I only

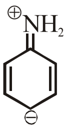
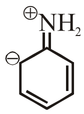
140. Which is maximum stable ?

(1) $CH_3\overset{\oplus}{C}H_2$ (2) 

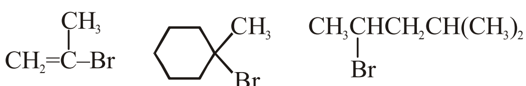
(3)  (4) $(CH_3)_2\overset{\oplus}{C}H$

141. Which is not the resonance structure of aniline?

(1)  (2) 

(3)  (4) 

142. Rank the following in order of decreasing rate of solvolysis with aqueous ethanol (fastest ⟶ slowest)

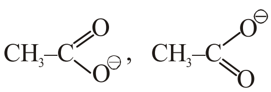


I II III

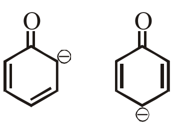
(1) II > I > III (2) I > II > III
(3) II > III > I (4) I > III > II

143. Pairs of related chemical species are given below. Which pair is not related through resonance ?

(1) $CH_3COCH_2COOC_2H_5$, $CH_3C(OH)=CHCOOC_2H_5$

(2) 

(3) 

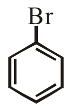
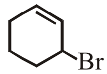
(4) 

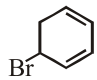
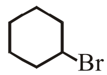
144. Which one of the following has all the effects, namely inductive, mesomeric and hyperconjugative?

(1) CH_3Cl
(2) $CH_3CH=CH_2$
(3) $CH_3CH=CH\overset{\overset{O}{\|}}{C}CH_3$
(4) $CH_2=CH-CH=CH_2$

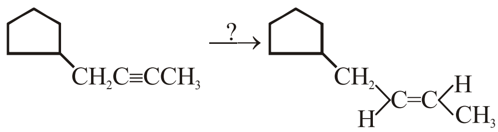
HYDROCARBON

145. Which of the following dehydrohalogenated to a maximum extent ?

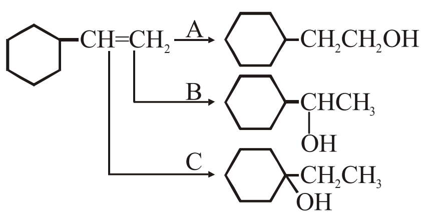
(1)  (2) 

(3)  (4) 

146. Reagent used to carry out following alkyne to alkene is

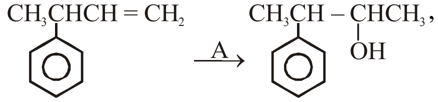


(1) Pd-C/H_2 (2) Na/NH_3
(3) Pd/H_2 (4) Ni/H_2

147. 

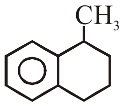
Schemes; A, B and C are :-

(1) simple hydration
(2) hydroboration oxidation, mercuration-demercuration, acidic hydration
(3) acidic hydration, hydroboration oxidation, mercuration-demercuration
(4) mercuration-demercuration, acidic hydration, hydroboration oxidation

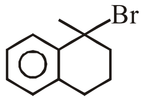
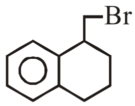
148. 

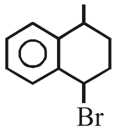
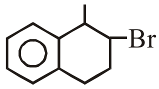
A will be

(1) H_2O/H^+
(2) $BH_3.THF/H_2O_2/OH^-$
(3) $Hg(OCOCH_3)_2.H_2O/NaBH_4$
(4) All are possible

149.  $\xrightarrow{NBS}$ (A) Major

Major product (A) is :-

(1) 
(2) 

(3) 
(4) 

150. Phenyl magnesium bromide reacts with methanol to give :-

(1) A mixture of anisole and Mg(OH)Br

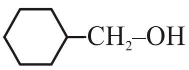
(2) A mixture of benzene and Mg(OMe)Br

(3) A mixture of toluene and Mg(OH)Br

(4) A mixture of phenol and Mg(Me)Br

151. The alcohol which reacts fast with conc. HCl to form alkyl halide at room temperature is :-

(1) CH_3–CH_2–OH
(2) Ph–CH_2–CH_2–OH

(3) Ph–$\underset{CH_3}{\overset{CH_3}{C}}$–OH
(4) 

152. The compound formed as a result of oxidation of ethyl benzene by $KMnO_4$ is

(1) Benzophenone

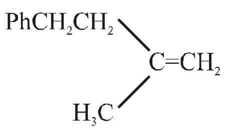
(2) Acetophenone

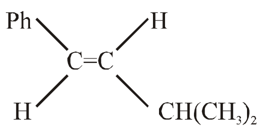
(3) Benzoic acid

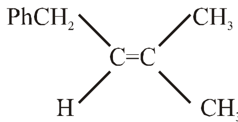
(4) Benzyl alcohol

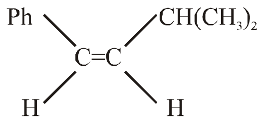
153. The main product of the following reaction is:

$PhCH_2CH(OH)CH(CH_3)_2 \xrightarrow{Conc.\ H_2SO_4/\Delta}$

(1) 

(2) 

(3) 

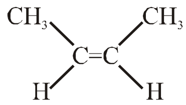
(4) 

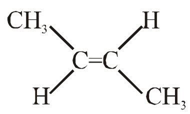
154. CH_3–$\underset{Cl}{CH}$–$\underset{Cl}{CH}$–CH_3 $\xrightarrow[(ii)NaNH2]{(i)Alc.KOH}$ A (major)

$\xrightarrow[Pd/CaCO_3]{H_2}$ B, what is end product (B)

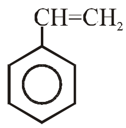
(1) CH_3CH_2–CH=CH_2

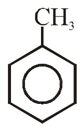
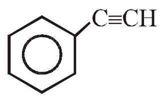
(2) CH_2–CH_2–CH_2–CH_3

(3) 

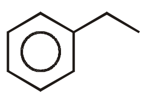
(4) 

155. Which of the following does not decolourise reddish brown solution of Br_2/CCl_4:

(1) 
(2) 

(3) 
(4) 

156.  +CH_3COCl $\xrightarrow{Anhyd.AlCl_3}$ A $\xrightarrow{x}$



Reagent x can be

(1) HI/Red P
(2) Zn-Hg/HCl

(3) $NH_2-NH_2/\bar{O}H$
(4) All of these

157. Which one of the following compounds will give white precipitate with Tollen's reagent?

(1) CH_3–$\underset{CH_3}{CH}$–CH_3
(2) CH_3–CH=CH_2

(3) CH_3–C≡CH
(4) CH_3–C≡C–CH_3

158. Acetylene on heating in the presence of red hot Fe tube gives:
(1) Benzene (2) Cyclooctatetraene
(3) Naphthalene (4) Anthracene

159. Which among the following alkenes will be most stable:
(1) Ethene
(2) 2-Methyl propene
(3) 2,3-Dimethyl-2-butene
(4) 2-butene

160. Which among the following compounds will give wurtz reaction?
(1) $CH_2=CH-NO_2$ (2) C_6H_5-Br
(3) $CH_2=CH-CH_2-Br$ (4) $HC\equiv C-COOH$

161. In the following reaction:

$$C_2H_2 \xrightarrow[HgSO_4/H_2SO_4]{H_2O} (x) \longrightarrow CH_3-CHO$$

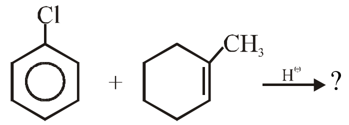
what is (x)
(1) CH_3-CH_2-OH (2) CH_3-O-CH_3
(3) CH_3-CH_2-CHO (4) $CH_2=CH-OH$

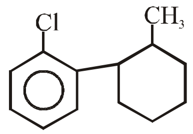
162. Anti-Markownikov's addition of HBr is not observed in:
(1) propene (2) 1-butene
(3) 1-pentene (4) 2-butene

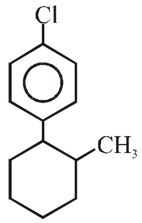
163. Propyne and propene can be distinguished by:
(1) Conc. H_2SO_4
(2) Br_2 in CCl_4
(3) Dil-$KMnO_4$
(4) Ammonical $AgNO_3$

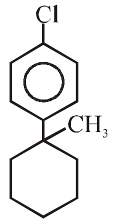
164. Which of these will not react with acetylene:
(1) NaOH
(2) Ammonical $AgNO_3$
(3) Na
(4) HCl

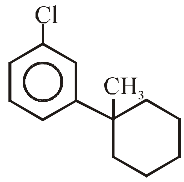
165. What is the product formed when acetylene reacts with hypochlorous acid:
(1) CH_3COCl (2) $ClCH_2CHO$
(3) Cl_2CHCHO (4) $ClCH_2COOH$

166. 

(1) 

(2) 

(3) 

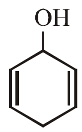
(4) 

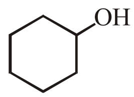
167. $CH_3-C(CH_3)(CH_3)-CH=CH-CH_3 \xrightarrow{H-Cl} ?$

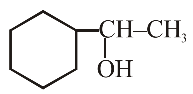
(1) $CH_3-C(CH_3)(CH_3)-CH_2-CH(Cl)-CH_3$
(2) $CH_3-C(Cl)(CH_3)-CH_2-CH(CH_3)-CH_3$
(3) $CH_3-C(CH_3)(CH_3)-CH(Cl)-CH_2-CH_3$
(4) $CH_3-C(Cl)(CH_3)-CH_2-CH(CH_3)-CH_3$

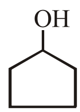
168. Debromination of d-2, 3-dibromobutane gives:
(1) Trans-2-butene (2) Cis-2-butene
(3) 1-butene (4) 2-butyne

169. Which is maximum reactive towards acid catalysed dehydration:

(1) 

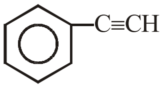
(2) 

(3) 

(4) 

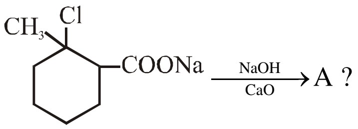
170. Which gives ketonic group after hydroboration oxidation :

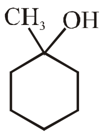
(1) $CH_3–CH_2–C≡CH$

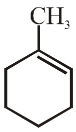
(2) 

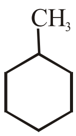
(3) $CH_3–C≡C–CH_3$

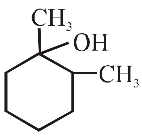
(4) $CH_3–CH(CH_3)–C≡CH$

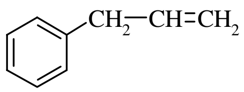
171. 

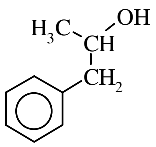
(1) 

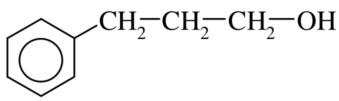
(2) 

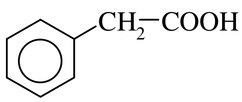
(3) 

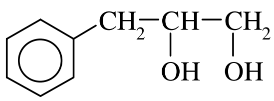
(4) 

172.  on oxymercuration-demercuration produces the major product

(1) 

(2) 

(3) 

(4) 

173. How many chiral compounds are possible on mono-chlorination of 2-Methyl butane?

(1) 6 (2) 8 (3) 2 (4) 4

174. Ozonolysis of an organic compound 'A' produces acetone and propionaldehyde in equimolar mixture. Identify A from the following compounds

(1) 2-methyl-1-pentene

(2) 1-pentene

(3) 2-pentene

(4) 2-Methyl-2-pentene

175. Arrange the following in decreasing order of reactivity towards electrophilic substitution reaction

Aniline (I) Acetanilide (II)

Phenol (III) Anilinium chloride (IV)

(1) II > I > III > IV (2) I > III > II > IV

(3) III > II > IV > I (4) IV > II > I > III

176. Which of the following is most basic

(1) Diphenylamine

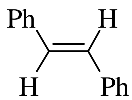
(2) Triphenyl amine

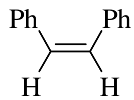
(3) p-Nitroaniline

(4) Benzylamine

177. $Ph–CH=CH–Ph \xrightarrow[CCl_4]{Cl_2} X \xrightarrow{2NaNH_2} Y \xrightarrow[BaSO_4]{H_2+Pd} Z$

Product Z is

(1) 

(2) 

(3) $Ph–C(Ph)=CH_2$

(4) $Ph–C≡C–Ph$

178. Which Intermediate is formed during addition of halogen on alkenes?

(1) Carbocation

(2) Carbanion

(3) cyclic halonium ion

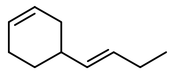
(4) halide ion

179. $CH_3–C≡CH \xrightarrow[dil.H_2SO_4]{HgSO_4}$ product

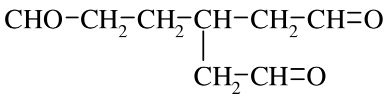
Product of the above reaction is

(1) an aldehyde (2) a ketone

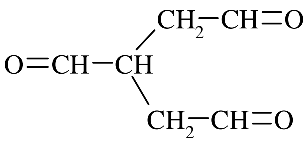
(3) an alcohol (4) a carboxylic acid

180.  $\xrightarrow[\text{(ii)}H_2O/Zn]{\text{(i)}O_3}$ Product; Product is

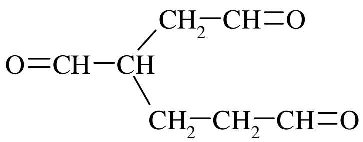
(1) $CH_3–CH_2–CH=O$ and



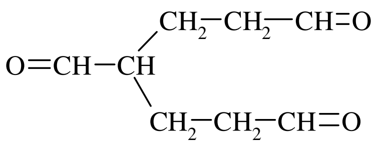
(2) $CH_3–CH_2–CH=O$ and



(3) $CH_3–CH_2–CH=O$ and



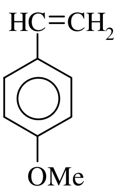
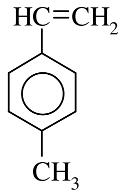
(4) $CH_3–CH_2–CH=O$ and

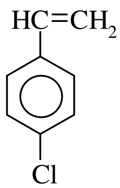
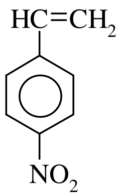


181. No. of structural isomeric alkenes (molecular formula=C_6H_{12}) which all give n-hexane on hydrogenation in presence of metal catalyst

(1) 2 (2) 3 (3) 4 (4) 5

182. Arrange the following towards their reactivity for electrophilic addition reaction (EAR)

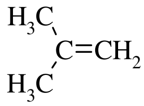
(I)  (II) 

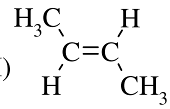
(III)  (IV) 

(1) I > III > II > IV (2) II > I > III > IV

(3) I > II > III > IV (4) II > III > I > IV

183. Arrange the following in order of heat of combustion.

(I) $CH_3–CH=CH_2$ (II) 

(III) 

(1) II > I > III (2) I > II > III

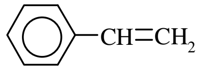
(3) II > III > I (4) III > II > I

184. Arrange the following towards their reactivity for acidic hydration?

(I) $CH_3O–CH=CH_2$

(II) $F–CH=CH_2$

(III) $CH_3–CH=CH_2$

(IV) 

(1) II > I > III > IV

(2) I > III > IV > II

(3) I > II > III > IV

(4) I > IV > III > II

185. Benzene + Acetylchloride $\xrightarrow{AlCl_3}$ product

Name of above reaction is

(1) Wurtz fittig reaction

(2) Gattermann reaction

(3) Friedel Craft reaction

(4) Shotten baumann reaction

186. Which of the following Alkane can not be prepared by hydrogenation of Olefine–

(1)Propane

(2) Iso butane

(3)Iso pentane

(4) Neo pentane

187. From how many alkyl halide we get Iso pentane on reaction with Zn + HCl (only structural isomers)

(1) 2 (2) 3

(3) 4 (4) 5

188. $C_2H_5Br + Na + CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}H}-Br \xrightarrow{\text{dry ether}}$ 'P' (mix. of major product)

Product P can't be –

(1) $CH_3-CH_2-CH_2-CH_3$

(2) $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{C}H}-CH_3$

(3) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}H}-\underset{\displaystyle CH_3}{\underset{|}{C}H}-CH_3$

(4) $CH_3-CH_2-CH_2-CH_2-CH_3$

189. $CH_3CH_2CH_2COOH$ $\xrightarrow{\text{Soda lime}}$ 'A' ; $\xrightarrow{\text{Red P + HI}}$ 'B'

Product 'A' & 'B' are–

(1) Isomer (2) Identical

(3) Homologue (4) Conformer

190. Methane can be prepared by–

(1) Kolbe's Electrolysis

(2) Wurtz reaction

(3) Carboxylic acid with soda lime

(4) Sabatier sendrence reaction

191. Kolbe's Electrolysis is completed by–

(1) Ionic followed by free radical mechanism

(2) Free radical followed by ionic mechanism

(3) Only ionic mechanism

(4) Only free radical mechanism

192. Iodination of methane can be carried out in presence of–

(1) HIO_3 (2) $HBrO_3$

(3) $HClO_3$ (4) HCl

193. Which of the following can be oxidised to alcohol by $KMnO_4$:–

(1) n-pentane (2) Iso pentane

(3) Neo pentane (4) propane

194. $CH_4 + O_2 \xrightarrow{Mo_2O_3}$ 'P' (major) Product 'P' is–

(1) CH_3OH (2) HCHO

(3) HCOOH (4) CH_3-CH_3

195. $CH_4 + H_2O \xrightarrow[1273K]{Ni}$ 'P' Product 'P' is–

(1) water gas (2) producer gas

(3) Synthesis gas (4) Natural gas

196. Alkene is known as olefins due to–

(1) It is oily in nature

(2) It gives oily liquid on reaction with Cl_2

(3) It is obtained from oily liquid

(4) It give oil with water

197. $CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}}=CH-C_6H_5$ $\xrightarrow{\text{HBr / Peroxide}}$ 'A' ; $\xrightarrow{\text{HBr}}$ 'B'

Product 'A' & 'B' are respectively–

(1) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle Br}{\overset{|}{C}}}}-CH_2-C_6H_5$ & $CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}H}-\underset{\displaystyle Br}{\underset{|}{C}H}-C_6H_5$

(2) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}H}-\underset{\displaystyle Br}{\underset{|}{C}H}-C_6H_5$ & $CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle Br}{\overset{|}{C}}}}-CH_2-C_6H_5$

(3) Both are $CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle Br}{\overset{|}{C}}}}-CH_2-C_6H_5$

(4) Both are $CH_3-\underset{\displaystyle CH_3}{\underset{|}{C}H}-\overset{\displaystyle Br}{\overset{|}{C}H}-C_6H_5$

198. $CH_3-C(CH_3)_2-CH=CH_2 \xrightarrow{\overset{+}{H}/H_2O}$ 'P' (major)

product 'P' is–

(1) $CH_3-C(CH_3)_2-CH_2-CH_2-OH$

(2) $CH_3-C(CH_3)_2-CH(OH)-CH_3$

(3) $CH_3-C(OH)(CH_3)-CH(CH_3)-CH_3$

(4) $CH_3-C(OH)(CH_3)-CH_2-CH_2-CH_3$

199. Which of the following compound gives CO_2 on reductive ozonolysis–

(1) $CH_2=CH-CH=CH_2$

(2) $CH_2=C=CH-CH_3$

(3) $CH_3-CH=CH-CH_3$

(4) $CH_2=CH-CH_3$

200. Propanal and pentan -3-one are ozonolysis product of–

(1) $CH_3-CH_2-CH=C(CH_2-CH_3)-CH_2-CH_3$

(2) $CH_3-CH=C(CH_2-CH_3)-CH_2-CH_3$

(3) $CH_3-CH_2-C(CH_3)=C(CH_3)-CH_3$

(4) $CH_3-CH=CH-CH(CH_3)-CH_2-CH_2-CH_3$

201. Benzene is obtained by all the following reactions except

(1) Decarboxylation of sodiumbenzoate

(2) Deoxygenation of phenol

(3) Reduction of Benzene diazonium chloride

(4) Catalytic hydrogenation of acetylene

202. Oxidation of which of the following arenes does not give benzoic acid with acidic $KMnO_4$

(1) Ethylbenzene (2) Cumene

(3) Mesitylene (4) Sec. Butylbenzene

203. When benzene is heated with acetic anhydride in presence of anhydrous $AlCl_3$, the product formed is –

(1) Benzoic acid

(2) Acetophenone

(3) Ethylphenyl ketone

(4) Benzophenone

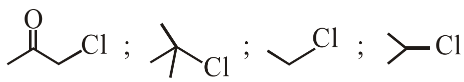
204. Which of the following acid on decarboxylation does not form an aromatic hydrocarbon

(1) Phenylacetic acid (2) Salycylic acid

(3) Ortho toluic acid (4) Benzoic acid

ALKYL HALIDES AND ARYL HALIDES

205. The relative reactivity of following halides towards S_N2 reaction follows the order :-



P Q R S

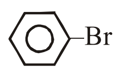
(1) Q > S > R > P

(2) P > S > R > Q

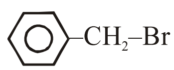
(3) S > R > Q > P

(4) P > R > S > Q

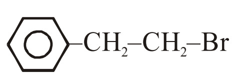
206. Rate of S_N1 reaction is :-



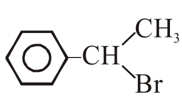
(P)



(Q)



(R)



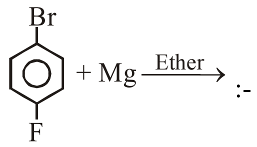
(S)

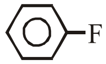
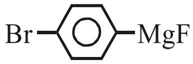
(1) S > Q > R > P

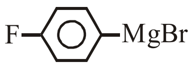
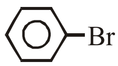
(2) S > R > P > Q

(3) P > Q > R > S

(4) S > R > Q > P

207.  :-

(1)  (2) 

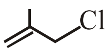
(3)  (4) 

208. Reactivity order of S_{N^1} is :-

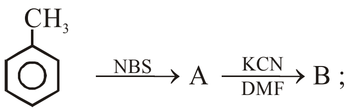
$H_3C–O–CH_2–Cl$ (P)

$H_3C–\underset{\displaystyle H}{N}–CH_2–Cl$ (Q)

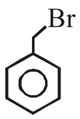
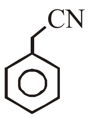
 (R)

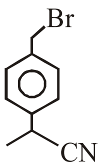
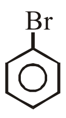
 (S)

(1) P > Q > R > S
(2) R > Q > P > S
(3) Q > P > R > S
(4) Q > P > S > R

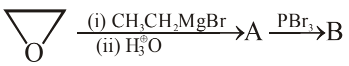
209. 

Product B is :-

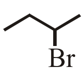
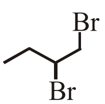
(1)  (2) 

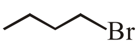
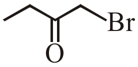
(3)  (4) 

210. Consider the following sequence of reaction.

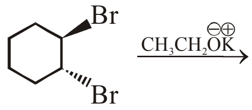


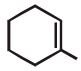
The product B is :

(1)  (2) 

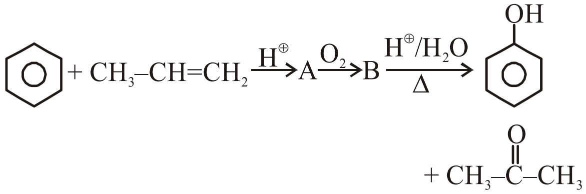
(3)  (4) 

211. The most probable product in the following reaction is :-

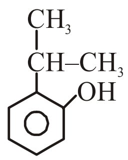
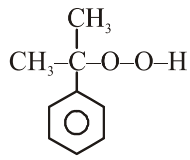


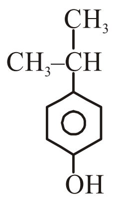
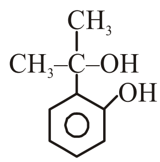
(1)  (2)  (3)  (4) 

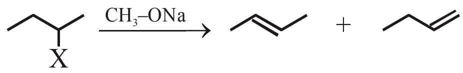
212. In following reaction



The structure of 'B' is:-

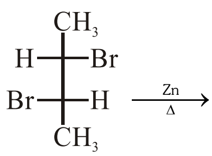
(1)  (2) 

(3)  (4) 

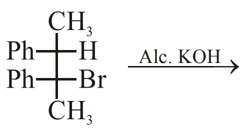
213. 

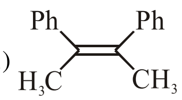
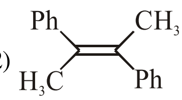
In the above reaction, maximum Hoffmann product will be obtained, where X is :

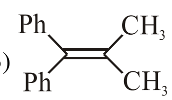
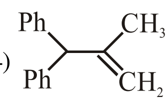
(1) –F (2) –Cl (3) –Br (4) –I

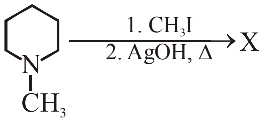
214.  Product is ?

(1) 2-Butyne (2) Cis-2-butene
(3) Trans-2-butene (4) 1-Butene

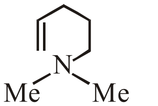
215.  Major product :-

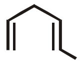
(1)  (2) 

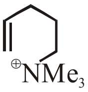
(3)  (4) 

216. 

Identify the major product X :

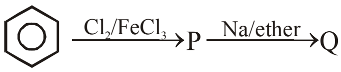
(1) 

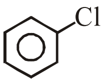
(2) 

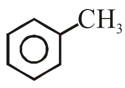
(3) 

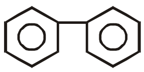
(4) 

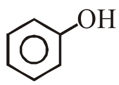
217. The end product (Q) in the following sequence of reaction is :-



(1) 

(2) 

(3) 

(4) 

218. Identify the products (A) and (B).

(i) $CH_3-\underset{\displaystyle Br}{CH}-CH_2-CH_3 \xrightarrow[\Delta]{C_2H_5ONa} (A)$

(ii) $CH_2=\overset{\displaystyle CH_3}{C}-CH_2CH_3 \xrightarrow[\text{Peroxide}]{HBr} (B)$

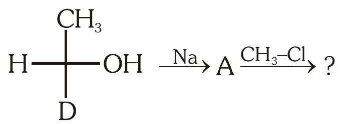
(1) (A) $\rightarrow CH_3CH_2-CH_2-CH_3$,
(B) $\rightarrow CH_3(CH_2)_2CH_3$

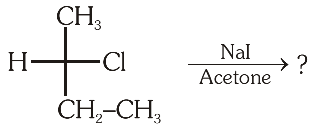
(2) (A) $\rightarrow CH_3CH_2-\underset{\displaystyle OC_2H_5}{C}(CH_3)_2$,
(B) $\rightarrow CH_3-\underset{\displaystyle Br}{\overset{\displaystyle CH_3}{C}}-CH_2CH_3$

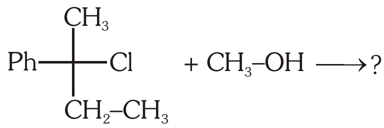
(3) (A) $\rightarrow CH_3-CH=CH-CH_3$,
(B) $\rightarrow BrCH_2-\underset{\displaystyle H}{\overset{\displaystyle CH_3}{C}}-CH_2CH_3$

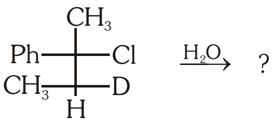
(4) (A) $\rightarrow CH_3CH_2-CH_2CH_2CH_3$,
(B) $\rightarrow CH_3-\underset{\displaystyle Br}{CH}-CH_2CH_3$

219. In which reaction racemic mixture is obtained as a major product ?

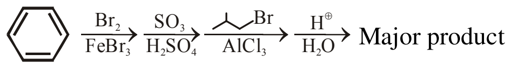
(1) 

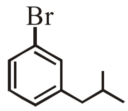
(2) 

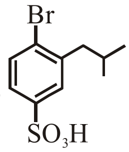
(3) 

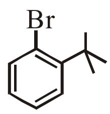
(4) 

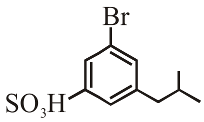
220. Give the major product from the following reaction sequence



(1) 

(2) 

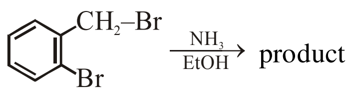
(3) 

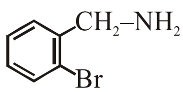
(4) 

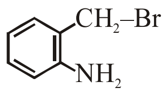
221. One mole of 1,2-dibromopropane on treatment with X moles of $NaNH_2$ followed by treatment with ethyl bromide give a pentyne. The value of X is :-

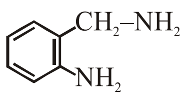
(1) One (2) Two
(3) Three (4) Four

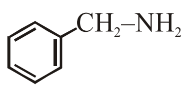
222. What is the major product obtained in the following reaction ?

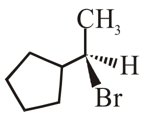


(1) 

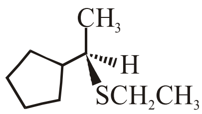
(2) 

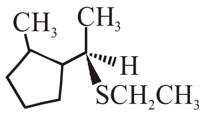
(3) 

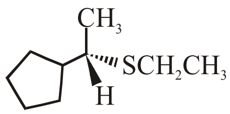
(4) 

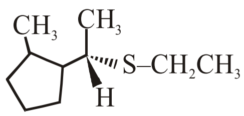
223.  $\xrightarrow{Na^{\oplus}\overset{\ominus}{S}CH_2CH_3}$ Product

Product is :-

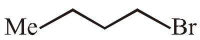
(1) 

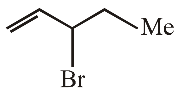
(2) 

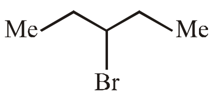
(3) 

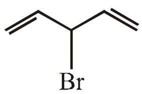
(4) 

224. Among the following alkyl bromide correct order of S_N1 reactivity is:

(I) 

(II) 

(III) 

(IV) 

(1) I > II > III > IV

(2) IV > II > III > I

(3) IV > III > II > I

(4) II > IV > III > I

225. The product of the reaction is

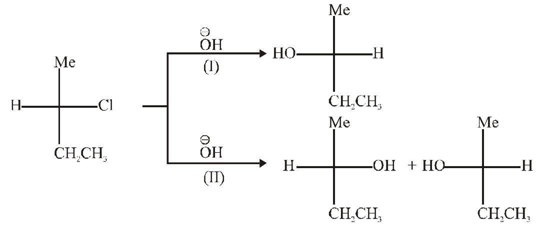
$$CH_3-CH_2-\overset{\overset{O}{\|}}{C}-OAg \xrightarrow[\Delta]{Br_2} \text{Major product}$$

(1) $CH_2=CH_2$

(2) CH_3-CH_2-Br

(3) $CH_3CH_2CH_2-Br$

(4) $CH_3-CH=CH-CH_3$

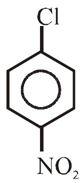
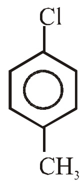
226. 

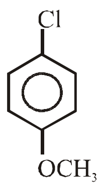
Reaction I[st] and II[nd] are:

(1) Both S_N1 (2) Both S_N2

(3) (I) S_N1, (II) S_N2 (4) (I) S_N2, (II) S_N1

227. Which of the following compound undergo hydrolysis most easily:

(1)  (2) 

(3)  (4) 

228. Consider the reaction:

$CH_3-CH_2-CH_2-Br+NaCN \rightarrow CH_3CH_2CH_2-C\equiv N + NaBr$

The **correct** statement is :-

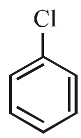
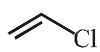
(1) The reaction will be fastest in water

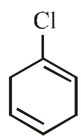
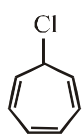
(2) The reaction will be fastest in N, N–dimethylformamide (DMF)

(3) Transition state of S_N2 is tetrahedral and sp^3 hybridized

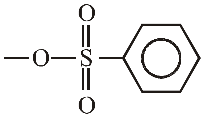
(4) If conc. of alkyl bromide is tripled and conc. of $\overset{\ominus}{C}N$ is reduced to half rate of S_N2 increased by 2 times

229. Which of the following compound will give curdy precipitate with $AgNO_3$ solution:

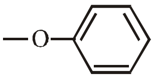
(1)  (2) 

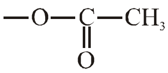
(3)  (4) 

230. Arrange the following group in order of their decreasing leaving ability:

(A) 

(B) —O—CH_3

(C) 

(D) 

(1) A > D > B > C (2) A > D > C > B
(3) A > B > C > D (4) D > C > B > A

231. $(CH_3)_2CHCl + NaI \xrightarrow{\text{Acetone}} (CH_3)_2CHI$

The above reaction is known as

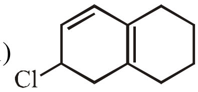
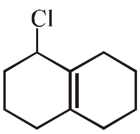
(1) Swart's reaction

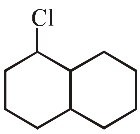
(2) Finkelsteins

(3) Fittig reaction

(4) Sabatier-senderence reaction

232. The correct order for E_2 reaction with alc. KOH will be:-

(i)  (ii) 

(iii) 

(1) i > ii > iii (2) iii > ii > i
(3) ii > i > iii (4) iii > i > ii

233. Which of the following reaction is correct:

(1) $CH_3CH_2-Br+KCN \longrightarrow CH_3CH_2-NC$

(2) $CH_3CH_2-Br+AgCN \longrightarrow CH_3CH_2-CN$

(3) $CH_3CH_2-Br + KNO_2 \longrightarrow CH_3-CH_2-N{\genfrac{}{}{0pt}{}{=O}{\rightarrow O}}$

(4) $CH_3-CH_2-Br+AgNO_2 \longrightarrow CH_3-CH_2-N{\genfrac{}{}{0pt}{}{=O}{\rightarrow O}}$

234. Which of the following compound is most reactive toward S_N2 displacement?

(1) $CH_3CH_2CH_2CH_2-Cl$

(2) $(CH_3)_3C-Cl$

(3) $CH_3-\underset{CH_3}{CH}-\underset{Cl}{CH}-CH_3$

(4) $CH_3-\underset{CH_3}{CH}-CH_2-Cl$

235. Which of the following statement is incorrect:-

(1) Boiling point of alkylhalide is R–I > R–Br > R–Cl > R–F

(2) Alkyl halide are soluble in organic solvent and less soluble in water

(3) Freons are used for aerosol propellant, refrigeration and air conditioning purpose.

(4) o and m - dichlorobenzene has higher melting point than those of p-isomer

236. Which of the following is correctly match:-

(A) Chloramphinicol → (1) iodine containing Hormone

(B) Thyroxine → (2) Chlorine containing antibiotic

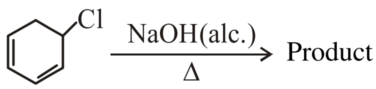
(C) Chloroquine→ (3) Treatment for malaria

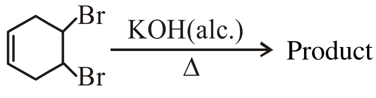
(D) Fluorinated compound (4) Potential blood substitute

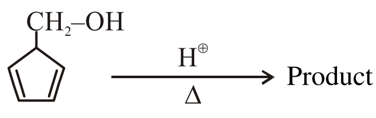
(E) Iodoform → (5) Antiseptic

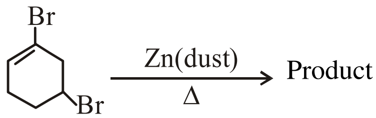
(1) A, B, C (2) B, C
(3) C, D, E (4) D, E

237. In which of the following reaction benzene is not form as major product

(1) 

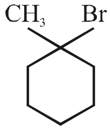
(2) 

(3) 

(4) 

238. Which one of the following will give racemised product in C_2H_5OH?

(1) $CH_3-CH(CH_3)-Br$ (2) $Ph-C(CH_3)_2-Cl$

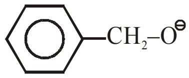
(3) $Ph-CH(CH_3)-Cl$ (4) 

239. Arrange the following group in order of their nucleophilic strength:-

(A) $CH_3-O^{\ominus}$

(B) $CH_3CH_2-S^{\ominus}$

(C) $CH_3-O-CH_2-O^{\ominus}$

(D) 

(1) A > B > C > D (2) D > B > C > A
(3) B > A > D > C (4) A > B > D > C

240. Which of the following molecule would have a carbon halogen bond most susceptible to nucleophilic substitution?
(1) 2-Flourobutane (2) 2-Chlorobutane
(3) 2-Bromo butane (4) 2-Iodobutane

241. Which of the following is most reactive towards nucleophilic substitution reaction?
(1) $CH_2=CH-Cl$ (2) C_6H_5-Cl
(3) $CH_3-CH=CH-Cl$ (4) $Cl-CH_2-CH=CH_2$

242. A S_N2 reaction at an asymmetric carbon of a compound always give:
(1) an enantiomer of the substrate
(2) a product with opposite rotation
(3) a mixture of diastereomer
(4) a single stereoisomers

243. The synthesis of alkyl fluoride is best accomplished by:
(1) Free radical fluorination
(2) Sandmeyer reaction
(3) Finkelstein reaction
(4) Swarts reaction

244. The reagent that brings about the conversion of(R)–butan-2-ol to (R)-2-chlorobutane is:-
(1) $SOCl_2$ (2) PCl_5
(3) PCl_3 (4) all of them

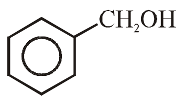
245. Which of the following is correct order of dipole moment:-
(1) $CH_3-F > CH_3-Cl > CH_3-Br > CH_3-I$
(2) $CH_3-Cl > CH_3-F > CH_3-Br > CH_3-I$
(3) $CH_3-Cl > CH_2Cl_2 > CCl_4 > CHCl_3$
(4) $CH_2Cl_2 > CH_3Cl > CHCl_3 > CCl_4$

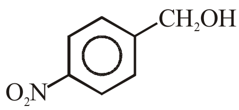
246. DDT is manufactured by the acid catalysed condensation of:-
(1) Chloroform and acetone
(2) Chloroform and nitric acid
(3) Chlorobenzene and chloral
(4) Chlorobenzene and chloroform

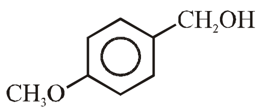
247. Which of the following compound is used as refrigerant?
(1) Urea (2) Acrolein
(3) CCl_2F_2 (4) $CH_2(OH)-CH_2(OH)$

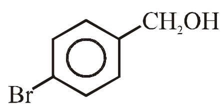
248. Which of the following is added to chloroform in a small quantity before is bottled for sale?
(1) CH_3OH (2) C_2H_5Cl
(3) C_2H_5OH (4) $AgNO_3$

249. Arrange the following compounds for their reactivity toward nucleophilic substitution with HBr?

(A) 

(B) 

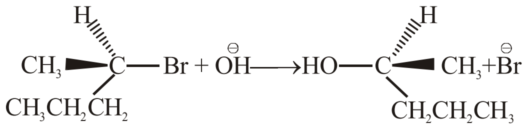
(C) 

(D) 

(1) A > B > C > D (2) C > A > D > B
(3) D > C > B > A (4) B > D > A > C

250. Which of the following is secondary alkyl halide?
(1) Isobutyl chloride (2) Isopentyl chloride
(3) Isopropyl chloride (4) Neopentyl chloride

251. The following reaction is taken place by which mechanism ?



(1) S_{N^1} (2) S_{N^2} (3) S_{N^i} (4) $S_{N^{AE}}$

252. Which of the following reaction occur most rapidly?

(1) $(CH_3)_2CH\ OH \xrightarrow{HBr}$

(2) $(CH_3)_2CHOH \xrightarrow{HI}$

(3) $(CH_3)_3C-OH \xrightarrow{HBr}$

(4) $(CH_3)_3C-OH \xrightarrow{HI}$

253. In S_N2 reactions, the correct order of reactivity for following compounds CH_3Cl, CH_3CH_2Cl, $(CH_3)_2CHCl$, $(CH_3)_3C–Cl$ is

(1) $CH_3CH_2Cl > CH_3Cl > (CH_3)_2CHCl > (CH_3)_3CCl$

(2) $(CH_3)_2CHCl > CH_3CH_2Cl > CH_3Cl > (CH_3)_3CCl$

(3) $CH_3Cl > (CH_3)_2CHCl > CH_3CH_2Cl > (CH_3)_3CCl$

(4) $CH_3Cl > CH_3CH_2Cl > (CH_3)_2CHCl > (CH_3)_3CCl$

254. Which of the following compounds will give a yellow ppt with iodine and alkali

(1) Acetamide (2) 3-Hydroxypentane

(3) Acetophenone (4) Methyl acetate

255. Which of the following undergoes fastest reaction with water ?

(1) $C_6H_5-\underset{\mid}{\underset{Cl}{CH}}-OMe$

(2) $C_6H_5-\underset{\mid}{\underset{Cl}{CH}}-CH_3$

(3) $C_6H_5-\underset{\mid}{\underset{Cl}{CH}}-CH_2CH_3$

(4) $C_6H_5-\underset{\mid}{\underset{Cl}{CH}}-C_6H_5$

256. Which of the following electrophile is formed in Reimer-Tiemann reaction?

(1) Carbocation (2) Carbanion

(3) Carbene (4) Free radical

257. Which of the following statement is not correct for nucleophilic substitution reaction?

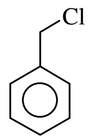
(1) correct decreasing order for S_N1 reactivity is $(CH_3)_3C–Cl > (CH_3)_2CHCl > CH_3CH_2Cl$

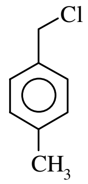
(2) Stereochemical inversion occurs in S_N2 reaction at reaction centre.

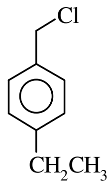
(3) Racemisation may occurs in S_N1 reaction.

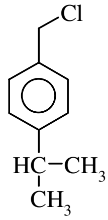
(4) Carbocation is formed during S_N2 reaction.

258. Reactivity order for S_N1 reaction

(I) 

(II) 

(III) 

(IV) 

(1) I > II > III > IV

(2) II > III > IV > I

(3) IV > III > II > I

(4) III > IV > II > I

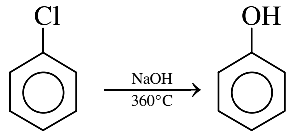
259. Major product of the following reaction

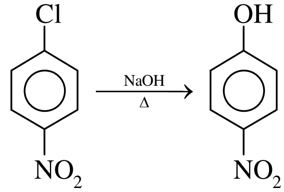
$$CH_3–CH_2–Br \xrightarrow{KCN} \xrightarrow[H_2O]{\bar{O}H} \xrightarrow[\Delta]{PCl_5} \text{product}$$

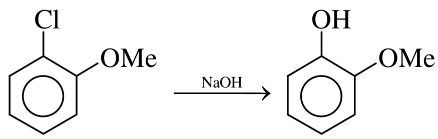
(1) $CH_3–CH_2–Cl$ (2) $H_3C-CH_2-\underset{\underset{O}{\parallel}}{C}-CH_3$

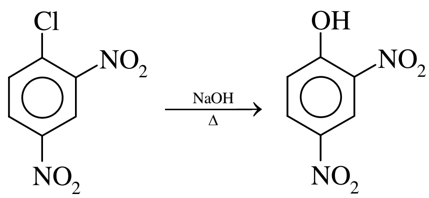
(3) $CH_3–CH_2–CN$ (4) $CH_3-\underset{\underset{O}{\parallel}}{C}-NH_2$

260. Which reaction is least feasible

(1) 

(2) 

(3) 

(4) 

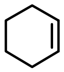
261. In which reaction halide is not obtained

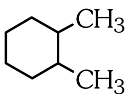
(1) R—OH + NaBr $\longrightarrow$

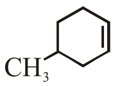
(2) R—OH + H —Br $\longrightarrow$

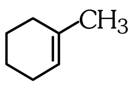
(3) R—OH + $SOCl_2$ $\longrightarrow$

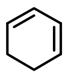
(4) R—OH + PCl_5 $\longrightarrow$

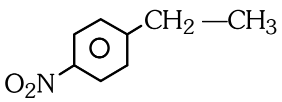
262.  + Br_2 $\xrightarrow{hv}$ A $\xrightarrow[\Delta]{CH_3Na}$ B ; 'B' is–

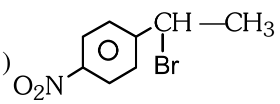
(1) 

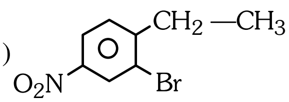
(2) 

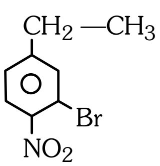
(3) 

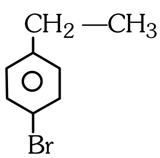
(4) 

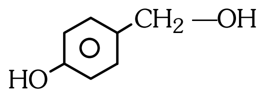
263.  $\xrightarrow{Br_2 / Fe}$?

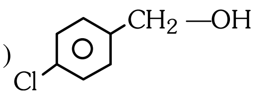
(1) 

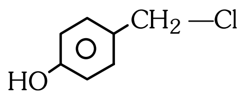
(2) 

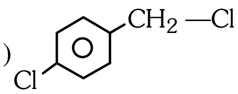
(3) 

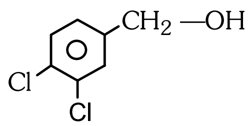
(4) 

264.  $\xrightarrow{HCl}$?

(1) 

(2) 

(3) 

(4) 

265. $CH_3-\underset{Cl}{CH}-\underset{CH_3}{CH}-CH_3$ + Alcoholic KOH $\xrightarrow{\Delta}$

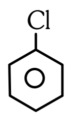
(A) major + B (minor). A and B respectively are:

(1) 3-methyl-but-1-ene, 2-methyl-but-2-ene

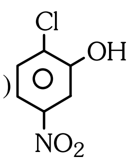
(2) 2-methyl-but-2-ene, 3-methyl-but-1-ene

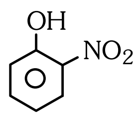
(3) 2-methyl-but-1-ene, 3-methyl-but-1-ene

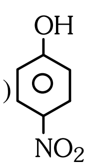
(4) 2-methyl-but-2-ene, 2-methyl-but-1-ene

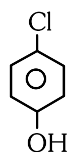
266.  $\xrightarrow[\text{conc } H_2SO_4]{\text{conc } HNO_3}$ A $\xrightarrow[\text{ii) } H^{\oplus}]{\text{i) NaOH/443K}}$ B

B as a major product is–

(1) 

(2) 

(3) 

(4) 

267. Which is insoluble in water

(1) D.D.T. (2) CHI_3

(3) CH_3-CH_2-Cl (4) All

OXYGEN CONTAINING COMPOUND-I

268. An ether is more volatile then alcohol having the same moleculer formula this is due to:-

(1) Inter moleculer H–bonding in ether

(2) Inter moleculer H-bonding in alcohols

(3) Dipolar character of ether

(4) Alcohol have resonating structure

269. Among the following compounds which can be dehydrated most easily is:-

(1) $CH_3-CH_2-CH(CH_3)-CH_2-OH$

(2) $CH_3-CH_2-CH_2-CH(OH)-CH_3$

(3) $CH_3-CH_2-C(CH_3)(OH)-CH_2-CH_3$

(4) $CH_3-CH_2-CH_2-CH_2-CH_2-OH$

270. Among the following the one that gives positive iodoform test upon reaction with I_2 and NaOH is:-

(1) $CH_3-CH_2-CH(OH)-CH_2-CH_3$

(2) $C_6H_5-CH_2-CH_2-OH$

(3) $CH_3-CH(CH_3)-CH_2OH$

(4) $PhCHOHCH_3$

271. In the following sequence of reaction:

$$CH_3CH_2OH \xrightarrow{Red\ P+I_2} A \xrightarrow[dry\ ether]{Mg} B \xrightarrow{HCHO} C \xrightarrow{H_2O} D$$

the compound D is :-

(1) Butanal

(2) n-butyl alcohol

(3) n-propyl alcohol

(4) propanal

272. n-propyl bromide reacts with KOH to form:-

(1) Propane

(2) Propene

(3) Propyne

(4) 1-propanol

273. When phenol is treated with bromine /H_2O, it gives:-

(1) O- and p-dibromo phenol

(2) 2, 3, 4-tribromophenol

(3) 2, 4, 6-tribromo phenol

(4) none

274. When CH_3-CH_2-COOH is reduced with $LiAlH_4$, the compound obtained will be:-

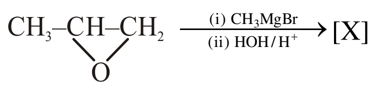
(1) CH_3-CH_2-COOH

(2) $CH_2=CH-CH_2-OH$

(3) $CH_3-CH_2-CH_2-OH$

(4) CH_3-CH_2-CHO

275. In the given reaction :



[X] will be :

(1) $CH_3-CH(CH_3)-CH_2-OH$

(2) $CH_3-CH(OH)-CH_2-CH_3$

(3) $CH_3-C(OH)(CH_3)-CH_3$

(4) $CH_2=CH-CH_3$

276. The fermentation of starch to give alcohol occurs mainly with the help of :

(1) O_2 (2) Air

(3) CO_2 (4) Enzymes

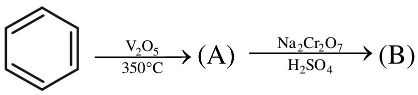
277. $A \xrightarrow[dil.\ H_2SO_4]{K_2Cr_2O_7} B \xrightarrow[ii)\ H_2O/H^{\oplus}]{i)\ CH_3MgI} CH_3-C(CH_3)(OH)-CH_3$

The reactant A is :

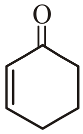
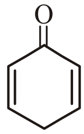
(1) $CH_3CHOHCH_3$ (2) CH_3COCH_3

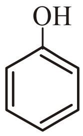
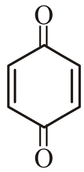
(3) C_2H_5OH (4) CH_3COOH

278. In the given reaction:



Product (B) is :

(1)  (2) 

(3)  (4) 

279. In the given reaction

$CH_3–CH=CH–CH_2–OH \xrightarrow{PCC}$ Product

Product is :

(1) $CH_3–CH_2–CH_2–CH_2–OH$
(2) $CH_3–CH_2–CH_2–CHO$
(3) $CH_3–CH=CH–CHO$
(4) $CH_3–CH=CH–COOH$

280. Reaction of phenol with dil. HNO_3 gives :

(1) p- and m- nitro phenol
(2) o- and p- nitro phenol
(3) picric acid
(4) o- and m- nitro phenol

281. In the given reaction

$CO + H_2 \xrightarrow[573–673K]{ZnO+Cr_2O_3}$ Product

(1) CH_3OH (2) $\begin{array}{l} CH_2–OH \\ | \\ CH_2–OH \end{array}$

(3) $\begin{array}{l} CHO \\ | \\ CHO \end{array}$ (4) $\begin{array}{l} COOH \\ | \\ COOH \end{array}$

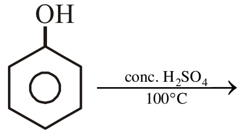
282. In the following series of chemical reaction. Identify Z :

$C_3H_7OH \xrightarrow[170°C]{Conc.\ H_2SO_4} X \xrightarrow[CCl_4]{Br_2} Y \xrightarrow[(II)\ 1\ eq.NaNH_2]{(I)\ Alc.KOH} Z$

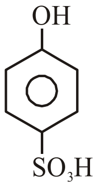
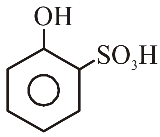
(1) $CH_3–\underset{NH_2}{\underset{|}{C}H}–\underset{NH_2}{\underset{|}{C}H_2}$ (2) $CH_3–\underset{OH}{\underset{|}{C}H}–\underset{OH}{\underset{|}{C}H_2}$

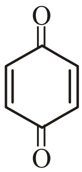
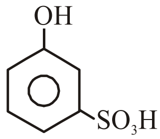
(3) $CH_3–\underset{OH}{\underset{|}{C}}=CH_2$ (4) $CH_3–C \equiv CH$

283. In following reaction sequence :



Major product is :

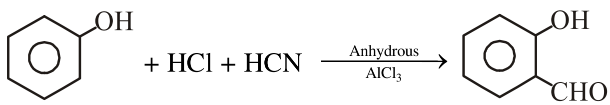
(1)  (2) 

(3)  (4) 

284. Methanol and ethanol are distinguished by the :

(1) Action of HCl (2) Iodoform test
(3) Lucas test (4) Sodium

285. The following reaction :



is known as :

(1) Perkin reaction
(2) Gattermann aldehyde synthesis
(3) Kolbe reaction
(4) Gattermann-Koch reaction

286. Which of the following give ether ?

(1) CH_3CH_2ONa and C_2H_5I
(2) $CH_3–CH_2–OH \xrightarrow[140°C]{H^{\oplus}}$
(3) $CH_3–CH_2–I \xrightarrow{dry\ Ag_2O}$
(4) All of the above

287. In the given reaction

$$C_6H_5-O-CH_2-CH_3 \xrightarrow{\text{Excess HI}/\Delta} [X] + [Y]$$

[X] and [Y] will respectively be :

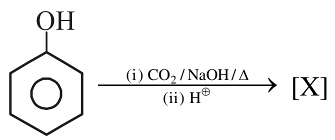
(1) C_6H_5I and CH_3CH_2I

(2) C_6H_5OH and CH_3-CH_2-I

(3) C_6H_5I and CH_3CH_2OH

(4) C_6H_5OH and $CH_2=CH_2$

288. In the given reaction :



Major product [X] is :

(1) Salicylic acid

(2) m-hydroxybenzoic acid

(3) Mixture of 1 and 2

(4) Salicylaldehyde

289. Rectified spirit contains (% by mass) :

(1) 94.6% C_2H_5OH (2) 95.6% C_2H_5OH

(3) 96.6% C_2H_5OH (4) 97.6% C_2H_5OH

290. Phenol reacts with PCl_5 to give mainly :

(1) p-chlorophenol

(2) meta dichlorobenzene

(3) o-and p-chlorophenol

(4) Triphenyl phosphate

291. In the given reaction

$$CH_3-CH_2-CH_2-O-CH_2-CH_3 \xrightarrow{HI} [X] \ \& \ [Y]$$

[X] and [Y] respectively be ;

(1) $CH_3CH_2CH_2OH$ and CH_3CH_2I

(2) $CH_3CH_2CH_2I$ and CH_3CH_2OH

(3) $CH_3CH_2CH_2I$ and $CH_2=CH_2$

(4) $CH_3-CH=CH_2$ and $CH_2=CH_2$

292. Phenolphthalein is obtained from condensation between :

(1) Phenol and succinic acid

(2) Phenol and phthalic anhydride

(3) Phenol and succinic anhydride

(4) Phenol and benzaldehyde

293. Which of the following is most acidic :

(1) Phenol

(2) Benzyl alcohol

(3) m-chlorophenol

(4) cyclohexanol

294. Ethylene glycol on oxidation with periodic acid gives :

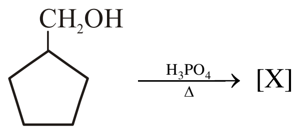
(1) Oxalic acid

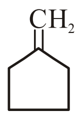
(2) Glyoxal

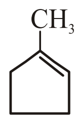
(3) Formaldehyde

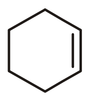
(4) Glycolic acid

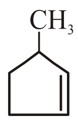
295. Identify the product [X]



(1) 

(2) 

(3) 

(4) 

296. Amongst the following alcohols which would react fastest with conc. HCl and $ZnCl_2$?

(1) 2-Methylbutan-1-ol

(2) 2-Pentanol

(3) 1-Pentanol

(4) 2-Methyl-2-butanol

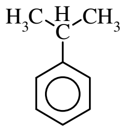
297. Alcohol is not formed as product in

(1) $CH_3-CH_2-NH_2 \xrightarrow{NaNO_2+HCl(aq)}$

(2) $CH_3-CHO \xrightarrow[(ii)H_2O]{(i)CH_3-MgBr}$

(3) $H_3C-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow{LiAlH_4}$

(4) $CH_3-CH_2-Cl \xrightarrow{KOH(aq)}$

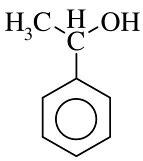
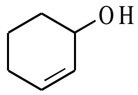
298.  $\xrightarrow[\text{catalyst}]{O_2,\Delta}$ P $\xrightarrow{\text{dil. } H_2SO_4}$ Q + Acetone

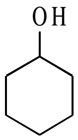
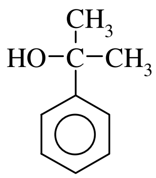
Q does not react with

(1) Na (2) NaOH

(3) $NaHCO_3$ (4) Zn,Δ

299. Minimum reactive towards dehydration in acidic medium

(1)  (2) 

(3)  (4) 

300. Order of boiling points is–

Pentan-1-ol, n-butane, pentanal, ethoxyethane

(a) (b) (c) (d)

(1) b > d > c > a

(2) a > c > d > b

(3) c > d > b > a

(4) d > a > b > c

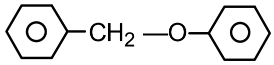
301. $CH_3CH_2-CH(CH_3)-CH_2-O-CH_2CH_3 + HI \rightarrow$?

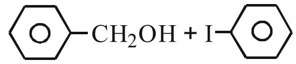
(1) $CH_3CH_2-CH(CH_3)-CH_2OH + CH_3CH_2I$

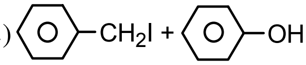
(2) $CH_3-CH_2-CH(CH_3)-CH_2I + CH_3CH_2OH$

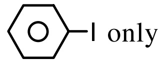
(3) $CH_3-CH_2-CH(CH_3)-CH_2I + CH_3-I$

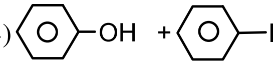
(4) CH_3-CH_2-OH only

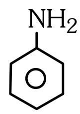
302.  + HI ⟶ ?

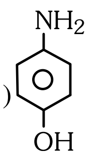
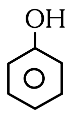
(1) 

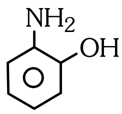
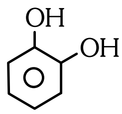
(2) 

(3) 

(4) 

303.  $\xrightarrow[+HCl]{NaNO_2}$ A $\xrightarrow[\text{boil}]{H_2O}$ B ; B is–

(1)  (2) 

(3)  (4) 

304. $CH_3-Br + NaO-C(CH_3)_2-CH_3 \longrightarrow$?

(1) $CH_3-O-C(CH_3)_2-CH_3$

(2) $CH_3-C(=CH_2)-CH_3$

(3) $CH_2=CH_2$

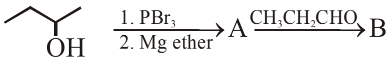
(4) $CH_3-O-CH(CH_3)-CH_3$

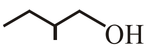
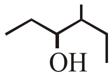
305. The product of the following reaction is :-

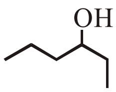
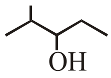
$(CH_3)_3C-CHOH-CH_3 \xrightarrow[ZnCl_2]{Conc.HCl}$

(1) $(CH_3)_3C-CHCl-CH_3$
(2) $(CH_3)_3C-CH_2Cl$
(3) $(CH_3)_2CCl-CH(CH_3)_2$
(4) $(CH_3)_2C=C(CH_3)_2$

306. Identify B in the following reaction :



(1)  (2) 

(3)  (4) 

307. $\underset{\displaystyle Br}{CH_3\underset{|}{C}HCH_3} \xrightarrow[\Delta]{alc\ KOH} A \xrightarrow{HBr/perodixe} B$

In the above reaction sequence, the final product is:

(1) 2-Bromopropane (2) 1-Bromopropane
(3) 3-Bromopropene (4) 2-Bromopropene

308. $H_3C-\underset{\diagdown O \diagup}{CH-CH_2} \xrightarrow[(ii)\ CH_3I]{(i)\ CH_3-C\equiv C^{\ominus}}$ Product ?

(1) $H_3C-\underset{\displaystyle OMe}{\underset{|}{C}H}-CH_2-CH=CH-CH_3$

(2) $H_3C-\underset{\displaystyle OMe}{\underset{|}{C}H}-CH_2-C\equiv C-CH_3$

(3) $H_3C-\underset{\displaystyle OH}{\underset{|}{C}H}-CH_2-C\equiv C-CH_3$

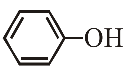
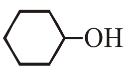
(4) $H_3C-\underset{\diagdown O \diagup}{CH-CH}-C\equiv C-CH_3$

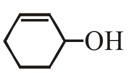
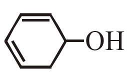
309. Arrange the following alcohols in order of increasing reactivity towards sodium metal.

(i) $(CH_3)_3C-OH$ (ii) $(CH_3)_2CH-OH$
(iii) CH_3CH_2OH

(1) (iii) < (ii) < (i) (2) (ii) < (i) < (iii)
(3) (i) < (ii) < (iii) (4) (iii) < (i) < (ii)

310. Dehydration of the following compounds in increasing order is:-

(I)  (II) 

(III)  (IV) 

(1) I < II < III < IV (2) II < III < IV < I
(3) I < III < IV < II (4) None of these

311. Complete the missing links.

$CH_3CHBrCH_3 \xrightarrow{alc.\ KOH} X \xrightarrow[Peroxide]{HBr} Y \xrightarrow{CH_3ONa} Z$

X, Y, Z are respectively ?

(1) $CH_3CH=CH_2$ $CH_3CH(Br)CH_2Br$ $CH_3CH(OH)CH_3$
(2) $CH_3CH=CH_2$ $CH_3CH_2CH_2Br$ $CH_3CH_2OCH_2CH_3$
(3) $CH_3CH=CH_2$ $CH_3CH(Br)CH_3$ $CH_3CH_2OCH_2CH_3$
(4) $CH_3CH=CH_2$ $CH_3CH_2CH_2Br$ $CH_3CH_2CH_2OCH_3$

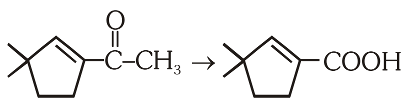
312. Cumene on reaction with oxygen followed by hydrolysis gives :-

(1) CH_3OH and $C_6H_5COCH_3$
(2) C_6H_5OH and $(CH_3)_2O$
(3) $C_6H_5OCH_3$ and CH_3OH
(4) C_6H_5OH and CH_3COCH_3

313. Match the column I with column II and mark the appropriate choice.

Column I		Column II	
(A)	Williamson's synthesis	(i)	$C_6H_5OH + CH_3COCl$ in presence of pyridine
(B)	ROR'	(ii)	$C_2H_5ONa + C_2H_5Br$
(C)	p-Nitrophenol	(iii)	Unsymmetrical ether
(D)	Acetylation	(iv)	Intermolecular hydrogen bonding

(1) (A) → (i), (B) → (iii), (C) → (ii), (D) → (iv)
(2) (A) → (iii), (B) → (i), (C) → (ii), (D) → (iv)
(3) (A) → (ii), (B) → (iii), (C) → (iv), (D) → (i)
(4) (A) → (iv), (B) → (i), (C) → (ii), (D) → (iii)

314. 

Suitable reagent for above conversion is

(1) $\xrightarrow[(ii)\ H^{\oplus}]{(i)\ Tollen's\ Reagent}$ (2) $\xrightarrow[(ii)\ H^{\oplus}]{(i)\ Benzoyl\ peroxide}$

(3) $\xrightarrow[(ii)\ H^{\oplus}]{(i)\ I_2/NaOH}$ (4) $\xrightarrow[\Delta]{H^{\oplus}/KMnO_4}$

315. Butan-2-one is obtained, if following compound is reacted with methyl magnesium bromide followed by hydrolysis ?

(1) CH_3-CHO (2) $CH_3-\underset{\displaystyle O}{\underset{||}{C}}-Cl$

(3) $CH_3-C\equiv N$ (4) $CH_3-CH_2-C\equiv N$

316. $CH_3-\underset{\underset{O}{\|}}{C}-CH_2-\underset{\underset{O}{\|}}{C}-H + \begin{matrix}CH_2-OH\\ |\\ CH_2-OH\end{matrix} \xrightarrow{H^{\oplus}} A$
1 eq

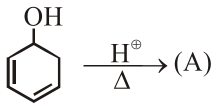
$\xrightarrow[H_2O]{LiAlH_4} B \xrightarrow{H_3O^{\oplus}} ?$

(1) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CHO$

(2) $CH_3-\underset{\underset{O}{\|}}{C}-CH_2-CH_2-OH$

(3) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CH_2-OH$

(4) $CH_3-CH=C\langle\begin{matrix}O-CH_2\\ \quad |\\ O-CH_2\end{matrix}$

317. 

Product (A) is :

(1)  (2) 

(3)  (4) 

OXYGEN CONTAINING COMPOUND-II

318. Cumene $\xrightarrow[Boil]{KMnO_4} A \xrightarrow{PCl_5} B \xrightarrow{C}$ Benzaldehyde

Find "C" :

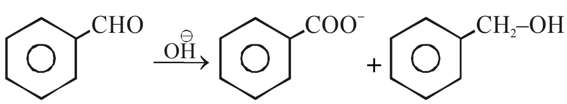
(1) $\dfrac{H_2/Pd-BaSO_4}{Quinolene}$

(2) $CO / HCl / AlCl_3$

(3) $CH_3NH_2 / HCl / AlCl_3$

(4) CrO_3 / Ac_2O

319. Which of the following correct set of name reaction :

(a) 

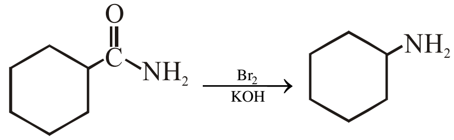
Cannizzaro reaction

(b) $CH_3-CH_2-CHO \xrightarrow[\Delta]{NaOH} CH_3-\underset{\underset{CHO}{|}}{C}=CH-CH_2-CH_3$

Aldol condensation

(c) $CH_3-CH_2-COOH \xrightarrow[Red\ P]{Cl_2} CH_3-\underset{\underset{Cl}{|}}{CH}-COOH$

HVZ reaction

(d) 

Hoffmann's bromamide

(1) a, b (2) b, c

(3) a, b, c (4) All

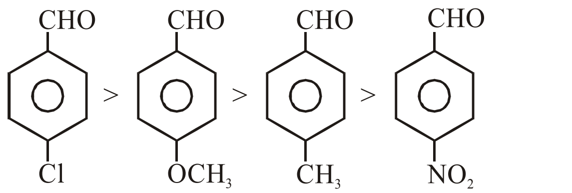
320. Which of the following is correct order :

(p) $CH_3-CHO > Ph-CHO > CH_3-\overset{\overset{O}{\|}}{C}-CH_3 > Ph-\overset{\overset{O}{\|}}{C}-Ph$

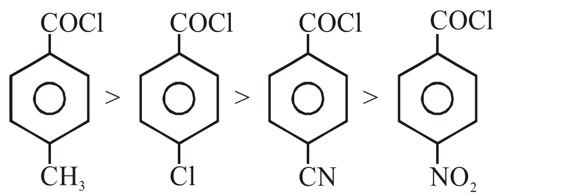
Reactivity in nucleophilic addition reaction(NAR)

(q) $Ph-\overset{\overset{O}{\|}}{C}-Cl > Ph-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-CH_3 > Ph-\overset{\overset{O}{\|}}{C}-OR > Ph-\overset{\overset{O}{\|}}{C}-NH_2$

Reactivity in hydrolysis

(r) 

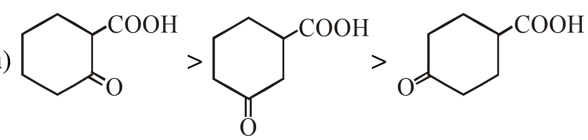
Reactivity in nucleophilic addition reaction(NAR)

(s) 

Reactivity in hydrolysis

(t) $HCHO > CH_3CHO > (CH_3)_2CO > Ph-CHO$

Reactivity towards Grignard reagent

(u) 

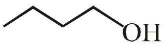
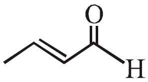
Rate of decarboxylation

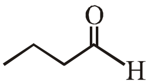
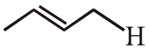
(1) p, q (2) r, s, t

(3) p, q, r, s (4) p, q, u

321. But-2-ene $\xrightarrow[2.\ Zn, H_2O]{1.\ O_3}$ P $\xrightarrow[\Delta]{OH^{\ominus}}$ Q $\xrightarrow[H_2(1eq)]{Pd/C}$ R

R is :

(1) 
(2) 

(3) 
(4) 

322. Glyoxal form by :

(a)  $\xrightarrow[2.\ Zn, H_2O]{1.\ O_3}$

(b) CH_3–CHO $\xrightarrow{SeO_2}$

(c) CH ≡ CH $\xrightarrow[CCl_4]{2Cl_2}$ $\xrightarrow{Ca(OH)_2}$

(d) CH ≡ CH $\xrightarrow{Hg^{2+}/dil.\ H_2SO_4}$

(1) a, b (2) a, c, d

(3) a, b, c (4) All

323. (a) Phthaldehyde $\xrightarrow{Conc.\ NaOH}$ Product

(Cannizzaro reaction)

(b) α-Methylcyclohexanone $\xrightarrow[\Delta]{NaOH}$ Product

(Aldol condensation)

(c) Toluene $\xrightarrow[H_3O^+]{CrO_2Cl_2/CS_2}$ Product

(Etard reaction)

(d) Benzene + CO + HCl $\xrightarrow{AlCl_3}$ Product

(Gattermann's Koch reaction)

Which of the following is correct set :

(1) a, c, d (2) a, b, d

(3) a, b, c (4) a, b, c, d

324. Which of the following is not correct reaction :

(Q) Malonic acid $\xrightarrow{\Delta}$ Acetic acid

(R) Succinic acid $\xrightarrow{\Delta}$ Succinic Anhydride

(S) Adipic acid $\xrightarrow{\Delta}$ Cyclopentanone

(T) Formic acid $\xrightarrow{\Delta}$ CO

(1) Q, R (2) R, S, T

(3) Q, T (4) Only T

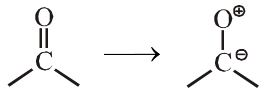
325. Which of the following statement is not true about carbonyl group :

(1) Carbon-oxygen bond is polarised due to higher electronegativity of oxygen

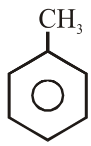
(2) Carbonyl carbon is electrophilic while oxygen is nucleophilic center

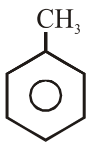
(3) Carbonyl group have more dipole moment than ether

(4) Polarity of carbonyl group as given



326. Which of the following reaction form benzene carbaldehyde as a product :

(a)  $\xrightarrow[H_3O^+]{CrO_2Cl_2/CS_2}$

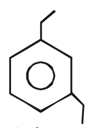
(b)  $\xrightarrow[(CH_3CO)_2O/H_3O^+]{CrO_3}$

(c)  $\xrightarrow[AlCl_3]{CO/HCl}$

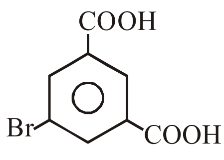
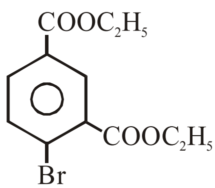
(d)  $\xrightarrow[(ii)\ H_2O]{(i)\ HCN/HCl,\ AlCl_3}$

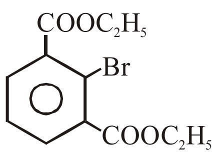
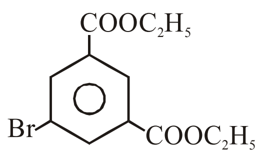
(1) a, b, c (2) a, b, d

(3) a, b (4) a, b, c, d

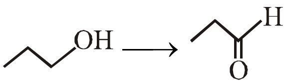
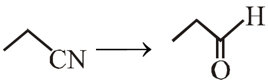
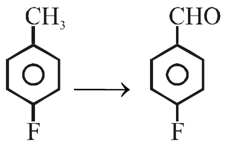
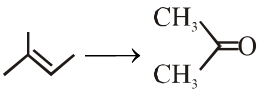
327.  $\xrightarrow{KMnO_4/H^+}$ A $\xrightarrow[FeBr_3]{Br_2}$ B $\xrightarrow[H^{\oplus}]{C_2H_5OH}$ C

C is :

(1) 
(2) 

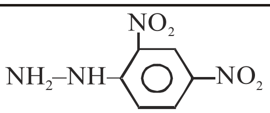
(3) 
(4) 

328. Which of the following is not correct matched:

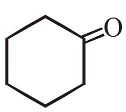
Reaction	*Reagent*
(1) 	PCC
(2) 	DIBAL-H
(3) 	$CrO_3/Ac_2O\ /H_3O^+$
(4)  + CH_3CHO	$KMnO_4/H^+$

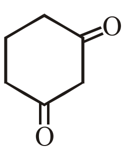
329. Which of the follwing product is not correctly matched :

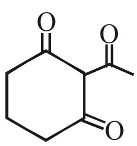
Carbonyl compound + Reagent ⟶ Product

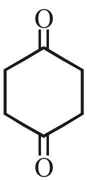
Reagent	Product
(1) NH_3	Imine
(2) NH_2OH	Oxime
(3) 	2,4-Dinitrophenyl hydrazone
(4) $NH_2-\overset{\overset{\large O}{\|}}{C}-NH_2$	Semicarbazone

330. Which of the compound have most acidic αH:

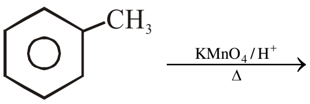
(1) 

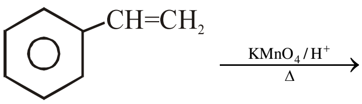
(2) 

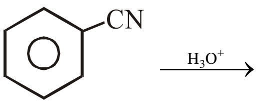
(3) 

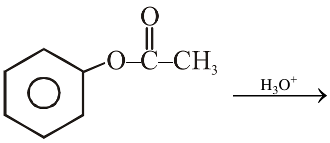
(4) 

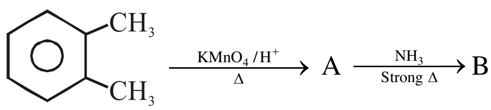
331. Benzoic acid is not formed in which reaction ?

(1) 

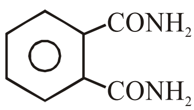
(2) 

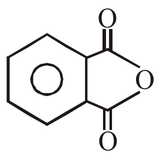
(3) 

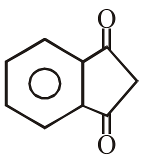
(4) 

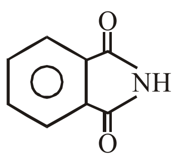
332. 

B is :

(1) 

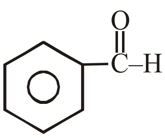
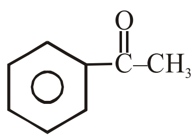
(2) 

(3) 

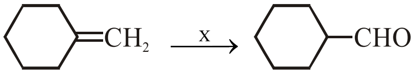
(4) 

333. How many compounds gives Cannizzaro reaction and aldol condensation respectively :

(1) $H-\overset{\overset{\large O}{\|}}{C}-H$; (2) $CH_3-\overset{\overset{\large O}{\|}}{C}-H$;

(3)  ; (4)  ;

(1) 2, 2
(2) 2, 3
(3) 1, 4
(4) 4, 1

334. 

X is :

(1) (a) BH_3, THF (b) H_2O_2 / OH^-

(2) H_3O^+

(3) (a) O_3 (b) Zn/H_2O

(4) (a) BH_3/THF, H_2O_2/OH^- (b) Anhydrous CrO_3

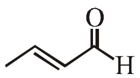
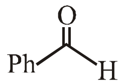
335. Which of the following set is correct :

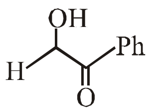
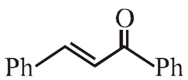
	Compound	Positive test
(a)	CH_3CHO	Fehling solution
(b)	Ph–CHO	Tollen's reagent
(c)	HCOOH	Benedict solution
(d)	Ph–CHO	Fehling solution

(1) a, b, c, d (2) a, b, c

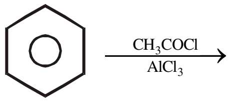
(3) a, b, d (4) All

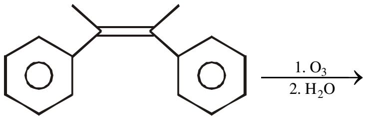
336. Which of the following compounds not shows positive Tollen's test :

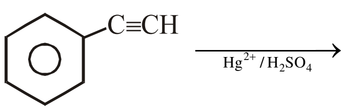
(1)  (2) 

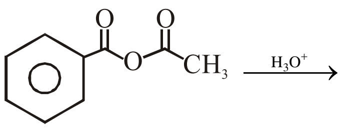
(3)  (4) 

337. Acetophenone not produced in which reaction :

(1) 

(2) 

(3) 

(4) 

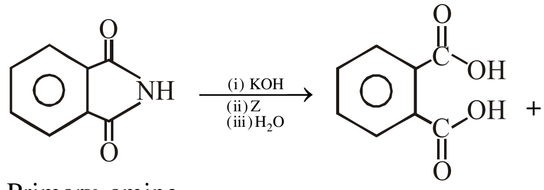
338. Which reaction shows decrease in number of carbon atoms in product :

(1) Decarboxylation reaction

(2) HVZ reaction

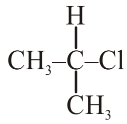
(3) Hoffmann's hypobromamide reaction

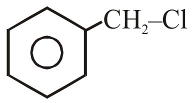
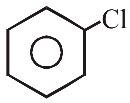
(4) 1 & 3 both

339. 

Primary amine

Z may not be :

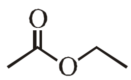
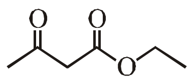
(1) CH_3–Cl (2) 

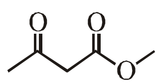
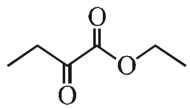
(3)  (4) 

340. A $\xrightarrow{H_3O^+}$ B + EtOH

B $\xrightarrow{\Delta}$ Acetone

A is :

(1)  (2) 

(3)  (4) 

341. CH_3–C≡N $\xrightarrow[2.\ H_3O^+]{1. SnCl_2/HCl}$ Product

Reaction is called :

(1) Stephen reaction for preparation of amine

(2) Stephen reaction for preparation of imine

(3) Stephen reaction for preparation of ketone

(4) Stephen reaction for preparation of aldehyde

342. Correct statement for nucleophilic addition of sodium bisulphite on carbonyl compounds is:

(1) This addition is highly sensitive to crowding, steric crowding increases, then addition decreases, so aldehyde more reactive than ketone in this reaction

(2) This reaction used for separation of aldehyde and ketone from a mixture containing some other compounds because addition product is a crystalline salt

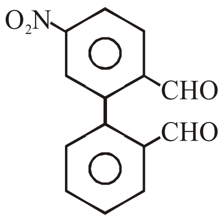
(3) This reaction can be reversed in presence of acid or base, so bisulphite addition product can be convert in to corresponding carbonyl compound.

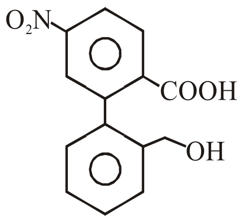
(4) All of these

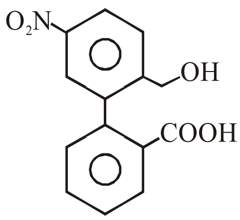
343. Which of the following compound does not form idoform with I_2 + NaOH or NaOI:

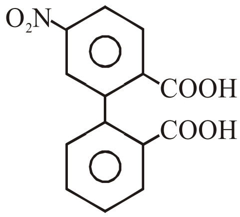
(1) $CH_3–CHO$ (2) $CH_3–CH_2–CHO$

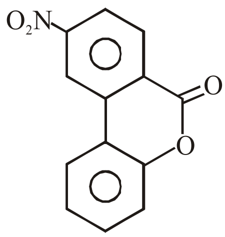
(3) $CH_3–CH_2–OH$ (4) $CH_3–\underset{\underset{O}{\parallel}}{C}–CH_3$

344.  $\xrightarrow[2.\ H^+]{1.\ OH^-}$ Product

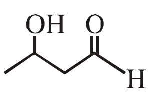
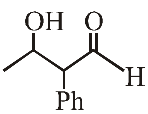
(1) 

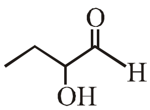
(2) 

(3) 

(4) 

345. Correct order of dehydration of aldol :

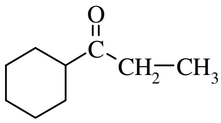
(I)  (II) 

(III) 

(1) II > I > III (2) I > II > III

(3) III > I > II (4) II > III > I

346. Which of the following compound can form hydrazone and will give iodoform test but will not give Tollen's test?

(1) 

(2) $H_3C–\underset{\underset{O}{\parallel}}{C}–CH_2–CH_3$

(3) $H_3C–\underset{\underset{O}{\parallel}}{C}–CH_2–CHO$

(4) $CH_3–CH_2–OH$

347. Propanal on treatment with dilute sodium hydroxide forms

(1) $CH_3CH_2CH_2CH_2CH_2CHO$

(2) $CH_3CH_2CH(OH)CH_2CH_2CHO$

(3) $CH_3CH_2CH(OH)CH(CH_3)CHO$

(4) CH_3CH_2COONa

348. Which of the following reaction will not result in the formation of carbon-carbon bond

(1) Friedel Craft's alkylation

(2) Reimer Tiemann reaction

(3) Cannizzaro reaction

(4) Wurtz reaction

349. Which of the following is most reactive towards nucleophilic attack at carbonyl group

(1) CH_3COCl (2) CH_3COOCH_3

(3) CH_3CONH_2 (4) $CH_3COOCOCH_3$

350. $CH_3CH_2COOH \xrightarrow{SOCl_2} A \xrightarrow[\Delta]{NH_3} B \xrightarrow[KOH]{Br_2} C$

Structure of compound C is

(1) $CH_3CH_2CH_2NH_2$

(2) $CH_3CH_2NH_2$

(3) $CH_3CH_2NHCH_3$

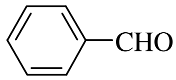
(4) $CH_3CH_2CONH_2$

351. Acetaldehyde and benzaldehyde can not be distinguished by

(1) Tollen's reagent (2) Fehling solution

(3) Benedict solution (4) Iodoform test

352. Which of the following compound does not give positive Fehling's test

(1) CH_3CHO (2) 

(3) $H_3C-\underset{\underset{O}{\|}}{C}-CH_3$ (4) 2 & 3 both

353. $RCH_2OH \xrightarrow{[o]} RCHO$

Which of the following oxidising agent is most suitable for above reaction?

(1) PCC (2) $K_2Cr_2O_7$

(3) $KMnO_4$ (4) All

354. $CH_3-C\equiv CH \xrightarrow[Hg^{+2}]{dil.H_2SO_4} X \xrightarrow[\Delta]{Ba(OH)_2} Y$; Y is

(1) $CH_3-CH_2-CH=CH-\underset{\underset{O}{\|}}{C}-CH_3$

(2) $CH_3-\overset{\overset{CH_3}{|}}{C}=CH-\underset{\underset{O}{\|}}{C}-CH_3$

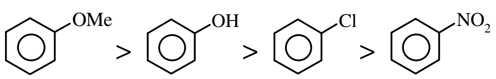
(3) $CH_3-\overset{\overset{CH_3}{|}}{C}=CH-CH_2-CH=O$

(4) $CH_3–CH_2$-CH=CH-CH_2–CH=O

355. Which of the following is correct

(1) CH_3–COCl > CH_3–COOCO–CH_3 > CH_3–$COOC_2H_5$ > CH_3–$CONH_2$ (reactivity towards hydrolysis)

(2) C_6H_5–CH=O > CH_3–CH=O > CH_3–CO–CH_3 (reactivity towards cyanohydrin formation)

(3) 

(reactivity towards electrophilic substitution reaction)

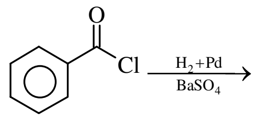
(4) $CH_3–CH_2–CH_2–CH_2$–OH > $H_3C-CH_2-\overset{\overset{OH}{|}}{CH}-CH_3$ > $(CH_3)_3$C–OH

(reactivity towards Lucas reagent)

356. Benzaldehyde and acetaldehyde can be distinguish by :

(1) Hinsberg reagent (2) Tollen's reagent

(3) Baeyer's reagent (4) Iodoform test

357. 

The above reaction is known as

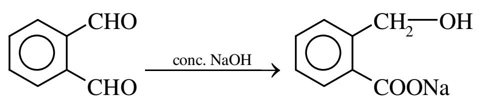
(1) Etard reaction

(2) Rosenmund's reduction

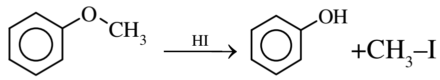
(3) Friedel craft reaction

(4) Clemmensen's reduction

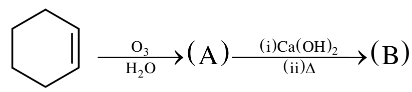
358. Which of the following reaction is not correct according to major product

(1) 

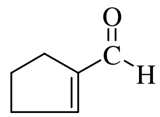
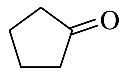
(2) $H_3C-CH_2-\underset{\underset{F}{|}}{CH}-CH_3 \xrightarrow[\Delta]{Alc.\ KOH} CH_3–CH=CH–CH_3$

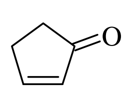
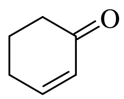
(3) 

(4) $CCl_3-\underset{\underset{O}{\|}}{C}-H \xrightarrow{NaOH} CHCl_3 + HCOONa$

359. 

Product B is

(1)  (2) 

(3)  (4) 

360. In which reaction racemic mixture is obtained as a product?

(1) Ph–CHO+HCN →

(2) $H_3C-\underset{\underset{C_2H_5}{|}}{CH}-CH_2-OH+SOCl_2 \xrightarrow{pyridine}$

(3) $H_3C-\underset{\underset{Cl}{|}}{CH}-CH_2-CH_3 + C_3H_7\overset{\ominus\oplus}{O}Na \xrightarrow[\Delta]{ROH}$

(4) All of these

361. Arrange the following in order of their reactivity towards CH_3MgBr?

(I) $CCl_3-\underset{\underset{O}{\|}}{C}-H$ (II) $H_3C-\underset{\underset{O}{\|}}{C}-CH_3$

(III) $H_3C-\underset{\underset{O}{\|}}{C}-H$ (IV) $Ph-\underset{\underset{O}{\|}}{C}-CH_3$

(1) I > IV > III > II (2) II > I > III > IV
(3) I > II > III >IV (4) I > III > II > IV

362. $\begin{matrix} CHO \\ | \\ CHO \end{matrix} \xrightarrow{NaOH}$ major product

(1) $\begin{matrix} CH_2\text{-}OH \\ | \\ CH_2\text{-}OH \end{matrix}$ (2) $\begin{matrix} COONa \\ | \\ COONa \end{matrix}$

(3) $\begin{matrix} CH_2\text{-}OH \\ | \\ COONa \end{matrix}$ (4) $\begin{matrix} CH_2\text{-}OH \\ | \\ CHO \end{matrix}$

363. Which of the following do not give aldol reaction with dil. NaOH
(1) CH_3CHO (2) CD_3CHO
(3) CH_3COCH_3 (4) $PhCHO$

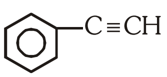
364. How many aldol product will be obtained in the following reaction (Structural only)

$CH_3CHO + PhCHO \xrightarrow{NaOH}$

(1) 2 (2) 3 (3) 4 (4) 1

365. Which of the following contains most acidic hydrogen
(1) CH_3CHO
(2) $CH_3COCH_2COCH_3$
(3) $CH_3COCH_2COOCH_3$
(4) CH_3COCH_3

366. CH_3CHO cna react with
(1) NH_2-NH_2 (2) $NH_2NH-\underset{\underset{O}{\|}}{C}-NH_2$
(3) $Ph-NH-NH_2$ (4) All

367. Which gives aldehyde on hydration by H_2SO_4 and $HgSO_4$
(1) $CH_3-C\equiv CH$
(2) $CH_3-C\equiv C-CH_2-CH_3$
(3) 

(4) $CH\equiv CH$

368. $CH_3-CH=CH-CH_2-CH_2-CN \xrightarrow[2.\ H_2O]{1.\ DIBAL-H}$?
(1) $CH_3(CH_2)_4\,CN$
(2) $CH_3(CH_2)_5\,NH_2$
(3) $CH_3-CH=CH-CH_2-CH_2-CHO$
(4) $CH_3\,(CH_2)_3-CH_2-CHO$

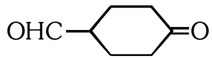
369. In which reaction ethyl phenyl ketone is obtained

(1) $C_6H_5-C\equiv N + Et\,MgBr \rightarrow A \xrightarrow[H^{\oplus}]{H_2O}$

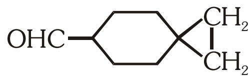
(2) $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-Cl + Ph_2Cd \rightarrow$

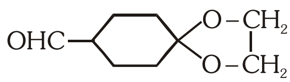
(3)  $+ CH_3-CH_2-\underset{\underset{O}{\|}}{C}-Cl \xrightarrow[AlCl_3]{Anhyd.}$

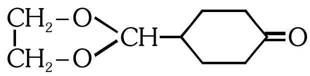
(4) All of them

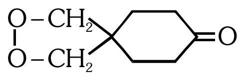
370. 

$+ \begin{matrix} CH_2-OH \\ | \\ CH_2-OH \end{matrix} \xrightarrow{H^{\oplus}}$ A 'A' is
(1 mole)

(1) 

(2) 

(3) 

(4) 

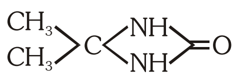
371. $CH_3-\underset{\underset{O}{\|}}{C}-CH_3 + NH_2-\underset{\underset{O}{\|}}{C}-NH-NH_2 \xrightarrow{H^{\oplus}}$
(semicarbazide)

major product is

(1) $CH_3-\underset{\underset{O}{\|}}{C}-CH=\underset{\underset{NH_2}{|}}{C}-NH-NH_2$

(2) $CH_3-\underset{\underset{CH_3}{|}}{C}=N-NH-\underset{\underset{O}{\|}}{C}-NH_2$

(3) $CH_3-\underset{\underset{CH_3}{|}}{C}=N-\underset{\underset{O}{\|}}{C}-NH-NH_2$

(4) 

372. $Ph-\underset{\underset{O}{||}}{C}-CH_3 \xrightarrow{NaOCl} A + \text{salt}$

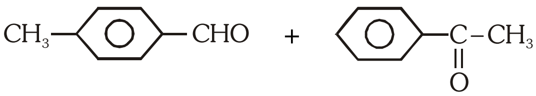
$A + CH_3-\underset{\underset{O}{||}}{C}-CH_3 \rightarrow B$ 'B' is

(1) $CH_3-\underset{\underset{O}{||}}{C}-CH=\underset{\underset{CH_3}{|}}{C}-Ph$

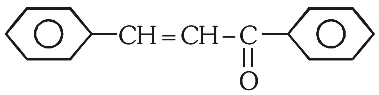
(2) $Ph-\underset{\underset{O}{||}}{C}-CH=CH-\underset{\underset{O}{||}}{C}-CH_3$

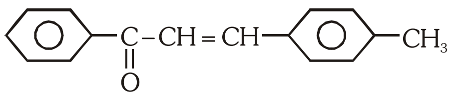
(3) $(CH_3)(H_3C)C(CCl_3)(OH)$

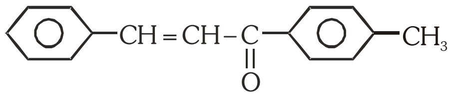
(4) $CH_3-\underset{\underset{Cl}{|}}{\overset{\overset{Cl}{|}}{C}}-CH_3$

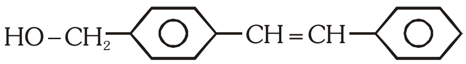
373. 

$\xrightarrow[\Delta]{\overline{O}H}$ product is

(1) 

(2) 

(3) 

(4) 

374. $CH_3-CH=CH-\underset{\underset{O}{||}}{C}-CH_3 + NH_2-OH \xrightarrow{H^{\oplus}}$

major product is

(1) $CH_3-\underset{\underset{NH_2}{|}}{CH}-\underset{\underset{OH}{|}}{CH}-\underset{\underset{O}{||}}{C}-CH_3$

(2) $CH_3-\underset{\underset{OH}{|}}{CH}-\underset{\underset{NH_2}{|}}{CH}-\underset{\underset{O}{||}}{C}-CH_3$

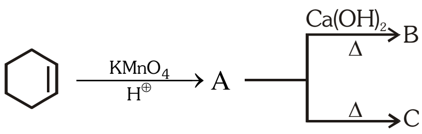
(3) $CH_3-CH=CH-\underset{\underset{CH_3}{|}}{C}=N-OH$

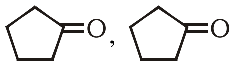
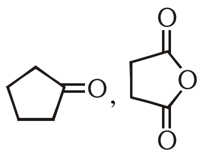
(4) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-\underset{\underset{O}{||}}{C}-CH_3$

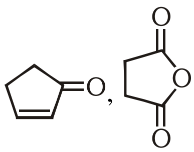
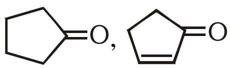
375. Give electrophilicity order of following carbonyl compounds

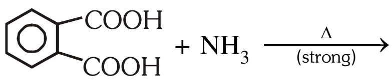
(I) Benzaldehyde
(II) p-Tolualdehyde
(III) P-Nitrobenzaldehyde
(IV) Acetophenone

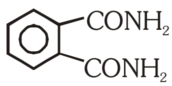
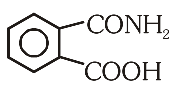
(1) I > IV > III > II (2) III > I > II > IV
(3) IV > II > I > III (4) II > IV > I > III

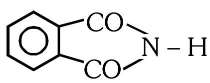
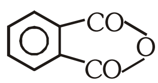
376.  B and C are

(1)  (2) 

(3)  (4) 

377.  A 'A' is

(1)  (2) 

(3)  (4) 

378. $CH_3CH_2COOH + Cl_2 \xrightarrow{Red P} A \xrightarrow[\Delta]{NaOH/CaO} B \xrightarrow[\Delta]{Alc.KOH} C$ 'C' is

(1) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$

(2) $CH_3-CH=CH_2$

(3) $CH_3-CH_2-\underset{\underset{O}{||}}{C}-OR$

(4) $CH_2=CH_2$

379 Carboxylic carbon is.................than carbonyl carbon:

(1) Less electrophilic (2) More electrophilic

(3) less nucleophilic (4) More nucleophilic

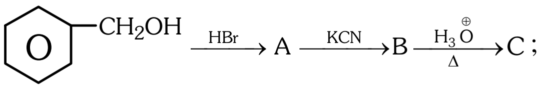
380 Hydrolysis of which of the following gives acid at fastest rate:

(1) $R-\overset{\overset{\displaystyle O}{||}}{C}-OR$

(2) $R-\overset{\overset{\displaystyle O}{||}}{C}-Cl$

(3) $R-\overset{\overset{\displaystyle O}{||}}{C}-NH_2$

(4) $R-\overset{\overset{\displaystyle O}{||}}{C}-O-\overset{\overset{\displaystyle O}{||}}{C}-R$

381 

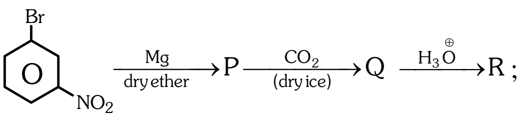
C is–

(1) Phenyl ethanoic acid

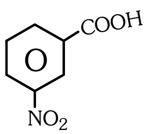
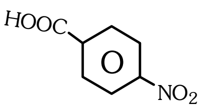
(2) Benzoic acid

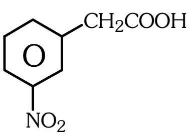
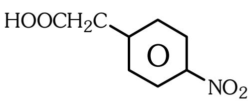
(3) Benzyl amine

(4) Aniline

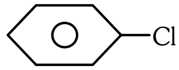
382 

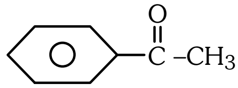
R is–

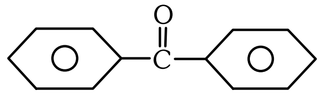
(1)  (2) 

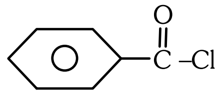
(3)  (4) 

383 $H_3C-COOH \xrightarrow[\Delta]{P_2O_5} A \xrightarrow{PCl_5} B \xrightarrow[AlCl_3]{C_6H_6} C$; C is–

(1) 

(2) 

(3) 

(4) 

384 Which of the following is not correctly matched

Compound	**Use**
(1) Ethanoic acid	Solvent
(2) Ester of benzoic acid	Perfumes
(3) Sodium benzoate	Preservatives
(4) Iodoform	Aneasthetic

385 Identify the end product:

$CH_3CH_2CH_2CHO \xrightarrow[\text{(ii) } H^{\oplus}]{\text{(i) Ammonical } AgNO_3}$

$X \xrightarrow{NaHCO_3} Y \xrightarrow[\Delta]{(NaOH+CaO)} Z$

(1) $CH_3-CH_2CH_2-COOH$

(2) $CH_3-CH_2-CH_2-CH_3$

(3) CH_3-CH_2-COOH

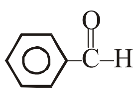
(4) $CH_3-CH_2-CH_3$

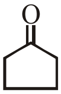
386. Which of the following compound when reacts with CH_3 MgBr formation of 3° alcohol take place?

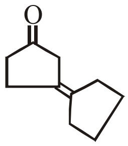
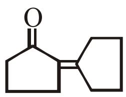
(1) $H-\overset{\overset{\displaystyle O}{||}}{C}-H$

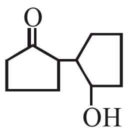
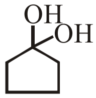
(2) $CH_3-\overset{\overset{\displaystyle O}{||}}{C}-H$

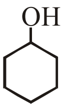
(3) $CH_3-\overset{\overset{\displaystyle O}{||}}{C}-CH_3$

(4) 

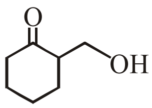
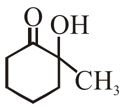
387.  $\xrightarrow[\Delta]{\text{dil. KOH}}$ (A); Product (A) is :-

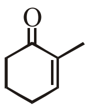
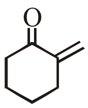
(1)  (2) 

(3)  (4) 

388.  $\xrightarrow{\text{PCC}}$ (A) $\xrightarrow[\text{dil. KOH}/\Delta]{\text{HCHO}}$ (B)

Give structure of (B) is :

(1)  (2) 

(3)  (4) 

389. $CH_3–CH=CH_2 \xrightarrow[\text{Reagent}]{\text{Baeyer's}} X \xrightarrow{HIO_4}$ product

Product is :

(1) $CH_3–CH=O$

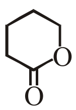
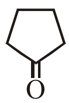
(2) $H_2C=O$

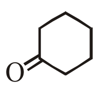
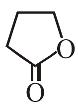
(3) $CH_3–CO–CH_3$

(4) (1) and (2) both

390.  $\xrightarrow{KMnO_4} X \xrightarrow{Ca(OH)_2} Y \xrightarrow{\Delta} Z$;

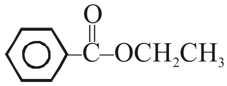
Z is

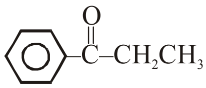
(1)  (2) 

(3)  (4) 

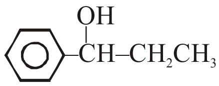
391. The final product (C) obtained in the reaction sequence is

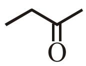
$CH_3CH_2COOH \xrightarrow{PCl_3} (A) \xrightarrow[AlCl_3]{C_6H_6} (B) \xrightarrow[\text{Base, heat}]{NH_2NH_2} (C)$

(1) 

(2) 

(3) 

(4) 

392.  $\xrightarrow[\text{(ii) y-moles } NaNH_2]{\text{(i) x}PCl_5} CH_3–C \equiv C–CH_3$

Sum of (x + y = ?)

(1) 2 (2) 3 (3) 4 (4) 5

393. Alkene (X) (C_5H_{10}) on ozonolysis gives a mixture of two compounds (Y) and (Z). Compound (Y) and (Z) gives positive Fehling's test, (Z) does not give idoform test. Compounds (X), (Y) and (Z) are :-

	X	Y	Z
(1)	$CH_2=C(CH_3)-CH_2CH_3$	HCHO	$CH_3–CO–CH_2CH_3$
(2)	$CH_3–CH=C(CH_3)–CH_3$	CH_3CHO	CH_3COCH_3
(3)	$CH_3CH_2CH=CH_2$	CH_3CH_2CHO	HCHO
(4)	$CH_3CH=CHCH_2CH_3$	CH_3CHO	CH_3CH_2CHO

394. Buta-1, 3-diene was subjected to ozonolysis to prepare aldehydes. Which of the following aldehydes will be obtained during the reaction?

(1) $\begin{matrix} CHO \\ | \\ CHO \end{matrix}$ + 2HCHO

(2) CH_3CHO + 2HCHO

(3) $CH_3CH_2CHO + CH_3CHO$

(4) $2CH_3CH_2CHO$

395. In following reactions identify the product (X).

$CH_3CHO + H-CHO \xrightarrow[\text{Heat}]{\text{dil.NaOH}} X$

(1) $CH_2=CH-\underset{\substack{|\\OH}}{CH}-COOH$

(2) $\underset{\substack{|\\OH}}{CH_2}-CH_2-COOH$

(3) $\underset{\substack{|\\OH}}{CH_2}-CH_2-CHO$

(4) $CH_2=CH-CHO$

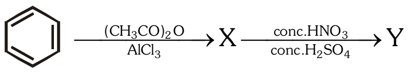
396. In the following sequence of reaction, the final product (Z) is :

$CH\equiv CH \xrightarrow{Hg^{2+}/\text{dil. }H_2SO_4} X \xrightarrow[H_2O]{CH_3MgX} Y \xrightarrow{[O]} Z$

(1) ethanal (2) propan-2-ol

(3) propanone (4) propan-1-ol

397. Identify the products (X) and (Y) in the given reaction:



(1) X = Acetophenone,
Y = m-Nitroacetophenone

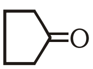
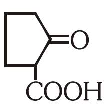
(2) X = Toluene,
Y = m-Nitroacetotoluene

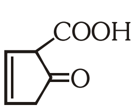
(3) X = Acetophenone,
Y = o and p-Dinitroacetophenone

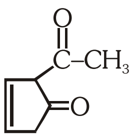
(4) X = Benzaldehyde,
Y = m-Nitrobenzaldehyde

398. $CH_3-O-\underset{\substack{||\\O}}{C}-CH_2-CH_2-CH_2-CH_2-\underset{\substack{||\\O}}{C}-O-CH_3$

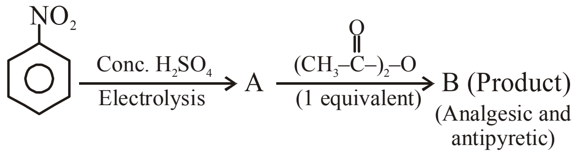
$\xrightarrow{CH_3-\overset{-}{O}\overset{+}{Na}} A \xrightarrow{H_2O} B \xrightarrow{\Delta} C$?

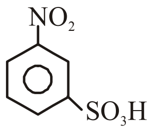
(1) 
(2) 

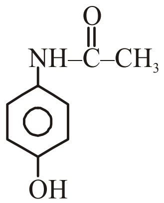
(3) 

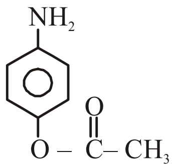
(4) 

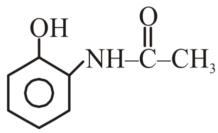
399. The B product of following reaction will be :-

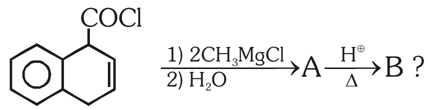


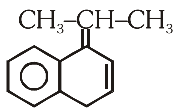
(1) 

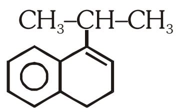
(2) 

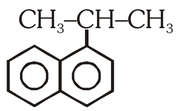
(3) 

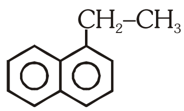
(4) 

400. 

(1) 

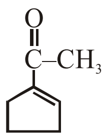
(2) 

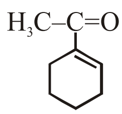
(3) 

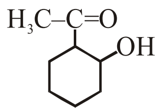
(4) 

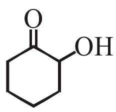
401. $CH_3-\overset{\substack{O\\||}}{C}-CH_2-CH_2-CH_2-CH_2-CH_2-\overset{\substack{O\\||}}{C}-H$

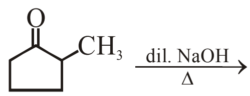
$\xrightarrow[\Delta]{\text{dil. NaOH}}$ (A) product is

(1) 

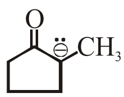
(2) 

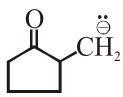
(3) 

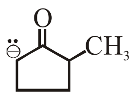
(4) 

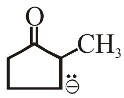
402. 

In the above Aldol condensation attacking nucleophile will be ?

(1) 

(2) 

(3) 

(4) 

403. Which of the following cannot undergo Intra molecular Cannizzaro Reaction ?

(1) $H-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-H$ (2) $H-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-Ph$

(3) $CH_3-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-CH_3$ (4) $H-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-D$

404. $H-\overset{O}{\overset{\|}{C}}-H \xrightarrow{OH^{\ominus}}$ product ?

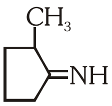
(1) $H-COO^{\ominus} + HCHO$

(2) $H-COO^{\ominus} + H-COO^{\ominus}$

(3) $HCOO^{\ominus} + CH_3OH$

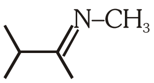
(4) $HCHO + CH_3OH$

405. Which final product gives iodoform test ?

(1)  $\xrightarrow[H^{\oplus}]{H_2O}$?

(2) $CH_3-CH_2-\underset{\underset{O}{\|}}{C}-O-\underset{\underset{CH_3}{|}}{C}=CH-CH_3 \xrightarrow{H_3O^{\oplus}}$?

(3) $Ph-C\equiv C-CH_3 \xrightarrow[\text{dil. } H_2SO_4]{Hg^{+2}}$?

(4)  $+ CH_3MgBr \xrightarrow{H_2O}$?

406. In which of the following group, each member gives positive iodoform test ?

(1) methanol, ethanol, propanone

(2) ethanol, isopropanal, methanal

(3) ethanal, ethanol, isopropyl alcohol

(4) propanal, 2-propanol, propanone

407.

$$CH_3-\overset{OH}{\overset{|}{C}H}-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-CH_3 \rightarrow CH_3-\overset{OH}{\overset{|}{C}H}-(CH_2)_3-CH_3$$

Above conversion can be achieved by :-

(1) Wolf-Kishner reduction

(2) Clemmensen reduction

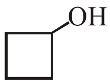
(3) $LiAlH_4$

(4) $NaBH_4$

408. Compound (X) C_4H_8O, which reacts with 2, 4-DNP derivative but gives negative haloform test is :-

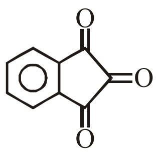
(1) $CH_3-\overset{O}{\overset{\|}{C}}-CH_2-CH_3$

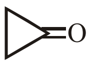
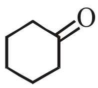
(2) $CH_3-\underset{\underset{CH_3}{|}}{C}H-CHO$

(3) 

(4) $CH_3-CH_2-\overset{OH}{\overset{|}{C}H}-CH_3$

409. Which of the following does not form a stable hydrate on addition of H_2O ?

(1) $Ph-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-Ph$ (2) 

(3)  (4) 

410. $Ph-\overset{OH}{\overset{|}{C}H}-CH_3 \xrightarrow{PCC} (A) \xrightarrow{NH_2-NH-\overset{O}{\overset{\|}{C}}-NH_2} (B)$

Product (B) is :-

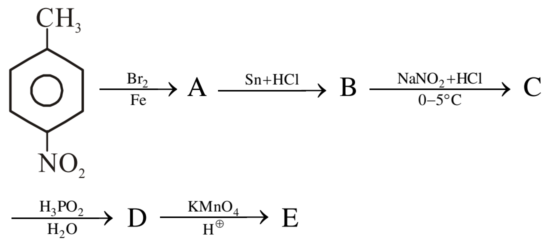
(1) $Ph-\overset{CH_3}{\overset{|}{C}}=N-\overset{O}{\overset{\|}{C}}-NH-NH_2$

(2) $Ph-\underset{\underset{CH_3}{|}}{C}=N-NH-\overset{O}{\overset{\|}{C}}-NH_2$

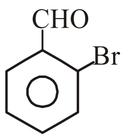
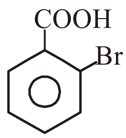
(3) $Ph-CH=N-\underset{\underset{CH_3}{|}}{N}-\overset{O}{\overset{\|}{C}}-NH_2$

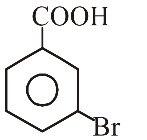
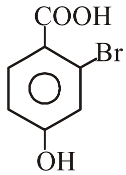
(4) $Ph-CH=N-\overset{O}{\overset{\|}{C}}-NH_2$

NITROGEN CONTAINING COMPOUND

411. 

End product E will be :

(1)  (2) 

(3)  (4) 

412. Which is Sandmeyer reaction among the following :

(1) $C_6H_5N_2Cl \xrightarrow[\Delta]{HBF_4} C_6H_5F$

(2) $C_6H_5N_2Cl \xrightarrow{Cu/HCl} C_6H_5Cl$

(3) $C_6H_5N_2Cl \xrightarrow{Cu_2Cl_2/HCl} C_6H_5Cl$

(4) $C_6H_5N_2Cl \xrightarrow{C_6H_6/NaOH} C_6H_5Cl$

413. Arrange the following in decreasing order of boiling point :

(I) $CH_3CH_2CH_2OH$ (II) $CH_3CH_2CH_2NH_2$

(III) $CH_3-N(CH_3)-CH_3$ (IV) $CH_3CH_2-NH-CH_3$

(V) $CH_3CH_2CH_2CH_3$

(1) I > II > IV > III > V

(2) III > IV > II > I > V

(3) I > V > II > III > IV

(4) II > I > III > IV > V

414. A compound reacts with Hinsberg's reagent and gives a product which is soluble in alkali solution (NaOH solution) compound is :

(1) $C_2H_5NH_2$ (2) $C_2H_5NHC_2H_5$

(3) $C_6H_5NHC_2H_5$ (4) $C_2H_5-N(C_2H_5)-C_2H_5$

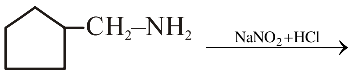
415. Which of the following compound cannot be formed by Gabriel phthalimide synthesis :

(1) 

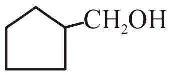
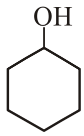
(2) 

(3) $(CH_3)_2CH-NH_2$

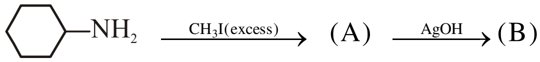
(4) $(C_2H_5)(CH_3)CH-NH_2$

416. 

Find out major product :

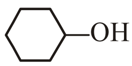
(1)  (2) 

(3)  (4) 

417. 

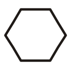
$\xrightarrow{\Delta}$ (C) + (D)

Product (C) and (D) will be :

(1)  + $(CH_3)_3N$

(2)  + $(CH_3)_3N$

(3)  + CH_3-OH

(4)  + CH_3-OH

418. $CH_3Cl \xrightarrow{KCN} A \xrightarrow[Hydrolysis]{Partial} B$; $B \xrightarrow{PCl_5} C$; $B \xrightarrow{LAH} D$

Products C and D are respectively :

(1) CH_3COCl and CH_3CH_2OH

(2) CH_3COCl and $CH_3CH_2NH_2$

(3) CH_3CN and $CH_3CH_2NH_2$

(4) CH_3CH_2Cl and CH_3CH_2OH

419. Which of the following conversion is incorrect:

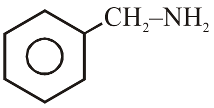
(1) $CH_3CN \xrightarrow[\text{or LAH}]{Na/C_2H_5OH} CH_2CH_2NH_2$

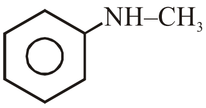
(2) $CH_3CONH_2 \xrightarrow{Br_2/KOH} CH_3CH_2NH_2$

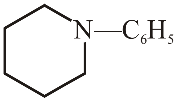
(3) $CH_3CONH_2 \xrightarrow{NaNO_2+HCl} CH_3COOH$

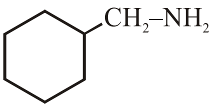
(4) $CH_3CONH_2 \xrightarrow{P_2O_5} CH_3CN$

420. Which of the following compound gives yellow oily liquid nitrosamine when react with nitrous acid:

(1) 

(2) 

(3) 

(4) 

421. Aniline can be obtained by:

(a) $C_6H_5COOH \xrightarrow[H^{\oplus}]{N_3H}$

(b) $C_6H_5CONH_2 \xrightarrow[\Delta]{Br_2/KOH}$

(c) $C_6H_5COOH \xrightarrow[\Delta]{NH_3}$

(d) $C_6H_5NO_2 \xrightarrow{Fe+H_2O}$

(1) a, b, c (2) a, b

(3) a, b, d (4) a, b, c, d

422. Aniline cannot be obtained by :

(1) $C_6H_5NO_2 \xrightarrow[\text{Electrolysis}]{\text{conc. } H_2SO_4}$

(2) $C_6H_5Cl \xrightarrow[\Delta]{NaNH_2}$

(3) $C_6H_5COCl \xrightarrow[(KOH)]{NaN_3}$

(4) $C_6H_5OH \xrightarrow[\text{Dry } ZnCl_2]{NH_3}$

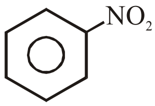
423. Aniline can be prepared by :

(1) $C_6H_5NO_2 \xrightarrow[\text{Electrolysis}]{\text{Conc. } H_2SO_4}$

(2) $C_6H_5NO_2 \xrightarrow[\text{Electrolysis}]{\text{Dil. } H_2SO_4}$

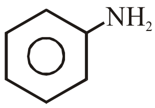
(3) $C_6H_5NO_2 \xrightarrow{\text{Aq.sodium arsenite}}$

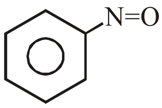
(4) $C_6H_5NO_2 \xrightarrow{Zn+NaOH(H_2O)}$

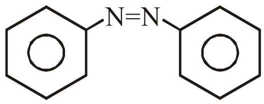
424. 

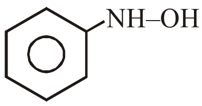
$\xrightarrow{Zn+NH_4Cl}$ A

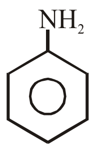
Compound A is :

(1) 

(2) 

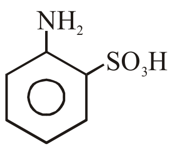
(3) 

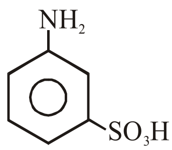
(4) 

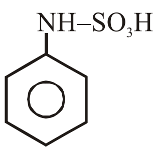
425. 

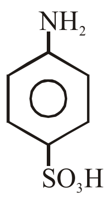
$\xrightarrow{H_2SO_4}$ A $\xrightarrow{453–473K}$ B

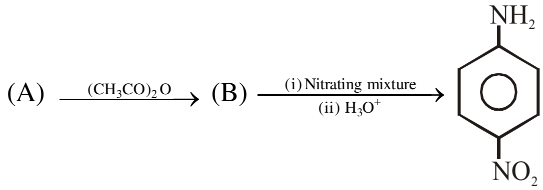
Compound B is :

(1) 

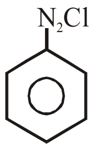
(2) 

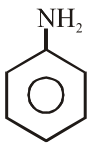
(3) 

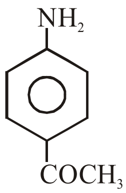
(4) 

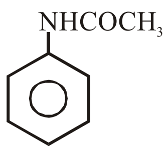
426. 

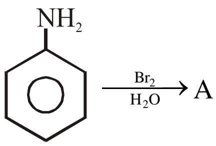
Find out (A) :

(1) 

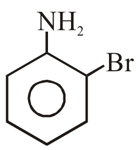
(2) 

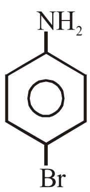
(3) 

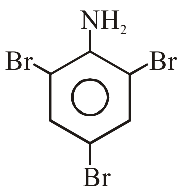
(4) 

427. 

Product (A) is :

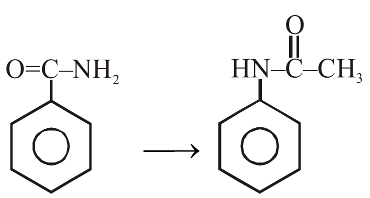
(1) 

(2) 

(3) 

(4) Both (1) and (2)

428. The reagents needed to convert :

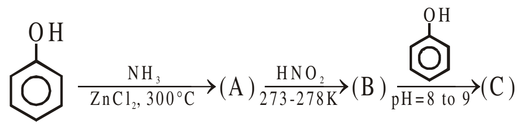


(1) KOH, Br_2 ; $LiAlH_4$

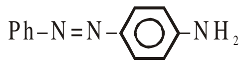
(2) KOH, Br_2 ; CH_3COCl

(3) KOH, Br_2 ; Ni,H_2 ; CH_3COCl

(4) HONO ; $(CH_3CO)_2O$

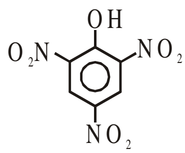
429. 

Final product ?

(1) 

(2) Ph–N=N–Ph

(3) 

(4) 

430. Ethylamine on heating with CS_2 in presence of $HgCl_2$ forms :

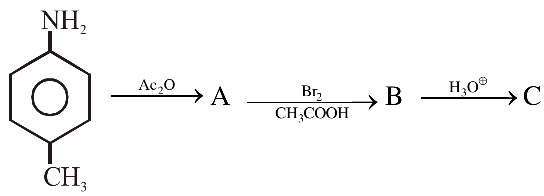
(1) C_2H_5CNS (2) C_2H_5NCS

(3) C_2H_5CNO (4) C_2H_5SH

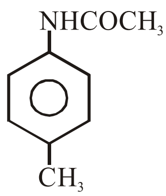
431. Leakage of which gas was responsible for the Bhopal gas tragedy in 1984 :

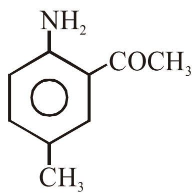
(1) CH_3NCO (2) CH_3NCS

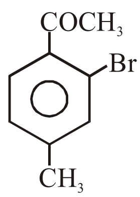
(3) CH_3CNO (4) CH_3CNS

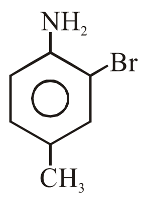
432. 

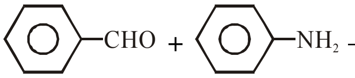
"C" would be :

(1) 

(2) 

(3) 

(4) 

433.  $\longrightarrow$ Product

Product A is called :

(1) Azo dye (2) Schiff reagent

(3) Schiff base (4) Glyptal (a polymer)

434. Benzoylation reaction of $-NH_2$ group is known as :

(1) Balz-Schiemann's reaction

(2) Schotten-Baumann's reaction

(3) Gabriel phthalimide reaction

(4) Friedel craft reaction

435. Aryl fluoride may be prepared from arene diazonium chloride using

(1) CuF/HF (2) Cu/HF

(3) HBF_4/Δ (4) $HBF_4/NaNO_2$, Cu

436. On heating an aliphatic primary amine with chloroform and ethanolic KOH, the organic compound formed is

(1) an alkyl cyanide (2) an alkyl isocyanide

(3) an alkanol (4) an alkanediol

437. $H_3C-\overset{\overset{\large O}{||}}{C}-NH_2 \xrightarrow{PCl_5} A \xrightarrow[H_2O]{LiAlH_4} B$

Final product B is

(1) A primary alcohol

(2) A primary amine

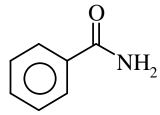
(3) A secondary amine

(4) A carboxylic acid

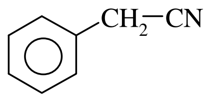
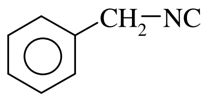
438. Which of the following compound gives a product with foul smell when reacted with chloroform in presence of a alkali?

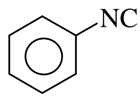
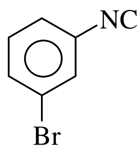
(1) $(CH_3)_2NH$ (2) CH_3NH_2

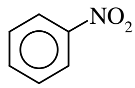
(3) $H_3C-\overset{\overset{\large O}{||}}{C}-NH_2$ (4) $H_3C-\overset{\overset{\large CH_3}{|}}{N}-CH_3$

439.  $\xrightarrow{Br_2+KOH(aq)} X \xrightarrow[\Delta]{CHCl_3+KOH} Y$;

Y is

(1)  (2) 

(3)  (4) 

440.  $\xrightarrow{Fe+HCl}$ Product is

(1) Phenyl hydroxylamine

(2) p-amino phenol

(3) azoxybenzene

(4) aniline

441. $R-CN \xrightarrow[HCl]{SnCl_2} \xrightarrow{H_3O^{\oplus}} R-CHO$

Above reaction is

(1) Mendius reaction

(2) Stephen's reaction

(3) Rosenmund reaction

(4) Perkin reaction

442. Which of the following has maximum Boiling point

(1) Methoxy ethane (2) n-Butane

(3) Propanal (4) Acetone

443. Aniline can not be prepared by Gabriel phthalimide synthesis due to–

(1) Bulky nature of Benzene Ring

(2) Aryl halide do not undergo nucleophilic substitution.

(3) Aryl halide do not undergo electrophilic substitution reaction

(4) Both (1) & (3)

444. Which one can give propanamine on reaction with $LiAlH_4$

(1)Propyl cyanide

(2)N-Methyl Propanamide

(3)Propane nitrile

(4)Both (1) & (3)

445. 'A' $\xrightarrow{Br_2+NaOH} CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-NH_2$

Find out the structure of compound 'A'

(1) $CH_3-\underset{\underset{CH_3}{|}}{CH}-\overset{\overset{O}{\|}}{C}-NH_2$

(2) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-\overset{\overset{O}{\|}}{C}-NH_2$

(3) $CH_3-\overset{\overset{CH_3}{|}}{CH}-\overset{\overset{O}{\|}}{C}-OH$

(4) $CH_3-CH_2-\overset{\overset{CH_3}{|}}{CH}-\overset{\overset{O}{\|}}{C}-NH_2$

446. Which of the following have maximum value of pK_b in Aqueous phase–

(1)N-Methyl aniline

(2)N,N-diethyl ethanamine

(3)Benzenamine

(4)Phenyl methanamine

447. Isocyanide test is used for detection of ?

(1) Primary amine

(2) Secondary amine

(3) Only aniline

(4) Tert. amine

448. Which one is soluble in NaOH :–

(1)N-Ethyl benzenesulphonamide

(2)N,N-diethyl benzenesulphonamide

(3)N-ethyl aniline

(4)N-methyl ethanamine

449. Benzendiazonium chloride + 'X' $\xrightarrow{H^{\oplus}}$ Yellow dye. 'X' in above reaction is

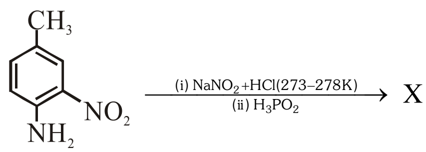
(1) Phenol

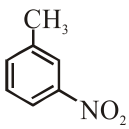
(2) Aniline

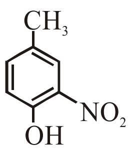
(3) Benzene

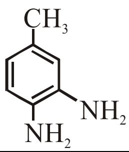
(4) Benzene sulphonic acid

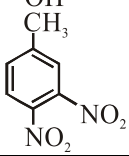
450. The structure of product X in the following reaction is :-



(1) 

(2) 

(3) 

(4) 

451. Identify X and Y in the reaction.

$CH_3-\underset{\underset{CH_3}{|}}{CH}-CONH_2 \xrightarrow{Br_2/NaOH} X \xrightarrow{HNO_2} Y$

(1) $X = CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2NH_2$;

$Y = CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-OH$

(2) $X = CH_3CH{=}CHNH_2$; $Y = CH_3CH{=}CHOH$

(3) $X = CH_3-\underset{\underset{CH_3}{|}}{CH}-NH_2$; $Y = CH_3-\underset{\underset{CH_3}{|}}{CH}-OH$

(4) $X = CH_3CH_2CH_2NH_2$; $Y = CH_3CH_2CH_2OH$

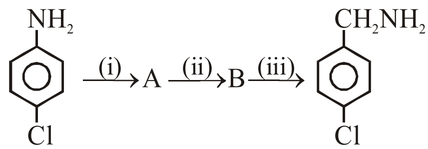
452. Identify 'Z' in the sequence :

$C_6H_5NH_2 \xrightarrow[273\ K]{NaNO_2 + HCl} X \xrightarrow{CuCN} Y \xrightarrow[Boil]{H^+/H_2O} Z$

(1) C_6H_5CN (2) $C_6H_5CONH_2$

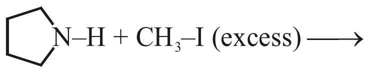
(3) C_6H_5COOH (4) $C_6H_5CH_2NH_2$

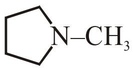
453. Mark the correct route of the conversion of p-chloroaniline to p-chlorobenzylamine.

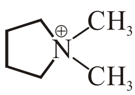


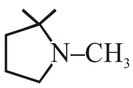
	(i)	**(ii)**	**(iii)**
(1)	Alkylation	KCN	H_2/Pt
(2)	Diazotisation	CuCN	H_2/Pt
(3)	Oxidation	H_2/Pt	Hydrolysis
(4)	Diazotisation	H_2O/H^+	Sn/HCl

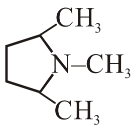
454. Which is the following compound is formed in the given reaction ?



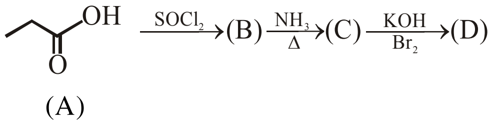
(1) 

(2) 

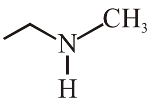
(3) 

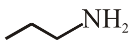
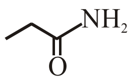
(4) 

455. In a set of reactions propionic acid yielded a compound D.

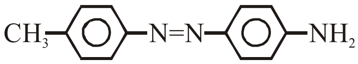


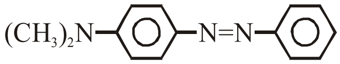
The structure of (D) would be :

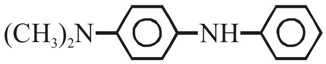
(1)  (2) 

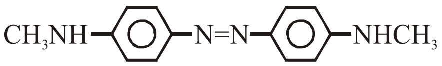
(3)  (4) 

456. Aniline when diazotised in cold and then treated with N, N-dimethylaniline gives a coloured product. The structure of this product is :-

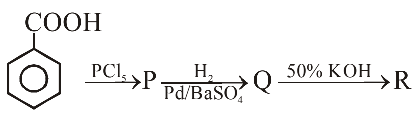
(1) 

(2) 

(3) 

(4) 

457. What is the end product in the following sequence of reactions ?



(1) Acid + Alcohol
(2) Ester
(3) Aldol
(4) Salt + Alcohol

458. A compound (X) with molecular formula C_3H_9N reacts with $C_6H_5SO_2Cl$ to give a solid which is insoluble in alkali. (X) is :-

(1) $CH_3CH_2CH_2NH_2$ (2) $CH_3-\ddot{N}(CH_3)-CH_3$

(3) $CH_3-NH-CH_2CH_3$ (4) $CH_3-CH(CH_3)-NH-Ph$

459. Which of the following compound will give Hoffmann bromamide reaction ?

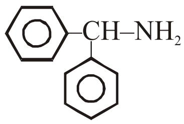
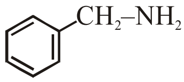
(1) $Ph-C(=O)-NH-CH_3$

(2) $CH_3-N(CH_3)-CH_3$

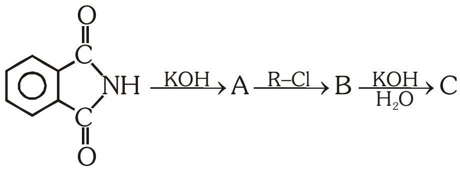
(3) $Ph-NH_2$

(4) $CH_3-CH_2-C(=O)-NH_2$

460. Which of the following compound will not give Carbylamine Reaction ?

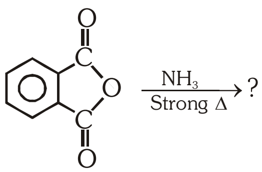
(1)  (2) 

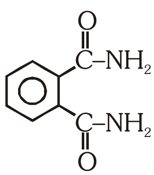
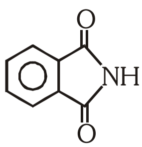
(3) $CH_3-C(CH_3)_2-NH_2$ (4) 

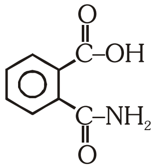
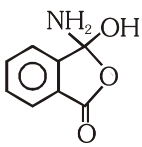
461. 

'C' can not be ?

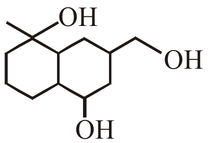
(1) $CH_3-CH_2-NH_2$ (2) $Ph-CH_2-NH_2$
(3) $(CH_3)_2CH-NH_2$ (4) $Ph-NH_2$

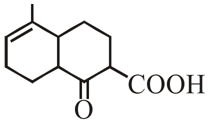
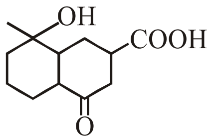
462. 

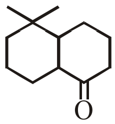
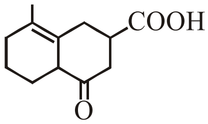
(1)  (2) 

(3)  (4) 

OXIDATION & REDUCTION

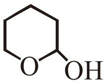
463.  $\xrightarrow{KMnO_4/H^{\oplus}}$ Product :-

(1)  (2) 

(3)  (4) 

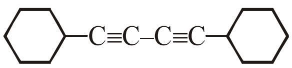
464. Which of the following compound will not give positive Tollens test :-

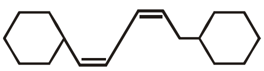
(1) CH_3CHO

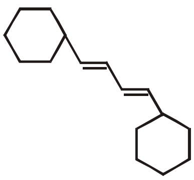
(2) 

(3) $CH_3CH_2CH_2CH_2CH(OH)–OCH_3$

(4) $CH_3–CH(OCH_3)_2$

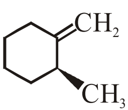
465.  $\xrightarrow{Li/NH_3(l)}$ product

(1) 

(2) 

(3) Both (1) and (2)

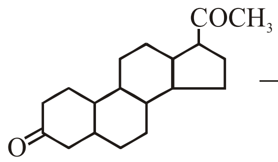
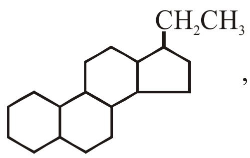
(4) None of these

466.  $\xrightarrow{H_2,Pd}$

Products of the above reaction will be :-

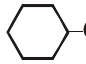
(1) Recemic mixture (2) Diastereomers

(3) Meso (4) Structural isomer

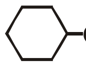
467.  $\xrightarrow{X}$ , X can be

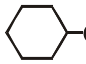
(1) $NH_2–NH_2/KOH$ (2) Zn–Hg/HCl

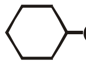
(3) Red P + HI (4) All

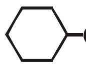
468. B $\xleftarrow{NaBH_4}$ –CH=CH–CH_2OH $\xrightarrow{H_2/Pd}$ A

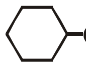
A and B are respectively :-

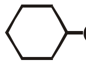
(1) –CH_2CH_2CHO,

–CH=CH–CH_2OH

(2) –$CH_2CH_2CH_2OH$,

–CH=CH–CH_2OH

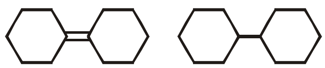
(3) –CH=CH–CHO in both case

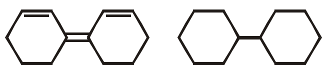
(4) –$CH_2CH_2CH_2OH$ in both case

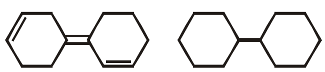
469. 2  $\xrightarrow[\Delta]{Conc.\ H_2SO_4}$ A $\xrightarrow{Pt/H_2}$ B

A and B respectively are :

(1) Both 

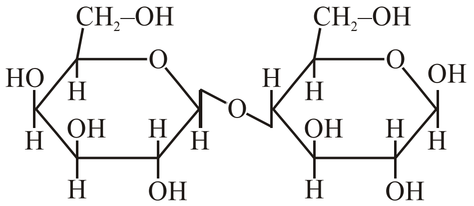
(2) 

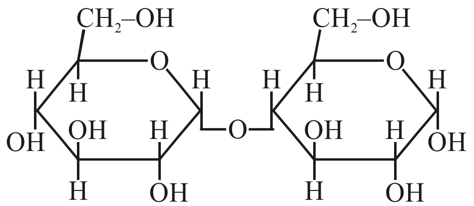
(3) 

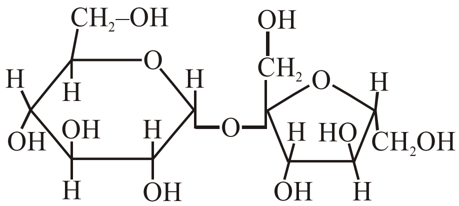
(4) 

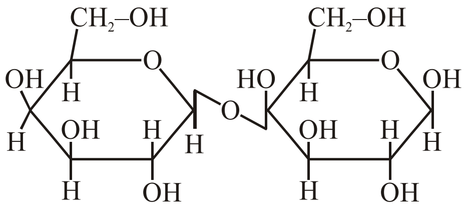
BIOMOLECULE, POLYMERS AND CHEMISTRY IN EVERYDAY LIFE

470. Which of the following disaccharide is a non-reducing sugar :

(1) 

(2) 

(3) 

(4) 

471. Lysine, $H_2N–(CH_2)_4–\underset{\displaystyle NH_2}{\underset{|}{C}H}–COOH$ is :

(a) α-amino acid

(b) Basic amino acid

(c) Amino acid synthesis in body

(d) β-Amino acid

(1) a, b, c (2) b, c (3) b, d (4) a, b

472. Which of the following reaction of glucose can be explained only by its cyclic structure :

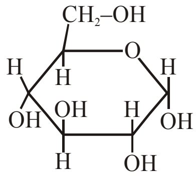
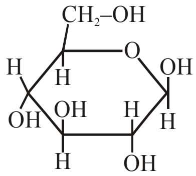
(1) Glucose forms pentaacetate

(2) Glucose reacts with hydroxylamine to form an oxime

(3) Pentacetate of glucose does not react with hydroxylamine

(4) Glucose is oxidised by Bromine water to gluconic acid

473. Relation between (A) and (B) is :

(A)  (B) 

(1) Enantiomers (2) Anomers

(3) Positional isomers (4) Functional isomers

474. $\begin{array}{l} CHO \\ | \\ (CH–OH)_4 \\ | \\ CH_2–OH \end{array} \xrightarrow{HIO_4}$ Product

(1) 5 HCHO +HCOOH

(2) 5 HCOOH + HCHO

(3) CH_3COOH + 5HCHO

(4) $5CH_3CHO$ + HCHO

475. Which is not correct match :

Polymer	*Monomers*
(1) Orlon	Acrylonitrile
(2) Nylon-6,6	Hexamethylene diamine and adipic acid
(3) Dacron	Ethylene glycol and terephthalic acid
(4) Caprolactum	Nylon-6, 10

476. Buna-S is a polymer of :

(1) Butadiene only

(2) Buta-1, 3-diene and styrene

(3) Styrene only

(4) Butadiene and acrylonitrile

477. Which one of the following is not a polyamide polymer :

(a) Nylon-6,6 (b) Nylon-6,10

(c) Glyptal (d) Dacron

(1) a, b (2) c, d

(3) a, c (4) a, b, c

478. Which of the following polymer is biodegradable?

(1) $(-CH_2-\underset{\underset{Cl}{|}}{C}=CH-CH_2-)_n$

(2) $(-CH_2-CH=CH-CH_2-CH_2-\overset{\overset{CN}{|}}{C}H-)_n$

(3) $(-O-\underset{\underset{CH_3}{|}}{C}H-CH_2-\underset{\underset{O}{||}}{C}-O-\underset{\underset{CH_2-CH_3}{|}}{C}H-CH_2-\underset{\underset{O}{||}}{C}-)_n$

(4) $(-\overset{\overset{H}{|}}{N}-(CH_2)_6-\overset{\overset{H}{|}}{N}-\overset{\overset{O}{||}}{C}-(CH_2)_4-\overset{\overset{O}{||}}{C}-)_n$

479. Which of the following show mutarotation:

(a) Glucose (b) Sucrose

(c) Maltose (d) Galactose

(1) a, b, c (2) b, c

(3) a, c, d (4) b, d

480. Glucose and mannose are :

(1) Anomers (2) Position isomers

(3) Functional isomers (4) Epimers

481. Invert sugar is :

(1) Sucrose (2) Fructose

(3) Glucose (4) Galactose

482. Glucose and fructose form identical osazones because :

(1) they are monosaccharide

(2) they are reducing sugar

(3) they are epimer

(4) their constitution differ only at C–1 and C–2

483. In purine nucleoside, C-1 of sugar form glycosidic linkage with which position of purine :

(1) 1 (2) 2

(3) 9 (4) 8

484. Which of the following is sulphur containing amino acid :

(a) Methionine (b) Cysteine

(c) Leucine (d) Glutamic acid

(1) a, b, c (2) a, b

(3) b, c (4) c, d

485. A metal present in vitamin B-12 is:

(1) Aluminium (2) Zinc

(3) Iron (4) Cobalt

486. Correct statement for Bakelite is :

(a) It is formed by phenol and formaldehyde

(b) It is crosslinked polymer

(c) It is thermosetting plastic

(d) It is natural polymer

(1) a, d (2) a, b, c

(3) a, b, d (4) c, d

487. Which of the following polymers can have strong intermolecular forces :

(a) Nylon (b) Dacron

(c) Rubber (d) Polyester

(1) a, b (2) a, d

(3) a, b, d (4) b, c, d

488. Match the column :

	Polymer		Commercial name
(i)	Polyester of glycol and phthalic acid	(a)	Novolac
(ii)	Copolymer of 1,3-butadiene and styrene	(b)	Glyptal
(iii)	Phenol and formaldehyde resins	(c)	Buna-S
(iv)	Polyester of glycol and terephthalic acid	(d)	Buna-N
(v)	Copolymer of 1,3-butadiene and acrylonitrile	(e)	Dacron

(1) i-b, ii-c, iii-e, iv-a, v-d

(2) i-b, ii-c, iii-a, iv-e, v-d

(3) i-e, ii-c, iii-a, iv-b, v-d

(4) i-e, ii-c, iii-b, iv-a, v-d

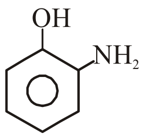
489. Match the column :

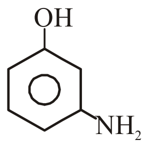
	Polymer		Linkage
(i)	Terylene	(a)	Glycosidic linkage
(ii)	Cellulose	(b)	Ester linkage
(iii)	Protein	(c)	Phosphodiester linkage
(iv)	RNA	(d)	Amide linkage

(1) i-b, ii-a, iii-c, iv-d (2) i-b, ii-a, iii-d, iv-c

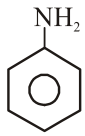
(3) i-d, ii-b, iii-a, iv-c (4) i-a, ii-b, iii-d, iv-c

490. Which of the following gives paracetamol on acetylation :

(1) 

(2) 

(3) 

(4) 

491. Aspirin is chemically :

(1) Methyl benzoate

(2) Ethyl salicylate

(3) Acetyl salicylic acid

(4) o-hydroxy benzoic acid

492. Artificial sweetner which is stable under cold conditions only is :

(1) Saccharin (2) Sucralose

(3) Aspartame (4) Alitame

493. Which of the following is food preservative :

(a) Salt of sorbic acid

(b) Sodium benzoate

(c) Salt of propanoic acid

(d) Methanol

(1) a, b, d (2) a, b, c

(3) c, d (4) a, d

494. Which of the following statements are correct about barbiturates :

(a) Hypnotics or sleep producing agents

(b) These are tranquilizers

(c) Non-narcotic analgesics

(d) Pain reducing without disturbing the nervous system

(1) a, b (2) a, b, c

(3) b, c (4) c, d

495. Which of the following compounds are administered as antacids :

(a) Sodium carbonate

(b) Sodium hydrogen carbonate

(c) Aluminium carbonate

(d) Magnesium hydroxide

(1) a, b (2) b, d (3) a, b, c (4) c, d

496. Amongst the following antihistamines which are antacids :

(a) Ranitidine

(b) Cimetidine

(c) Aspirin

(d) Paracetamol

(1) a, b (2) a, b, c (3) b, c (4) c, d

497. Veronal and luminal are derivatives of barbituric acid which are :

(a) Tranquilizers

(b) Non-narcotic analgesic

(c) Antiallergic drugs

(d) Neurologically active drugs

(1) a, d (2) a, b (3) a, c (4) b, c

498. Bithional is generally added to soaps as an additive to function as an :

(1) Dryer

(2) Antiseptic

(3) Softner

(4) Antibiotic

499. Match the column :

	Column-I Structure		Column-II Type of detergents
(i)	$CH_3(CH_2)_{16}COO(CH_2CH_2O)_n\,CH_2CH_2OH$	(a)	Cationic detergent
(ii)	$C_{17}H_{35}COO^-Na^+$	(b)	Anionic detergent
(iii)	$CH_3-(CH_2)_{10}CH_2OS\overset{\ominus}{O}_3\overset{\oplus}{Na}$	(c)	Nonionic detergent
(iv)	$\left[CH_3(CH_2)_{15}-\underset{CH_3}{\overset{CH_3}{N}}-CH_3\right]^{\oplus}\overset{\ominus}{Br}$	(d)	Soap

(1) i-a, ii-b, iii-c, iv-d

(2) i-a, ii-b, iii-d, iv-c

(3) i-c, ii-d, iii-b, iv-a

(4) i-c, ii-d, iii-a, iv-b

500. The species which can best serve as an initiator for the cationic polymerisation is

(1) HNO_3 (2) $AlCl_3$

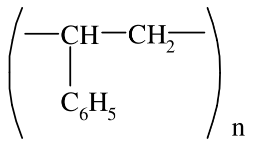
(3) BuLi (4) $LiAlH_4$

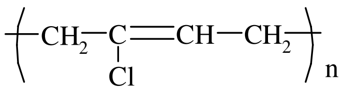
501. Which artificial sweetener contains chlorine?

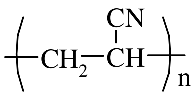
(1) Aspartame (2) Saccharin

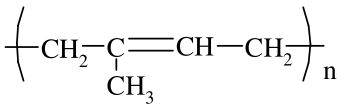
(3) Sucralose (4) Alitame

502. Which of the following structures represents Neoprene polymer

(1) 

(2) 

(3) 

(4) 

503. Which of the following hormones contains iodine

(1) Thyroxine (2) Insulin

(3) Testosterone (4) Adrenaline

504. α-D-(+)glucose and β-D-(+)glucose are

(1) position isomers (2) anomers

(3) enantiomers (4) conformers

505. Glucose does not give positive test with

(1) 2,4-DNP (2) Tollen's reagent

(3) Fehling's solution (4) All

506. Which of the following pair does not show correct relationship?

(1) Isoprene : Natural rubber

(2) Nylon : Polyamide

(3) Nylon-6 : Adipic acid

(4) Ziegler Natta catalyst : $TiCl_4 + (C_2H_5)_3Al$

507. Incorrect match is

(1) Nylon-6 : Caprolactum

(2) Buna-N : Elastomer

(3) Alitame : Artificial sweetener

(4) Tranquilizer : Ranitidine

508. Correct statement for α-D-glucose is

(1) Can show mutarotation

(2) Does not give 2,4-DNP test

(3) Shows reducing property with Tollen's reagent

(4) All of these

509. Which of the following is not correctly matched

(1) Nylon-6 polyamide

(2) Buna-S Co-polymer

(3) Bakelite Thermoplastic

(4) Polythene Addition polymer

510. Which of the following is not correctly matched

(1) Antacid-Ranitidine

(2) Sucrose-Reducing sugar

(3) Antihistamine-Seldane

(4) Lactose-Milk sugar

511. Which of the following is not a monosaccharide?

(1) Glucose (2) Fructose

(3) Cellulose (4) D-Ribose

512. Proteins give white precipitate with Millon's reagent, which is –

(1) Mercurous and mercuric nitrate in HNO_3

(2) Mercurous and mercuric chloride in HCl

(3) Mercurous and mercuric chloride in HNO_3

(4) None

513. The end product of protein digestion is –

(1) Peptides (2) Peptones

(3) Protones (4) α–Amino acids

514. "Kwashiorkor" is a disease caused by the deficiency of

(1) Vitamins

(2) Hormones

(3) Blood

(4) Essential amino acids

515. Which carbohydrate is used in silvering of mirrors

(1) Sucrose (2) Cellulose

(3) Starch (4) Glucose

516. Natural rubber is also called as

(1) cis-1, 4-polyisoprene

(2) trans-1, 4-polyisoprene

(3) cis-1, 4-neoprene

(4) trans-1, 4-neoprene

517. Buna-N is copolymer of

(1) 1, 3-butadiene and Vinyl chloride

(2) 1, 3-Butadiene and acrylonitrile

(3) chloroprene + vinyl chloride

(4) chloroprene and acrylonitrile

518. PHBV is a polymer of
(1) 3-Hydroxy butanoic acid and 3-Hydroxy pentanoic acid
(2) 2-Hydroxy butanoic acid and 2-Hydroxy pentanoic acid
(3) 4-Hydroxy butanoic acid and 4-Hydroxy pentanoic acid
(4) 3-Hydroxy butanoic acid and 2-Hydroxy pentanoic acid

519. Which of the following is prepared by condensation poylmerisation ?
(1) Styrene (2) Nylon-6,6
(3) Teflon (4) Rubber

520. Novolac is
(1) Linear condensation product of phenol and formaldehyde
(2) Cross linked condensation product of urea and formaldehyde
(3) Cross linked condensation product of phenol and formaldehyde
(4) Linear condensation product of urea and formaldehyde

521. Hydrolysis of oil by caustic alkali is called –
(1) Esterification (2) Saponification
(3) Acetylation (4) Carboxylation

522. Which of the following oil can be used for the preparation of soap by alkaline hydrolysis
(1) Paraffin oil (2) Coconut oil
(3) Kerosine oil (4) Turpentine oil

523. Hydrolysis of fats and oils yields –
(1) Dihydric alcohol
(2) Trihydric alcohol
(3) Esters
(4) Monohydric alcohol

524. Which of the following detergent belong to ABS–group –
(1) Sodium Lauryl sulphate
(2) Sodium dodecyl benzene sulphonate
(3) Cetyltrimethyl ammonium bromide
(4) Pentaerythrityl stearate

525. Which of the following statement is not true –
(1) Some disinfectants can be used as antiseptic at low concentration
(2) Sulphapyridine is a synthetic antibacterial
(3) Ampicillin and Amoxycillin are natural antibiotic
(4) Aspirin is analgesic and antipyretic both.

526. Which of the following medicines are the derivatives of malonyl urea. (Barbituric acid)
(1) Antibiotics (2) Antiseptics
(3) Analgesics (4) Tranquilizers

527. Which of the following medicines is used in the treatment of Typhoid –
(1) Penicillin (2) Bithional
(3) Norethindrone (4) Chloramphenicol

528. Dettol consists of –
(1) Cresol + ethanol
(2) Xylenol + terpineol
(3) Chloroxylenol + Terpineol
(4) None of these

529. Which is incorrect statement –
(1) Boric acid in dilute aqueous solution is weak antiseptic for eye.
(2) Chloramphenicol is used in the treatment of typhoid, eye infection and pneumonia.
(3) Broad spectrum antibiotics are effective against both gram positive and gram negative bacterias.
(4) Bithional is used as antifertility drug.

530. Which chemical is resposible for inflammation in the tissue and causes pain ?
(1) Hydrochloric acid (2) Histamine
(3) Prostaglandine (4) All of these

531. Which is incorrect statement ?
(1) Bithional is added to soap to impart antiseptic properties
(2) Sulphanilamide is used in the formation of sulpha drugs.
(3) Aspartame is the most successful and widely used artificial sweetener
(4) Soframycin is used as an disinfectant.

ENVIRONMENTAL CHEMISTRY AND QUALITATIVE AND QUANTITATIVE ANALYSIS

532. Which of the following is biodegradable pollutant?
(1) Nuclear waste (2) DDT
(3) Vegetable matter (4) Plastic

533. The region of atmosphere where human beings live?
(1) Troposphere (2) Stratosphere
(3) Biosphere (4) Exosphere

534. Stratosphere contains :
(1) N_2 (2) O_2
(3) O_3 (4) All of these

535. Ozone is a toxic gas, yet its presence is essential in the atmosphere because :
(1) It protects from harmful UV radiation
(2) It protects from infrared radiation
(3) It maintains temperature of earth to optimum level
(4) It reduces poisonous gases to non-poisonous gases

536. Which of the following is not categorised as particulate pollutant ?
(1) Dust (2) Smoke
(3) Smog (4) Carbon dioxide

537. What is DDT among the following?
(1) Non-biodegradable pollutant
(2) Green house gas
(3) A fertilizer
(4) Biodegradable pollutant

538. Major contributor of global warming is:
(1) Carbon dioxide
(2) Carbon monooxide
(3) Methane
(4) Chlorofluoro carbon

539. Which of the following is not a green house gas ?
(1) CH_4 (2) O_3
(3) CO (4) H_2O vapour

540. BOD value less than 5 ppm indicates as water sample to be :
(1) rich in dissolved oxygen
(2) poor in dissolved oxygen
(3) highly polluted
(4) not suitable for aquatic life

541. Acid rain is due to increase in atmospheric concentration of :
(1) Ozone and dust (2) CO_2 and CO
(3) SO_3 and CO (4) SO_2 and NO_2

542. Ozone in the stratosphere is depleted by :
(1) C_7H_{16} (2) CF_2Cl_2
(3) $C_6H_6Cl_6$ (4) C_6F_6

543. Taj Mahal is threatened by pollution from ?
(1) Sulphur dioxide (2) Hydrogen sulphide
(3) Chlorine (4) Carbon dioxide

544. Eutrophication causes reduction in :
(1) Dissolved hydrogen
(2) Dissolved oxygen
(3) Dissolved salts
(4) All of these

545. If BOD value of water is 20 ppm then what can we conclude regarding water ?
(1) Polluted water
(2) Less polluted water
(3) Pure water
(4) Highly polluted water

546. Which one of the following is not a common component of photochemical smog ?
(1) O_3 (2) $\underset{\text{Acrolein}}{CH_2=CH-CHO}$
(3) $CH_3-\underset{\underset{O}{||}}{C}-O-O-NO_2$ (4) CFC
Peroxyacetyl nitrate (PAN)

547. Which of the following statement is false for classical smog ?
(1) It occurs in cool, humid climate
(2) It is composed of smoke, fog and SO_2
(3) It is reducing in nature
(4) It is oxidising in nature

548. What is range of pH of acid rain ?
(1) More than 5.6 (2) In between 6 to 7
(3) Less than 5.6 (4) In between 6.5 to 7.5

549. Which of the following statements is wrong ?
(1) Ozone can oxidise sulphur dioxide present in the atmosphere to sulphur trioxide
(2) Chloro fluoro carbons are supposed to be main cause of ozone layer depletion
(3) Ozone is produced in upper stratosphere by action of UV rays
(4) Ozone is not responsible for green house effect

550. Which of the following condition show polluted environment ?
(1) Eutrophication
(2) BOD is 4 ppm
(3) pH of rain water is 5.6
(4) Amount of CO_2 in the atmosphere is 0.03%

551. The prescribed upper limit concentration of lead in drinking water is about :
(1) 30 ppb (2) 70 ppb
(3) 50 ppb (4) 90 ppb

552. The consequence of global warming may be :
(1) Increase in average temperature of earth
(2) Melting of himalyan glaciers
(3) Increased BOD
(4) Both (1) and (2)

553. Which of the following statements about photochemical smog is wrong ?
(1) Photochemical smog is formed through photochemical reaction involving solar energy
(2) Plantation of some plants like pinus helps in controlling photochemical smog
(3) Photochemical smog has high concentration of oxidising agent
(4) Photochemical smog does not cause irritation to throat and eyes

554. Which of the following block delivery of oxygen to organs and tissues ?
(1) CO (2) CO_2 (3) SO_2 (4) O_3

555. Blue baby syndrome is caused by excess of :
(1) $F^{\ominus}$ (2) $NO_3^{\ominus}$ (3) Pb (4) SO_4^{2-}

556. Green chemistry deals with study of ?
(1) Plant physiology
(2) Reactions involved in synthesis of chlorophyll
(3) Synthesis of chemical compound using green light
(4) Existing knowledge and priciples of chemistry and other sciences to reduce the adverse impact on environment

557. Which of the folllowing are primary precursors of photochemical smog ?
(1) CO_2 and PAN
(2) NO_2 and hydrocarbons
(3) O_3 and PAN
(4) NO_2 and PAN

558. Which of the following belong to secondary air pollutant ?
(1) CO
(2) Hydrocarbon
(3) Peroxyacetyl nitrate (PAN)
(4) NO

559. In Duma's method for estimation of nitrogen 0.4 gm of an organic compound gave 60 ml of nitrogen collected at 300 K temperature and 720 mm pressure. Calculate the percentage composition of nitrogen in the compound : (Aqueous tension at 300 K = 20 mm)
(1) 16.72% (2) 15.93%
(3) 15.72% (4) 7.46%

560. Carbon and hydrogen are estimated in organic compounds by :
(1) Kjeldahl's method (2) Duma's method
(3) Liebig's method (4) Carius method

561. Lassaigne's test for the detection of nitrogen will fail in case of :
(1) NH_2CONH_2
(2) $NH_2CONHNH_2.HCl$
(3) $NH_2NH_2.HCl$
(4) $C_6H_5NHNH_2.2HCl$

562. In a Lassaigne's test for sulphur in the organic compound with sodium nitroprusside solution the violet colour formed is due to :
(1) $Na_4[Fe(CN)_5NOS]$
(2) $Na_3[Fe(CN)_5S]$
(3) $Na_2[Fe(CN)_5NOS]$
(4) $Na_3[Fe(CN)_6]$

563. During Lassaigne's test, N and S both present in an organic compound changes into :
(1) Na_2S and NaCN
(2) NaSCN
(3) Na_2SO_4 and NaCN
(4) Na_2S and NaCNO

564. The purpose of boiling sodium extract with conc. HNO_3 before testing for halogen is :
(1) to make solution acidic
(2) to make solution clear
(3) to convert Fe^{+2} to Fe^{+3}
(4) to convert NaCN to HCN and Na_2S to H_2S so that they do not interfere with $AgNO_3$

565. In Duma's method and Kjeldahl's method, respectively nitrogen present is estimated as :
(1) N_2, NH_3 (2) NH_3, N_2
(3) NO_2, NH_3 (4) N_2, N_2

566. The sodium extract of an organic compound on acidification with acetic acid and addition of lead acetate solution gives a black precipitate. The organic compound contains :
(1) Nitrogen (2) Halogen
(3) Sulphur (4) Phosphorous

567. When N and S both are present in an organic compound, the sodium extract with $FeCl_3$ gives :
(1) Green colour
(2) Blue colour
(3) Yellow colour
(4) Red colour

568. During Lassaigne's test nitrogen containing organic compound when fused with sodium metal forms 'X' while sulphur containing organic compound when fused with sodium metal forms 'Y'. Then identify X and Y :
(1) X = NaCN ; Y = Na_2S
(2) X = NaNC, Y = Na_2S
(3) X = $NaNO_2$, Y = Na_2SO_4
(4) X = NaCN ; Y = Na_2SO_4

569. In Lassaigne's test, the organic compound is fused with sodium metal as to :
(1) Hydrolyse the compound
(2) Form a sodium derivative
(3) Burn the compound
(4) convert nitrogen, sulphur or halogen if present into soluble ionic sodium salt

570. Aniline is separated from aniline-water mixture by :
(1) Simple distillation
(2) Fractional distillation
(3) Steam distillation
(4) Distillation under reduced pressure

571. o-nitrophenol and p-nitrophenol are separated by :
(1) Steam distillation
(2) Simple distillation
(3) Fractional distillation
(4) Distillation under reduced pressure

572. Glycerol is separated from spent lye by :
(1) Steam distillation
(2) Vacuum distillation
(3) Sublimation
(4) Simple distillation

573. Two volatile and miscible liquids can be separated by fractional distillations into pure components under the condition when :
(1) They have low boiling points
(2) The difference in boiling point is large
(3) The boiling points of two liquids are close to each other
(4) They do not form azeotropic mixture

574. A mixture contain four solid organic compounds A, B, C and D. On heating only C changes from solid to vapour state directly. C can be separated from rest in the mixture by :
(1) Distillation
(2) Sublimation
(3) Fractional distillation
(4) Crystallisation

575. Simple distillaton technique can be used to separate ?
(1) Chloroform and Aniline
(2) Ether and Toluene
(3) Hexane and Toluene
(4) All of these

576. Mixture of acetone and methyl alcohol can be separated by :

(1) Simple distillation

(2) Fractional distillation

(3) Sublimation

(4) Chromatography

577. In column chromatography, the moving phase is :

(1) The substances that are to be separate

(2) Solvent (Eluant)

(3) Adsorbent

(4) Mixture of eluant and substance to be separated

578. The principle involved in paper chromatography is :

(1) Adsorption (2) Partition

(3) Solubility (4) Volatility

579. A mixture of naphthalene and benzoic acid can be separated by :

(1) Chromatography

(2) Sublimation

(3) Fractional crystallisation

(4) Distillation

580. A sample of 0.5 g of an organic compound was analysed using Kjeldahl's method. The ammonia evolved was absorbed in 50 ml of 0.5 M H_2SO_4. the unused acid after neutralisation by ammonia consumed 80 ml of 0.5N NaOH. Then calculate percentage of nitrogen in organic compound :

(1) 28 (2) 42 (3) 56 (4) 26

581. In Kjeldahl's method used for estimation of nitrogen, ammonia evolved from 0.6 g of sample of organic compound neutralised 20 ml of 1 N H_2SO_4. Then calculate % of nitrogen in that compound :

(1) 37.33 (2) 46.67

(3) 45.77 (4) 43.33

582. In the estimation of sulphur by carius method 0.480 g of organic compound give 0.699 g of Barium sulphate. The percentage of sulphur in the compound is (Atomic masses ; Ba = 137, S = 32, O =16)

(1) 15% (2) 35%

(3) 20% (4) 30%

583. 1.4 g of an organic compound was digested according to Kjeldahl's method and the ammonia evolved was absorbed in 60 mL of M/10 H_2SO_4 solution. The excess sulphuric acid requires 20 mL of M/10 NaOH solution for neutralisation. The percentage of nitrogen in the compound is

(1) 3 (2) 5

(3) 24 (4) 10

584. In Duma's method of estimation of nitrogen 0.35 g of an organic compound gave 55 ml of nitrogen collected at 300 K temp. and 715 mm pressure. The percentage of nitrogen in compound is (aqueous tension at 300 K = 15 mm)

(1) 15.45 (2) 16.45

(3) 17.45 (4) 14.45

585. Which compound is responsible for the pH of normal rain water 5.6 ?

(1) SO_2 (2) NO_2

(3) CO_2 (4) All

586. Which of the following concentration of metals in drinking water is not in prescribed limits –

(1) Fe 0.2 ppm

(2) Cu 3.0 ppm

(3) Cd 2.0 ppm

(4) Zn 5.0 ppm

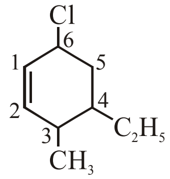
ORGANIC CHEMISTRY ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**2**	**2**	**3**	**4**	**3**	**3**	**4**	**3**	**4**	**2**	**2**	**3**	**3**	**2**	**1**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**3**	**4**	**2**	**2**	**2**	**2**	**3**	**3**	**4**	**4**	**2**	**1**	**2**	**1**	**3**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**3**	**3**	**4**	**3**	**4**	**4**	**2**	**1**	**4**	**4**	**3**	**3**	**2**	**1**	**2**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**3**	**2**	**2**	**4**	**4**	**2**	**2**	**4**	**2**	**2**	**2**	**3**	**1**	**2**	**2**
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	**3**	**2**	**2**	**2**	**3**	**1**	**1**	**4**	**1**	**3**	**4**	**3**	**4**	**1**	**1**
Que.	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Ans.	**1**	**3**	**3**	**3**	**2**	**3**	**3**	**4**	**2**	**4**	**2**	**3**	**2**	**4**	**3**
Que.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Ans.	**3**	**2**	**3**	**4**	**3**	**3**	**3**	**2**	**2**	**1**	**3**	**3**	**2**	**2**	**3**
Que.	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	**1**	**1**	**1**	**2**	**3**	**2**	**1**	**3**	**3**	**2**	**4**	**4**	**3**	**1**	**3**
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Ans.	**3**	**3**	**3**	**3**	**2**	**1**	**1**	**2**	**4**	**3**	**2**	**3**	**2**	**3**	**1**
Que.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Ans.	**1**	**1**	**1**	**2**	**3**	**1**	**3**	**1**	**3**	**3**	**2**	**2**	**3**	**1**	**2**
Que.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Ans.	**3**	**3**	**2**	**3**	**3**	**4**	**3**	**1**	**3**	**3**	**4**	**4**	**4**	**1**	**3**
Que.	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	**3**	**1**	**2**	**1**	**3**	**2**	**1**	**4**	**4**	**2**	**4**	**2**	**3**	**2**	**3**
Que.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
Ans.	**2**	**3**	**4**	**4**	**3**	**4**	**3**	**4**	**3**	**3**	**1**	**1**	**2**	**2**	**3**
Que.	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210
Ans.	**2**	**1**	**3**	**2**	**1**	**4**	**3**	**2**	**2**	**4**	**1**	**3**	**3**	**2**	**3**
Que.	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225
Ans.	**2**	**2**	**1**	**2**	**2**	**1**	**3**	**3**	**3**	**3**	**3**	**1**	**3**	**2**	**2**
Que.	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240
Ans.	**4**	**1**	**2**	**4**	**2**	**2**	**1**	**4**	**1**	**4**	**3**	**4**	**3**	**3**	**4**
Que.	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255
Ans.	**4**	**4**	**4**	**1**	**2**	**3**	**3**	**3**	**2**	**3**	**2**	**4**	**4**	**3**	**1**
Que.	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270
Ans.	**3**	**4**	**2**	**3**	**3**	**1**	**4**	**2**	**2**	**2**	**3**	**4**	**2**	**3**	**4**
Que.	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285
Ans.	**3**	**4**	**3**	**3**	**2**	**4**	**1**	**4**	**3**	**2**	**1**	**4**	**1**	**2**	**2**
Que.	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300
Ans.	**4**	**2**	**1**	**2**	**4**	**1**	**2**	**3**	**3**	**3**	**4**	**3**	**3**	**3**	**2**
Que.	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315
Ans.	**1**	**2**	**2**	**1**	**3**	**2**	**2**	**2**	**3**	**1**	**4**	**4**	**3**	**3**	**4**
Que.	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330
Ans.	**1**	**2**	**1**	**4**	**4**	**3**	**1**	**4**	**4**	**4**	**4**	**4**	**4**	**4**	**3**
Que.	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345
Ans.	**4**	**4**	**1**	**4**	**2**	**4**	**4**	**4**	**4**	**2**	**4**	**4**	**2**	**1**	**1**

Que.	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360
Ans.	**2**	**3**	**3**	**1**	**2**	**1**	**4**	**1**	**2**	**1**	**4**	**2**	**2**	**2**	**1**
Que.	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375
Ans.	**4**	**3**	**4**	**1**	**2**	**4**	**4**	**3**	**4**	**3**	**2**	**3**	**2**	**3**	**2**
Que.	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390
Ans.	**1**	**3**	**4**	**1**	**2**	**1**	**1**	**2**	**4**	**4**	**3**	**2**	**4**	**4**	**2**
Que.	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405
Ans.	**3**	**2**	**4**	**1**	**4**	**3**	**1**	**1**	**2**	**3**	**2**	**3**	**3**	**3**	**2**
Que.	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420
Ans.	**3**	**1**	**2**	**4**	**2**	**2**	**3**	**1**	**1**	**1**	**2**	**2**	**3**	**2**	**2**
Que.	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435
Ans.	**2**	**1**	**2**	**4**	**4**	**2**	**3**	**2**	**3**	**2**	**1**	**4**	**3**	**2**	**3**
Que.	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450
Ans.	**2**	**2**	**2**	**3**	**4**	**2**	**4**	**2**	**3**	**2**	**3**	**1**	**1**	**2**	**1**
Que.	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465
Ans.	**3**	**3**	**2**	**2**	**2**	**2**	**4**	**3**	**4**	**4**	**4**	**2**	**2**	**4**	**2**
Que.	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480
Ans.	**2**	**4**	**2**	**2**	**3**	**4**	**3**	**2**	**2**	**4**	**2**	**2**	**3**	**3**	**4**
Que.	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495
Ans.	**1**	**4**	**3**	**2**	**4**	**2**	**3**	**2**	**2**	**3**	**3**	**3**	**2**	**1**	**2**
Que.	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510
Ans.	**1**	**1**	**2**	**3**	**2**	**3**	**2**	**1**	**2**	**1**	**3**	**4**	**4**	**3**	**2**
Que.	511	512	513	514	515	516	517	518	519	520	521	522	523	524	525
Ans.	**3**	**1**	**4**	**4**	**4**	**1**	**2**	**1**	**2**	**1**	**2**	**2**	**2**	**2**	**3**
Que.	526	527	528	529	530	531	532	533	534	535	536	537	538	539	540
Ans.	**4**	**4**	**3**	**4**	**3**	**4**	**3**	**1**	**4**	**1**	**4**	**1**	**1**	**3**	**1**
Que.	541	542	543	544	545	546	547	548	549	550	551	552	553	554	555
Ans.	**4**	**2**	**1**	**2**	**4**	**4**	**4**	**3**	**4**	**1**	**3**	**4**	**4**	**1**	**2**
Que.	556	557	558	559	560	561	562	563	564	565	566	567	568	569	570
Ans.	**4**	**2**	**3**	**3**	**3**	**3**	**1**	**2**	**4**	**1**	**3**	**4**	**1**	**4**	**3**
Que.	571	572	573	574	575	576	577	578	579	580	581	582	583	584	585
Ans.	**1**	**2**	**3**	**2**	**4**	**2**	**4**	**2**	**2**	**1**	**2**	**3**	**4**	**2**	**3**
Que.	586														
Ans.	**3**														

ORGANIC CHEMISTRY SOLUTIONS

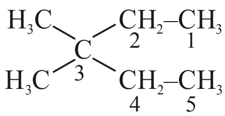
NOMENCLATURE

5. 

Numbering decide by lowest set/locant rule

10. $H_5C_6-\overset{\overset{O}{\|}}{\underset{1}{C}}-\underset{2}{CH_2}-\underset{3}{CH_3}$

1-Phenylpropan-1-one

12. 

3,3-Dimethylpentane

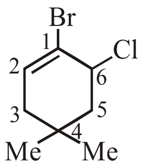
14. $\underset{5}{H_3C}-\overset{\overset{O}{\|}}{\underset{4}{C}}-\underset{3}{CH_2}-\underset{\underset{\underset{1}{CN}}{2|}}{CH}-CH_2-CH_3$

2-Ethyl-4-oxopentanenitrile

Priority $-\overset{\overset{O}{\|}}{C}- < -C\equiv N$

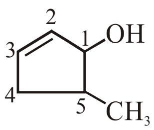
15. $H_3C-\underset{\underset{CH_3}{4\ |\ 3}}{CH}-\overset{\overset{O}{\|}}{\underset{2}{C}}-\overset{\overset{O}{\|}}{\underset{1}{C}}-O-CH_2-Cl$ (PFG is ester)

Chloromethyl 3-methyl-2-oxobutanoate

20.  Follow lowest set rule

23. $H_3C-CH_2-NH-\overset{\overset{O}{\|}}{C}-H$ (PFG is amide)

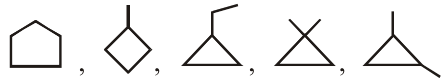
N–Ethylmethanamide

25. 

Priority order for numbering

PFG > Multiple bond > side chain/Subsituents

ISOMERISM

27. 

5 structural isomers

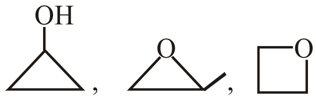
30. Dou = 1 $\Rightarrow$ one π bond or one cycle

a) π bonded

$CH_3-CH_2-\overset{\overset{O}{\|}}{C}-H$, $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$, $CH_3-CH=\overset{\overset{OH}{|}}{CH}$

$CH_2=CH-\overset{\overset{OH}{|}}{CH_2}$, $H_3C-\overset{\overset{OH}{|}}{C}=CH_2$, $H_2C=CH-OCH_3$

b) Cyclic



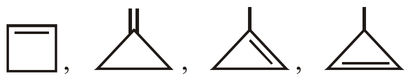
Total 9 structural isomers

31. C_4H_6 DOU = 2 $\Rightarrow$ a) Two π bond
b) One π + one cycle
c) Two cycle

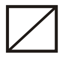
a) Two π bond

C–C–C≡C, C–C≡C–C, C=C–C=C,
C=C=C–C

b) One π + one cycle



c) Two cycle

 (Total 9 structures)

32. $C_3H_9N \Rightarrow$ DOU = Zero (Only σ bond)

$H_3C-CH_2-CH_2-NH_2$, $H_3C-\overset{\overset{NH_2}{|}}{CH}-CH_3$,

$CH_3-CH_2-NH-CH_3$, $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{N}}$

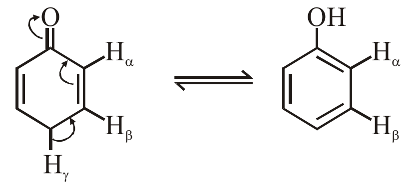
Total 4-structures

33. $H-\overset{\overset{O}{\|}}{C}-O-CH_2-CH_2-CH_3$, $H-\overset{\overset{O}{\|}}{C}-O-CH\begin{smallmatrix}CH_3\\CH_3\end{smallmatrix}$

$H_3C-\overset{\overset{O}{\|}}{C}-O-CH_2-CH_3$, $H_3C-CH_2-\overset{\overset{O}{\|}}{C}-O-CH_3$

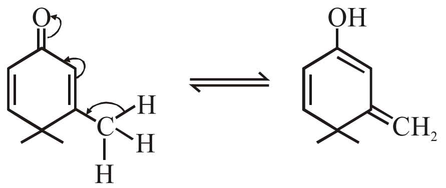
Total 4-structures

34. Only H_γ is in conjugation with $-\overset{O}{\overset{\|}{C}}-$ group
So it gives enol form

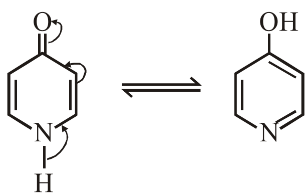


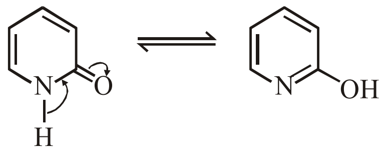
35. 1) & 2) have double bond at bridge head position in enol form, which can't possible

3) not have conjugate H with respect to $-\overset{O}{\overset{\|}{C}}-$

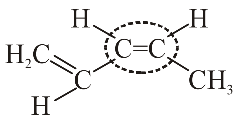
4) 

36. a & c having acidic H on nitrogen, which is in conjugation with CO, so they can show tautomerism

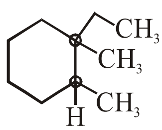
a) 

c) 

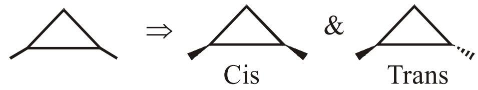
39. In general Stability order Keto > Enol
So II > I, III
and mores substituted alkene is more stable then less substituted alkene so III > I
and over all II > III > I

43. 

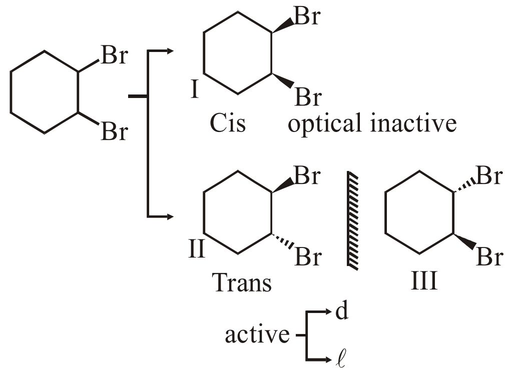
Different groups on double bonded carbon so show GI

45. II) 

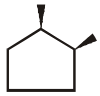
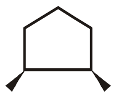
Different groups on at least two carbons of ring

47. 

Two GI

51. 

Total 3 stereo isomers

54. 2)  or 

Meso form optical inactive with two chiral centres

57. $H_3C-CH=CH-\underset{|}{C}H-CH_3$ (with OH on the CH)

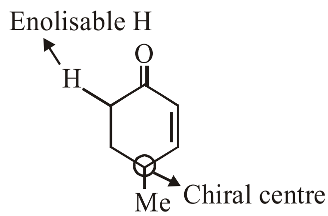
$$H_3C-CH=CH-\underset{\underset{OH}{|}}{CH}-CH_3$$

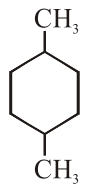
Two stereogenic units without symmetry
n = 2
$2^2 = 4$

58. $CH_3-CH_2-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-H$, $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-\overset{O}{\overset{\|}{C}}-H$,

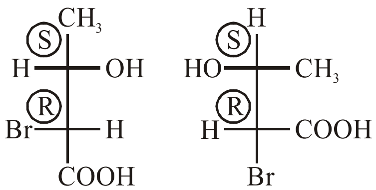
$CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-\overset{O}{\overset{\|}{C}}-H$, $H_3C-\overset{CH_3}{\overset{|}{\underset{\underset{CH_3}{|}}{C}}}-\overset{O}{\overset{\|}{C}}-H$

Total 4 structures

59. 

60. 

Not have chiral centre and its Cis and trans forms are optically inactive

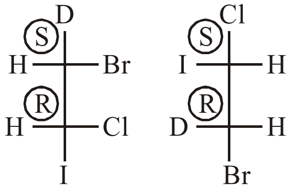
67. 

Same constitution and have same configuration of chiral centers So Identical.

68. $H_3C-CH=CH-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH=CH-C_2H_5$

3 stereogenic units without symmetry

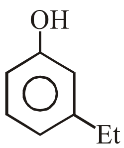
n = 3 $\Rightarrow$ $2^3 = 8$

69. 

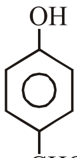
Opposite configuration at identical chiral centres So Enantiomers.

GOC-I

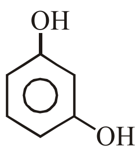
80. Acidic strength $\propto$ – M, –I $\propto \dfrac{1}{+M,+I}$

(1) 

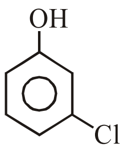
(+I)

(2) 

(–M)

(3) 

(–I)

(4) 

(–I)

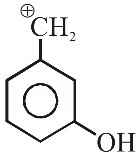
–I $\Rightarrow$ Cl > OH

A.S. Order 2 > 4 > 3 > 1

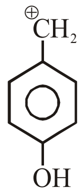
pK_a order 1 > 3 > 4 > 2 $\qquad pK_a \propto \dfrac{1}{A.S.}$

84. Stability of carbocation $\propto$ +M, +H, +I

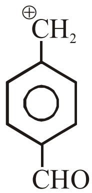
$\propto \dfrac{1}{-M,-H,-I}$

(A) 

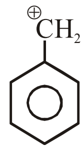
(–I)

(B) 

(+M)

(C) 

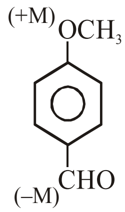
(–M)

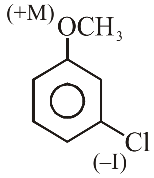
(D) 

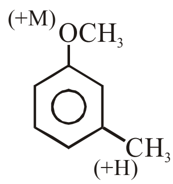
No effect

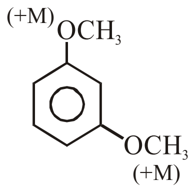
Stability order B > D > A > C

92. ESR reactivity $\propto$ E.D.G. $\propto \dfrac{1}{EWG}$

(1) 

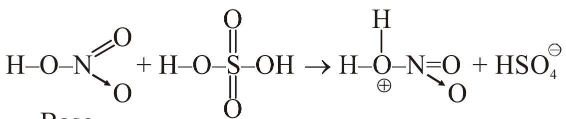
(2) 

(3) 

(4) 

So 4 > 3 > 2 > 1

93. Nitrating mixture



Base Acid

102. o-Nitrophenol is less acidic than H_2CO_3

So it is insoluble in $NaHCO_3$

109. Acidic strength RCOOH > H_2CO_3 > Phenol

So RCOOH gives effervescence with $NaHCO_3$ but phenol not gives.

114. a) Greater the resonance, greater will be C=C B.L.

So C = C B.L. $\Rightarrow$ II > I

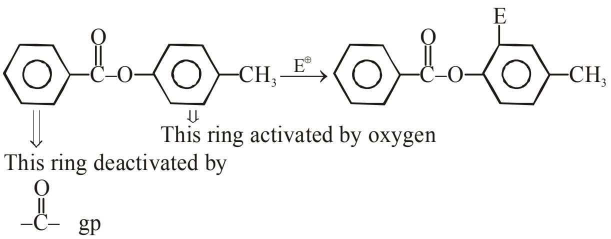
b) Greater the hyperconjugation, greater will be C=CB.L.

So C = C B.L. $\Rightarrow$ IV > III

c) Resonance effect > Hyperconjugation

So overall II > I > IV > III

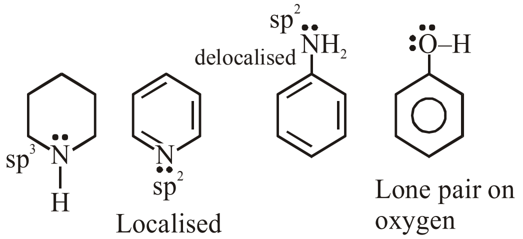
116.



So electrophile link at ortho position of activated ring because para position has blocked.

118. Acidic strength $-SO_3H > -COOH$

137.



lone pair availability $\propto \frac{1}{\text{E.N.}}$

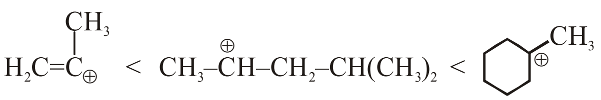
E.N. $\Rightarrow N_{sp^3} < N_{sp^2}$

Localised e^- pair more available than delocalised.

So basic strength order I > II > III > IV

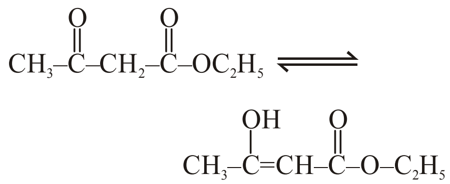
138. In option 1 Br link with vinylic carbon so C–Br bond has partial double bond character by resonance and it is difficult to break this bond for S_N reaction.

142. Reaction proceed via S_N1 mechanism so rate check by stability of carbocation.



So rate II > III > I

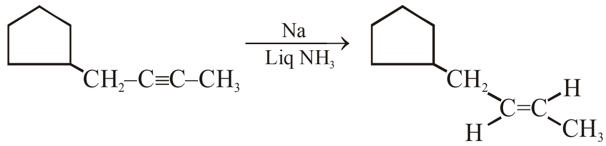
143. (1)



Both form, form by delocalisation of H So they are Tautomers of each other.

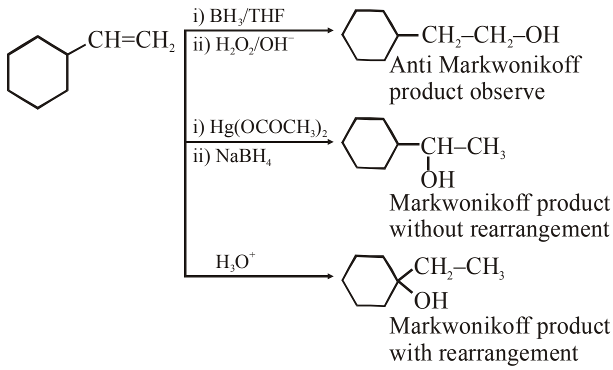
HYDROCARBON

146.

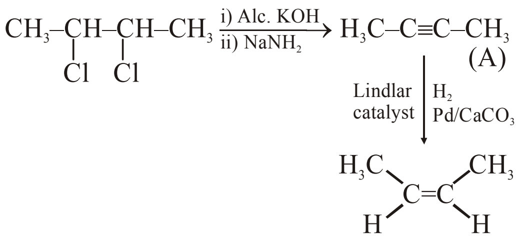


Anti addition

147.

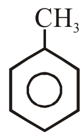


154.



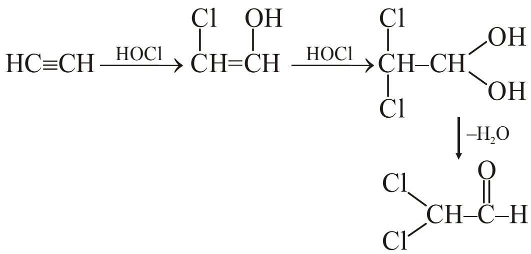
Syn addition

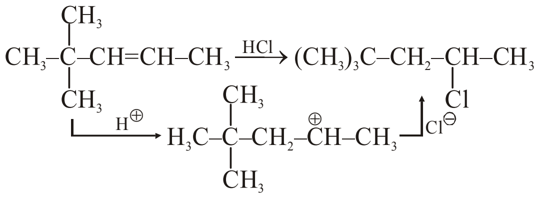
155.



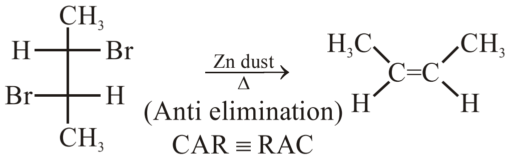
does not gives EAR or ESR with Br_2/CCl_4 so can't decolourise Br_2/CCl_4

165.

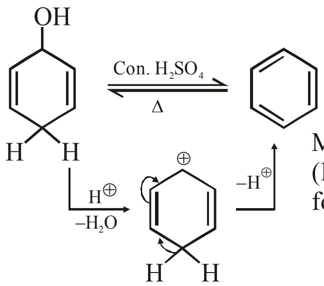


167. 

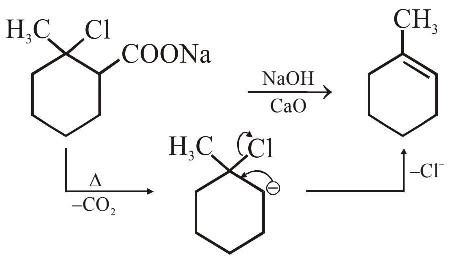
(More stable carbocation can't rearrange into less stable carbocation)

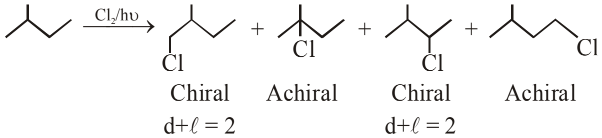
168. 

d-2,3-Dibromobutane (O Active isomer)

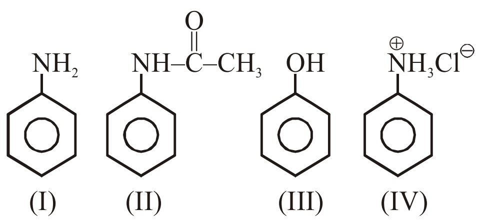
169. 

Most stable product (Reaction shift in forward direction)

171. 

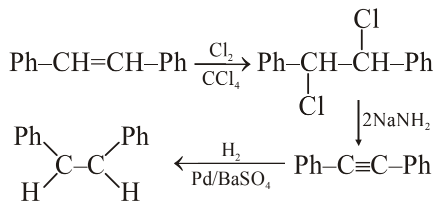
173. 

Total 4 chiral product

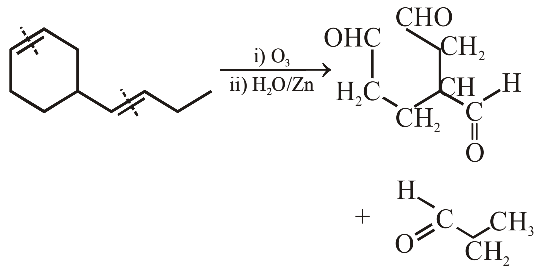
175. 

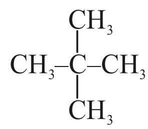
$$ESR \propto EDG\ (-NH_2 > -OH > -NH-\overset{O}{\overset{\|}{C}}-CH_3)$$

$$\propto \frac{1}{EWG}\left(\overset{\oplus}{N}H_3\right)$$

177. 

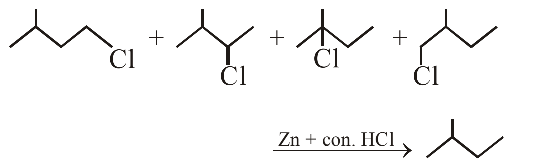
(Syn addition)

180. 

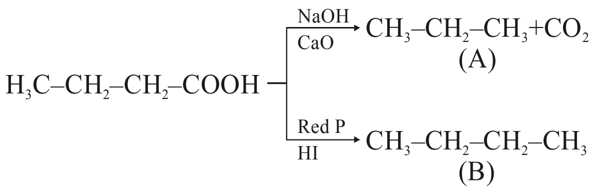
186. 

Neopentane

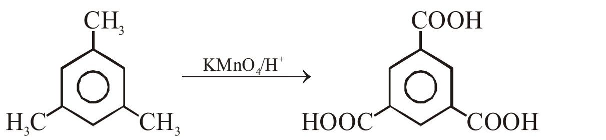
No corresponding alkene is possible.

187. 

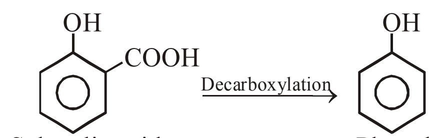
Isopentane

189. 

A & B are homologue

202. 

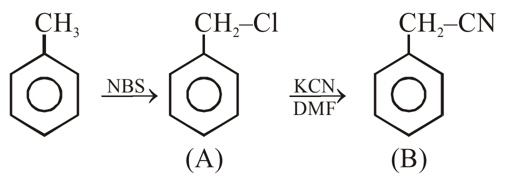
Mesitylene Benzene-1,3,5-tricarboxylic acid

204. 

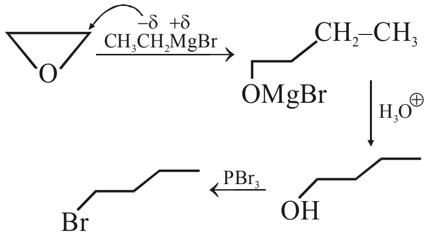
Salycylic acid Phenol (Hydrocarbon derivative)

ALKYL HALIDES AND ARYL HALIDES

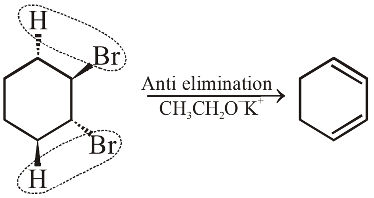
209.



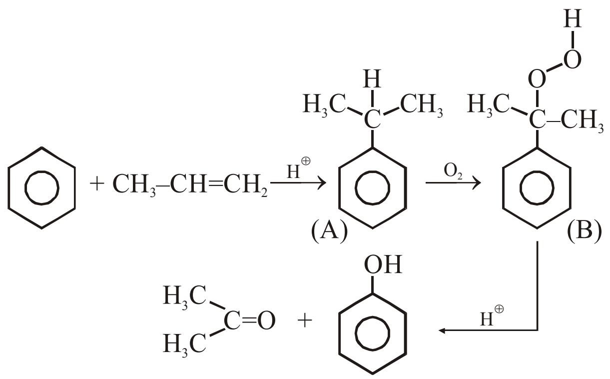
210.



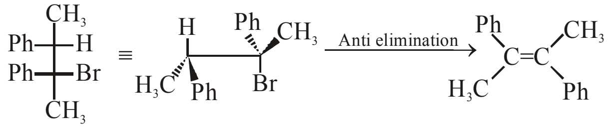
211.



212.

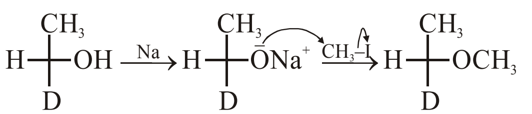


215.



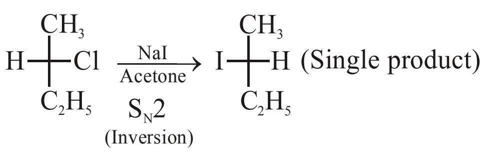
219.

(1)

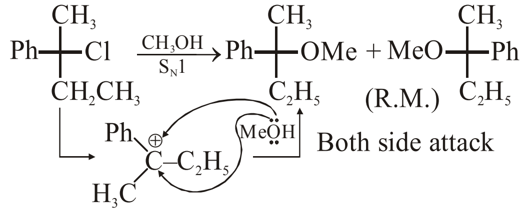


Single product

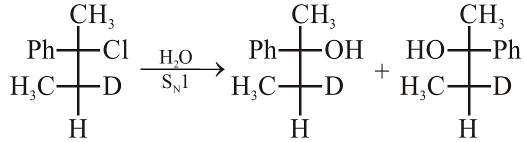
(2)



(3)

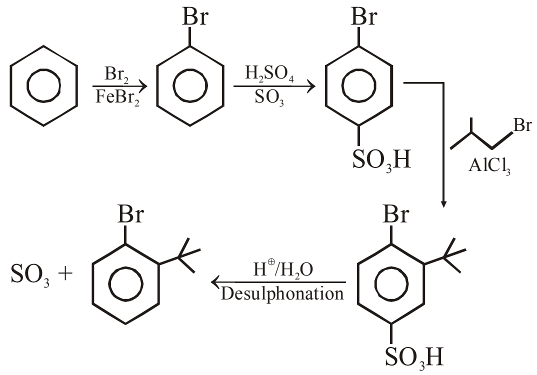


(4)

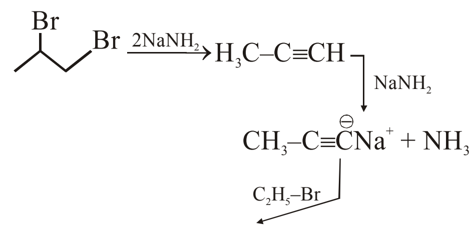


(Diastereomeric mixture)

220.



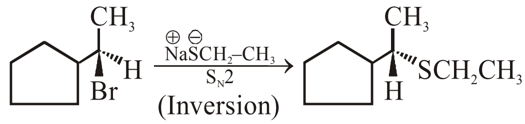
221.



$NaBr + CH_3–C{\equiv}C–CH_2–CH_3$

X = 3 (3 mole $NaNH_2$)

223.



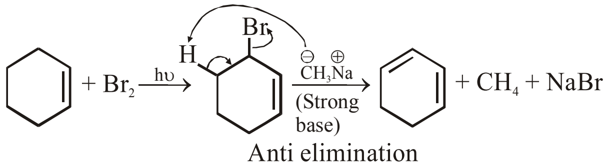
227. Electron withdrawing group ($–NO_2$) enhance rate of S_N in Aryl halide.

254. $Ph–\overset{\overset{O}{\|}}{C}–CH_3 \xrightarrow{NaOH + I_2} Ph–\overset{\overset{O}{\|}}{C}–O^-Na^+ + CHI_3$

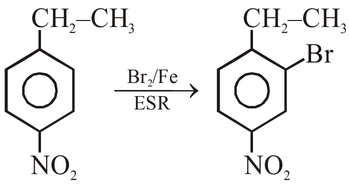
Acetophenone — Yellow ppt

259. $CH_3–CH_2–Br \xrightarrow{KCN} CH_3–CH_2–CN \xrightarrow{\overset{\ominus}{OH} / H_2O} CH_3–CH_2–\overset{\overset{O}{\|}}{C}–NH_2 \xrightarrow[\Delta]{PCl_5} CH_3–CH_2–CN$

262.



263.



OXYGEN CONTAINING COMPOUNDS-I

270.

$$Ph\text{–}\underset{}{\overset{OH}{\overset{|}{C}H}}\text{–}CH_3 \xrightarrow[{[O]}]{I_2/\overset{\ominus}{O}H} Ph\text{–}\overset{O}{\overset{\|}{C}}\text{–}CH_3 \xrightarrow{I_2/\overset{\ominus}{O}H} CHI_3 + PhCO\overset{\ominus}{O}$$

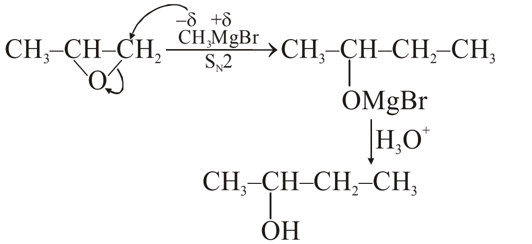
271.

$$CH_3\text{–}CH_2\text{–}OH \xrightarrow{P+I_2} CH_3\text{–}CH_2\text{–}I \xrightarrow[Ether]{Mg} CH_3\text{–}\overset{-\delta}{C}H_2\overset{+\delta}{MgI}$$

$$\downarrow H\text{–}\overset{O}{\overset{\|}{C}}\text{–}H$$

$$CH_3\text{–}CH_2\text{–}CH_2\text{–}OH \xleftarrow{H_2O} CH_3\text{–}CH_2\text{–}CH_2\text{–}OMgI$$

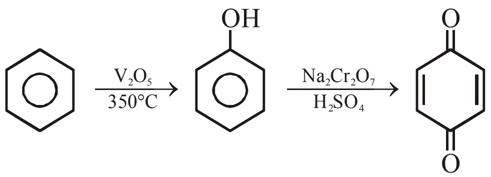
275.



277.

$$CH_3\text{–}\overset{OH}{\overset{|}{C}H}\text{–}CH_3 \xrightarrow[\text{dil } H_2SO_4]{K_2Cr_2O_7} H_3C\text{–}\overset{O}{\overset{\|}{C}}\text{–}CH_3 \xrightarrow[H_2O]{CH_3MgI} H_3C\text{–}\overset{CH_3}{\overset{|}{\underset{OH}{\underset{|}{C}}}}\text{–}CH_3$$

278.

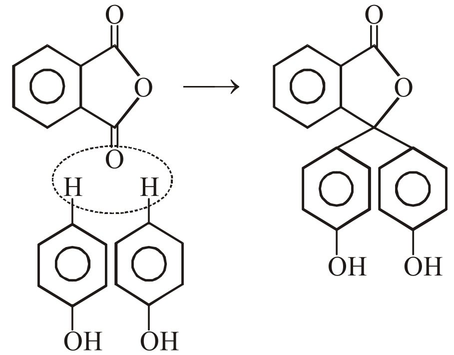


282.

$$C_3H_7OH \xrightarrow[170°C]{\text{Conc. } H_2SO_4} CH_3\text{–}CH{=}CH_2 \xrightarrow{Br_2/CCl_4} CH_3\text{–}\overset{Br}{\overset{|}{C}H}\text{–}\overset{Br}{\overset{|}{C}H_2} \xrightarrow{\text{i) Alc KOH; ii) } NaNH_2} CH_3\text{–}C{\equiv}CH$$

$$\left(\begin{array}{c} CH_3\text{–}CH_2\text{–}CH_2\text{–}OH \\ \text{or} \\ CH_3\text{–}\underset{OH}{\underset{|}{C}H}\text{–}CH_3 \end{array}\right)$$

292.

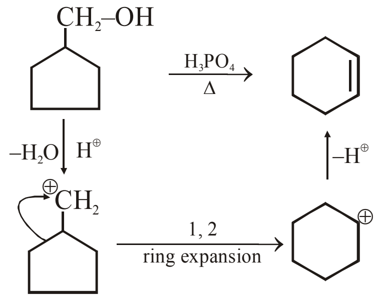


Phenol + phthalic anhydride

294.

$$\begin{array}{l} HO\,CH_2\text{–}OH \\ HO\,CH_2\text{–}OH \end{array} \xrightarrow{HIO_4} H\text{–}\overset{O}{\overset{\|}{C}}\text{–}H + H\text{–}\underset{O}{\underset{\|}{C}}\text{–}H$$

295.

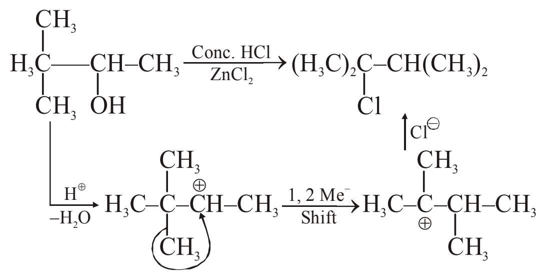


301.

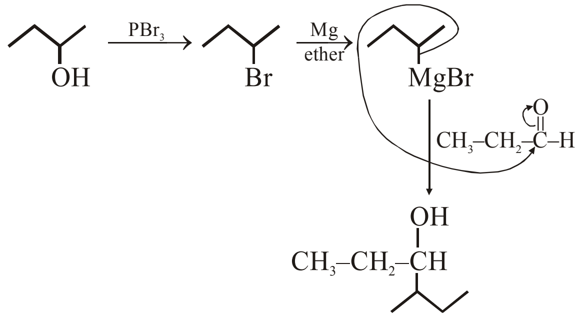
$$H_3C\text{–}CH_2\text{–}\underset{CH_3}{\underset{|}{C}H}\text{–}CH_2\text{–}O\text{–}CH_2\text{–}CH_3 + HI \xrightarrow[S_N2]{\overset{\ominus}{I}} CH_3\text{–}CH_2\text{–}\underset{CH_3}{\underset{|}{C}H}\text{–}CH_2\text{–}OH + CH_3\text{–}CH_2\text{–}I$$

Less sterically hidered side

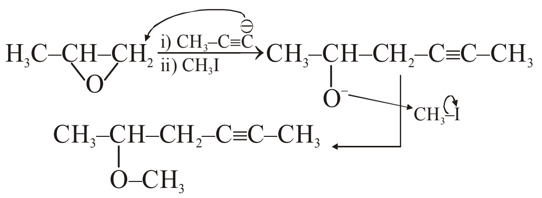
305.



306.



308.

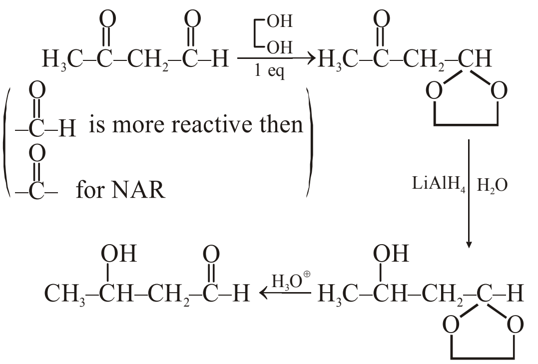


315.

$$CH_3-CH_2-C\equiv N \xrightarrow{CH_3MgI} CH_3-CH_2-C(CH_3)=NMgI$$

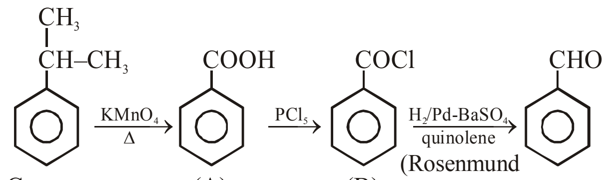
$$\xrightarrow{H_2O/H^{\oplus}} CH_3-CH_2-C(CH_3)=O + NH_3 + Mg(I)(OH)$$

316.



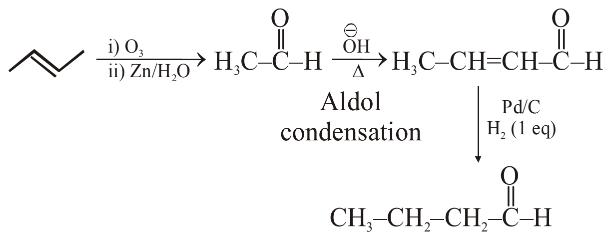
OXYGEN CONTAINING COMPOUNDS II

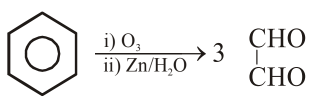
318.



Cumene (A) (B)

321.



322. a) 

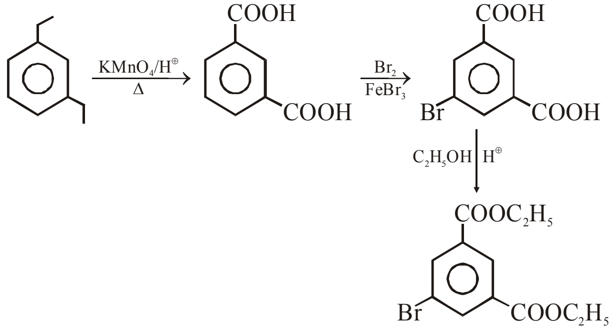
b) $CH_3-CHO \xrightarrow{SeO_2} HC(=O)-C(=O)-H$

c) $HC\equiv CH \xrightarrow[CCl_4]{2Cl_2} Cl_2CH-CHCl_2 \xrightarrow{Ca(OH)_2} Cl_2C=CHCl$

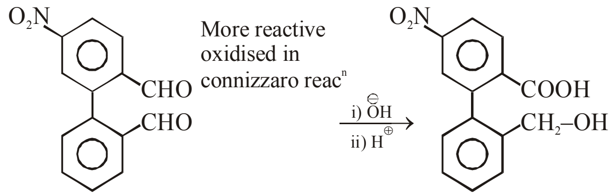
Westron Westrosol

d) $HC\equiv CH \xrightarrow[\text{dil } H_2SO_4]{Hg^{+2}} CH_2=CH-OH \rightleftharpoons CH_3-C(=O)-H$

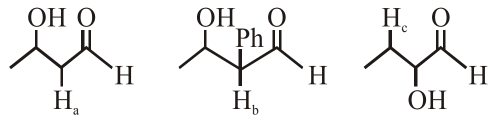
327.



344.



345.



(I) (II) (III)

Acidic strength $H_b > H_a > H_c$

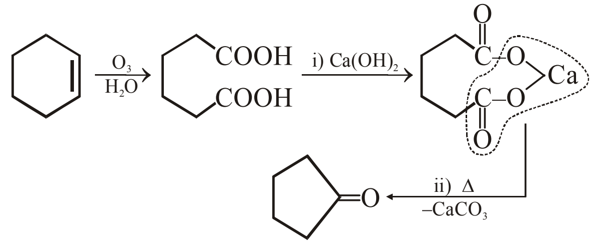
Order of dehydration II > I > III

354. $H_3C-C\equiv CH \xrightarrow[Hg^{+2}]{\text{dil } H_2SO_4} CH_3-C(=O)-CH_3 \xrightarrow[\text{Aldol condensation}]{Ba(OH)_2,\ \Delta} (H_3C)_2C=CH-C(=O)-CH_3$

Mesityl oxide

356. Benzaldehyde not have $CH_3-\overset{O}{\overset{\|}{C}}-$ group so it can't gives Iodoform test.

359.



364.

$$CH_3-\overset{H}{\overset{|}{C}}=O + H_2CH\overset{O}{\overset{||}{C}}-H \xrightarrow[\text{aldol}]{\text{Self}} CH_3-CH=CH-\overset{O}{\overset{||}{C}}-H$$

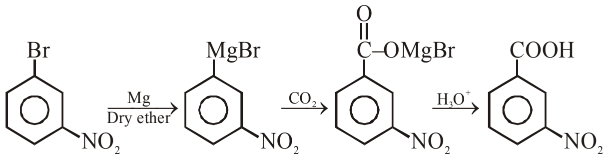
$$Ph-\underset{H}{\underset{|}{C}}=O + H_2CH\underset{O}{\underset{||}{C}}-H \xrightarrow[\text{aldol}]{\text{Cross}} PhCH=CH-\underset{O}{\underset{||}{C}}-H$$

372. $Ph-\overset{O}{\overset{||}{C}}-CH_3 \xrightarrow{NaOCl} \underset{(A)}{CHCl_3} + \underset{(Salt)}{Ph-\overset{O}{\overset{||}{C}}-ONa}$

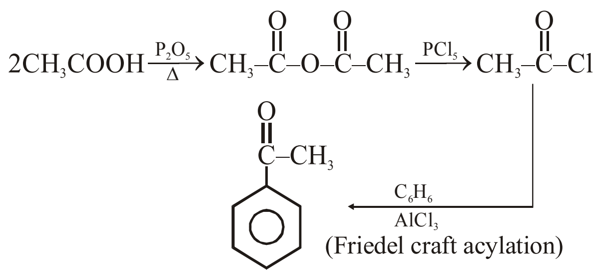
$$(H_3C)_2C=O + \underset{(A)}{CHCl_3} \longrightarrow (H_3C)_2C(OH)CCl_3$$

Chloretone

382.



383.



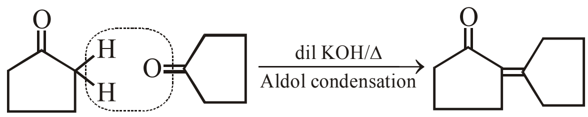
385.

$$CH_3-CH_2-CH_2CHO \xrightarrow[\text{ii) } H^{\oplus}]{\text{i) Ammonical } AgNO_3} CH_3CH_2-CH_2-COOH$$

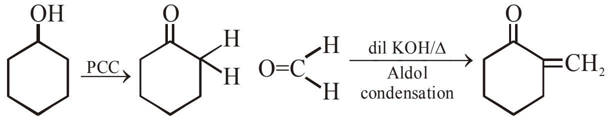
(Tollen's reagent)

$$\xrightarrow{NaHCO_3} CH_3CH_2CH_2COONa \xrightarrow[\Delta]{NaOH/CaO} CH_3CH_2CH_3$$

387.



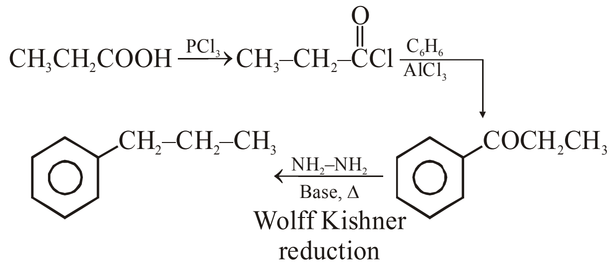
388.



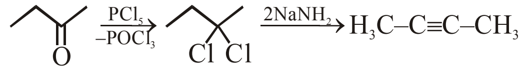
389. $CH_3-CH=CH_2 \xrightarrow[\text{reagent}]{\text{Baeyer's}} CH_3-CH(OH)-CH_2-OH \xrightarrow{HIO_4}$

$$2H_2O + H-\underset{H}{\underset{|}{C}}=O + H-\underset{CH_3}{\underset{|}{C}}=O$$

391.

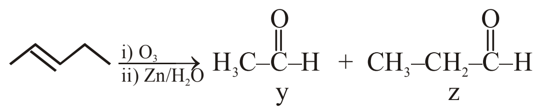


392.



$x = 1$, $y = 2$ so $x + y = 3$

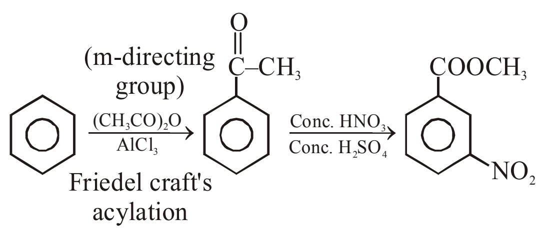
393.



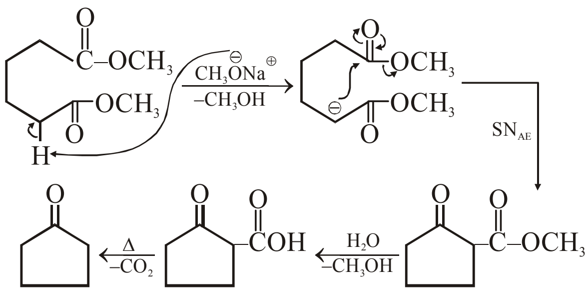
	y	z
Fehling test	✓	✓
Iodoform test	✓	✗

396. $HC\equiv CH \xrightarrow[\text{dil } H_2SO_4]{Hg^{+2}} CH_3-\overset{O}{\overset{||}{C}}-H \xrightarrow[H_2O]{CH_3MgX} CH_3-\overset{OH}{\overset{|}{C}H}-CH_3 \xrightarrow{[O]} CH_3-\overset{O}{\overset{||}{C}}-CH_3$

397.

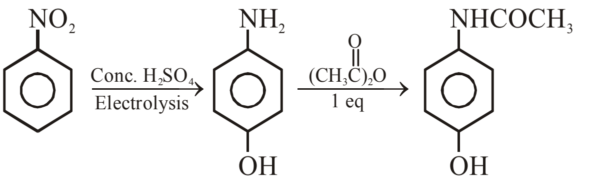


398.



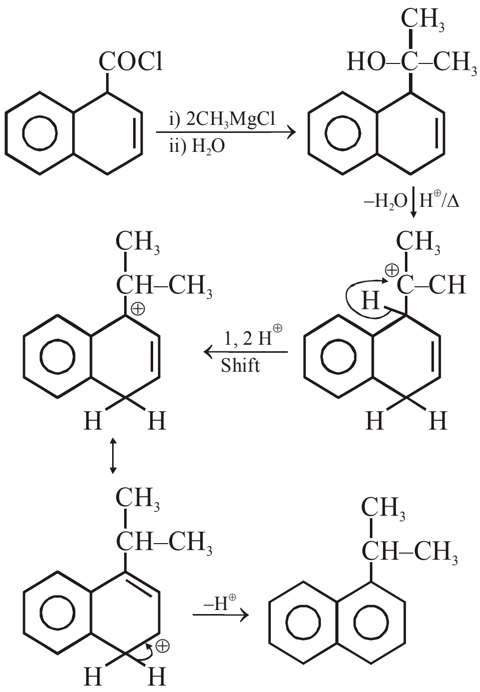
β-keto acid

399.

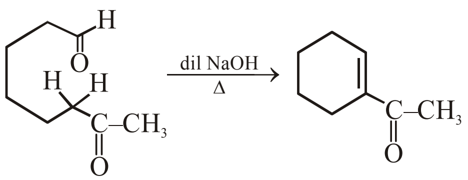


Paracetamol
(Analgesic and
antipyretic)

400.



401.



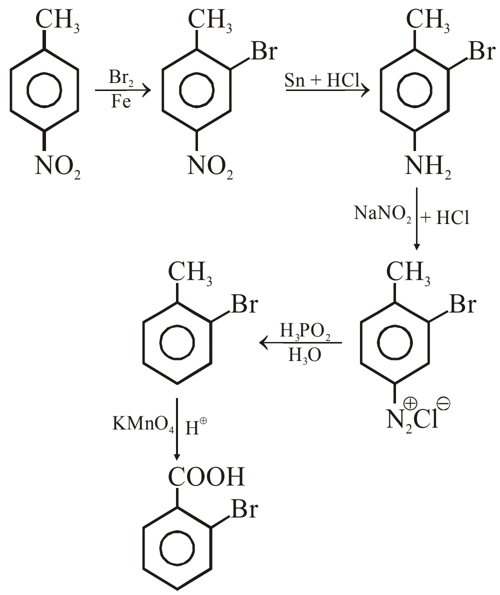
405.

$$CH_3-CH_2-\underset{\underset{O}{\parallel}}{C}-O-\underset{\underset{CH_3}{|}}{C}=CH-CH_3 \xrightarrow{H_3O^+} CH_3-CH_2-\overset{\overset{O}{\parallel}}{C}-OH + CH_3-CH_2-\overset{\overset{O}{\parallel}}{C}-CH_3$$

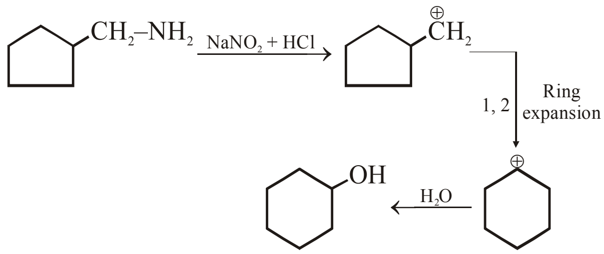
Gives Iodoform test

NITROGEN CONTAINING COMPOUNDS

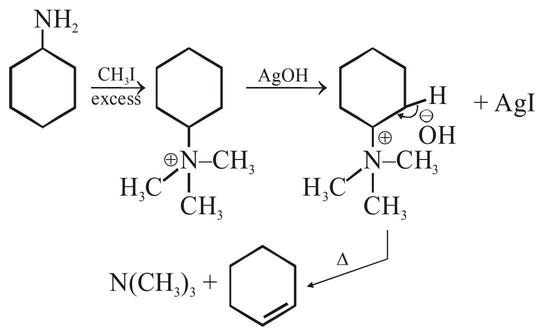
411.

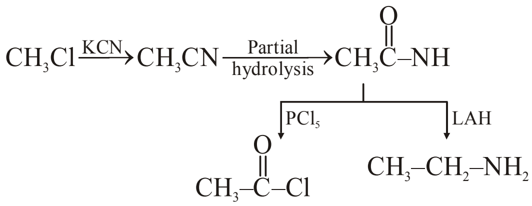


416.



417.



418. 

421. a) $C_6H_5COOH \xrightarrow[H^{\oplus}]{N_3H} C_6H_5NH_2$

(Schmidt reaction)

b) $C_6H_5CONH_2 \xrightarrow[\Delta]{Br_2/KOH} C_6H_5{=}NH_2$

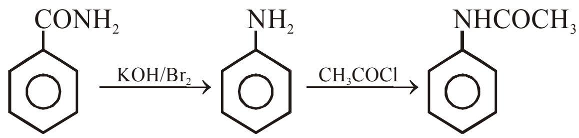
(Hoffmann bromamide reaction)

c) $C_6H_5COOH \xrightarrow[\Delta]{NH_3} C_6H_5CONH_2$

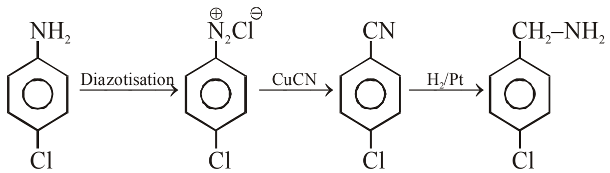
d) $C_6H_5NO_2 \xrightarrow{Fe+H_2O} C_6H_5NO$

(Nitrosobenzene)

428.



453.



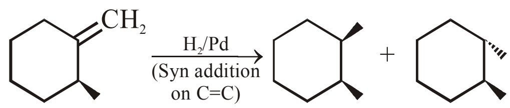
458. Sec amine (CH_3–NH–CH_2–CH_3) form a solid with Hinsberg's reagent which is insoluble in alkali.

461. Aromatic primary amine (C_6H_5–NH_2) can't form by Gabriel phthalimide synthesis because Ph–Cl is not good substrate for S_N reaction.

463.

F.G.	$KMnO_4/H^+$
Primary alcohol	Carboxytic acid
Sec. alcohol	Ketone
Tertiary alcohol	No effect

464. Option 4 is acetal which can't convert into aldehyde in basic medium so it can't gives positive Tollen's test.

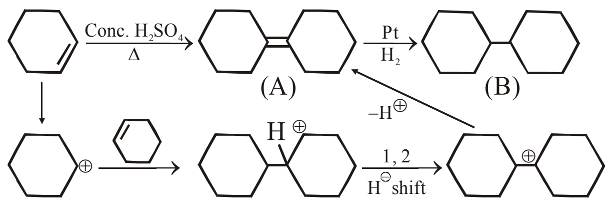
466. 

Diastereomers

468.

F.G.	$NaBH_4$	H_2/Pd
Alcohol	×	×
>C=C<	×	–CH–CH–

469.



BIOMOLECULES POLYMERS AND CHEMISTRY IN EVERY LIF

470. In option 3, both nucleus are in acetal form so they can't reduce Fehling solution, Tollen's reagent.

478. (3) PHBV is biodegradable polymer.

559. $PV = \frac{WN_2}{28}RT \qquad P = 720 - 20 = \frac{700}{760}\text{Atm}$

V = 60 ml = 0.060 L

T = 300 K

$$W_{N_2} = \frac{700}{760} \times \frac{0.060 \times 28}{0.0821 \times 300} \qquad W_{compound} = 0.4 \text{ gm}$$

$$W_{N_2} = 0.062 \text{ gm}$$

$$\%N = \frac{0.062}{0.4} \times 100 = 15.72\%$$

580. Total number of eq of base in acid base titration = Total number of eq of acid

So $Meq_{NH_3} + Meq_{NaOH} = Meq_{H_2SO_4}$

$Meq_{NH_3} + 0.5 \times 80 = 0.5 \times 2 \times 50$

$Meq_{NH_3} = 50 - 40$

$Meq_{NH_3} = 10$ (x factor = 1)

$W_N = 10 \times 10^{-3} \times 14$

$$\%N = \frac{0.14}{0.50} \times 100 = 28\%$$

581. $Meq_{NH_3} = Meq_{H_2SO_4}$

Meq_{NH_3} = 20 × 1 = 20 Meq

$Mmol_{NH_3} = Meq_{NH_3} = Mmol_N = 20$

W_N = 20 × 10^{-3} × 14

$\%N = \dfrac{0.28}{0.6} \times 100 = 46.67\%$

582. Mole of $BaSO_4$ = Mole of S = $\dfrac{0.699}{233}$

= 0.003 moles

W_S = 0.003 × 32 = 0.096 gm

$\%S = \dfrac{0.48}{0.096} \times 100 = 20\%$

583. $Meq_{NH_3} + Meq_{NaOH} = Meq_{H_2SO_4}$

$Meq_{NH_3} + \dfrac{1}{10} \times 1 \times 20 = \dfrac{1}{10} \times 2 \times 60$

Meq_{NH_3} = 12 – 2 = 10 Meq

$Meq_{NH_3} = Mmol_{NH_3} = Mmol_N = 10$

W_N = 10 × 10^{-3} × 14 gm

$\%N = \dfrac{0.14}{1.4} \times 100 = 10\%$

584. P = 715 – 15 = $\dfrac{700}{760}$ Atm

V = 55 ml = 0.0552

T = 300 K

$W_{Compound}$ = 0.35 gm

$PV = \dfrac{W_{N_2}}{28} \times RT$

$W_{N_2} = \dfrac{700}{760} \times \dfrac{0.055 \times 28}{0.0821 \times 300}$

$W_{N_2} = 0.057$ gm

$\%N = \dfrac{0.057}{0.35} \times 100 = 16.45\%$